HAND — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 103 chapters (0 from TO notes) –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

ACUTE SUBCLAVIAN ARTERY THROMBOSIS

– Hand · 1 / 103 –

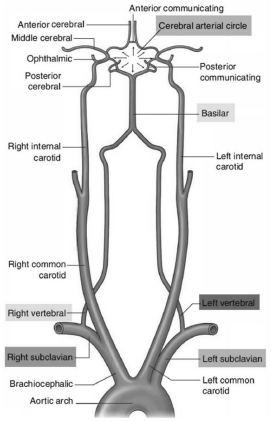
1 Basics

Epidemiology & risk factors

Definition	acute UE ischaemia from **proximal occlusion at subclavian artery** level
Incidence	rare, <1%
Risk factors	**aortic arch abnormality**, **aortic arch syndrome**, **trauma**, arterial catheterisation, pre-existing stenosis/atheroma at subclavian ostium
Patterns	slow occlusion (atherosclerosis) · **blue finger syndrome** (repeated microemboli) · acute global ischaemia (large emboli)

Anatomy

Left subclavian	**3rd branch of aortic arch**
Right subclavian	branch of **brachiocephalic (innominate)** artery
Vertebral arteries	arise from subclavian → join as **basilar** → cerebellum → risk of **vertebrobasilar insufficiency / basilar stroke**



Subclavian & vertebral branching

2 Mechanism

Commonest cause of UE thromboembolism	**atrial fibrillation** (80% of UE thromboembolism)
Cardiac embolism	up to **93%** of acute UE ischaemia; AF → loss of atrial contraction → stasis → thrombus
Other cardiac sources	cardiomyopathy, valvular disease, ASD, aortic arch abnormality; endocarditis → septic emboli
Hand arterial occlusion (general)	usually **post-traumatic ulnar artery**; venous occlusion → hypercoagulable state

– – Active space – –

3 Presentation

History	sudden onset, no inciting event
Symptoms	**arm pain**, **paraesthesia**, nausea, exertional pain/fatigue
Exam	**cyanotic/pale, cold** limb; **pulses absent** (axillary, brachial, radial, ulnar); **neuro exam often normal**
Non-occlusive	pulses absent but **Doppler signal present** (monophasic)
BP	differs between arms

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
Arteriography	**Recommended diagnostic test**; allows **percutaneous treatment** at same sitting
CT angiography	Fast; rules out other causes
MRA	cannot treat at diagnosis
Doppler US	Signal may persist in non-occlusive disease
Echocardiography	Rules out **cardiac embolus** after treatment



Arch angiogram, subclavian thrombus, stent with vertebral protection balloon

Labs

Hypercoagulable panel	rule out underlying disorder; labs usually unremarkable

5 Management

- **Acute subclavian thrombosis**
  - Symptomatic limb ischaemia → **percutaneous or open revascularisation**
    - percutaneous → **thrombolysis / PTA / aspiration / stent**
    - open → **embolectomy-thrombectomy / bypass**
  - Adjunct (all) → **IV + oral anticoagulation**

Arterial thromboembolism – treatment & anticoagulation

Percutaneous	**intra-arterial thrombolysis**, **PTA**, **manual aspiration**, **stenting** (protect vertebral artery with balloon)
Open	embolectomy/thrombectomy, bypass grafting
Medical	heparin, aspirin, clopidogrel, IV thrombolysis, GP IIb/IIIa antagonists; limited role alone if limb-threatening
✗ Wrong option: aortic root replacement	for **ascending thoracic aortic dissection**, not subclavian disease
AF prophylaxis	**anticoagulants** ✗ Wrong options: antiplatelets = mechanical valves · statin = ischaemic cardiomyopathy · antibiotics = endocarditis · β-blocker = rate only

AF anticoagulation (AHA/ACC/HRS 2014)	Factors
High risk (any 1 → anticoagulate)	Prior stroke/TIA, systemic thromboembolism, **age ≥75**
Moderate (>1 → anticoagulate)	Age 65–74, ♀, **HTN**, **DM**, heart failure, arterial disease

6 Complications

Limb ischaemia	gangrene, limb loss; **delayed diagnosis** → amputation
Basilar stroke	prolonged ischaemia, or **iatrogenic emboli** if vertebrobasilar system unprotected during endovascular Rx
Prognosis	favourable with prompt Rx; untreated → stroke + amputation

How it's asked

"Absent pulses + Doppler signal – treatment EXCEPT"	→ **aortic root replacement** ✗ Wrong options: thrombolysis, PTA, aspiration, stent
"Best diagnostic test"	→ **arteriography** (diagnose + treat)
"Commonest cause of UE thromboembolism"	→ **atrial fibrillation**
"AF patient – reduce hand thromboembolism"	→ **anticoagulants**
"Feared complication of delay / endovascular Rx"	→ **basilar stroke**

– – Active space – –

AIN COMPRESSIVE NEUROPATHY

– Hand · 2 / 103 –

1 Basics

Epidemiology & aetiology

Eponym	**Kiloh-Nevin syndrome** = AIN compression (TO) (✗ Wrong options: PIN compression (TO))
Incidence	rare; **<1%** of median neuropathies
Commonest cause	**transient neuritis (Parsonage-Turner)**; non-traumatic AIN palsy usually = **neuritis** (viral, autoimmune)
Parsonage-Turner (TO)	**leading differential of AIN syndrome; not a compression neuropathy** · triggers: viral illness, **vaccination**, pregnancy, trauma, surgery, stress · **pain first → weakness as pain settles** · whole plexus, part (C5–C6) or AIN only · **high-dose steroids if diagnosed within 72 h** · full recovery not assured
Parsonage-Turner	viral **brachial neuritis → bilateral AIN signs** (TO); preceded by **intense shoulder pain + viral prodrome**
Compression sites	**deep head of pronator teres** (most common anatomical, TO) · **FDS arch** (TO) · **edge of lacertus fibrosus** (TO) · **Gantzer's muscle** = accessory head FPL (TO) · thrombosed radial/ulnar/AI artery · accessory FDS→FDP muscle · FCRB, palmaris profundus
Surgical finding (Park)	commonest compressive structure at exploration = **fibrous band of FDS**; 4/11 had **no compression**

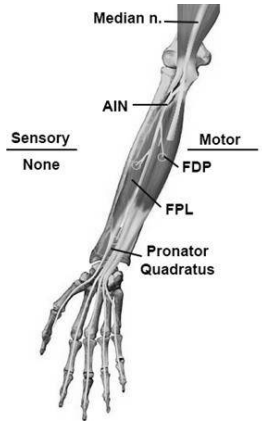


Gantzer's muscle

AIN motor distribution (FPL, FDP index)

Origin	**terminal motor branch of median n.**; ~**4 cm distal to medial epicondyle** (Orthobullets) · **5–8 cm distal to lateral epicondyle** (OB/Orthobullets); dorsoradial aspect of median n.
Course	between FDS & FDP → between **FPL & FDP** → on **interosseous membrane** with AI artery → **pronator quadratus** → wrist capsule (radiocarpal, intercarpal, DRUJ)
Innervates (3)	**FPL** · **FDP index + middle** · **pronator quadratus**
Sensory	**none** cutaneous (articular branches only)

Not AIN	**pronator teres** (median, before AIN leaves) · FCR (median) · FCU (ulnar) · FDB (foot, medial plantar)



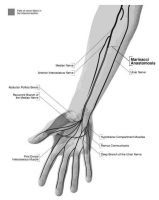
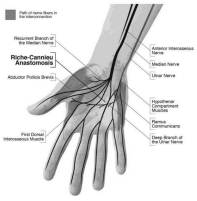
AIN lineage

Martin-Gruber anastomosis

MGA	**motor** connection **median/AIN → ulnar** in **proximal 1/3 forearm**; sensory spared; **most common forearm anomalous innervation**
Prevalence	**15%** (Orthobullets) · **15–40%** (OB) · pooled **19.5%**, 57.6% from AIN (meta-analysis); more often **right**
Supplies	**adductor pollicis + dorsal interossei** (intrinsics) → complete AIN palsy may show **intrinsic weakness**
High ulnar laceration + intrinsics work	**Martin-Gruber**
MGA type II	**most common**; median fibres → **1st dorsal interosseous** → median laceration = median deficit **+ cannot abduct index**; ulnar sensation preserved

– – Active space – –

MGA II on NCS usually **normal** (1st DI not routinely tested) → careful exam

Anastomosis	Site	Direction	Clue / prevalence
 **Martin-Gruber**	Forearm	Median/AIN → ulnar (motor)	High ulnar cut, intrinsics work · median cut, no index abduction
 **Marinacci**	Forearm	Ulnar → median ("reverse MG")	**Rarest** (0.7% / 4%); proximal median lesion with **APB/thenar preserved**
 **Riche-Cannieu**	Hand	Recurrent median ↔ deep ulnar (motor ± sensory)	55%; **preserved thenar function**; "all-ulnar hand"
 **Berrettini**	Hand	Common digital nn. (3rd–4th), **sensory**	>80%; injured at CTR → ↓ 3rd web sensation; no motor

Not median–ulnar **Linburg-Comstock** = FPL–FDP index **tendinous** connection · **anconeus epitrochlearis** = cubital tunnel compression

3 Presentation

Hallmark **motor deficit without sensory loss** (TO); **normal median sensory exam**

OK sign **cannot make OK sign** = Kiloh-Nevin sign (TO) → weak **thumb IP flexion (FPL)** + **index DIP flexion (FDP)**

Function (TO) loss of **fine motor skills (writing, pinching)** · PQ hard to test (PT still pronates) · **history of shoulder pain first → Parsonage-Turner**

Pain vague deep forearm pain, often **absent**; if present lasts **2–3 wk**

PQ test weak resisted pronation with **elbow maximally flexed** (neutralises pronator teres)

Tenodesis **intact tenodesis** = tendons intact → distinguishes from **FPL attritional rupture** (RA)

Severe forearm atrophy



Inability to flex thumb IP and index DIP

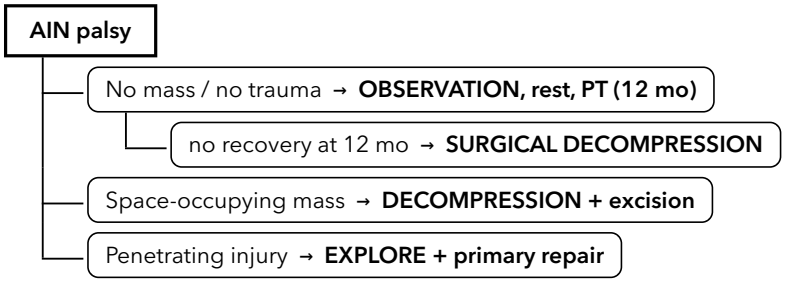
4 Diagnosis & Classification

Diagnosis	**clinical + EMG/NCS**

Key radiology findings

MRI	if **mass** suspected; T2/STIR ↑ signal FPL, FDP, PQ → **PQ oedema = most reliable**
EMG	**fibrillations, positive sharp waves** in **FPL + FDP index/middle**; rules out proximal lesion, grades severity, tracks recovery
Differentials	flexor tendon rupture · proximal lesion (C-spine, plexus) · **pronator syndrome** · CTS · conversion disorder (not if objective EMG findings)

5 Management



AIN syndrome – diagnosis & observation (neuritis)

First line	**observation, rest, PT** in all without a mass; treatment mostly **non-surgical** (TO); splinting at 90° (Rodner) · massage, forearm stretches
Recovery	begins **3–12 mo**; full by **18 mo** (mean ~9 mo) → "**gradual improvement over 1 y**" · "**persistent sx <2 y then spontaneous resolution**"
Better prognosis	**age <40** (faster recovery)
Not indicated	steroid injection (no data) · CTR · early tendon/nerve transfer

– – Active space – –

Operative criteria

Indication	Threshold
Clear **space-occupying mass**	Early decompression
Failed nonop	**12 mo** (Orthobullets, Chi & Harness) · **6 mo** in complete palsy (Sood & Burke)
Penetrating injury	Explore + **end-to-end repair**
Age <40	Surgery controversial (uniform nonop response)

Surgical success **~75%**

Decompression lazy-S proximal volar forearm; release superficial head PT, **lacertus fibrosus**, FDS arch, **Gantzer's**, ligate crossing vessels; elbow ROM day 2

6 Complications

Persistent AIN deficit rare (~5–10 cases) → **tendon transfer: brachioradialis → FPL**

X vs Y

	AIN syndrome	Pronator syndrome	PIN syndrome	FPL rupture
Sensory	**Normal**	Radial 3½ + thenar	Normal	Normal
Motor	FPL, FDP-I/M, PQ	±	**Thumb extension/retropulsion**, finger ext	FPL only
Pain	Absent / 2–3 wk	**Proximal forearm**	Lateral elbow	–
Tenodesis	**Intact**	Intact	Intact	**Lost**
Release	Gantzer's, FDS arch	**Lacertus fibrosus**	**Arcade of Frohse**	–

How it's asked

"Kiloh-Nevin – all true EXCEPT"	→ **compression of PIN** (TO)
"Parsonage-Turner + weak pronation, no sensory loss"	→ **AIN syndrome → observation** ✗ Wrong options: Gantzer release
"Fibs in FPL + FDP-I after flu, no numbness – next step?"	→ **observation, improves over 1 y**
"Weak thumb IP flexion, normal 2PD, normal MRI – course?"	→ **resolves spontaneously <2 y**
"Stab proximal forearm, no index DIP flexion – what else fails?"	→ **FPL** ✗ Wrong options: pronator teres, FCR
"High ulnar laceration, intrinsics still work"	→ **Martin-Gruber**
"Antecubital laceration, median deficit + no index abduction"	→ **Martin-Gruber type II**

"Most reliable MRI sign"	→ **PQ oedema**
"Persistent AIN palsy – transfer?"	→ **brachioradialis → FPL**

– – Active space – –

– – Active space – –

APERT SYNDROME

– Hand · 3 / 103 –

1 Basics

FGFR2 mutation & clinical features

Synonym	**acrocephalosyndactyly type 1**: cranium + hands + feet
Incidence	rare; **1/80,000** live births
Features	**bilateral complex syndactyly** hands & feet (**index, middle, ring** most affected) · **symphalangism** · **craniosynostosis** → flat skull, broad forehead (**acrocephaly**) · **hypertelorism** · glenoid hypoplasia · radioulnar synostosis
Cognition	normal → moderately disabled
Feet	predictable dysmorphism + **progressive fusions** through skeletal maturity

Genes & lab markers

Gene	Condition
FGFR2	**Apert** (AD, most new cases **sporadic**)
✗ Wrong option: FGFR3	Achondroplasia
✗ Wrong option: COL2A1 (chr 12)	SED congenita
✗ Wrong option: SEDL	X-linked SED tarda
✗ Wrong option: fibrillin	Marfan
✗ Wrong option: SHH	Radial longitudinal deficiency (TAR, Holt-Oram, VACTERL)
✗ Wrong option: PMP22	Charcot-Marie-Tooth
✗ Wrong option: COL1A1	Osteogenesis imperfecta

3 Presentation

Face	**dysmorphic**, flattened features, **wide-set eyes**
Rosebud hand	complex syndactyly; **index, middle, ring share a common nail**



Apert facies



Rosebud hands

4 Diagnosis & Classification

Key radiology findings

X-ray **complex syndactyly** (bony fusion of central digits)



Rosebud hand X-ray

5 Management

Apert hand

- ~1 y → **release BORDER digits**
- ~1.5 y → digit reconstruction: central 3 → **2 digits**

How it's asked

"Gene in Apert?"	→ **FGFR2** ✗ Wrong options: FGFR3, COL2A1, SEDL, fibrillin
"Flat face + wide-set eyes + fused fingers"	→ **Apert** ✗ Wrong options: MED, cleidocranial, Noonan, achondroplasia

– – Active space – –

"Infant rosebud hands — mutation?"	→ **FGFR2** ✗ Wrong options: SHH, PMP22, COL1A1
"Digits most affected"	→ **index, middle, ring**

ATYPICAL MYCOBACTERIUM INFECTIONS

– Hand · 4 / 103 –

1 Basics

Epidemiology & organisms

Incidence rare; **0.05 / 100,000 / y**

Who **marine workers**, fishermen; **immunocompromised** host (most have some immunosuppression)

Site **hand & wrist in 50%**

Incubation avg **2 wk** (up to 6 mo); time to diagnosis often **>1 y** (missed Dx)

Prior steroid injection **poorer outcome**

Organism	Habitat	Culture
M. marinum	**Most common**; stagnant fresh/salt water, **aquariums**, fish nets	**Löwenstein-Jensen at 30–32 °C**
M. kansasii	Soil	–
M. terrae	**Soil**	**37 °C**
M. avium intracellulare	**Terminal AIDS** (also non-HIV)	**Room temperature**

Mycobacterium marinum

Classic fisherman/aquarium cut → dorsal hand swelling weeks later → grows on **Löwenstein-Jensen at 30 °C** after ~2 wk → ***M. marinum***

Other aquatic pathogens *Vibrio* spp., *Aeromonas hydrophila*

Fishmonger's disease (erysipeloid)

Organisms (TO) commonest: ***Streptococcus iniae*, *Vibrio vulnificus*, *M. marinum*** · others: ***Erysipelothrix rhusiopathiae***, *Aeromonas hydrophila*, *Vibrio damsela*, *Proteus vulgaris*, *Morganella morganii*, *Lactococcus garvieae*, *M. abscessus*, *Streptococcus halichoeri*

Mechanism (TO) penetration by sharp instruments or **fish bony spines**; immunosuppressed → **high morbidity & mortality**

NOT a cause (TO) ***Pseudomonas aeruginosa***

Compound palmar ganglion (TB tenosynovitis)

Definition (TO) **tuberculous tenosynovitis** of wrist/hand flexor tendons; **radial + ulnar bursae**

Shape (TO) **hourglass / dumb-bell** swelling **proximal & distal to flexor retinaculum** (across wrist crease) → **positive cross-fluctuation**

– – Active space – –

Bursal anatomy (TO)	ulnar bursa communicates with radial bursa in **80%** → **horseshoe** tenosynovitis
Spread (TO)	**little finger → ulnar bursa → Parona's space** (more commonly associated) · thumb → radial → ulnar bursa
Causes (TO)	**TB** (commonest comorbidity) · **RA** · infection · SLE · gout · PVNS of tendon sheath · fungal
NOT a cause (TO)	**malignancy**
Diagnosis (TO)	clinical, **always confirmed by histopathology**
Treatment (TO)	progresses to gross destruction → **excision without delay**

3 Presentation

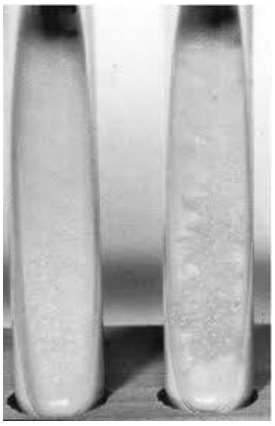
Early	**papules, ulcers, nodules** on hand; cutaneous rash; often **single nodule → lymphatic spread**
Late	**tenosynovitis, septic arthritis, osteomyelitis**
Look-alike	indistinguishable from TB infection



M. marinum nodules and ulcers

4 Diagnosis & Classification

Diagnosis	clinical + **culture** (cultures & sensitivities = key)
Histology	**granulomas ± acid-fast bacilli** on AFB stain
Culture	**Löwenstein-Jensen**; *M. marinum* **30–32 °C** · MAI room temp · *M. terrae* 37 °C

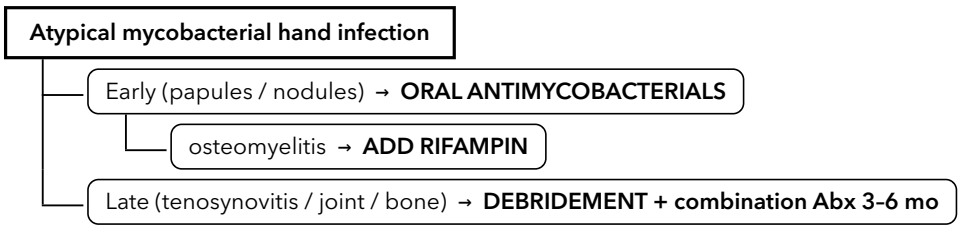


M. marinum on Löwenstein-Jensen agar

Differentials herpetic whitlow, fungal infection, psoriasis

– – Active space – –

5 Management



Oral agents **ethambutol, tetracycline, TMP-SMX, clarithromycin, azithromycin**; **+ rifampin** if osteomyelitis

Mortality **32%** with sepsis / untreated

X vs Y

Organism	Classic exposure
M. marinum	Fish nets, aquarium; LJ 30 °C
M. terrae	Soil; 37 °C
Sporothrix schenckii	**Thorn** (rose gardener) – fungus
Bartonella henselae	**Cat scratch**; epitrochlear lymphadenopathy; endocarditis
Eikenella corrodens	**Human bite**; Gram −ve
Erysipelothrix / S. iniae / Vibrio	Fishmonger's disease

How it's asked

"Fisherman cut, LJ medium at 30 °C"	→ ***M. marinum*** ✗ Wrong options: *M. terrae*, *Sporothrix*, *Bartonella*, *Eikenella*
"MAI culture temperature"	→ **room temperature**
"Add for osteomyelitis"	→ **rifampin**
"Fishmonger's disease NOT caused by"	→ ***Pseudomonas aeruginosa*** (TO)
"Compound palmar ganglion – true"	→ **more commonly with little finger infection** (TO)
"Dumb-bell wrist swelling, cross-fluctuation – comorbidity?"	→ **tuberculosis** (TO)
"Compound palmar ganglion NOT seen in"	→ **malignancy** (TO)

– – Active space – –

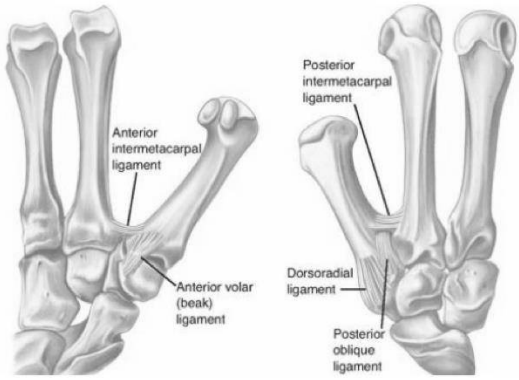
BASE OF THUMB FRACTURES

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1 Basics

Epidemiology & anatomy

Incidence	**80% of thumb fractures involve the MC base**; commonest pattern = **extra-articular epibasal**
CMC joint	**saddle-shaped**, trapezium + 1st MC base; flex–ext ~53°, abd–add ~42°, very limited axial rotation
Volar beak ligament	= **anterior oblique ligament (AOL)**; trapezial tuberosity → volar edge 1st MC → **keeps volar-ulnar Bennett fragment attached to trapezium**
Dorsoradial ligament	dorsoradial trapezial tubercle → dorsal MC base; important for CMC stability but does not hold Bennett fragment
Do not hold the fragment	anterior/posterior intermetacarpal ligaments (no trapezial attachment), posterior oblique ligament



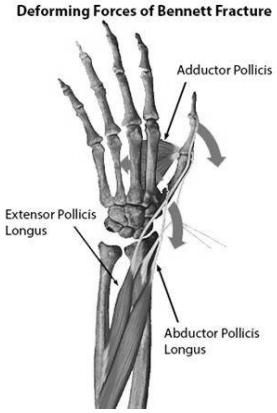
CMC ligaments: AOL (volar beak) holds the Bennett fragment

Deforming forces (APL, adductor pollicis, EPL) & reduction manoeuvre

Muscle	Nerve	Pull on shaft fragment
Abductor pollicis longus (main force, TO)	**PIN**	**Proximal, dorsal, radial**
Adductor pollicis	**Ulnar**	**Supination + adduction** (ulnar angulation)
Extensor pollicis longus	PIN	Proximal, dorsal, radial; adds ulnar angulation (less important)
EPB	PIN	Small contribution

Primary deforming pair	**APL + adductor pollicis** (OB, TO) · three-muscle answer = **APL + AdP + EPL** (Orthobullets) – read the options
Nerves of deforming muscles	**ulnar + PIN** ✗ Wrong options: AIN · recurrent median
Not deforming	**APB**, FPB, opponens (thenar, recurrent median), FPL, PQ, **brachioradialis** (distal radius)

Shaft displacement (TO)	Bennett: shaft **lateral, proximal, dorsal**; Rolando: diaphyseal fragment **usually not displaced**
Reduction manoeuvre	**longitudinal traction + palmar abduction + pronation** + pressure on MC base



Deforming forces on Bennett fracture

2 Mechanism

Mechanism	**axial load on partially flexed (adducted) thumb**, e.g. clenched-fist punch (TO)
Pathoanatomy	imperfect reduction → ↑ joint contact pressure → **early arthritis**; excessive angulation → **MCP hyperextension deformity** (malunion)

3 Presentation

Findings	base-of-thumb pain, swelling, ecchymosis, CMC tenderness, painful ROM

4 Diagnosis & Classification

Key radiology findings

Views	**Roberts view** (true AP: full pronation, dorsum of thumb on cassette) + **true lateral (hand pronated 30°, beam 15° distal)** + oblique
Best view	**30° pronated view**
Bennett	small MC base fragment still articulating with trapezium
Rolando	**Y sign** = base split into volar + dorsal fragments
Traction view	Rolando / comminuted pattern
CT	complex patterns, fragment detail

– – Active space – –



Rolando Y sign

Classification of 1st MC base fractures

Type	Description	Image
Extra-articular oblique	Oblique line, joint spared	
Extra-articular transverse	Transverse line, joint spared	
Bennett	**Partial intra-articular**, palmar-ulnar fragment; oblique fracture-dislocation	
Rolando	**Complete intra-articular**, **Y / T / V-shaped**, 3-part (TO)	
Comminuted	Severely comminuted complete intra-articular	

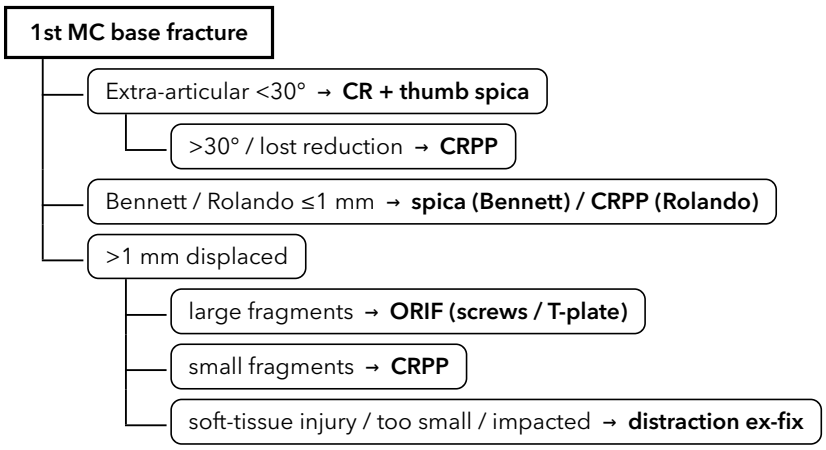
Rolando (TO) **intra-articular** fracture base 1st MC
✗ Wrong options: extra-articular · "adductor longus" · commonest thumb fracture-dislocation

Treatment criteria

Pattern	Accept / nonop	Operate
Extra-articular	**<30° angulation** after CR	>30° or can't hold in spica → **CRPP**
Bennett	**<1 mm** step-off / displacement	**>1 mm** → CRPP or ORIF
Rolando	–	<1 mm → CRPP; >1 mm → ORIF / ex-fix; comminution dictates strategy

Open reduction threshold **>1 mm** displacement (Orthobullets, Soyer) · **>2 mm** step-off despite CR (Carlsen) – read the options

5 Management



Nonop CR (traction, palmar abduction, pronation) + **thumb spica**; >3 wk old with no motion at fracture → accept step-off + cast

Small avulsion (TO) CR + thumb spica **6 wk**; unstable → closed/open reduction + K-wires + spica **4–6 wk** (TO)

CRPP transverse extra-articular → **transarticular** K-wire; oblique → **intermetacarpal** K-wire; risk **loss of reduction**

ORIF **volar Gedda–Moberg approach**; thenar reflected, longitudinal capsulotomy; screws or T-plate; risk **SBRN injury**, wound problems if oedema

Bennett (TO) single volar-ulnar fragment held by **anterior oblique ligament**; MC displaces proximal + dorsal → pin MC to trapezium · fragment need not be pinned if **< 2 mm step** after reduction

Rolando (TO) Y/T comminuted → traction, ex-fix or **mini T-plate** · bone graft often needed

Ex-fix two 3 mm pins dorsoradial MC + two in radius ± 2nd MC; risk pin-site infection

Key outcome factor **anatomic articular congruity** > fixation method; reduction predicts **radiographic** arthritis
✗ Wrong options: symptomatic

– – Active space – –

6 Complications

Post-traumatic CMC arthritis	risk ↑ with comminution, major step-off, multiple small fragments
Malunion	MCP hyperextension deformity, long-term disability
Poor prognosis	Bennett, Rolando, comminuted, delayed treatment · good: acute, extra-articular

X vs Y

	Bennett	**Rolando**
Articular	**Partial** intra-articular	**Complete** intra-articular
Shape	Oblique, single volar-ulnar fragment	**Y / T / V**, volar + dorsal fragments
Held by	AOL (volar beak)	AOL holds volar fragment
Shaft	Displaced proximal/dorsal/radial	Usually not displaced (TO)
X-ray sign	Fragment stays with trapezium	**Y sign**

How it's asked

"Primary deforming forces on Bennett / Rolando?"	→ **APL + adductor pollicis** ✗ Wrong options: adductor pollicis longus + abductor pollicis
"Three deforming muscles overcome by reduction?"	→ **APL, adductor pollicis, EPL** ✗ Wrong options: APB · brachioradialis
"Nerves to the deforming muscles?"	→ **ulnar + PIN** ✗ Wrong options: AIN · recurrent median
"Structure holding fragment to trapezium?"	→ **anterior oblique (volar beak) ligament** ✗ Wrong options: dorsoradial · intermetacarpal · posterior oblique
"Reduction manoeuvre?"	→ **traction + palmar abduction + pronation**
"True of Rolando?"	→ **intra-articular fracture base 1st MC** (TO)
"Best radiographic view?"	→ **30° pronated true lateral / Roberts view**

BASILAR THUMB ARTHRITIS

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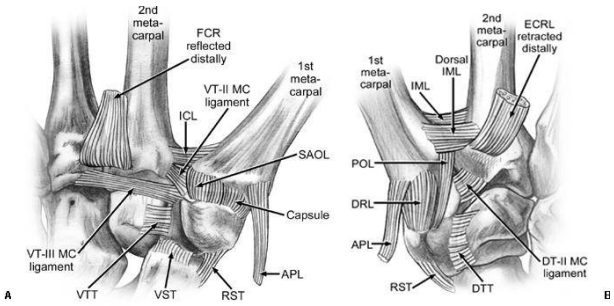
1 Basics

Epidemiology

Frequency in hand	**DIP > thumb CMC > PIP > MCP** (2nd most common hand OA)
Who	**postmenopausal** ♀, Caucasian; 40% ♀ / 25% ♂ >75 y
Second patient group	young **manual labourer**, post-traumatic (old intra-articular #) → fusion candidate
Associated	**CTS in up to 50%** · **MCP hyperextension** · STT OA (62% at trapeziectomy)

Anatomy

Joint	trapeziometacarpal (TMC) = **biconcave saddle**; CMC joint reactive force = **13× pinch force**
Anterior oblique (volar beak) lig	**primary stabiliser vs subluxation**; palmar tubercle trapezium → ulnar MC base; **its attenuation = cause of OA**
Dorsoradial lig	**strongest/thickest**; primary restraint to **dorsal dislocation**
Intermetacarpal lig	2nd MC base → 1st MC base; restrains radial translation
FCR	runs in palmar groove of trapezium (used in LRTI)

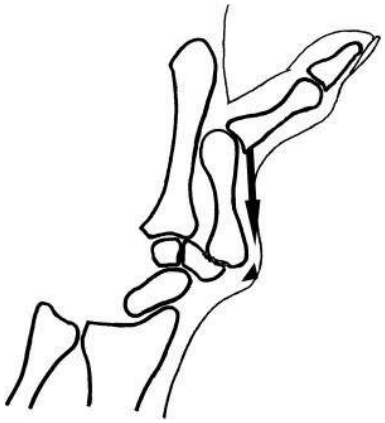


Ligament anatomy

2 Mechanism

Beak lig attenuation → dorsoradial subluxation of MC base → cartilage wear	
Z (zig-zag) deformity	painful abduction avoided → **MC adduction + palmar flexion** → 1st web contracture → **compensatory MCP hyperextension** (EPB pull) → ± IP flexion
NOT a sequela	**CMC extension deformity** (EXCEPT question)
Why MCP hyperext matters	left untreated → deforming force → recurrent MC subluxation after CMC surgery

– – Active space – –



Z deformity forces: arrowhead = MC base subluxation, arrow = EPB vector

3 Presentation

History	pain at thumb base, ↓ **pinch/grip**, worse **opening jars**, key pinch
Exam	TTP over CMC · **grind test** (axial load + circumduction) · **shear test** (circumduction + translation) · crepitus
Late signs	**MC adduction**, 1st web contracture, **fixed MCP hyperextension (Z)**



Z deformity of thumb

4 Diagnosis & Classification

Key radiology findings

Finding / view	Fact asked
Roberts view	True AP of TMC: thumb flat, **hyperpronated**, beam on trapezium
Standard series	AP + lateral hand
OA signs	Joint space narrowing, sclerosis, osteophytes (**<2 mm = II, >2 mm = III**), dorsoradial subluxation of MC base
STT involvement	= **Eaton IV (pantrapezial)** → contraindicates CMC fusion
MCP hyperextension	Seen on lateral / clinical → measure angle (drives MCP treatment)
STT OA + **DISI**	Lateral: extended lunate, volar-tilted capitate → STT fusion, not distal pole excision
After trapeziectomy	**MC subsidence** (~25%) → no clinical effect



MCP hyperextension on radiograph

– – Active space – –

Eaton–Littler classification

Eaton–Littler radiographic; **stage IV = pantrapezial (STT)** → decides fusion vs not; historical value, limited clinical utility

Stage	Radiograph	Key decider	Image
I	**Joint widening** (pre-arthritis, laxity)	Osteotomy / volar lig recon	
II	Narrowing, sclerosis, **osteophytes <2 mm**	Osteotomy if stable; fusion (labourer)	
III	Marked narrowing, **osteophytes >2 mm**	Trapeziectomy; **fusion if young labourer**	
IV	**Pantrapezial – STT involved**	Trapeziectomy (± prox ⅓ trapezoid); **no fusion**	

Differentials **C6 radiculopathy** (neck → thumb/index paraesthesia) · **de Quervain** (Finkelstein +) · **STT OA** (TTP just proximal to TMC) · SNAC / radioscaphoid OA (on X-ray)

5 Management

Nonoperative care – splinting & injections

1st line	NSAIDs + **thumb spica / night splint**
2nd line	**steroid injection** (moderate evidence)
RCT trap	**saline = steroid = hylan** (all ≈ 3 mo relief; Heyworth) → answer "all equally effective"
Hyaluronic acid	not indicated (no better than placebo/steroid)
Thumb MCPJ OA (post-RCL insufficiency, labourer)	1st: **spica 4–6 wk + NSAIDs** → fail → **MCP fusion** (5–20° flex, 15° pronation); no arthroplasty in lig-deficient thumb/labourer

Operative algorithm by stage

Failed nonop

- Stage I–II, stable joint, MCP hyperext <10° → **1st MC dorsal closing-wedge EXTENSION osteotomy**
  (unstable/hypermobile Stage I → volar lig recon with FCR)
- Stage II–III, young male heavy labourer, NO STT OA → **CMC ARTHRODESIS**
- Stage III–IV (most patients) → **TRAPEZIECTOMY ± LRTI / suspension**
  (Stage IV → + excise proximal ⅓ trapezoid)

Operative criteria

Procedure	Indication	Contraindication
Arthroscopic debridement	Early stage (I–II)	Stage III–IV
Volar lig recon (FCR)	**Stage I, hypermobile/unstable**	Established OA
1st MC extension osteotomy	**Stage I–II**, stable joint	**Hypermobile/fixed subluxed CMC · MCP hyperext >10°**
CMC arthrodesis	**Stage II–III, young (<50) male heavy labourer**, post-traumatic	**STT (pantrapezial) OA**
Trapeziectomy ± LRTI	**Stage I–IV** (most patients, older ♀)	Young heavy labourer (relative)
+ excise proximal ⅓ trapezoid	**Stage IV** / ST-trapezoid OA (62%)	–
Prosthetic arthroplasty	not recommended	Lig-deficient thumb, labourer
MCP fusion	MCP hyperext **>30–40°**, not correctable, MCP OA	–

Osteotomy contraindications	**hypermobile / fixed subluxed CMC** · **MCP hyperext >10°** (volar joint stays overloaded)
Osteotomy result	unloads volar joint; 93% improved at 7 y
Arthroscopic debridement	early stage only; portals **1R** (radial to APL) & **1U** (ulnar to EPB)
Prosthetic arthroplasty	not recommended; silicone → **silicone synovitis**; pyrocarbon → subluxation (shallow cup)
Denervation	resect 4 nerves; pain ≈ trapeziectomy/fusion; ROM > fusion

Trapeziectomy ± LRTI – equivalence, technique & complications

Key fact	**trapezial excision is the essential step**; **no technique superior** to trapeziectomy alone
LRTI vs trapeziectomy alone	**no difference** in pain, grip, **key pinch**, ROM, DASH, subsidence → answer "**no benefit of LRTI**"
Complications	LRTI/Weilby **> complications** than simple trapeziectomy (OR 4.3)
Tendon for LRTI	**FCR** most common; alternatives **ECRL** or **APL** ✗ Wrong options: FPL

– – Active space – –

FCR avulsed intra-op (trapezium already out)	**LRTI with ECRL** ✗ Wrong options: fusion (no trapezium left) · implant
Revision after FCR LRTI	use **APL**
Subsidence	~**25%** (21% rest, +11% pinch) → **no effect on outcome → observe**
Hematoma arthroplasty	trapeziectomy alone ± pin; simple, cheap, = LRTI
Wagner approach – radial artery	**most at risk**: dorsoradial capsule **proximal to trapezium** · **least at risk**: level of **1st MC base / TMC joint**
Other nerve at risk	superficial radial nerve



Subsidence after trapeziectomy → observe

MP hyperextension – MP fusion/capsulodesis with CMC reconstruction

Must address at CMC surgery	else adduction contracture + recurrent subluxation

MCP hyperextension	Treatment
<10°	Nothing
10–20°	**Pin MCP in 25–35° flexion × 3–4 wk** ± EPB transfer
20–40°	**Volar capsulodesis** or sesamoidesis
>40° (or >30° in some stems), not passively correctable, MCP OA	**MCP arthrodesis**

Mild (<30°) untreated	1-y outcome **same** as no hyperextension (key pinch, grip, abduction) → "no significant difference"
Classic answer	end-stage CMC + MCP hyperext 40° → **CMC resection arthroplasty + MCP fusion**
Failed LRTI + MCP 60°	**CMC revision stabilisation (APL) + MCP fusion** ✗ Wrong options: pinning alone

CMC arthrodesis – young labourers & contraindications (pantrapezial arthritis)

Indication	**young (<50) male heavy labourer**, stage II–III, post-traumatic → **preserves grip**, stable

Contraindication	**STT / pantrapezial OA (stage IV)** ✗ Wrong options: radiocarpal OA · chronicity · post-traumatic
Position	35° radial abduction · 30° palmar abduction · 15° pronation
Main complication	**nonunion 8–21%** (~12–17%); also STT/MCP OA (mostly asymptomatic)
Downside	↓ ROM, **can't lay hand flat**
Avoid in labourer	trapeziectomy/LRTI (↓ grip, no salvage), arthroplasty (bone loss)



CMC fusion

STT arthritis – STT arthrodesis

STT OA	16% of population, ♀ >50; pain with pinch; tender proximal to TMC; associated **type II lunate**
STT OA + DISI (extended lunate, volar-tilted capitate)	**STT arthrodesis** (lowest revision risk)
Avoid distal pole excision with DISI	worsens midcarpal instability → revision
Avoid CMC arthrodesis for STT OA	STT not addressed, ↑ shear



STT OA with DISI on lateral

6 Complications

After trapeziectomy	MC subsidence (asymptomatic → observe); radial artery; superficial radial n.

– – Active space – –

If subsidence symptomatic & FCR used	LRTI with **ECRL or APL**
After fusion	nonunion; adjacent STT/MCP OA
After silicone implant	**silicone synovitis**, implant fracture/loosening
Prognosis	OA in one joint of a row predicts OA in other joints of same row

X vs Y

	Trapeziectomy ± LRTI	**CMC arthrodesis**	**1st MC osteotomy**
Patient	Older ♀, low demand	**Young male labourer**	Early, stable joint
Stage	I–IV (mainly III–IV)	**II–III**	**I–II**
Absolute contraindication	–	**STT OA**	**Hypermobile CMC, MCP hyperext >10°**
Signature complication	**Subsidence (ignore)**	**Nonunion**	Osteotomy nonunion
Grip	↑	**Preserved best**	Preserved

	Basilar thumb OA	**STT OA**	**De Quervain**	**C6 radiculopathy**
Tender point	TMC joint	Just **proximal** to TMC	1st dorsal compartment	Neck
Test	**Grind**	Injection/X-ray	**Finkelstein**	Spurling
Surgery	Trapeziectomy / fusion	**STT fusion** (if DISI)	1st compartment release	ACDF

Thumb deformity	**MCP**	**IP**
CMC OA (Z deformity)	**Hyperextension**	± Flexion
Lupus (hitchhiker)	Flexion	Hyperextension

How it's asked

"Compared with trapeziectomy alone, which gives superior pain relief / key pinch?"	→ **None of the above**
"Which injection gives greatest improvement?"	→ **All equally effective** (saline = steroid = hylan)
"Radiograph 1 y after trapeziectomy shows subsidence – next?"	→ **Observation**
"FCR avulsed during LRTI – next?"	→ **ECRL LRTI**
"35-y mechanic / rancher, TMC OA, no STT"	→ **CMC arthrodesis**; complication asked → **nonunion**
"Contraindication to CMC arthrodesis?"	→ **Pantrapezial (STT) arthritis**

"Contraindication to osteotomy?"	→ **CMC hypermobility / MCP hyperextension >10°**
"All are sequelae EXCEPT"	→ **CMC extension deformity**
"Patient with STT OA + DISI — surgery that avoids secondary procedure?"	→ **STT arthrodesis**
"MCP hyperextension 9.8°/ <30° untreated — outcome?"	→ **No significant difference**

– – Active space – –

Stem clue: deformity with web contracture + MCP hyperextension → TMC OA
✗ Wrong options: hypoplasia · median neuropathy · lupus · tendon rupture

– – Active space – –

BLOOD SUPPLY TO HAND

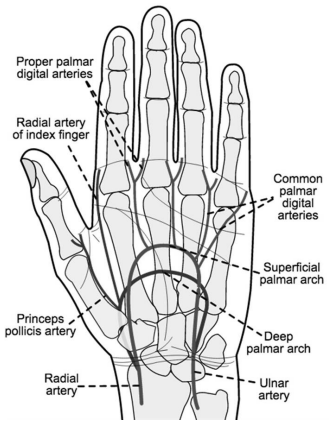
– Hand · 7 / 103 –

1 Basics

Source arteries

Radial artery	between **brachioradialis & FCR** → dorsum via **snuffbox** (between FCR and APL/EPB) → gives **superficial palmar branch** → dives between **2 heads of 1st dorsal interosseous** → **deep palmar arch**
Ulnar artery	under FCU; **lateral (radial) to ulnar nerve** at wrist → **Guyon's canal** → lies on TCL
Supplemental	anterior interosseous, posterior interosseous, median artery (occasional)

Vascular anatomy – deep palmar arch & digital neurovascular relationship



Superficial and deep palmar arches

	Superficial arch	Deep arch
Main supply	**Ulnar artery**	**Deep (terminal) branch of radial artery**
Minor supply	Superficial palmar br. of radial a.	Deep palmar br. of ulnar a.
Depth	Deep to palmar fascia	**Deep to flexor tendons**, on MC bases & interossei, under oblique adductor pollicis
Position	**Distal**	**Proximal** (1 fingerbreadth proximal to superficial)
Surface marking	Line across palm from distal edge of fully abducted thumb	1 fingerbreadth proximal to that line
From Kaplan's cardinal line	15 mm distal	7 mm distal
From distal wrist crease	50 mm distal	40 mm distal
Complete	**80%**	**97%**
Direct digital branch	**Ulnar digital a. of little finger**	**Radial digital a. of index**
Other branches	Deep branch (to deep arch), **common palmar digital aa.**	**Princeps pollicis** (between 1st DI & adductor pollicis), common digital aa. to 2nd–4th webs

Deep arch main contributor — **deep dorsal (terminal) branch of radial a.**
✗ Wrong options: superficial palmar br. radial · princeps pollicis · ulnar a.

Wrist course of radial a. — between radial collateral ligament & 1st dorsal compartment → between heads of 1st DI → anastomoses with deep palmar br. of ulnar

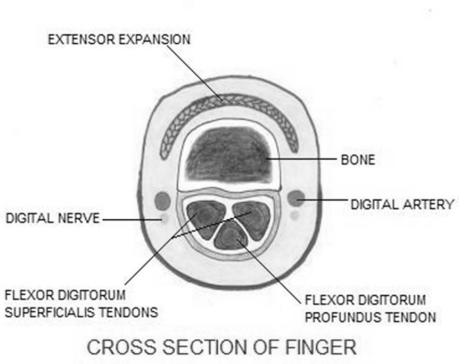
Digital arteries

Origin — common digital aa. from **superficial arch** → divide into proper digital at web spaces → dorsal branches distal to PIP

Palm — digital artery **volar** to nerve

Finger — **artery DORSAL to nerve** (nerve palmar: pads feel, nail refill)
✗ Wrong options: artery palmar · nerve dorsal

Ligament relations — NV bundle **volar to Cleland's** (not in Dupuytren's) · Grayson's volar (involved in Dupuytren's)



Digital artery dorsal to digital nerve in the finger

– – Active space – –

Digit	Dominant digital artery
Rule	**Side closer to midline (median side)**
Index	**Ulnar**
Little	**Radial**
Middle / ring	Ulnar / radial (less obvious)

Dorsal arteries & veins

Dorsal carpal arch — from posterior interosseous + dorsal perforating br. of anterior interosseous → **dorsal metacarpal aa.** (flaps; 1st & 2nd most consistent)

Veins — deep = **venae comitantes**; superficial dorsal → basilic & cephalic

4 Diagnosis & Classification

Allen test

Purpose — patency of collateral (dual) circulation before **radial artery cannulation/ harvest** (thrombosis risk) (TO)

Positive Allen (TO) — colour returns quickly = **adequate dual supply → indication** for cannulation

Negative Allen (TO) — pallor persists = **no patent dual supply → contraindication** to radial cannulation

	Original Allen (TO)	Modified Allen (TO)
Hands	**Both hands**, one artery at a time	**One hand** at a time
Method	Clench both fists 1 min, compress **both radial aa.** simultaneously, open → repeat with ulnar	Raise hand, clench 30 s, compress **radial + ulnar together**, open (blanched) → **release ulnar**
Normal	Pale → pink quickly	Pink **within 6 s**

How it's asked

"Main contributor to deep palmar arch, between heads of 1st DI?"	**→ deep dorsal (terminal) branch of radial artery**
"Main supply of superficial arch?"	**→ ulnar artery**
"Artery–nerve relation in the finger?"	**→ artery dorsal to nerve**
"True about Allen test?"	**→ negative = no patent dual supply** (TO) ✗ Wrong options: positive is a contraindication · original compresses radial + ulnar of one hand · modified compresses them one by one
"Dominant digital artery of index?"	**→ ulnar**
"Which arch is more complete?"	**→ deep (97%)** vs superficial 80%

BOUTONNIERE DEFORMITY

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1 Basics

Anatomy – extensor mechanism

Central slip	central part of extensor hood → inserts **base of middle phalanx** → **extends PIP**
Lateral bands	deep head dorsal interossei + volar interossei → insert **base of distal phalanx** → **extend DIP**
Lumbricals	origin **FDP** → insert on **lateral bands**
Triangular ligament	spans the 2 lateral bands dorsally → **prevents volar subluxation**
Transverse retinacular ligament	**prevents dorsal subluxation** of lateral bands

Epidemiology

Rheumatoid	up to **50% of RA** patients → boutonniere in ≥1 digit
Definition	**Zone III** extensor injury → **PIP flexion + DIP hyperextension** ("buttonhole": PP head buttons through extensor)

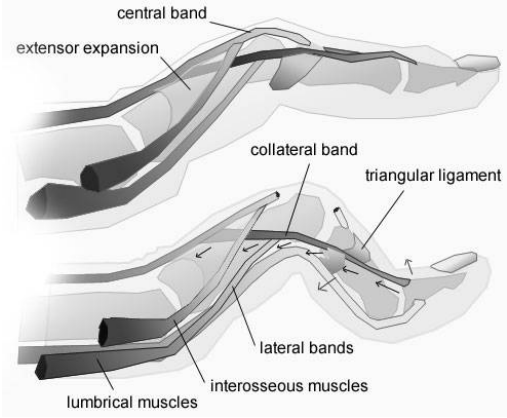
2 Mechanism

Pathophysiology – central slip injury & lateral band relocation

Causes of central slip failure	**laceration** · **traumatic avulsion (jammed finger)** · **PIP synovitis / capsular distension (RA)**
Primary trigger (TO)	**attenuated central slip + volar displacement of lateral bands** ✗ Wrong options: dorsal subluxation of lateral bands · ORL shortening · volar plate rupture
Chronic injury to	**central slip** of extensor tendon ✗ Wrong options: terminal tendon → mallet · sagittal band → MCP extensor subluxation · volar plate → swan neck · FDP insertion → jersey

Step	Event	Effect
1	**Central slip rupture/attenuation**	EDC extension of PIP lost
2	**Triangular ligament attenuates**	Lateral bands free to slide volarly
3	**Lateral bands drop volar to PIP axis**	Become **PIP flexors** (TO); all extensor force → **DIP hyperextension**

– – Active space – –



Boutonniere pathoanatomy

Lumbricals	unopposed pull through lateral bands → flex PIP + extend DIP
Pseudo-boutonniere	PIP flexion contracture **without DIP extension**

3 Presentation

Deformity	**PIP flexion + DIP hyperextension**; RA patient or jammed/lacerated finger



Boutonniere clinical

4 Diagnosis & Classification

Diagnostic test – Elson

Elson test	**most reliable** for central slip injury **before deformity evident**
Technique	PIP 90° over table edge → extend middle phalanx against resistance
Positive	**weak PIP extension + rigid DIP** (lateral bands fire)
Negative (intact slip)	**DIP floppy**



Elson test

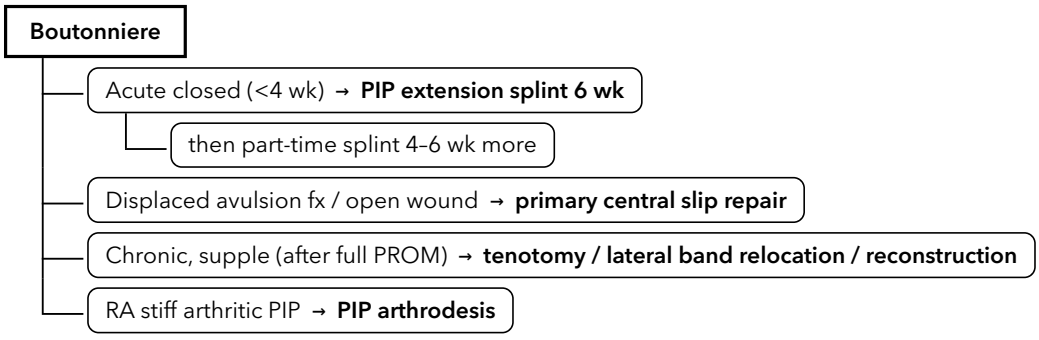
Key radiology findings

X-ray	**not required** for diagnosis; shows **avulsion fracture** of middle phalanx base → ORIF/repair

Rheumatoid staging (TO)

Stage	Findings	Surgery
I (mild)	**Passively correctable** (PIP extends, DIP flexes)	**Dolphin-Nalebuff terminal tenotomy**
II (moderate)	**Fixed PIP flexion contracture**, joint space preserved	Matev lateral band transfer / central slip reattachment
III (severe)	Fixed PIP + **radiographic destruction** or severe soft-tissue contracture	**PIP arthrodesis** (most reliable)

5 Management



Nonoperative

Splint	**PIP in full extension 6 wk** (continuous) + part-time 4–6 wk; **acute closed <4 wk**
In splint	**active DIP flexion/extension** → prevents **ORL contracture**
Traumatic vs RA (TO)	traumatic responds well to splinting; **RA tissue "sick" → splinting less successful**, progresses

– – Active space – –

Operative criteria

Procedure	Indication
Primary central slip repair	**Acute displaced avulsion fx** · open wound needing I&D (ORIF of attachment fractures successful)
Lateral band dorsal relocation	**~40° active flexion contracture with full passive extension**
Terminal tendon tenotomy (modified Fowler / **Dolphin**)	Chronic, after FROM; PIP flexible but can't flex DIP; **never central slip tenotomy**
Matev (TO)	One lateral band → central slip; other extends DIP
Secondary reconstruction	Tendon graft, Littler, Matev, triangular ligament reconstruction
PIP arthrodesis	**RA**, painful stiff arthritic PIP; **stage III** (TO)
Silastic arthroplasty (TO)	Occasionally ulnar digits; high recurrence

PIP fusion angles (TO) **ring/small 40–50°**; **index 25–30°**

RA outcome (TO) boutonniere surgery **less predictable than swan-neck**

X vs Y

(TO)	Swan-neck	Boutonniere
PIP	Hyperextension	**Flexion**
DIP	Flexion	**Hyperextension**
Primary defect	**Volar plate / FDS**	**Central slip**
Lateral bands	Subluxate **dorsally**	Subluxate **volarly**

Structure injured	Deformity
Central slip	**Boutonniere**
Terminal extensor tendon	Mallet
Sagittal band	Extensor subluxation at MCP
Volar plate (chronic)	Swan neck
FDP insertion avulsion	Jersey finger

How it's asked

"Primary trigger of boutonniere cascade?"	→ **attenuated central slip + volar lateral bands** (TO) ✗ Wrong options: dorsal lateral bands · ORL
"Surgery relocates lateral bands dorsally — which deformity?"	→ **boutonniere** ✗ Wrong options: swan neck · mallet · jersey · lumbrical plus
"Chronic injury to which structure → boutonniere?"	→ **central slip** ✗ Wrong options: terminal tendon · sagittal band · volar plate · FDP

"Detect central slip injury before deformity?"	→ **Elson test** (rigid DIP + weak PIP extension)
"Acute closed central slip injury – treatment?"	→ **PIP extension splint 6 wk, DIP free**
"Stage III RA boutonniere?"	→ **PIP arthrodesis** (TO)
"Tenotomy for chronic boutonniere?"	→ **terminal tendon (Dolphin/Fowler)** ✗ Wrong options: central slip

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CAMPTODACTYLY

– Hand · 9 / 103 –

1 Basics

Definition & epidemiology

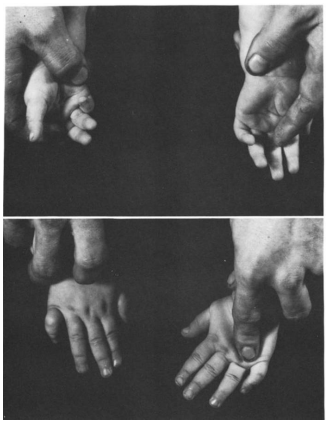
Definition (TO)	Greek "**bent finger**"; **nontraumatic flexion deformity of the PIP joint**, classically **small finger**
Prevalence	**<1%**
Laterality	unilateral 33% · **bilateral 66%** (often **asymmetric** (TO))
Other digits (TO)	occasionally; incidence & severity ↑ toward **ulnar side** of hand
Genetics	mostly **sporadic**; familial = **AD, incomplete penetrance / variable expressivity**
Isolated case	usually idiopathic → **no genetic workup**; Type III / multi-digit → syndromes (e.g. arthrogryposis)

Etiology

Commonest causes	**abnormal lumbrical** insertion/origin · **abnormal FDS** insertion (adherent, hypoplastic, tight)
Less common	central slip, extensor hood, volar plate, skin/subcutaneous contracture
Lumbrical theory (TO)	lumbrical inserts into lumbrical canal/FDS instead of extensor hood → like an **intrinsic-minus digit** (Jones, Marmor, Lankford)
Other TO theories	FDS tightness / "slow retraction" of flexors · short FDP · faulty dorsal extensor aponeurosis
Unifying concept (TO)	**final common expression of flexion–extension imbalance at PIP** ("virtually every structure implicated", Smith & Kaplan)

3 Presentation

Exam	**PIP flexion deformity, small finger**; flexible or fixed; **painless**, no swelling/erythema; normal strength, sensation, perfusion
Neighbouring joints	DIP & MCP usually normal; untreated → compensatory **MCP hyperextension** (commonest adjacent)
Course	progressive; **rapid worsening in growth spurts**
Function	rarely compromised; often unnoticed



Camptodactyly small finger PIP

4 Diagnosis & Classification

Diagnosis

Diagnosis	**clinical**
Onset (TO)	congenital (♂ = ♀) or **adolescent** (early teens, ♀, gradual)

Key radiology findings

X-ray	often **normal** early (articular congruity maintained)
Late X-ray	↓ **P1 head convexity** · volar subluxation + flattening of **P2 base**

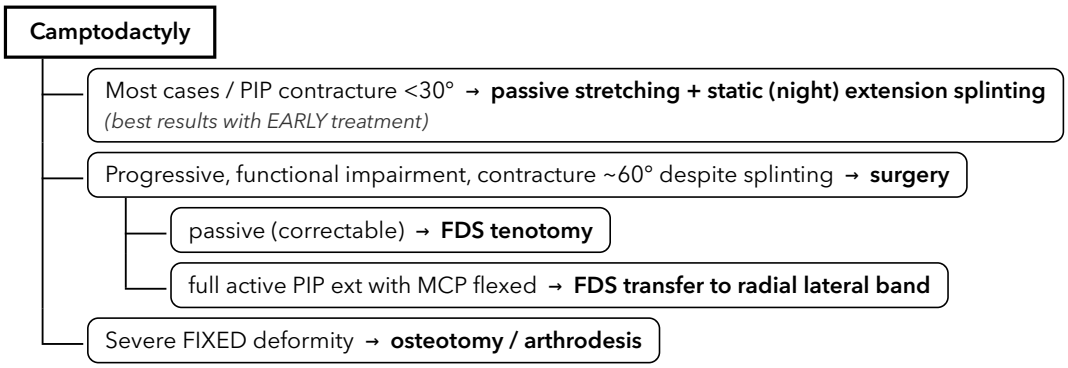
Benson classification

Type	Features	Treatment	Image
I (infantile)	Isolated small finger, **infancy**, ♂ = ♀; **most common** (Orthobullets)	**Stretching/splinting**	
II (adolescent)	Same features, **adolescence**; ♀ **>** ♂	If **full active PIP extension with MCP flexed → FDS transfer to radial lateral band**	
III (syndromic)	**Birth**, severe, **multiple digits**, syndrome	Nonop; after maturity if functional deficit → **osteotomy / fusion**	–
Kirner's	Small finger **distal phalanx** volar-radial curve (apex dorsal-ulnar); preadolescent ♀; bilateral; no deficit	–	

Commonest type – conflict	**Type I/infantile** (Orthobullets) · **adolescent type more common** (TO) – read the options

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5 Management



Treatment – stretching & extension splinting

First line — **night extension splinting + stretching** (intrinsic-plus splint) → always try first, even mild deformity
✗ Wrong options: observation alone: shorter duration → better result

Mild (<30°) — stretching ± splint → expect resolution (Rhee: **20° → 1°** over 5 mo, ≥20 sessions/day, children <3 y)

Severe (>60°) — stretching helps but slower (Rhee: **75° → 28°**, up to 2 y) → may need **FDS tenotomy** if bothersome

(✗ Wrong options: **Traps** – complete correction in 1–2 mo) · **serial casting 90% for >50°** · progressive contracture of all joints · routine bilateral tenotomy
✗ Wrong options: not effective for significant deformity

Serial casting – conflict — not effective for significant deformity (OB) · listed among nonop options for **mild** cases (TO)

TO nonop list — active exercises, dynamic splints, serial casts, passive stretching (mild cases)

Surgery

Indication — **progressive** deformity → functional impairment; failed splinting

Principle — address **all** abnormal anatomy; rebalance PIP flexion–extension

Deforming factor (TO)	Procedure
FDS contracture	**FDS tenotomy**
Anomalous lumbrical	Excise or reinsert into **lateral band** + volar Z-plasties + active post-op splinting
Flexor–extensor imbalance	Volar release + **FDS transfer through lumbrical canal to extensor aponeurosis**
Extreme	**FDP lengthening + PIP capsulotomy**

Outcomes — **inconsistent / variable**; complications: persistent PIP contracture & MCP hyperextension, residual stiffness, **P1 head remodelling** blocking extension

X vs Y

Differential (TO)	Distinguishing point
Camptodactyly	**PIP flexion**, painless, slow, no trauma
Clinodactyly	Deviation in **radioulnar plane**
Boutonnière	Trauma, swelling/pain, **DIP hyperextension**
Dupuytren	**MCP contracture**, palmar dimpling/fascia
Trigger digit	Triggering; affects **MCP**
Congenital absent extensor mechanism	Affects **MCP**, not PIP

Clinodactyly (TO) radial/ulnar curvature, usually small finger, **Down syndrome** · symptomatic → closing/opening wedge osteotomy

How it's asked

"3-y-old, bilateral asymmetric fixed little-finger PIP flexion, painless, no trauma, normal DIP/MCP"	→ **camptodactyly** ("compatodactyly" in TO) ✗ Wrong options: clinodactyly, Dupuytren, absent extensor mechanism
"PIP 25° one side, 60° other – counsel"	→ **stretching fixes 25°; 60° slower ± FDS tenotomy**
"5-y-old, isolated small finger, limited passive extension – next step"	→ **night extension splint + stretching** ✗ Wrong options: observation · genetic workup · FDS transfer first
"Full active PIP extension with MCP flexed"	→ **FDS transfer to radial lateral band**
"Severe fixed deformity"	→ **osteotomy / arthrodesis**
"Commonest causes"	→ **abnormal lumbrical or FDS insertion**

– – Active space – –

CARPAL INSTABILITY NONDISSOCIATIVE (CIND)

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1 Basics

Definition & epidemiology

CIND (TO) instability **between carpal rows** (midcarpal) or **between radius and carpus** (radiocarpal) **without intrinsic (SLIL/LTIL) tear**; each row intact, **extrinsic ligament** insufficiency

Not CIND ligament disruption within a single row (= CID) · trans-scaphoid perilunate (= CIC) · DR malunion tilt alone

Types **midcarpal** (proximal ↔ distal row; **most common CIND**, TO) · **radiocarpal** (radius ↔ proximal row; "inferior arc injury")

Incidence rare, **<1 per 100,000/y**

Associations SL/LT injury, **acute CTS**, compartment syndrome

Extrinsic ligament anatomy

Ligament	Origin → insertion	Role (TO)	Failure
Radioscaphocapitate (RSC)	Radial styloid → scaphoid waist + capitate	Fulcrum / sling for scaphoid	Scaphoid instability
Long radiolunate (LRL)	Radial styloid → volar lunate + triquetrum	**Primary stabiliser vs ulnar translocation**	**Ulnar drift of carpus** (RA)
Short radiolunate (SRL)	Ulnar notch rim → volar-ulnar lunate	Lunate anchor	Lunate dislocation
Ulnolunate / ulnotriquetral	TFCC → lunate / triquetrum	Ulnar column	**VISI**
Arcuate (triquetrohamatocapitate)	Palmar midcarpal	Midcarpal kinematics	**Midcarpal instability**

Space of Poirier (TO) weak area between RC and LRL over capitolunate joint → carpus dissociates here in perilunate (Mayfield II); lunate stays on SRL

2 Mechanism

Midcarpal lax **palmar arcuate ligament** + dorsal radiocarpal ligaments (TO); usually **no trauma**, **generalised laxity**

Radiocarpal high energy; ulnar translation = **global extrinsic rupture**; ± radial/ulnar styloid fractures

Commonest cause of radiocarpal instability **distal radius malunion** (Orthobullets) · ulnar translocation commonly **RA / RSC disruption** (TO) – read the options

Radiocarpal instability mechanism (TO)	failure of **extrinsic volar ligaments** → proximal row translates **volarly** · midcarpal instability → failure of midcarpal stabilisers → **capitate translation → VISI**
Dorsal radiocarpal instability (TO)	DR malunion with **loss of volar tilt**

3 Presentation

Symptoms	wrist **giving way**, painful/painless subluxation, **clunk**
Midcarpal shift test (TO)	stabilise forearm, **axial + palmar load**, wrist **neutral → ulnar deviation** (pronation) → painful **"catch-up clunk"** as proximal row snaps **VISI (flexed) → neutral/extended**

(✗ Wrong options: **Distractor exams** – radial pain with ulnar deviation (**Finkelstein**)) · tenderness distal to ulnar styloid (**fovea sign**) · clunk radial deviation + scaphoid pressure (**Watson**) · lunate pain with dorsal pressure (**Kienböck**)

4 Diagnosis & Classification

Key radiology findings

Static X-ray	AP + lateral; often **normal or mild VISI**
Dynamic fluoroscopy / cineradiography	**diagnostic** (TO); sudden jump of proximal row **flexion → extension** with ulnar deviation
Stress views (TO)	**volar / dorsal drawer** → detect midcarpal instability
Ulnar translation	**>50% of lunate width** translated ulnarly off the lunate fossa



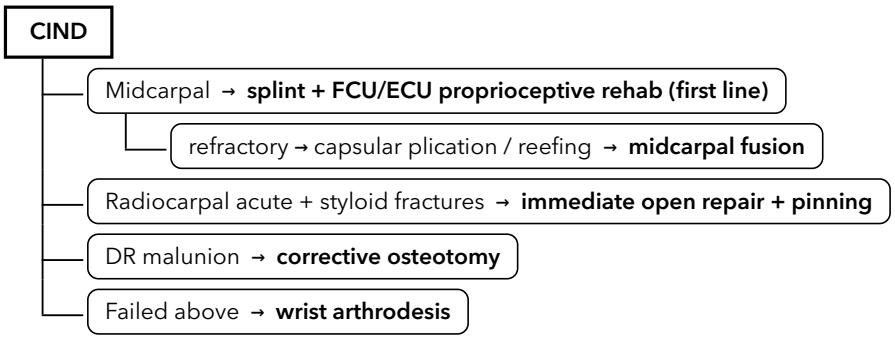
Ulnar translation >50% lunate width

Carpal instability classification

Type	Lesion	Example
CID (dissociative)	**Intrinsic** ligament, within a row	SL dissociation
CIND	**Extrinsic**, between rows / radius–carpus	**Midcarpal instability**, ulnar translocation
CIC (complex)	CID + CIND	Perilunate injury
CIA (adaptive)	Extracarpal cause	Madelung, DR malunion

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5 Management



Nonop	first line; **midcarpal most amenable to splinting**; rehab FCU + ECU (dynamic stabilisers, TO), activity modification, NSAIDs
Midcarpal surgery (TO)	thermal capsulorrhaphy (controversial, capsular necrosis) · open plication of arcuate ligament / dorsal capsule · reefing (Apergis: 13/14 good–excellent) · **limited midcarpal arthrodesis (4-corner / triquetrohamate)** · total wrist fusion if intractable
Midcarpal fusion	**preferred over ligament reconstruction** and for late diagnosis failing nonop; **20–35% loss of motion**
Radiocarpal	immediate open repair/reduction/pinning if **ulnar translation + styloid fractures**; **late repair poor**; **ligament reconstruction poor long term**
Ulnar translocation in RA (TO)	radiocarpal or total wrist arthrodesis
Radiocarpal arthritis + functional midcarpal joint	**radioscapholunate (RSL) fusion + distal scaphoid excision** (↑ ROM)
Wrist arthrodesis	salvage; radiocarpal fusion → **55–60% loss of motion**
Not for isolated radiocarpal arthritis	4-corner fusion (radiocarpal motion still painful) · capitolunate fusion (midcarpal OK) · TWA (young manual labourer, fails) · total fusion (salvage only)

6 Complications

Prognosis	**volar dislocation worse than dorsal**; late repair and ligament reconstruction poor

X vs Y

	Midcarpal CIND	**Radiocarpal CIND**
Frequency	**Most common**	Less common
History	No trauma, lax	High energy / DR malunion / RA
Sign	**Catch-up clunk** on ulnar deviation	Ulnar translation >50% lunate
Key ligament	Palmar arcuate	**LRL** / RSC
Surgery	Plication → **midcarpal fusion**	Open repair, osteotomy, RSL / wrist fusion

How it's asked

"Defining feature of CIND?"	→ **instability between rows / radius–carpus without intrinsic tear** (TO)
"Clunk on ulnar deviation, proximal row jumps flexion → extension, no SL/LT dissociation?"	→ **CIND – midcarpal instability** (TO)
"Most diagnostic exam for midcarpal instability?"	→ **clunk with axial + palmar load, neutral → ulnar deviation** ✗ Wrong options: Watson · Finkelstein · fovea
"Primary volar restraint to ulnar translocation?"	→ **long radiolunate ligament** (TO) ✗ Wrong options: RSC · SRL · ulnolunate
"Radioscaphoid + radiolunate arthritis, preserve motion?"	→ **RSL fusion + distal scaphoid excision**
"Diagnostic investigation?"	→ **dynamic fluoroscopy**

– – Active space – –

CARPAL TUNNEL SYNDROME

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1 Basics

Epidemiology & associations

Commonest	**most common compressive neuropathy**; 0.1–10% population; ♀:♂ **3:1**; 40–60 y; bilateral up to 70%
Commonest causes (TO)	**idiopathic > hypothyroidism > pregnancy**
Risk factors	♀, **obesity**, **pregnancy**, **hypothyroidism**, **RA**, DM, **smoking**, vibration, acromegaly, amyloidosis, **hyperparathyroidism**, CRF, OCP, DRF malunion; **children → mucopolysaccharidosis**
NOT associated (TO)	**Addison's disease**
Compression risk factors (TO list)	DM · hypothyroidism · acromegaly · **excessive alcohol** · tobacco · obesity · **hereditary motor sensory neuropathy**
Patient-reported tools (TO)	CTS questionnaire · **DASH** · Michigan Hand Outcomes Questionnaire · **PROMIS** · Patient Specific Functional Scale
Pregnancy CTS (TO)	usually **resolves after delivery**
Diabetics (TO)	**worse surgical outcome** than idiopathic

Anatomy – tunnel contents, TCL attachments, recurrent motor branch & anastomoses

Contents	median nerve + **FPL** + 4 FDS + 4 FDP (9 tendons); **not FCR** (own sheath, trapezial groove) · not ulnar n.
FDS layering	FDS 3rd & 4th **volar** to FDS 2nd & 5th
Most radial structure	FPL
TCL (roof) attachments	radial: **scaphoid tubercle + trapezium**; ulnar: **hook of hamate + pisiform**
Narrowest	level of **hook of hamate** · **~2 cm from proximal (leading) edge** = site of morphological nerve change (TO)
Superficial to flexor retinaculum (TO)	**ulnar nerve + artery** · **palmar cutaneous branch of median n.**

Roof zone (TO)	Structure
Proximal	**Deep forearm fascia**
Central	**TCL**
Distal	**Aponeurosis of thenar + hypothenar muscles**

Median nerve at wrist (TO)	**~30 fascicles** · **motor fascicles volar** (→ recurrent branch) · sensory radial, ulnar, dorsal · **94% sensory fibres** → sensory changes first
Floor	proximal carpal row
Pressures	normal 2.5 mmHg (neutral) → 30 (flexion); **20 mmHg → venous congestion**; **60–80 → arteriolar arrest**; CTS 30–110

Recurrent motor branch — **APB, OP, superficial FPB** (+ lumbricals 1–2 via digital nn.); arises radial/central

Recurrent branch variant	%
Extraligamentous	50 (Orthobullets) · **46–90%** (TO)
Subligamentous	30 · **31%** (TO)
Transligamentous (at risk)	**20** · **23%** (TO)
From **ulnar border** of median n. (TO)	Rare, **highest risk** at CTR
On top of TCL (TO)	Very rare

Other variants (TO) — origin from **ulnar border** of median n. · branch **lying on top of TCL** · **Lanz grade III = high division** proximal to tunnel (persistent median artery / aberrant muscle) · **grade IV** = thenar branch proximal to tunnel / accessory branches

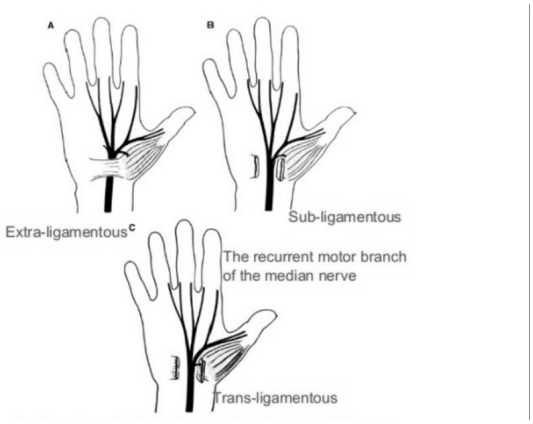
Surgical rule — **cut TCL far ulnar** (radial cut → transects transligamentous branch → **loss of opposition**)

Palmar cutaneous branch — arises **~5 cm (4–8)** proximal to crease, **ulnar side of FCR / between PL & FCR**, **superficial to TCL** → sensation **thenar eminence** → **spared in CTS**

Median nerve laceration vs CTS — laceration proximal to wrist adds **thenar eminence numbness** (palmar cutaneous br.)

Spared muscles in CTS — **FPL, PQ, FDP index** (AIN, proximal); **adductor pollicis** (ulnar)

Weak muscle in CTS — **APB** (only purely median-distal-to-TCL); inserts **radial base of thumb proximal phalanx**



Recurrent motor branch variants

Median–ulnar connection	Site	Direction	Clinical clue
Martin-Gruber	Forearm	Median/AIN → ulnar	Intrinsics spared with high ulnar lesion
Marinacci	Forearm	Ulnar → median	Reverse MG
Riche-Cannieu	**Hand**	**Deep ulnar → recurrent median**	**"All-ulnar hand"**: thenar strength **preserved** despite median laceration / severe CTS
Berrettini	Hand	Sensory, common digital	no motor sparing

– – Active space – –

2 Mechanism

Two compression sites (TO) **proximal edge of TCL** → wrist flexion ↑ compression → **Phalen +** · **hook of hamate** → **hourglass nerve** → **Durkan +, Phalen –**

Compression stage (TO)	Histopathology	Symptoms	Abnormal test
Early	Blood-nerve barrier breakdown, **endoneurial oedema**	**Intermittent** (nocturnal) paraesthesia	**Provocative manoeuvres**
Intermediate	Intraneural connective-tissue thickening; **demyelination**	**Constant** paraesthesia ± weakness	**Thresholds** (Semmes-Weinstein, vibration)
Late	**Axonal degeneration**	Numbness + **atrophy**	**2PD**

Fibres affected first (TO) heavily myelinated **A-beta** (touch, pressure) · pain fibres last → numb but **protective sensation kept** (no trophic ulcers/burns)

Night symptoms (TO) wrist flexion in sleep + fluid shift → **endoneurial oedema** · later daytime (pinch, flexion) → constant

Idiopathic **inflamed synovium** = commonest cause; ↑ pressure → ↓ intraneural blood flow

Lab & histology markers

Histology **oedema, fibrosis, vascular sclerosis**; scattered lymphocytes; ± amyloid; no acute inflammation

Secondary (space-occupying) ganglion, tumour, **gout tophus** (nodule worse after alcohol) → **tissue diagnosis + rheumatology**
✗ Wrong options: splint · excision first

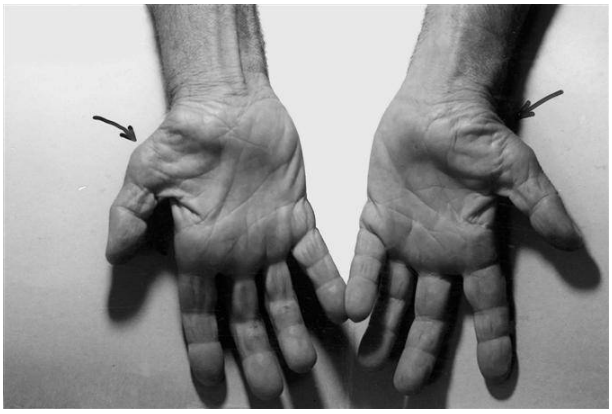
Acute CTS after DRF/carpal dislocation/ORIF → **urgent release**

3 Presentation

Symptoms numbness/tingling **radial 3½ digits**, **night waking**, clumsiness, ↓ grip

Late **thenar atrophy**, weak APB

Sensation spared thenar eminence (palmar cutaneous br.)



Thenar atrophy

4 Diagnosis & Classification

Diagnosis – provocative tests, CTS-6 & electrodiagnostics

Diagnosis **clinical**; EDS not required (AAOS: **limited** rec)

Test	Fact asked
Durkan (carpal compression, 30 s)	**Most sensitive provocative test**
Semmes-Weinstein monofilament	**Most sensitive sensory test / early CTS** (single fibre threshold)
Hand diagram (self-administered)	**Most SPECIFIC (76%)** (TO ×3)
Tinel	High specificity (TO)
Phalen (wrist flexion 60 s)	Less sensitive than Durkan
2-point discrimination	Positive if fails at **5 mm**; innervation density test; best for **nerve regeneration after repair**
Scratch collapse	Supportive

Nocturnal paraesthesia (TO) **pathognomonic** of CTS · also fixed wrist positions (newspaper, mouse)

Phalen vs Tinel Phalen (60 s) **more sensitive + specific than Tinel** (TO, median compression lecture) · Tinel high specificity (TO, other source) – read the options

Scratch collapse (TO) elbows 90°, patient resists ER → scratch over compression site → **sudden loss of ER power** = positive

High sensitivity (TO) Durkan, hand diagram, night pain, Semmes-Weinstein

High specificity (TO) **hand diagram, Tinel**

Four-item rule (TO) **abnormal hand diagram + abnormal Semmes-Weinstein + positive Durkan + night pain** → probability ↑ · **all four negative → probability extremely low**

NOT CTS tests **Murphy** (TO) · **Wartenberg** (cubital tunnel) · **Froment** (ulnar) · **OK sign** (AIN) · hypothenar atrophy (ulnar)

Diagnostic criteria

Clinical criteria median-territory numbness · **nocturnal** numbness · thenar weakness/ atrophy · Tinel + · Phalen + · **2PD loss >5 mm** (= CTS-6 items)

CTS-6 max 26; **≥12 → 80% probability**; **>5 → 25%**

CTS-6 item	Points
Numbness mainly median territory	3.5
Nocturnal numbness	4
Thenar atrophy/weakness	5
Phalen +	5
Loss of 2PD (>5 mm)	4.5
Tinel +	4

– – Active space – –

High/low pretest probability	EDS **won't change probability** → treat clinically
EDS role	confirm (work comp), **prognosis** (**greater latency → slower recovery**, TO); AAOS: obtain if surgery planned

NCS (latency/velocity)	EMG (muscle)
Distal sensory latency >3.5 ms	↑ insertional activity
Distal motor latency >4.5 ms	**Positive sharp waves**
Velocity <52 m/s (less specific)	**Fibrillations**, fasciculations
Side-to-side asymmetry >1 m/s (TO)	↓ motor recruitment, CRDs
Only **large myelinated fibres** tested	= **axonal loss → severe**

Cut-off conflict	**DSL >3.5 / DML >4.5 ms** (CTS note, TO, OB) · **DSL >3.2 / DML >4.3 ms** (Orthobullets NCS note) – read the options
Abnormal motor latency + normal SNAP (TO)	rule out **C8 radiculopathy** · **anterior horn cell disorder** · **isolated recurrent motor branch compression**
NCV after CTR (TO)	**does not return to normal** despite full clinical recovery (retrograde degeneration, incomplete remyelination)
EMG EXCEPT trap	**distal sensory latency** is NCS, not EMG

Clinical severity (TO)	Features
Mild	Failed splint/NSAIDs; no sensory/motor loss; **normal 2PD**, normal EMG
Moderate	Daytime symptoms; **2PD > 15 mm** (TO – verify: abnormal 2PD usually > 5–6 mm); weak abduction, **no wasting**; sensory slowing on NCV
Severe	Constant; **APB wasting**; sensory + motor NCV abnormal → **no role for conservative care**

AAEM severity	Findings
Mild	Sensory only
Moderate	Sensory + motor latency ↑
Severe	**Denervation (fibs, ↓ CMAP)** → early surgery

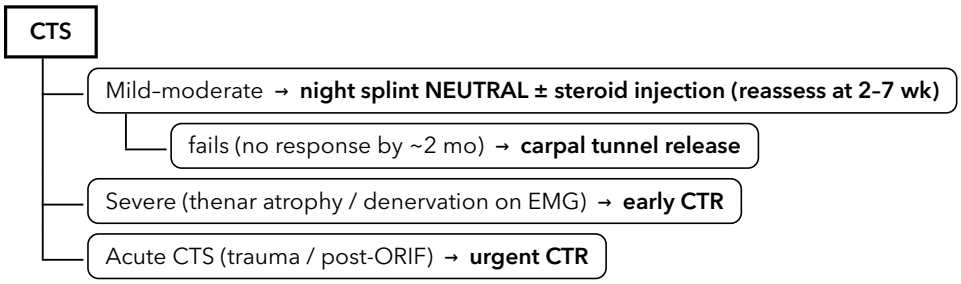
Key radiology findings

Modality	Fact asked
Ultrasound	Median nerve **CSA >10 mm²** at **pisiform / proximal TCL** · **US preferred over MRI** (higher resolution, real time, shows thenar wasting; MRI may be normal early) (TO)
MRI (TO)	Nerve thickened + loss of fascicular pattern **proximal** to retinaculum, narrowed under it; compare areas, no absolute cut-off
X-ray	**Not routine** (not needed for diagnosis)
MRI / CT	Only if **space-occupying lesion** suspected (ganglion, tumour, **gout tophus**)
NCS/EMG	See table above: **DSL >3.5 ms · DML >4.5 ms · CV <52 m/s**; fibs/PSW = severe

Differentials	AIN syndrome (motor only), **pronator syndrome** (thenar numbness + forearm pain), ulnar tunnel, **C6 radiculopathy** (steroid injection helps discriminate)

– – Active space – –

5 Management



Nonoperative care – splint position & steroid injection

Splint	**neutral** (night); **30° extension (cock-up) ↑ pressure** (avoid)
AAOS strong	splints ✓, steroid injection ✓, **surgery > nonop at 6 & 12 mo**, no post-op immobilisation, no magnets
AAOS moderate	NSAIDs no benefit; oral steroids ✓ · **limited** – endoscopic short-term benefit
not appropriate	B6, phonophoresis, bumetanide, PT alone
Steroid injection	**80% improve, 20% of these symptom-free at 1 y** (Green: 81%, mean 3.3 mo)
Injection = diagnostic marker (TO)	relief confirms CTS vs cervical / proximal cause
Injection response	good relief → **good surgical prognosis**; no relief → poorer outcome (but still may benefit)
Best responders to nonop (TO)	symptoms **< 1 year** · manage thyroid / diabetes
Dose/agent	no difference (methylpred = triamcinolone; dose ≠ duration)
With neck pain	injection still indicated (diagnostic + partial relief)
Time rule	effective nonop works in **2–7 wk**; failed 8 wk splint → **CTR** ✗ Wrong options: more splinting

– – Active space – –

Surgical release – indications, guidelines, open vs endoscopic & prognosis

Operative criteria

Indication for CTR	Note
Failed nonop (splint ± injection)	No response in **2–7 wk** → operate
Severe CTS: thenar atrophy / **denervation on EMG**	**Early CTR** (limit further denervation)
Acute CTS (DRF, carpal dislocation, post-ORIF)	**Urgent** release
Classic Hx + exam despite splints	Elective CTR (EDS won't change probability)
Revision	Failure to improve → **incomplete release** commonest → **open** revision

Avoid endoscopic (TO) revision · anatomic variant · prior tendon surgery/scar · limited wrist extension

Open (Kaplan's cardinal line) 1st web → hook of hamate = **superficial palmar arch**; incision in line with **ulnar border of long finger**; stay ulnar

Kaplan perpendiculars (TO) from **radial border of middle finger** → meets Kaplan line at **recurrent motor branch** · from **ulnar border of ring finger** → **ulnar nerve + artery**

Incision (TO) **~6 mm ulnar to thenar crease**, in line with **radial border of extended ring finger** · **Kaplan line = DISTAL limit** · hook of hamate: from pisiform **1.5 cm distal + 1.5 cm radial**

![Kaplan cardinal line – distal limit of incision](../../TO notes/attachments/TOn-notes-compression-neuropathies-median-nerve-compression-sep--p06-f017.jpg)

Open release (TO) incision **ulnar to palmaris longus** (radial → palmar cutaneous br.) · release forearm fascia + TCL + distal aponeurosis (TO) · antebrachial fascia release of no benefit (Orthobullets) – read the options

Not in primary CTR (TO) neurolysis · epineurotomy · synovectomy

Contraindications to CTR (TO) **untreated hypothyroidism** · **uncontrolled diabetes** · other metabolic neuropathies

Endoscopic techniques (TO) **Chow = two-portal** · **Agee = single-portal** · landmarks: **index–pisiform line**, **4th metacarpal line**

No benefit internal neurolysis, **tenosynovectomy** (idiopathic), antebrachial fascia release, Guyon release (decompressed anyway)

Lengthen TCL only if flexor tendon repair done (allows wrist flexion splint)

Endoscopic vs open endoscopic: ↓ **early pain/analgesia** (main advantage), earlier RTW, better early key pinch, ↓ scar tenderness; **long term = open**

Endoscopic cons complications **more frequent + more serious** (TO) · learning curve, **incomplete release** (commonest complication), ↑ revision, ↑ nerve injury risk

Endoscopic contraindications (TO) **revision/recurrent CTS** · anatomic variant (APB wasting without sensory change) · previous tendon surgery/scar · **limited wrist extension**

Recovery **pinch 6 wk**; **grip 100% of pre-op by 12 wk (3 mo)**

Persistent symptoms at 1 y **2% moderate · 20% severe**

Poor prognosis	**severe EDS (EMG denervation)** = strongest predictor; DM; poor injection response; male
Good prognosis	**night symptoms**, relief with injection
Worst EDS option	**DML >4.5 AND DSL >3.5** (e.g. 5.0 / 4.0)

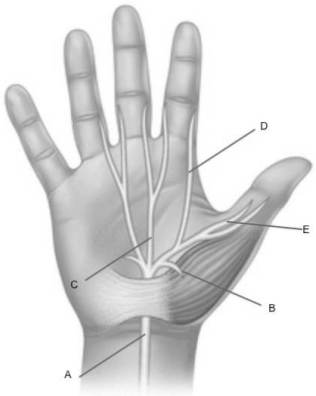
Perioperative care & complications

Antibiotics	**not indicated** for clean elective hand surgery → **standard prep only**
Analgesia	**ibuprofen + paracetamol only** for soft-tissue procedures (CTR, trigger finger, cubital tunnel)
Post-op	neutral night splint 2–3 wk for comfort; ROM/nerve glides immediately; strengthen after 4 wk

6 Complications

Complication	Key numbers / fact
Scar tenderness	**Most common** (19–61%); massage
Pillar pain	Thenar/hypothenar ache; 41% → 25% → 6% (1, 3, 12 mo)
Incomplete TCL release	**Commonest cause of failure/revision**; ↑ with endoscopic
Recurrence	1% 5 y, 1.6% 10 y
Recurrent motor branch	Open 3% / endo 0.7%; transligamentous; → repair or **opponensplasty (FDS ring → APB, Bunnell)**; EIP/PL alternatives · TO list: FDS (**Bunnell, Royle-Thompson**), EIP, **ADM (Huber)**, **PL (Camitz)**
Palmar cutaneous branch	**Open 19%** / endo 2.4%; painful neuroma
Proper digital n. index	→ **lumbrical weakness** (index MCP flexion + PIP extension weak)
Arterial arch	Endo 0.02%

Revision CTR 25% complete relief, 25% none



Nerve diagram: injury to D = proper digital nerve index → lumbrical weakness

– – Active space – –

X vs Y

	CTS	Pronator syndrome	AIN syndrome	C6 radiculopathy
Sensory	Radial 3½ digits, **night**	Radial 3½ **+ thenar eminence**	**None** (pure motor)	Thumb/index + neck
Motor	APB/OP	FPL, FDP-I, PQ ± thenar	**FPL, FDP-I, PQ (OK sign)**	Biceps/wrist ext
Provocative	Durkan, Phalen, Tinel at wrist	Resisted pronation; Tinel forearm	–	Spurling
Night sx	**Yes**	No	No	±

	Open CTR	Endoscopic CTR
Early pain, RTW, key pinch	Worse	**Better**
Scar tenderness	↑	↓
Incomplete release / revision	↓	↑
Long-term	=	=
Revision cases	✓	**No (contraindicated)**

How it's asked

"Most specific test?"	→ **Hand diagram** (TO)
"Most sensitive provocative?"	→ **Durkan**
"Compression at hook of hamate – which test negative?"	→ **Phalen** (Durkan positive, hourglass nerve) (TO)
"Abnormal motor latency, normal SNAP – rule out?"	→ **C8 radiculopathy / anterior horn cell disease / isolated recurrent branch compression** (TO)
"Most sensitive early sensory?"	→ **Semmes-Weinstein**
"Not a test for CTS"	→ **Murphy** / Wartenberg
"CTS not common in"	→ **Addison's** (TO)
"Not true"	→ **endoscopic better in recurrent cases** ✗ Wrong options: – open is used
"Contents EXCEPT"	→ **FCR**
"TCL attaches to"	→ **trapezium + hook of hamate**
"Recurrent branch cut – deficit & how?"	→ **opposition lost; cut TCL radially**
"Median laceration but can oppose thumb"	→ **Riche-Cannieu**
"Laceration vs CTS – extra finding"	→ **thenar eminence numbness**
"Which muscle weak in CTS?"	→ **APB** ✗ Wrong options: FPL, PQ, FDP-I, adductor pollicis

"Cock-up splint at 30°"	→ **↑ pressure** (use neutral)
"Counsel on injection"	→ **80% improve / 20% last 1 y**
"Splinted 8 wk, still symptomatic"	→ **CTR**
"Main advantage of endoscopic"	→ **↓ post-op pain**
"Pathognomonic symptom of CTS?"	→ **nocturnal paraesthesia** (TO)
"Distal limit of open CTR incision?"	→ **Kaplan cardinal line (superficial palmar arch beyond)** (TO)
"Recurrent motor branch variant at highest risk?"	→ **origin from ulnar border of median nerve** (TO)
"Grip at 3 mo"	→ **100%**
"Persistent sx at 1 y, severe CTS"	→ **20%**
"Predicts poor outcome"	→ **EMG/EDS severity**
"Reduce SSI"	→ **standard prep; no antibiotics**
"Weak index MCP flexion + PIP extension after CTR"	→ **proper digital nerve to index (lumbrical)**
"Volar wrist nodule worse after alcohol + CTS"	→ **gout → tissue Dx + rheumatology**

– – Active space – –

– – Active space – –

CLEFT HAND

– Hand · 12 / 103 –

1 Basics

Definition & epidemiology

Synonyms	**lobster-claw deformity**, ectrodactyly, split hand
Definition	**central longitudinal deficiency** → absence of ≥1 central digits (index, middle, ring) of hand or foot
Incidence	**rare (1:10,000–1:90,000)**
Sex	♂:♀ **5:1** (Orthobullets + TO)
Site	hands, **usually bilateral**; often feet too (**cleft foot**)
Absent metacarpals	present in cleft hand → separates from **symbrachydactyly**
Small finger	**always present** on ulnar side

Pathogenesis

AER defect	wedge-shaped degeneration of **central part of apical ectodermal ridge** → loss of signalling → progress zone stops proliferating → central ray defect
Swanson	**type I failure of formation** (longitudinal arrest) of central ray

Genes & inheritance

	Typical cleft hand	Atypical cleft hand (TO)
Shape	**V-shaped** central cleft	**U-shaped**
Laterality	Bilateral	Unilateral
Feet	Cleft foot common	–
Nature	Genetic	Form of **symbrachydactyly** (absent/syndactylous/hypoplastic digits)
Inheritance	See below	**Sporadic, not inherited**; few syndromes

Inheritance (typical)	**autosomal dominant**, reduced penetrance (**70%**) (Orthobullets, OB) · **X-linked dominant**, incomplete penetrance (TO) – read the options; OB pedigree answer = **AD**
Severity	inherited forms **worsen each generation**; sporadic cases = de novo mutation
Gene	deletions/inversions/translocations of **7q → split hand–split foot syndrome**
Counselling	affected families → **genetic counselling**
Associations	**EEC syndrome** (ectrodactyly–ectodermal dysplasia–cleft lip/palate), **sensorineural hearing loss**, syndactyly, polydactyly

3 Presentation

History ± family history

Symptoms aesthetic > functional limitation

Exam absent/short **central (3rd) ray** · ± absent radial digits · ± **syndactyly of ulnar digits** · ± feet



Typical V-shaped cleft hand

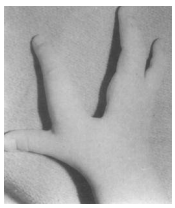
4 Diagnosis & Classification

Key radiology findings

Views AP, lateral, oblique **bilateral hands** + **feet** if involved

– – Active space – –

Manske & Halikis classification (thumb web)

Type	Web	Characteristics	Image
I	**Normal**	Thumb space not narrowed	
IIA	Mildly narrowed	Thumb space mildly narrowed	
IIB	Severely narrowed	Thumb space severely narrowed	
III	Syndactylized	Thumb + index syndactylized, web obliterated	
IV	Merged	Index ray suppressed, thumb web merged with cleft	
V	Absent	Thumb suppressed, ulnar rays remain, no thumb web	

Most accepted Manske & Halikis (TO)

Spot diagnosis (TO) missing middle ray + **normal thumb–index web → type I**

5 Management

Priorities

Thumb web **greater priority** than central cleft

Sequence thumb reconstruction **should not precede cleft closure** (compromises skin flaps)

Cleft closure local tissue flaps from cleft + **close intermetacarpal space**

Thumb/web options web deepening, tendon transfer, rotational osteotomy, toe-to-hand transfer

Treatment by Manske type

Type	Web surgery	Central cleft
I	**None** (normal web, no impairment) (TO)	**Cleft closure + reduce intermetacarpal space** ± excise polydactylous bone (TO)
IIA	**4-flap Z-plasty** (TO)	Dorsal/palmar cleft flaps + reduce IM space
IIB	Cleft flaps transposed into thumb web	Close with web procedure; IM space reduced by **ligament reconstruction**
III	**Priority to web**: cleft flaps + thumb–index syndactyly release + skin graft	Local pedicle flaps; ligament reconstruction
IV	Usually none (less impairment); **MCP arthrodesis**, excise excess skin	–
V	**None**; consider **free toe-to-hand transfer** → converts to type IV	–

Conflict – type I **observation** for types I & IV (Orthobullets) · **no web surgery but cleft closure + IM reduction** for type I (TO) – read the options

Operative indication (Orthobullets) types **IIA, IIB, III, V** → web + thumb + cleft reconstruction

Prognosis

Function depends on **1st web space involvement**

Classic line **aesthetically displeasing, not functionally limiting**

X vs Y

	Cleft hand	Symbrachydactyly
Metacarpals	**Absent**	Present (rudimentary nubbins)
Shape	V	U ("atypical cleft")
Laterality	**Bilateral**	Unilateral
Feet	Involved	Not
Inheritance	Familial	Sporadic

How it's asked

"Pedigree for cleft hand inheritance?"	→ **autosomal dominant** ✗ Wrong options: AR, X-linked recessive, X-linked dominant, mitochondrial
"Missing middle ray, normal web – type and treatment?"	→ **Manske type I: no web surgery; cleft closure + reduce intermetacarpal space** (TO)

– – Active space – –

"Type IIA web surgery?"	→ **4-flap Z-plasty** (TO)
"Which comes first, thumb or cleft?"	→ **thumb web has priority, but thumb reconstruction must not precede cleft closure**
"Feature separating cleft hand from symbrachydactyly?"	→ **absent metacarpals**, bilateral, feet involved
"Associated syndrome?"	→ **EEC syndrome**

CONGENITAL CLASPED THUMB

– Hand · 13 / 103 –

1 Basics

Definition & epidemiology

Definition	**flexion-adduction deformity of thumb** persisting **beyond 3rd–4th month** of life; often **bilateral**
Sex	♂:♀ ≈ **2.5:1**
Predisposing	consanguinity, family history

Pathoanatomy

Key lesion	**attenuation/absence of EPB** (commoner, mild cases) **or EPL, or both**
Also	1st web contracture · adductor pollicis / 1st dorsal interosseous contracture · global MCP instability · abnormal MCP cartilage

Genes & lab markers

Inheritance	**autosomal dominant**, variable expressivity; may be **sporadic**

Associations

Syndromes	**arthrogryposis** · digitotalar dysmorphism · **Freeman-Sheldon** · X-linked **MASA**
Lower limb	**congenital vertical talus**, bilateral **CTEV**
Upper limb	flexion deformities of the four fingers

3 Presentation

History	persistent flexion-adduction **beyond 3–4 mo**, usually bilateral; family history
Examination	type of clasped thumb (passive extension/abduction possible?) + associated anomalies



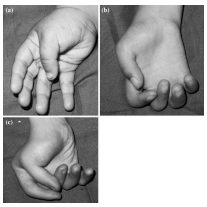
Congenital clasped thumb

- - Active space - -

4 Diagnosis & Classification

Diagnosis **clinical**

Tsuyuguchi classification

Type	Features	Image
I – Supple	Thumb **can be passively abducted & extended**; no other digital anomaly	
II – With contracture	**Cannot passively extend/abduct**; ± other digital anomalies	
III – Rigid	With **arthrogryposis** + marked soft-tissue deficit	

5 Management

Congenital clasped thumb (all types)

- First line → **SERIAL SPLINTING + STRETCHING 3–6 MO (start ~6 mo of age)**
  - Type I with EPL or EPB present → **good definitive result**
- Fails / residual deficit → **SURGERY (delay to 3–5 y)**
  - Type I–II, residual active extension deficit → EIP → **EPL TRANSFER**
  - Type II–III, web deficit / MCP contracture → **THUMB RECONSTRUCTION**
  - Joint deformity not corrected by release + transfer → **ARTHRODESIS**

First-line treatment – serial splinting

All types **serial splinting of thumb 3–6 mo** = first line
✗ Wrong options: observation (will not correct)

Timing begin **~6 months** of age

Good results **Type I** when **one of EPL/EPB present**

Poor results Type I with **both EPL & EPB absent**; **Type II & III**

Operative options

Procedure	Indication
EIP → EPL transfer	**Type I or II** with residual deficit of active extension
Thumb reconstruction	Failed nonop; **1st web soft-tissue deficit (Type III)**; **Type II/III with significant MCP contracture**
MCP arthrodesis	**Last resort**: severe deformity / MCP instability not overcome by skin release + transfer

Reconstruction age **3–5 years**

Web widening dorsal rotational advancement flap · **4- or 5-flap Z-plasty**

Web deepening	release thenar origins from TCL · release 1st MCP capsule
Tendons	**FPL Z-lengthening** in forearm; **EPB + EPL absent → tendon transfer**; **isolated EPB absence → usually no transfer**

– – Active space – –

6 Complications

List	cosmetic appearance · **MCP instability** · reduced thumb function

X vs Y

	Clasped thumb Type I	**Type II**	**Type III**
Passive extension/ abduction	Yes	No	No (rigid)
Association	None	± digital anomalies	**Arthrogryposis**
Splinting result	**Good**	Poor	Poor
Usual surgery	EIP → EPL if residual lag	Web reconstruction ± transfer	Web reconstruction ± arthrodesis

How it's asked

"8-mo-old, restricted passive thumb extension/ abduction – first line?"	→ **serial splinting** ✗ Wrong options: observation · web reconstruction · tendon transfer · MCP arthrodesis
"Commonest deficient tendon?"	→ **EPB**
"Syndrome with rigid Type III?"	→ **arthrogryposis**
"Type I–II with residual active extension deficit?"	→ **EIP → EPL transfer**
"Age for reconstruction?"	→ **3–5 y**

– – Active space – –

CONGENITAL RADIAL HEAD DISLOCATION

– Hand · 14 / 103 –

1 Basics

Epidemiology & pathoanatomy

Incidence	rare, **<1 per 100,000**
Direction	**posterior ~70%** > anterior 15% = lateral 15%
Laterality	**bilateral in majority**; unilateral congenital form can occur (Agnew)
Pathoanatomy	**failure of capitellum development** → loss of contact pressure → abnormal radial head → altered PRUJ → abnormal ulna
Radius	**bowed**, appears **too long vs ulna**
Associations	other congenital anomalies; **radial longitudinal deficiency**, **proximal radioulnar synostosis**; syndromes; familial

3 Presentation

Symptoms	often **asymptomatic**, **painless**; radial head prominence (posterolateral)
ROM	limited, **especially extension & supination**; usually functional

4 Diagnosis & Classification

Diagnostic criteria – congenital vs traumatic

Clinical	Radiographic
Bilateral	**Dome-shaped (convex) radial head**, long narrow neck
Present at birth, familial	**Hypoplastic / absent capitellum**
Other congenital anomalies	**Short ulna / long radius**, bowed radius
No trauma history	Deficient trochlea, prominent medial epicondyle
Irreducible by closed methods	Grooving distal radius; anterior curvature of posterior ulnar outline

Table I: Criteria for diagnosing congenital dislocation of radial head[3]

McFarland's criteria	
1	Relatively short ulna or long radius
2	Hypoplastic or absent capitellum
3	Partially defective trochlea
4	Prominent ulnar epicondyle
5	Groove in distal radius
6	Dome shaped radial head with long narrow neck
Mardam-Bey and Ger criteria	
1	Bilateral involvement
2	Familial history
3	Concomitant congenital anomalies
4	No history of trauma
5	Not reducible by closed methods
6	Dislocation seen at birth

CRHD diagnostic criteria

Key radiology findings

Radiocapitellar line — line along radius long axis should intersect capitellum on all views → misses = dislocation

X-ray — radial head **posterior to capitellum**; **convex "dome" radial head**; long, bowed radius

Next step (suspected CRHD) — **X-ray contralateral elbow** → confirms bilateral



Posterior CRHD, convex radial head

5 Management

CRHD

- Asymptomatic / functional ROM → **OBSERVATION**
- Pain / restricted motion / cosmesis (usually adulthood) → **RADIAL HEAD RESECTION**
  (treat each elbow on its own symptoms)

Diagnosis (bilateral, vs traumatic) & management – observation vs late excision

First line — **observation**
✗ Wrong options: closed reduction · open reduction ± osteotomy · child-abuse workup without other findings

Radial head resection — adult, symptomatic → **↓ pain**, improves **forearm rotation**; no change in flex-ext or carrying angle (Bengard)

Resection downside — **>25% wrist pain** post-op

– – Active space – –

Avoid open reduction	high **redislocation** (even with ulnar/radial osteotomy or annular ligament reconstruction) — hypoplastic capitellum, dome head, long radius
Avoid radial head arthroplasty	redislocation (undeveloped capitellum, non-collinear head)
Not interposition arthroplasty	for ulnohumeral arthritis, not radial head disease
Not physiotherapy	no gain in pain/ROM

X vs Y

	Congenital RHD	Traumatic RHD
Laterality	**Bilateral**	Unilateral
Radial head	**Convex / dome**	Normal concave
Capitellum	**Hypoplastic**	Normal
Trauma Hx	None	Present
Closed reduction	**Difficult / irreducible**	–

How it's asked

"Atraumatic nontender posterolateral elbow bump, dome radial head — next step?"	→ **X-ray contralateral elbow**
"7-y with convex radial head, vague fall, lacks 20° extension — treatment?"	→ **Observation** ✗ Wrong options: abuse workup · reduction
"Bilateral CRHD, right painful + stiff, left minimal loss — plan?"	→ **resect right radial head + nonop left**
"Why not open reduction?"	→ **redislocation**
"Commonest direction?"	→ **posterior**
"Complication after radial head excision?"	→ **wrist pain (>25%)**

CONGENITAL RADIAL ULNAR SYNOSTOSIS

– Hand · 15 / 103 –

1 Basics

Aetiology & associations (failure of segmentation, XXY)

Embryology	single cartilaginous anlage divides **distal → proximal** into radius + ulna in **7th week** in utero; proximal ends share common perichondrium
Cause	**failure of segmentation / differentiation** → proximal bony bridge ✗ Wrong options: failure of formation · trauma · transverse forearm deficiency
Demographics	♂ **>** ♀ **(3:2)**; **~60% bilateral** (TO: "more than half")
Bilateral	**~60%** (Orthobullets, TO lecture 1) · **80%** (TO lecture 2) – read the options
Family history	**~20%**; familial cases **autosomal dominant**
Associated syndromes (~30%)	**Apert** · Carpenter · **arthrogryposis** · mandibulofacial dysostosis · **Klinefelter**
Associated anomaly	posterior **radial head dislocation**
Associations (TO)	+ **Poland syndrome**

Genes & lab markers

Chromosomes	**sex chromosome duplication → Klinefelter 47 XXY** (most common chromosomal association)

✗ Distractor karyotype	Associations
45 XO (Turner)	Malignant hyperthermia, sexual infantilism, scoliosis, **short 4th MC**, webbed neck
Trisomy 21	C1–2, hip, patellofemoral instability, scoliosis
Trisomy 18	Cardiac defects; 5% survive 1 y
Partial 15 deletion (Prader-Willi)	Obese hypotonia, insatiable appetite, hip dysplasia, scoliosis

3 Presentation

Symptoms	**painless**; usually **asymptomatic**, noticed by parents/teachers
Age at diagnosis	average **6 y** (unilateral may go unnoticed until adolescence)
Exam	**fixed forearm pronation** (average **30°**); **elbow flexion/extension preserved**; no active or passive rotation
Compensation	**shoulder abduction → for lost pronation** · **shoulder adduction → for lost supination** · wrist hypermobility
Tasks	keyboard/tabletop ← pronation deficit; eating, washing face, catching ball ← supination deficit

– – Active space – –

Toddler clue rotates elbow in front of body to bring bottle to mouth; abducts shoulder to pick up blocks

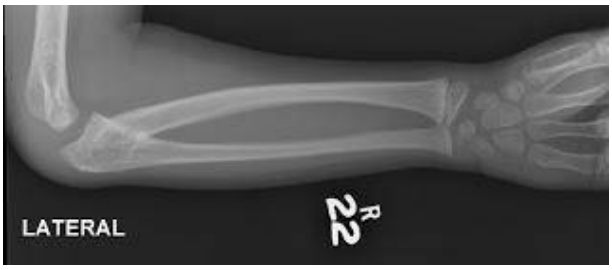
4 Diagnosis & Classification

Key radiology findings

Views AP + lateral forearm and elbow; **X-ray confirms**
✗ Wrong options: CT · MRI · arthrocentesis (no extra info)

Findings **proximal radioulnar synostosis** · **radius wide and bowed** · **ulna narrow and straight** · radial head ± dislocated/malformed

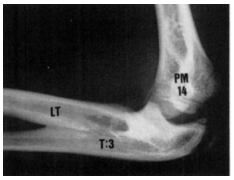
MRI (TO) shows **fibrous synostosis**



Proximal radioulnar synostosis

Fall + fixed pronation synostosis is congenital, not from acute trauma 3 days ago

Cleary & Omer classification

Type	Synostosis	Radial head	
1	**Cartilaginous** (no osseous)	Normal, reduced	
2	Osseous	Normal, reduced	
3	**Long osseous**	**Hypoplastic, posteriorly dislocated** (TO)	
4	**Short osseous**	**Mushroom-shaped, anteriorly dislocated**	

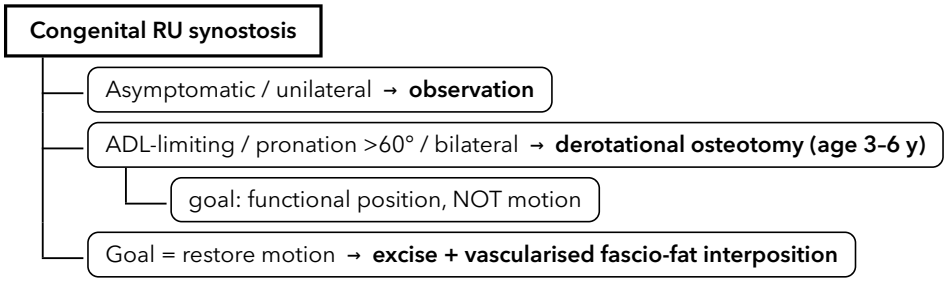
Cleary vs Wilkie (TO) Cleary & Omer = the better classification

Mital classification (TO) — **I fusion without radial head dislocation · II fusion with radial head dislocation**

Wilkie classification (TO)

Type	Features
1	Medullary canals of radius + ulna joined; usually **bilateral**; radial head **not** dislocated
2	Radial head **dislocated** (anterior or posterior); usually **unilateral**

5 Management



Management – observation vs osteotomy/mobilisation indications

Default — **observation** (most cases; surgery rarely needed) – 14-mo-old with ↓ supination → **observe**

Slight pronation — well tolerated (keyboard position)

Operative criteria

Indication	Type
Deformity **limiting activities** (sport, hygiene, eating)	Absolute
Pronation >60°	Relative
Bilateral	Relative (✓ answer over 45° pronation, 30° supination, age <3, affected sibling)

Optimal age — **3–6 y** (average ~5 y)

Derotational osteotomy (procedure of choice)

Goal — **functional resting position**; does **not** restore motion

Sites — at synostosis (proximal diaphysis; narrow space → tightness, loss of correction, NV risk, **PIN palsy**) · **distal to synostosis at different levels** (distributes rotation → safer) · radius distal diaphysis alone

– – Active space – –

Timing immediate · **delayed at 10 days** (Lin: percutaneous drill osteotomy + cast, manipulate in plaster at 10 d) (TO) · gradual with **Ilizarov** (**lowest NV complications**)

Position	Orthobullets	OB explanation	TO
Unilateral	**0–30° pronation**	30° pronation	0–30° pronation
Bilateral – dominant	**0–15° pronation**	10–20° pronation	20° pronation
Bilateral – nondominant	**Neutral**	Neutral	Neutral

Interposition procedures abandoned (TO) **double-level osteotomy preferred** (vs through / distal to synostosis) · goal functional position, rotation does not return

Age (TO)	Fixation after osteotomy
< 5 yr	Cast
5–10 yr	**Elastic nail**
> 10 yr	**Plate**

Old teaching nondominant in 10–15° (Orthobullets) / 20° (TO) **supination** → no longer recommended (keyboards, phones)

Bilateral older OB goal one forearm modest pronation, other modest supination

Synostosis excision (mobilisation)

Goal restore active rotation; gain usually **slight**, unsatisfactory results

Prevent recurrence **interpose material between radius and ulna**: **vascularised fascio-fat graft** (vascularised > free fat) · free/pedicled fat, muscle, Gelfoam, Micropore

Do not prevent recurrence soft-tissue reconstruction (capsule, annular ligament, biceps) · post-op casting · post-op radiation (not described) · **anconeus interposition**

6 Complications

Complication	Key fact
Recurrence of synostosis	**~100%** with excision alone or **anconeus** interposition; **0%** (negligible, TO) with vascularised fascio-fat graft
Loss of correction	Casting after osteotomy → **15–20°** (Orthobullets) · up to 20° (TO, Lin method)
Compartment syndrome	Up to **36%**; corrections **>60°**; ± prophylactic fasciotomy
Nerve palsy	**PIN** (proximal osteotomy), AIN, radial; most resolve in 3 mo

Osteotomy complications (TO) compartment syndrome · **nonunion** · **elbow stiffness** · **recurrence of deformity by remodelling**

How it's asked

"Fixed pronation, delayed milestones, obese uncoordinated boy — karyotype?"	→ **47 XXY** ✗ Wrong options: 45 XO · trisomy 21/18 · 15 deletion
"Proximal RU synostosis is caused by?"	→ **failure of segmentation** ✗ Wrong options: formation · trauma · transverse deficiency · always needs surgery
"Indication for future surgery?"	→ **bilateral involvement** ✗ Wrong options: 45° pronation · 30° supination · age <3 · sibling
"Excision to restore motion — step to prevent recurrence?"	→ **interposition between radius and ulna** ✗ Wrong options: soft-tissue reconstruction · radiation · casting
"14-month-old, ↓ supination, abducts shoulder — next step?"	→ **observation** ✗ Wrong options: CT · MRI · aspiration · reduction
"Long osseous synostosis, hypoplastic posteriorly dislocated radial head — Cleary & Omer type?"	→ **Type 3** (TO)
"Short synostosis, mushroom radial head dislocated anteriorly?"	→ **Cleary Type 4**
"Shoulder abduction compensates for?"	→ **loss of pronation**

– – Active space – –

CONGENITAL TRIGGER THUMB

– Hand · 16 / 103 –

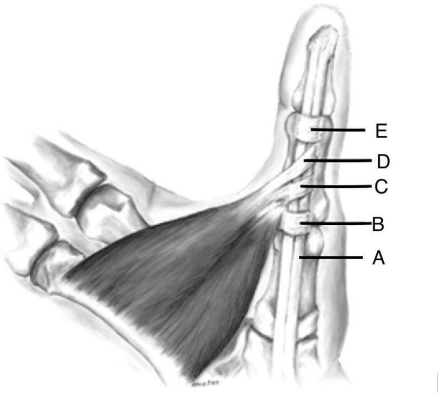
1 Basics

Epidemiology & pathoanatomy

Prevalence	**3 per 1,000** by age 1; ♂ **=** ♀; **25% bilateral**
Separate entity	distinct from adult acquired trigger thumb (and from pediatric trigger finger)
"Congenital" = misnomer	usually **acquired / developmental**; not noted at birth; rare AD reports (variable penetrance)
Pathoanatomy	**thickened FPL** (collagen degeneration + synovial proliferation) → ↑ diameter vs **A1 pulley** → abnormal gliding

Thumb pulleys

Label (OB217410)	Structure	Release?
A	FPL tendon	No – cutting → loss of IP & MCP flexion
B	**A1 pulley**	✓ **site of triggering**
C	Av pulley	No
D	Oblique pulley	No – cutting → **bowstringing**
E	A2 pulley	No – no role

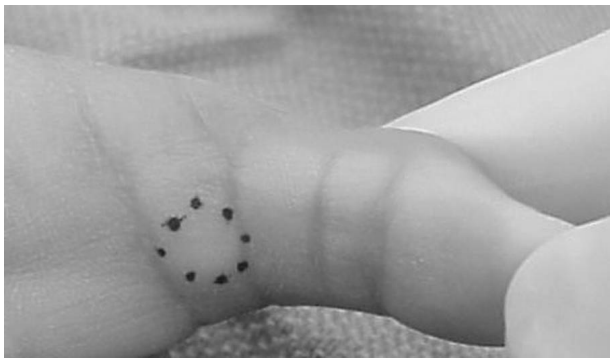


Thumb pulleys: B = A1

3 Presentation

Typical	**fixed flexion deformity of thumb IP joint** (not triggering), painless; rare trauma or family Hx
Notta's node (TO)	prominent **FPL tendon nodule** at **base of thumb / over MCP** = **pediatric trigger thumb**

Exam	loss of passive IP extension, **unchanged by wrist/MCP flexion**; no intrinsic IP contracture; NV intact; ± MCP hyperextension ✗ Wrong options: FPL shortening
Natural history	triggering → **fixed contracture**; spontaneous resolution unlikely **after age 2**



Notta's node outlined

4 Diagnosis & Classification

Key radiology findings

Diagnosis	**clinical**; imaging not required
X-ray	only if **trauma history** (AP/lateral hand ± thumb views) → **usually normal** (IP flexion, no osseous lesion)

5 Management

Pediatric trigger thumb

- Flexible, <2 y → **passive extension exercises + intermittent extension SPLINTING**
- Triggering (not locked), <1 y → **OBSERVE ~6 mo**
- Fixed >12 mo / locked 6 mo at age 2 / failed nonop → **OPEN A1 PULLEY RELEASE**

Diagnosis (Notta's node) & A1 pulley release timing

First line, flexible, 1-y-old (TO)	**passive extension exercises + intermittent extension splinting** ✗ Wrong options: release · exercises alone

– – Active space – –

Splinting (TO) — **more successful than observation / exercises alone**; IP hyperextension, block MCP hyperextension; **6–12 wk** (several months; not 30 days); high drop-out

Treatment	Resolution
Passive exercises / observation <2 y (TO)	**30–60%**
Spontaneous after 2 y	**<10%**
Splinting + exercises (TO)	**50–60%**
Nonop after 3 y (OB)	**50%**
Baek: no treatment, 48 mo	60%
A1 release	**65–95%** (Orthobullets) · **95%** (OB)

Delay to age 2 — **no sequelae**
✗ Wrong options: permanent contracture · motor delay · hand dominance change (handedness ~4 y)

Operative criteria

Source	Indication for A1 release
Orthobullets / TO	**Fixed deformity >12 mo** or failed nonop
Shah	Fixed >12 mo; consider surgery **after 2 y**
Marek	**Locked for 6 mo at age 2 → surgery**; triggering <1 y → observe 6 mo
OB explanation	**>3 y** or failed splinting / stretching
Dittmer	**Stage IV → immediate surgery** (4.6× surgery risk); stage II–III observe ≥1 y

Technique — **open** release (gold standard); transverse incision MCP flexion crease; **protect radial digital nerve**; watch nodule glide on passive IP extension

Avoid percutaneous — incomplete release, radial digital n. injury, FPL laceration; no advantage (GA needed anyway)

Not indicated — nodule resection · volar plate release · FPL lengthening · steroid injection (difficult, no benefit)

6-y fixed IP flexion + nodule — **complete A1 release**
✗ Wrong options: no treatment (won't resolve)



Intra-op A1 pulley

6 Complications

Digital nerve injury	high risk (close to tendon/A1)
Bowstringing	from **oblique pulley release**
Others	wound (scar contracture, abscess, infection), IP flexion deficit; Marek: no major complications/recurrence

X vs Y

	Pediatric trigger thumb	Camptodactyly	Clasped thumb	Blauth I hypoplasia
Digit	**Thumb IP**	**Little finger** PIP	Thumb	Thumb
Nodule	**Notta's node**	–	–	–
Treatment	**A1 release**	Stretching	–	**Nonop**

How it's asked

"Notta's node characteristic of?"	→ **pediatric trigger thumb** (TO) ✗ Wrong options: RLD, synostosis, constriction ring
"First treatment, flexible trigger thumb, 1-y girl?"	→ **passive exercises + intermittent extension splinting** (TO)
"2-y fixed IP flexion, nodule, splinted 6 mo – diagnosis & Rx?"	→ **pediatric trigger thumb; open A1 release** ✗ Wrong options: nodule resection · percutaneous
"Delay surgery from 13 mo to 2 y causes?"	→ **no sequelae**
"6-y fixed thumb IP flexion + nodule?"	→ **complete release of proximal annular (A1) pulley**
"IP not passively extendable, unchanged with wrist flexion?"	→ **A1 release** ✗ Wrong options: FPL lengthening · volar plate · 30-day splint
"Which structure to release (labelled pulleys)?"	→ **B = A1** ✗ Wrong options: oblique → bowstringing
"Structure in intra-op photo?"	→ **A1 pulley**

– – Active space – –

CONSTRICTIVE RING SYNDROME (STREETER'S DYSPLASIA)

– Hand · 17 / 103 –

1 Basics

Epidemiology

Synonyms	amniotic band syndrome (ABS), Streeter's dysplasia, **early amnion rupture sequence**
Incidence	**1:15,000** (Orthobullets, OB) · **1:10,000** "rare" stated true (TO) – read the options
Sex	M = F
Site	**hands/fingers 80%**, >90% distal to wrist; distal > trunk; **central digits** (long 28%, ring 27%, index 23%); average **3 limbs**
Leg bands (TO)	mostly **zone 2** (knee–ankle); tibia/fibula fractures may occur at band level (TO) ✗ Wrong options: zone 1
Risk factors	**low birth weight (<2500 g)**, **prematurity (<37 wk)**, maternal drugs, trauma in pregnancy, attempted 1st-trimester termination; abnormal gestational history **60%** ✗ Wrong options: late gestation / high birth weight
Mortality	infant 4.6%, perinatal 12.7%

Genes & inheritance

Inheritance	**sporadic, not hereditary** ✗ Wrong options: AD/AR/X-linked

2 Mechanism

Extrinsic theory (most accepted)	ruptured amnion → fibrous strands **wrap limb circumferentially**
Distal to band	lymphatic + NV compression → **lymphoedema**, altered circulation, neuropathy, amputation

3 Presentation

Associated conditions (clubfoot, cleft lip/palate) & sporadic inheritance

Exam	**normal anatomy proximal to ring**; band **perpendicular** to digit/limb axis
No amputation	look for secondary syndactyly, bony fusion, proximal sinus tracts
Clubfoot	**commonest associated foot condition**; **rigid**; casting poor (6% success, 77% need band Z-plasty first)
Clubfoot rate	12–56% (OB) · up to 25% (OB) · ~⅓ (Foulkes) · 38% (Koskimies)

Associated	**clubfoot**, **cleft lip/palate**, craniofacial defects, **cardiac defects**, **encephalocele** (off-midline), **acrosyndactyly**, pseudarthrosis, LLD, hemihypertrophy, renal, haemangioma, **lymphoedema** (TO: non-orthopaedic)
Anterolateral tibial bowing	listed as a manifestation (OB) · "not associated / NF1" (OB, 2 questions) — read the options
NOT associated	**bronchopulmonary dysplasia** (prematurity) · **complex syndactyly** (TO) · polydactyly · vertical/oblique talus · tarsal coalition · pes planovalgus · cutis aplasia · renal aplasia · dural ectasia
Syndactyly type (TO)	**simple incomplete, fenestrated** (slits proximally), bones not fused



Acrosyndactyly with fenestrations

4 Diagnosis & Classification

Key radiology findings

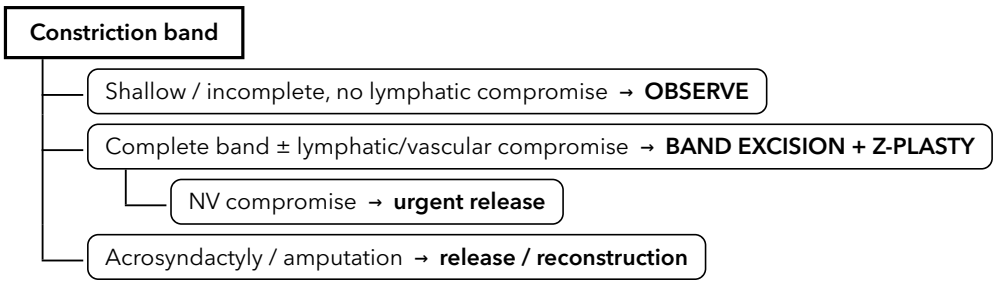
Antenatal US	diagnosis possible at **end of 1st trimester**; chorioamniotic membrane separation → bands
Diagnosis	clinical at birth

– – Active space – –

Patterson classification

Type	Description	Treatment	Image
I	**Simple constriction ring**	Observe; **release if circulation compromised**	
II	Ring + distal deformity (hypoplasia, **lymphoedema**)	**Circumferential Z-plasties**	
III	Distal fusion (**acrosyndactyly**)	**Syndactyly release** (early)	
IV	**Amputation**	Reconstruction (bone lengthening, web deepening)	

5 Management



Treatment – band release & Z-plasty

Indication	**complete circumferential band interfering with lymphatic drainage → excision + Z-plasty** (one-stage possible)
Trap	incomplete bands not affecting lymphatics → do not resect
Acrosyndactyly + distal oedema	**early** release → better prognosis
Clubfoot with band	**Z-plasty band release before** clubfoot correction
Follow-up	to skeletal maturity (contracture, growth disturbance)
Prognosis	depends on band location & severity

How it's asked

"Streeter band of leg — EXCEPT"	→ **most in zone 1** (TO) ✗ Wrong options: ; zone 2
"Not a feature of Streeter's dysplasia"	→ **complex syndactyly** (TO)
"Associated foot condition"	→ **clubfoot (CTEV)** ✗ Wrong options: vertical talus, coalition, equinovalgus
"Associated EXCEPT"	→ **bronchopulmonary dysplasia**
"Inheritance"	→ **sporadic**
"Associated condition"	→ **cleft palate**
"Cleft palate + Ponseti-treated clubfoot + syndactyly — expect"	→ **amniotic band syndrome of hand**
"True statement"	→ **complete band blocking lymphatics** → **excision + Z-plasty** ✗ Wrong options: trunk > limbs · late gestation/high BW · tibial bowing

– – Active space – –

CUBITAL TUNNEL SYNDROME

– Hand · 18 / 103 –

1 Basics

Epidemiology

Frequency	**2nd most common** upper-limb compressive neuropathy (after CTS); ~30/100,000/y
Demographics	♂ **>** ♀ (♀ present younger); ↑ with age
Associations	cubitus varus/valgus, medial epicondylitis, burns, elbow contracture release
Children	rare; trauma or repetitive use; nonop works in up to 50%

Anatomy – ulnar nerve course & cubital tunnel

Ulnar nerve **medial cord (C8–T1)** → posteromedial to brachial artery → pierces IM septum at **arcade of Struthers 8 cm proximal** to medial epicondyle → behind medial epicondyle (cubital tunnel) → **between 2 heads of FCU** → between FCU & FDP → **superficial to TCL** at wrist

Cubital tunnel	Structure
Roof	**FCU fascia + Osborne's ligament** (cubital tunnel retinaculum; medial epicondyle → olecranon)
Floor	**Posterior bundle (posterior oblique ligament) of MCL** + transverse band + capsule
Walls	Medial epicondyle + olecranon

✗ Floor distractors Osborne (= roof) · anterior/posterior band of anterior bundle UCL · transverse bundle

MUCL bundles **anterior bundle** = primary valgus stabiliser (anterior band isometric; posterior band tight in flexion) · **posterior bundle** = greatest tension change, **tight in flexion**, **floor of tunnel** · transverse = no stability (doesn't cross joint)

Compression sites (Osborne's ligament, anconeus epitrochlearis)

Site (proximal → distal)	Note
Arcade of Struthers	Hiatus in medial IM septum, 8 cm proximal
Medial IM septum / medial head triceps	
Medial epicondyle	Nonunion, osteophyte
Osborne's ligament–MCL (cubital tunnel)	
Anconeus epitrochlearis	Accessory muscle medial epicondyle → olecranon, **replaces Osborne's**
Two heads of FCU / aponeurosis	= humeral head forms **ulnar border of flexor-pronator mass**
Deep flexor-pronator aponeurosis / FDS proximal edge	

Most common site **between 2 heads of FCU** (Orthobullets, OB) · **Osborne's ligament** (one OB explanation) – read the options

Anconeus epitrochlearis ~**10% of population** · **13–30%** of surgically treated CuTS; static/dynamic compression; usually found at surgery → **excise + in situ decompression ± transposition**; intra-op: muscle over nerve posterior to medial epicondyle

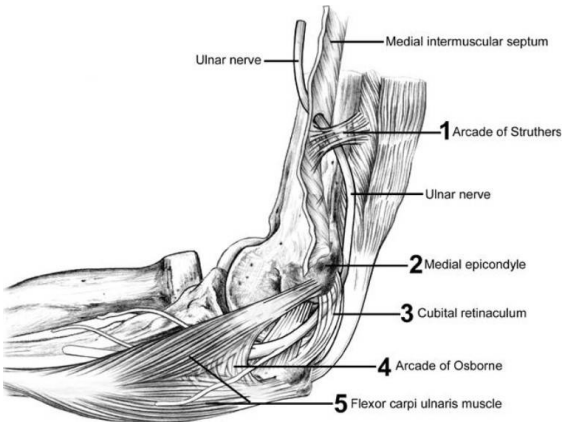


Anconeus epitrochlearis intra-op

External causes fracture/medial epicondyle nonunion, osteophytes, **HO**, tumours, ganglia

NOT ulnar nerve compression sites	Nerve
Ligament of Struthers (supracondylar process → medial epicondyle)	**Median** (pronator syndrome)
Supracondylar process (1%)	Median
Lacertus fibrosus	Median
Arcade of Frohse + FLEAS (fibrous bands over radial head, leash of Henry, ECRB edge, supinator)	**PIN / radial**
Ulnar artery thrombosis/aneurysm	Ulnar at **Guyon zone III** (no dorsal sensory loss)

– – Active space – –



Compression sites

2 Mechanism

Compression + traction	elbow flexion ↑ tension; superficial nerve; **subluxation = ↑ traction**
Exacerbating	**phone use (flexion)**, repetitive flexion + valgus (throwers); **night symptoms** (sleeping flexed)

Tardy ulnar nerve palsy (TO)

Cause (TO)	**nonunion of paediatric lateral condyle fracture → cubitus valgus → chronic stretch**; also medial epicondyle nonunion, **varus** malunion of supracondylar fracture, arthritis, cysts, recurrent dislocation, Monteggia
Onset (TO)	delayed; months to **up to 25 years** after injury; even minimal valgus can cause it
Pathology (TO)	tension neuritis + intraneural fibrosis; V-kinking, FCU fibrous band
Treatment (TO)	**surgical**: **anterior transposition** ± corrective osteotomy (osteotomy only if elbow complaint)
Trap (TO)	ulnar nerve supplies **adductor pollicis**, **not abductor pollicis brevis** (median)

3 Presentation

Sensory	paraesthesia **small + ulnar ring + ulnar dorsal hand**
Motor (intrinsics)	adductor pollicis, deep FPB, interossei, lumbricals 3–4 → **weak grasp** (MCP flexion) · **weak pinch** (up to 70% lost) · ↓ index abduction (1st DI)
Inspection	interosseous + **1st web atrophy**, ring/small **clawing**, **subluxation** over medial epicondyle on flexion–extension

Clinical signs (Wartenberg, Froment)

Sign	Finding	Mechanism
Froment	**Thumb IP flexion** during key pinch	**FPL (AIN/median)** compensates weak **adductor pollicis**
Jeanne	Thumb **MCP hyperextension** + adduction on key pinch	**EPL (radial)** compensates
Wartenberg	**Small finger abducted**	Weak **3rd palmar interosseous** + small lumbrical; **accessory slip of EDM (radial n.) to abductor digiti minimi tendon**
Masse	Flat palmar arch, lost ulnar hand elevation	Weak ODQ
Pollock	Can't flex DIP ring/small	Weak **FDP** (extrinsic → proximal lesion)



Froment sign

Froment = which nerve?	**ulnar** (cubital tunnel or Guyon's)
Wartenberg wrong explanations	✗ Wrong options: tetanic ADM (ADM is ulnar too) · RCL insufficiency · unopposed FDP (→ clawing) · Martin-Gruber
Wartenberg causes (TO)	ulnar palsy (commonest) · **avulsion of 3rd volar interosseous insertion** · **cervical cord compression** · mild hemiparesis ("digiti quinti sign") · hemiplegic migraine ✗ Wrong options: **median nerve palsy**
Wartenberg (TO)	unopposed EDM + EDC (radial) abduct small; more pronounced **in extension**
Provocative	**Tinel over cubital tunnel** · **elbow flexion test >60 s** · direct compression

4 Diagnosis & Classification

Diagnostic criteria – electrodiagnostics

EMG/NCS	diagnosis + **prognosis**; **NCV <50 m/s across elbow**; ↓ SNAP + CMAP amplitude
Repeat EDS	not when clinical diagnosis clear; only for new inconsistent symptoms

Key radiology findings

X-ray elbow	look for **osteophyte / HO** (→ changes procedure: medial epicondylectomy), cubitus valgus/varus, lateral condyle nonunion

– – Active space – –

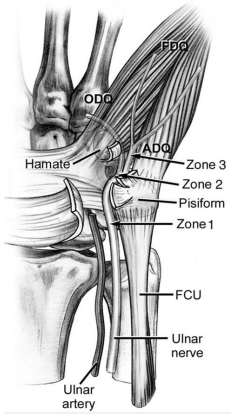
McGowan & Dellon classification

Type	Findings
1	Subjective sensory; no objective 2PD loss or atrophy
2A	Sensory + **weak pinch/grip, no atrophy**
2B	Sensory + **atrophy**, intrinsic strength ≤3
3	**Profound atrophy** + sensory disturbance

Ulnar tunnel (Guyon's canal) compression

Dorsal ulnar sensory branch arises **proximal** to Guyon → **dorsal hand spared** in Guyon/distal lesions

Guyon zone	Content	Deficit
I (proximal to bifurcation; palmar carpal ligament)	Mixed	Sensory volar ring/small + intrinsic weakness; dorsal spared
II (deep branch, hook of hamate)	Motor	**Intrinsic weakness only**
III (superficial branch, pisohamate ligament)	Sensory	**Volar ring/small numbness, normal motor**; ulnar artery thrombosis/aneurysm (**cyclist**)



Guyon zones

5 Management

Cubital tunnel syndrome

- Mild → **night extension splint 45° + NSAIDs + activity mod**
  - fails (~3 wk adequate trial) → **surgery**
- Stable nerve, idiopathic → **in situ decompression**
- Subluxing nerve in thrower / young / failed in situ / poor bed → **anterior transposition**
- Osteophyte/HO, thin patient with subluxation → **medial epicondylectomy**

Nonoperative

Regimen NSAIDs, activity modification, **night splint 45° extension, forearm neutral**; effective **~50%**

Not indicated — **daytime splinting** · **cortisone injection** (nerve risk) · continuing night splint another 6 wk in thrower

Operative criteria

Procedure	Indications
In situ decompression	Failed nonop; **before denervation**; idiopathic, stable nerve
Anterior transposition (subcut / intramuscular / submuscular)	**Failed in situ** · **throwing athlete** · **poor nerve bed (tumour, osteophyte, HO)** · **unstable/subluxing nerve** (snapping), esp. young · tardy palsy
Medial epicondylectomy	Visible symptomatic subluxing nerve · **thin patient** · **osteophyte/ HO** on X-ray

In situ decompression vs anterior transposition

In situ = transposition outcomes — **no difference** (motor or sensory); in situ **fewer complications** → choice for idiopathic

Subluxation — not a contraindication to in situ (Keiner) · but **throwers/snapping nerve → transposition**

Two "snaps" — ulnar nerve + **medial triceps** dislocating → transposition

Subcutaneous transposition — not for idiopathic (↑ complications, equal outcomes)

In situ technique — 4 cm incision between olecranon + epicondyle; release **Osborne's + FCU fascia** distally, triceps–IM septum fascia proximally; **avoid circumferential dissection** (devascularisation, hypermobility); endoscopic option

Transposition technique — circumferential dissection, **excise medial IM septum**, secure with subcut tissue / fascial sling / within or under flexor-pronator mass



Ulnar nerve transposition with fascial sling

Medial epicondylectomy — oblique osteotomy; **preserve MCL insertion** (risk medial instability)

Prognosis

Good — **80–90%** good if intermittent + no denervation; **ulnar nerve instability (subluxation) in young = good prognostic factor**

Poor — **intrinsic muscle atrophy** (strongest)

NCV — returns to baseline within ~2 wk; distal latency recovers better with early decompression

– – Active space – –

6 Complications

Medial antebrachial cutaneous nerve injury

MABC crosses field **~3 cm (3.1) distal** to medial epicondyle; minor branches at/proximal → **painful neuroma**, posteromedial elbow pain, **↓ sensation ulnar (anteromedial) forearm**

Medial brachial cutaneous crosses **7 cm proximal** to epicondyle

✗ Deficit distractor	Nerve
Can't flex thumb IP	AIN
Can't extend thumb IP	PIN
Can't hold fingers abducted	Ulnar
↓ radial forearm sensation	Lateral antebrachial cutaneous

Recurrence inadequate decompression, scarring, tethering at IM septum/FCU; **higher than after CTR**

Transposition risk of **new point of compression**

X vs Y

	Cubital tunnel	Guyon's canal (ulnar tunnel)
Clawing	**Less** (FDP also weak)	More
Dorsal ulnar hand sensation	**Lost**	**Spared**
Extrinsics (FDP ring/small, FCU)	**Weak**	Normal
Tinel / flexion test	**Elbow**	Wrist

	Cubital tunnel	C8 radiculopathy
Sensory	Ring + small	C8 dermatome
Motor	Ulnar intrinsics only	**+ APB, FPB, OP, lateral lumbricals** (median C8)
Shoulder abduction	–	**Relieves** symptoms

How it's asked

"Thrower, medial elbow pain, small-finger paraesthesia – structure?"	→ **anconeus epitrochlearis** ✗ Wrong options: supracondylar process · ulnar artery · lacertus · arcade of Frohse
"Ulnar nerve compression sites EXCEPT"	→ **ligament of Struthers** (median)
"Idiopathic CuTS failed splint/NSAIDs, no subluxation, no osteophyte – treatment & site?"	→ **in situ release; FCU aponeurosis**
"Floor of cubital tunnel?"	→ **posterior bundle of UCL** ✗ Wrong options: Osborne = roof

"Froment sign + EMG compression proximal to wrist — release?"	→ **border of flexor-pronator mass (FCU heads)** ✗ Wrong options: FLEAS radial tunnel structures
"Black-dot muscle over nerve behind epicondyle?"	→ **anconeus epitrochlearis**
"Wartenberg sign not caused by"	→ **median nerve palsy** (TO)
"Froment sign — nerve?"	→ **ulnar**
"Why is little finger abducted in ulnar palsy?"	→ **accessory EDM slip to ADM tendon**
"Surgery that minimises complications?"	→ **simple in situ decompression**
"In situ vs subcutaneous transposition with subluxation?"	→ **no difference in outcome**
"Good prognostic factor, adolescent gymnast?"	→ **ulnar nerve subluxation** ✗ Wrong options: atrophy = poor
"Snapping medial elbow + Jeanne sign, ex-pitcher?"	→ **decompression + anterior transposition**
"Javelin thrower, two snaps, failed 6 wk splint?"	→ **release + anterior transposition** ✗ Wrong options: more splinting · daytime splint · cortisone
"Painful paraesthesia near incision after transposition?"	→ **medial antebrachial cutaneous** nerve
"Nerve cut 3 cm distal to epicondyle — deficit?"	→ **↓ ulnar forearm sensation**
"Medial cord branch at risk 3 cm distal?"	→ **MABC (label C)**
"Cyclist, volar ring/small numbness, normal strength, dorsal spared?"	→ **superficial ulnar branch at pisohamate ligament (zone III)**
"False about tardy ulnar palsy nerve"	→ **supplies APB** (TO)

– – Active space – –

DE QUERVAIN'S TENOSYNOVITIS

– Hand · 19 / 103 –

1 Basics

Epidemiology

Definition	**stenosing tenosynovitis of 1st dorsal compartment** (APL + EPB)
Incidence	very common, ~1/1000/y
Demographics	♀ > ♂ **(6×, OB)**; **30–50 y** (Orthobullets) · 5th–6th decade (OB); **dominant wrist**
Risk factors	**overuse** (golf, racquet sports), post-traumatic, **postpartum / lactating** (classic: mother of 2-month-old)
Occupation	**overuse syndrome**; typist, bowler, cell-phone use (TO, OB) · **no causal link with occupational risk factors** (OB) – read the options

Pathology

Pathoanatomy	thickened, swollen **extensor retinaculum** → ↑ tendon friction
Not inflammatory	**NOT an inflammatory process**; **mucopolysaccharide** accumulation (Orthobullets) · **myxoid degeneration** + sheath thickening (TO)
Tendon mechanics (TO)	typist's graph = **hysteresis / energy-dissipation curve** of cyclically loaded tendon

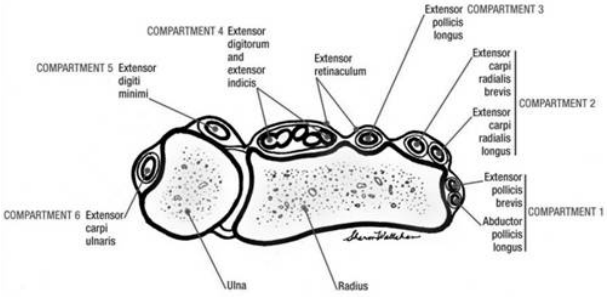
Viscoelastic term (TO)	Meaning
Creep	↑ deformation under **constant load**
Stress relaxation	↓ stress under **constant deformation**
Hysteresis	Unloading curve ≠ loading curve; area = energy lost as heat
Cyclic loading	Curve shifts right, hysteresis ↓ → reproducible = **pseudo-elasticity**
Toe region	**Un-crimping** of collagen fibrils (low stiffness) → linear → yield/failure

Anatomy – APL/EPB & injection site

Dorsal compartment	Contents	Condition
1	**APL, EPB**	**De Quervain's**
2	ECRL, ECRB	**Intersection syndrome** (1st/2nd crossing)
3	EPL	–
4	EIP, EDC	–
5	EDM	Vaughan-Jackson
6	ECU	Snapping ECU

Nerve	APL & EPB innervated by **PIN** ✗ Wrong options: radial n. proper · AIN · median · deep ulnar

EPB	**more dorsal** than APL · **thinner** · single tendon · **absent in 7%** · often in **own subsheath** (dorsal) → missed at release; separate EPB compartment ↑ failure of nonop
APL	**thicker**, usually **≥2 slips**, may have own fibro-osseous compartment
Separate compartment for EPB or APL	**24–34%** of wrists
1st compartment	**most numerous tendon variations** in upper limb
Radial artery	dives **deep** to 1st compartment distal to scaphoid ✗ Wrong options: superficial to APL
Not EPL	3rd compartment, never in 1st



Extensor compartments

3 Presentation

Symptoms	gradual **radial-sided wrist pain**, worse gripping/lifting with wrist neutral
Tenderness	1st compartment **at radial styloid** / **1–2 cm proximal to radial styloid**
Motion	usually normal; pain with resisted radial deviation
NV exam	**normal**

Test	Manoeuvre	Note
Finkelstein	Examiner grasps thumb, quickly **ulnar-deviates** hand → pain at radial styloid	More indicative of **EPB > APL**; TO and OB also apply the name to the fist-around-thumb version
Eichhoff	Thumb clenched in fist + wrist ulnar-deviated → pain, **relieved when thumb extended**	Often confused with Finkelstein

Bowler/typist (TO)	radial wrist pain ↑ by **thumb flexion + ulnar deviation → De Quervain's** ✗ Wrong options: tennis/golfer's elbow · lunate AVN · UCL injury · scaphoid insufficiency #

4 Diagnosis & Classification

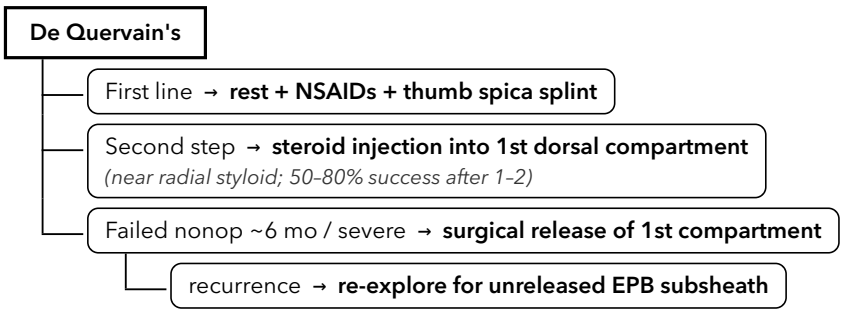
Key radiology findings

Diagnosis	**clinical**; advanced imaging not needed
X-ray	AP + lateral; **usually not indicated**; rule out **thumb CMC (basilar) arthritis**, **carpal arthritis**
Ultrasound	thickened retinaculum over 1st compartment tendons

– – Active space – –

Differentials thumb CMC OA, intersection syndrome, FCR tendinitis

5 Management



Nonoperative

Injection site	**1st dorsal compartment near radial styloid** ✗ Wrong options: CMC joint · carpal tunnel · A1 pulley · 1st/2nd crossing
Steroid vs splint	**steroid superior**; adding splint/NSAID after injection does not improve outcome
Immobilisation	high failure rate, **except lactating women** (nonop highly effective)
Septum between APL and EPB (TO)	may need **injection on both sides** of the septum (MRI/US shows it)
TO conservative list	activity modification, NSAIDs, ice massage, thumb spica, **steroid into APL/EPB sheath** → release if fails

Surgical release

Incision	radial-based, proximal to wrist; transverse
Release on DORSAL side of sheath → prevents **volar subluxation** of tendons	
Protect	**superficial radial sensory nerve**
Release all	APL slips + **EPB subsheath**

Recurrence – unreleased EPB subsheath

Classic	recurrent pain, **+ Finkelstein**, tender 1st compartment, **– Tinel**, normal skin → **failure to decompress EPB subsheath** ✗ Wrong options: neuroma: Tinel – · CRPS: no skin change · EPL/APB: not in 1st compartment
Prognosis	most resolve nonop; **high recurrence rate**

6 Complications

Complication	Clue
Superficial radial nerve injury / neuroma	**+ Tinel**; no positive Finkelstein
Incomplete release → recurrence	**Unreleased EPB / separate APL compartment**
Volar tendon subluxation	Avoided by dorsal-side release
CRPS	Skin changes

X vs Y

	De Quervain's	Intersection syndrome	Thumb CMC OA	Snapping ECU
Site	Radial styloid	**More proximal + more dorsal**	CMC joint	Ulnar wrist
Structures	APL + EPB	**1st crossing 2nd** (APL/EPB over ECRL/ECRB)	Basal joint	ECU (6th)
Clue	Finkelstein +	Crepitus, pain on **resisted wrist extension**	**Grind test**, older	Snap pronation → supination

How it's asked

"Postpartum radial wrist pain — inject where?"	→ **1st dorsal compartment near radial styloid**
"True during 1st compartment release?"	→ **EPB more dorsal than APL** ✗ Wrong options: EPB thicker · EPB multiple slips · radial artery superficial · EPL subsheath
"Exam finding that localises to APL + EPB?"	→ **radial pain on Finkelstein** ✗ Wrong options: resisted extension = intersection · Phalen = CTS · grind = CMC OA
"Recurrence 3 mo after release, Finkelstein +, Tinel –?"	→ **unreleased EPB subsheath**
"Nerve supplying affected tendons?"	→ **PIN**
"Bowler: pain ↑ thumb flexion + ulnar deviation?"	→ **De Quervain's** (TO)
"Typist, dorsolateral wrist pain + hysteresis curve?"	→ **De Quervain's** (TO)
"Which side of sheath to incise?"	→ **dorsal** (prevents volar subluxation)

– – Active space – –

DEEP SPACE & COLLAR BUTTON INFECTIONS

– Hand · 20 / 103 –

1 Basics

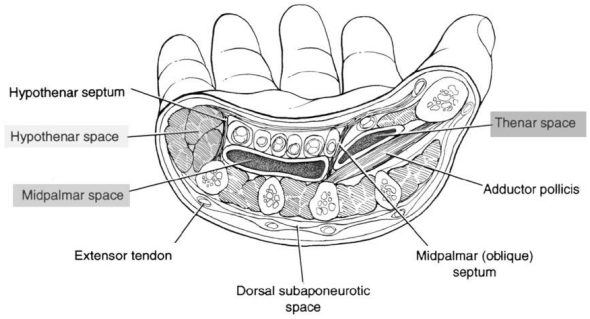
Definitions

Deep space infection **thenar** (**most commonly infected**) · hypothenar · **midpalmar** (rare)

Collar button abscess **web space** abscess between fingers, usually from a **volar puncture**

Anatomy – deep spaces

Space	Boundaries
Thenar	Potential space **palmar to adductor pollicis**, dorsal to flexor tendons; fascial septum from midpalmar
Midpalmar	**Dorsal and radial to hypothenar space**
Hypothenar	Palmar to 5th MC, dorsal-radial to hypothenar fascia, ulnar to hypothenar septum



Deep spaces of the hand

Space (TO)	Dorsal	Volar	Radial	Ulnar
Thenar	Adductor pollicis fascia, volar 2nd MC, 1st dorsal interosseous fascia	Index flexor sheath, radial palmar fascia	Adductor pollicis fascia + palmar fascia at thumb P1	**Midpalmar (oblique) septum** (3rd MC → palmar fascia)
Midpalmar	Fascia over 2nd/3rd volar interossei; periosteum 3rd–5th MC	Flexor sheaths middle, ring, little; palmar aponeurosis	Midpalmar oblique septum	**Hypothenar septum** (5th MC → palmar fascia)
Hypothenar	Periosteum 5th MC, deep hypothenar fascia	Palmar fascia, superficial hypothenar fascia	Hypothenar septum	Hypothenar muscle fascia

Superficial spaces (TO) **dorsal subcutaneous** · **dorsal subaponeurotic** · **interdigital web space** → less defined, can mimic deep infections

![Thenar space and its boundaries](../../TO notes/attachments/TOn-notes-acute-infections-in-the-hand-mar-2026-p06-f016.jpg)

Ulnar & radial bursae communications

Ulnar bursa (TO)	proximal continuation of **little finger (5th) sheath**
Radial bursa (TO)	proximal continuation of **thumb (FPL) sheath**
Extent (TO)	**2–3 cm above wrist**
Communications (TO)	ulnar ↔ **radial bursa** · ± **Parona's (forearm) space** · not **4th (ring) finger sheath**

2 Mechanism

Collar button spread	volar puncture → **dorsal through sinus in superficial transverse metacarpal ligament**; proximal spread **blocked by palmar aponeurosis**
Organisms	Staph & Strep; **CA-MRSA ~30%** of surgically treated hand infections, ↑ yearly; only independent risk factor **IV drug use**

3 Presentation

History ± penetrating trauma; pain, swelling

Space	Pain on flexion of
Thenar	**Thumb**
Hypothenar	**Small finger**
Midpalmar	**Middle, ring, small**

Thenar & midpalmar	**loss of palmar concavity**
Collar button	web-space tenderness + **abduction of adjacent digits (V-sign)**
Hourglass swelling (TO)	web-space pus tracks volar + dorsal with little in between · finger **ABDUCTED → not seen in dorsal subcutaneous infection** (no volar pus)
Chronic web-space infection (TO)	**"barber's abscess"**

4 Diagnosis & Classification

Key radiology findings

X-ray	if **foreign body** suspected
MRI	defines **extent** of infection

5 Management

Deep space / collar button infection

- I&D + IV antibiotics (standard of care)
  - Collar button → **DORSAL + VOLAR longitudinal incisions, avoid web skin ± Penrose drain**

– – Active space – –

Deep space drainage (TO)

Thenar	volar and/or dorsal incisions (combined if extensive) · protect **recurrent motor branch of median n.** + **princeps pollicis a.** · avoid **transverse dorsal 1st-web incision** (web contracture) (TO)
Hypothenar	incision **in line with ulnar border of ring finger** · protect **deep motor branch of ulnar n.** (TO)

Collar-button abscess drainage

Incisions	**two (dorsal + palmar)**, **avoid web space** → prevents **scar contracture & loss of abduction**

(✗ Wrong options: **Traps** – Brunner incisions) · incision across web space · oral antibiotics alone



Dorsal + volar incisions with Penrose drain

How it's asked

"Child, palmar abrasion, web space swelling, digits abducted – incisions?"	→ **dorsal + palmar, avoiding web** ✗ Wrong options: Brunner · web incision
"Ulnar bursa does NOT communicate with"	→ **4th finger sheath** (TO) ✗ Wrong options: 5th sheath, Parona, radial bursa
"Most commonly infected deep space"	→ **thenar**
"Chronic web-space infection?"	→ **barber's abscess** (TO)
"Thenar space drainage – structure to protect?"	→ **recurrent motor branch of median nerve** (TO)
"Pain with middle/ring/small flexion + loss of concavity"	→ **midpalmar space**

DIGITAL ARTERY ANEURYSM

– Hand · 21 / 103 –

1 Basics

Definition permanent dilation with **50% ↑ in normal diameter**

Incidence rare; **<1 / 100,000**

Type	Cause	Wall	Shape
True aneurysm	**Blunt** trauma	Weakened arterial wall	**Uniform**
False (pseudoaneurysm)	**Penetrating** trauma	Organised haematoma + fibrous wall	**Sac-like**
Non-traumatic	Inflammatory, atherosclerotic	–	–

Palmar arch pseudoaneurysm after laceration or **endoscopic CTR**

3 Presentation

Symptoms slow-growing **painful mass** after hand trauma; **paraesthesia** (digital nerve compression)

Pulsatile **~50%**

Digit **thumb > index > ring**

4 Diagnosis & Classification

Key radiology findings

X-ray usually normal; only if destructive lesion suspected

Doppler US / CT angiography pre-op confirmation: size, location, **thrombus**, **collateral circulation**

MRA false aneurysm of superficial palmar arch



MRA: false aneurysm of superficial palmar arch

– – Active space – –

Differentials	AV fistula, neurilemmoma, foreign body granuloma, ganglion, epidermoid cyst

5 Management

Digital / palmar artery aneurysm

- Small, asymptomatic → **OBSERVATION + analgesia**
- Symptomatic + adequate collaterals → **EXPLORATION + LIGATION (prox & distal) / excision**
- Symptomatic + inadequate collaterals → **EXCISION + INTERPOSITION GRAFT**

Thrombin injection (US-guided)	reported for more proximal hand/wrist lesions
Repair	end-to-end anastomosis or **autogenous vein/arterial interposition graft**
Not appropriate	aspiration, compression bandage, steroid injection, elevation/observation (symptomatic false aneurysm)

6 Complications

Complications	digital ischaemia, chronic pain

How it's asked

"Pulsatile palmar mass + paraesthesia after laceration, MRA shown – treatment?"	→ **surgical exploration** ✗ Wrong options: aspiration, compression, steroid, observation
"Blunt vs penetrating"	→ **blunt = true (uniform) · penetrating = false (sac-like)**

DIGITAL COLLATERAL LIGAMENT INJURY

– Hand · 22 / 103 –

1 Basics

Incidence	rare, <1 per 100,000/yr
Anatomy	collaterals on lateral aspect of **DIP, PIP, MCP**; key to **pinch stability**
Index RCL	**critical for pinch**; deficient → MCPJ deviates **ulnarly** on pinch; 5.8 mm wide × 6.4 mm long

2 Mechanism

Jammed finger	dorsal/volar dislocation tears one or both collaterals; direct blow

3 Presentation

Symptoms	joint pain, swelling, deformity; **pinch instability** after pain settles
Exam	varus/valgus stress; **endpoint** decides grade

4 Diagnosis & Classification

Key radiology findings

X-ray	PA, lateral, oblique ± **varus/valgus stress views**
MRI	if exam equivocal



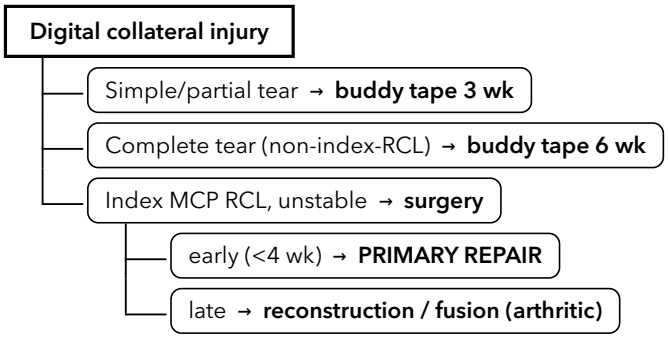
MRI collateral ligament

– – Active space – –

Index RCL grading

Grade	Exam	Treatment
I	Tender, stable	Immobilise **4–6 wk**
II	Laxity, **definite endpoint**	Immobilise **4–6 wk** → excellent
III	**No endpoint** = unstable	Early (<4 wk): **repair** (or strict immobilisation); late: **reconstruction / fusion**

5 Management



Index MCP radial collateral ligament – repair

Unstable index RCL	**best outcome = early surgical repair**
Traps	RCL not key to pinch · ✗ Wrong options: stable → surgery (immobilise instead) · early vs late equivocal (late = poor) · early **reconstruction** (reconstruction = late, irreparable)
Late presentation	universally poor; chronic instability → OA

Bowler's thumb (ulnar digital nerve neuropathy)

Bowler's thumb (TO)	**neuropathy of ulnar digital nerve of thumb** from friction of ball
Pathology (TO)	**perineural fibrosis** ✗ Wrong options: true neuroma
Presentation (TO)	pain/tenderness **1st web space**, numbness/paraesthesia along medial thumb, tender small mass, hyperaesthesia
Treatment (TO)	conservative: rest, pressure relief, avoid activity; refractory → **neurolysis, neurectomy, transposition**
Not (TO)	thumb UCL injury · mallet thumb/index · 1st MC neck fracture

How it's asked

"Index MCP, no endpoint on ulnar stress, weak pinch – true?"	→ **early surgical repair best**
"Stable collateral injury"	→ **immobilise / buddy tape** ✗ Wrong options: surgery
"Bowler's thumb is"	→ **ulnar digital nerve neuropathy of thumb** (TO)

– – Active space – –

DIP AND PIP JOINT ARTHRITIS

– Hand · 23 / 103 –

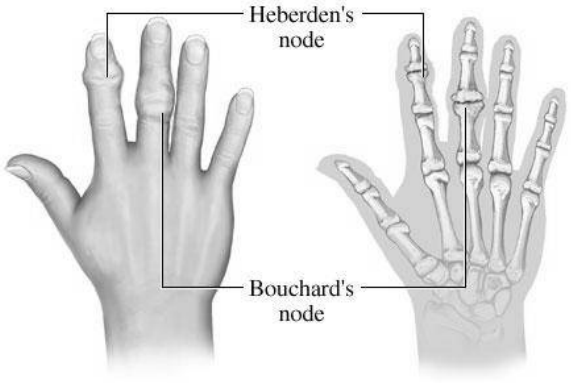
1 Basics

Epidemiology

Commonest hand OA	**DIP > thumb CMC > PIP > MCP**
DIP	**highest joint forces in hand**; ≥60% of >60-y-olds show DIP OA
PIP OA prevalence	**~17% ♀ · 13% ♂**

Forms of arthritis

Form	Key features
Primary OA – DIP	**Heberden nodes** (osteophytes), **mucous cysts**, nail ridging/splitting/deformity/loss of gloss
Primary OA – PIP	**Bouchard nodes**; contracture from ligament fibrosis
Erosive OA	DIP > PIP; **middle-aged ♀, 10:1**; intermittent inflammatory episodes; RA-like synovium but **not systemic**; self-limiting yet destructive



Heberden (DIP) & Bouchard (PIP) nodes

3 Presentation

Primary OA	pain, deformity; PIP → **flexion contracture**, ↓ motion, swelling
Mucous cyst	dorsal DIP mass, proximal to nail fold; nail ridging/deformity; → draining sinus → **septic arthritis**

– – Active space – –



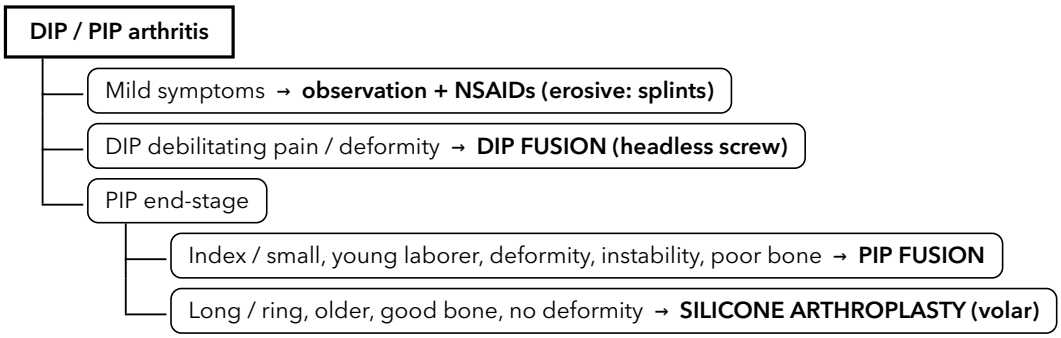
Mucous cyst with nail ridging

4 Diagnosis & Classification

Key radiology findings

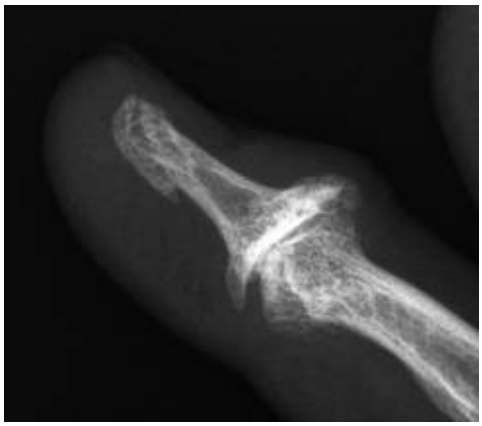
Views	AP, lateral, oblique hand
OA	joint space narrowing DIP/PIP, **dorsal osteophytes** (under mucous cyst)
Erosive OA	cartilage destruction, osteophytes, subchondral erosion → **"gull wing" deformity**
Diagnosis	history + exam + radiographs

5 Management



Mucous cyst – excision with osteophyte removal

First line	observation (**20–60% resolve**) (Orthobullets) · rarely resolve, usually progress → nail deformity (OB)
Aspiration ± steroid	**>40% recurrence**; 60% complete resolution (Rizzo)
Surgery	**excision of cyst + osteophyte (bone spur) removal → 0% recurrence**, 100% resolution; key = **osteophytes MUST be debrided**
Surgery indications	significant pain, impending rupture, **repeated drainage**
Not needed	skin excision/graft (lesion independent of skin; flap controversial), nail plate removal, collateral/extensor resection, steroid injection, infectious workup
DIP fusion instead	**pain with forcible joint movement** + advanced arthritis



Mucous cyst over DIP spur

DIP arthrodesis

Indication debilitating pain/deformity; **DIP OA of lesser digits** (e.g. ring DIP marked narrowing) → arthrodesis

Headless compression screw **highest fusion rate**; nonunion 10% (Orthobullets) · union 95% (Kocak), 94–97% (Cox, Matsumoto)

Digit	DIP fusion position
Index, long	**Extension**
Ring, small	**10–20° flexion**

DIP arthrodesis complication – screw length

Screw error	Result
Too long	Prominent tip → **fingertip hypersensitivity**
Too wide	**Fracture, nailbed deformity**
Headless	↓ prominence ✓
Cannulated vs not	No difference

PIP arthroplasty (silicone, pyrocarbon) vs arthrodesis – indications & outcomes

PIP option	Indication
Collateral excision + volar plate release + osteophyte excision	**Predominant contracture, minimal/no joint arthritis**
Arthrodesis	**Border digits (index, small)**; long/ring with **angulation/rotation, ligament instability, poor bone stock**; **young laborer**
Silicone arthroplasty	**Long & ring**, good bone stock, no deformity, older
Pyrocarbon arthroplasty	Needs **intact collaterals + volar plate** (unconstrained)
Resection arthroplasty	Recurrent infection, bone loss
Ray resection	Salvage only
Vascularised toe PIP transfer	Young; not in clotting disorder (factor V Leiden)

Index PIP relative CI to arthroplasty (**ulnar-deviating forces in key pinch**); arthroplasty → 4.3× complications, no ROM/pain gain vs fusion (Vitale)

– – Active space – –

Long finger	may also be a relative CI (Milone)
Young + post-traumatic + rotation	**arthrodesis** (rotation = collateral insufficiency)
Silicone & collaterals (conflict)	RCL should be intact for pinch (Orthobullets) · linked silicone **not dependent** on collaterals/volar plate → choice for **collateral-deficient ring PIP** (OB)
Silicone outcomes	**reliable pain relief (98%)**; **no significant ↑ ROM, grip, DASH**; arc ~55°; 60% deformation, 40% failure (Bales); **<60 y → ~25% revision**
Approach + implant	**silicone via volar approach → best ROM, least extensor lag, lowest revision (6% vs 18%** surface/dorsal; = 33%); **surface (metal-on-plastic) → highest revision**; dorsal = historic gold standard
Pyrocarbon PIP	**high complication rate** → rarely used; **↓ arc (57→31°)**; squeaking, loosening, dislocation, contracture; 34% reoperation, 51% subsidence, 36% unstable (Wagner)
Pyrocarbon MCP	**low complications**, mild ↑ arc (62→76°), pain relief; squeaking reported in both MCP & PIP
PIP fusion technique	headless screw (highest union), tension band, or plate

PIP fusion angle	Value
Orthobullets	**Index 30° · long 35° · ring 40° · small 45°**
OB (radial → ulnar)	**40–55°** increasing
Arauz (index)	**0–20°** → natural pinch posture



Post-traumatic index PIP OA

Erosive OA

Treatment	splints + NSAIDs (tolerable) → **fusion** for intolerable deformity (same angles)

X vs Y

	Silicone PIP	Pyrocarbon PIP	Surface (metal-plastic)
Constraint	Linked, single piece	**Unconstrained**	–
Needs collaterals	No (OB)	**Yes**	–
Complications	Low	**High**	**Highest revision**
ROM	No significant gain	↓	Worse than volar silicone

Thumb / digit scenario (OB218081)	Best surgery
Ring DIP marked narrowing	**Arthrodesis**
Thumb CMC stage II	Trapeziectomy ± LRTI
Thumb CMC hypermobile, 14° hyperextension, mild OA	**Metacarpal osteotomy**
Index PIP contracture, no OA	Collateral excision / volar plate release
Ring PIP OA, stable	Silicone arthroplasty

How it's asked

"Tender dorsal DIP mass + spur – best treatment?"	→ **remove bone spur + cyst** ✗ Wrong options: aspiration >40% recurrence · fusion
"Advantage of excision + debridement over aspiration?"	→ **↓ recurrence**
"Important component treating mucous cyst?"	→ **excise DIP osteophytes** ✗ Wrong options: skin · nail · collateral
"Repeated drainage from DIP lesion – surgery?"	→ **mass excision + osteophyte removal**
"Middle finger PIP arthroplasty – approach/implant?"	→ **silicone via volar: best ROM + lowest revision**
"28-y index PIP post-traumatic OA + rotation?"	→ **PIP arthrodesis** ✗ Wrong options: pyrocarbon (collaterals deficient) · toe transfer (factor V Leiden)
"True about pyrocarbon arthroplasty?"	→ **PIPJ: high complication rate** ✗ Wrong options: ↑ ROM · no squeaking
"Best treated with arthrodesis?"	→ **ring DIP OA with marked narrowing**
"Best for PIP arthrodesis?"	→ **44-y laborer, index PIP OA**
"True of silicone PIP arthroplasty?"	→ **no significant ↑ ROM** ✗ Wrong options: ↑ grip/DASH · dorsal better
"Fingertip hypersensitivity after DIP fusion?"	→ **excessive screw length** ✗ Wrong options: diameter → fracture/ nailbed

– – Active space – –

DOG AND CAT BITES

– Hand · 24 / 103 –

1 Basics

Epidemiology

Frequency **dog (90%) > cat (10%) > human**; dog known to victim 90%

Demographics ♂; **2–19 y**; **upper > lower extremity**

Serious / fatal large aggressive dogs, small children, head & neck bites

Mechanism by animal

	Dog	Cat
Teeth	Blunt, powerful (>450 lb/in²)	**Small, sharp** → punctures **seal immediately**
Injury	Crush, avulsion, tear → **structural damage** (nerve, vessel, joint)	Penetrate **joints, bone, flexor sheath** → septic arthritis, OM
Infection risk	Lower	**Higher**

Pasteurella multocida & microbiology

Polymicrobial most, with **>1 anaerobe**; median 5 isolates/culture

Most common isolate (dog & cat) **Pasteurella** (gram-neg facultative anaerobic pleomorphic coccobacillus)

	Dog	Cat
Pasteurella share	**50%**	**70–80%** (Orthobullets) · **75%** (OB)
Species	***P. canis***	***P. multocida***, *P. septica*
Others	S. aureus, α-strep, Corynebacterium, Bacteroides, Moraxella, Neisseria	Similar

Pasteurella culture tell lab; special media; **~1 wk** to grow

Cat Pasteurella intense pain + swelling within **48 h**

Capnocytophaga canimorsus (dog) rare; fatal in **splenectomy/immunocompromised** (30–60% mortality); sepsis, endocarditis, meningitis, DIC

Rabies rhabdovirus; dogs, raccoons, **bats**, foxes

Infection risk factors

Wound hand, foot, major joint; puncture or crush

Host **delay >12 h**, **age >50**, steroids, **asplenia**, alcoholism, DM, vascular disease, oedema

3 Presentation

History	animal type, time since bite, comorbidities
Exam	puncture depth, crush, NV status, **joint penetration**; photograph wounds

4 Diagnosis & Classification

Key radiology findings

X-ray indications	crush, suspected fracture, **foreign body** (tooth)
MRI	not for flexor tenosynovitis (clinical Dx, delays treatment)
Culture	only if infected; **deep aerobic + anaerobic** ✗ Wrong options: wound swab → skin flora

5 Management

Dog / cat bite

- All → **irrigation (>150 mL saline, 18-19G) + tetanus ± rabies Ig/vaccine**
- Cat / hand / deep / >8 h / immunocompromised / DM → **AMOXICILLIN-CLAVULANATE**
- Abscess / tenosynovitis / septic joint / FB / devitalised tissue → **FORMAL I&D**

Antibiotics	**amoxicillin/clavulanate** (covers P. multocida); alternatives cefuroxime, ceftriaxone
Rabies prophylaxis	any suspicion (unprovoked attack, bizarre behaviour) → **HDCV + human rabies Ig**; povidone-iodine irrigation if high risk
Immobilise + elevate	

Operative criteria

Formal surgical debridement if
Crush / devitalised tissue, foreign body
Bites to **pulp, nail bed, flexor sheath, deep palmar spaces, joints**
Tenosynovitis, **septic arthritis**, abscess

Cat bite flexor tenosynovitis – drainage

Cat bite + FTS (3 days, fever)	**open I&D of flexor sheath** ± drain + pathogen-directed Abx
Early / mild FTS	catheter irrigation: incision **proximal to A1 + distal to A4**, 18G angiocatheter
Severe / >1 wk / necrosis	open debridement (mid-axial or Brunner) → amputation
Nonop (splint, elevation, IV Abx)	only if **<48 h** → I&D if no improvement in 12–24 h; **cat bites often need surgery**

– – Active space – –

Avoid radial bursa irrigation for long finger	long finger sheath **does not connect** to radial bursa
Outcome	stiffness possible despite prompt treatment



Cat bite FTS, long finger

Cat-scratch disease (Bartonella) – epitrochlear lymphadenitis

Organism	***Bartonella henselae***
Course	self-limited **6–12 wk** untreated; papule/pustule at inoculation (25–60%) 3–10 d → lymphadenopathy 1–2 wk later
Nodes	**epitrochlear**, axillary, cervical, inguinal; tender, may suppurate → **soft-tissue tumour simulator** (medial elbow mass)
Histology	**lymphoid tissue + multiple necrotising granulomas** (only other granulomatous option = M. avium)
Labs/imaging	↑ ESR/CRP; MRI mass + oedema in lymphatic drainage area; serum Bartonella test
Diagnostic criteria (3 of 4)	cat contact · positive CSD skin test · characteristic node lesions · negative workup for other lymphadenopathy
Treatment	**antibiotics** (azithromycin, ciprofloxacin, doxycycline) ✗ Wrong options: chemo/radiation · observation

X vs Y

Organism	Association
Pasteurella canis	Dog bite
Pasteurella multocida	Cat bite
Capnocytophaga	Dog bite, asplenic → sepsis
Bartonella henselae	Cat scratch, epitrochlear node, granulomas
Borrelia burgdorferi	Lyme
Mycobacterium avium	Immunocompromised / HIV

How it's asked

"Most common organism, dog & cat bites?"	→ **Pasteurella** ✗ Wrong options: Strep, Staph, Bacteroides, Moraxella
"Cat bite 3 days ago, Kanavel signs, fever — next?"	→ **open I&D of flexor sheath** ✗ Wrong options: MRI · swab · radial bursa catheter
"9-y medial elbow mass, biopsy granulomas — next?"	→ **antibiotics** (cat-scratch disease)
"Epitrochlear mass, necrotising granulomas — organism?"	→ **Bartonella henselae**
"Antibiotic of choice for bite?"	→ **amoxicillin-clavulanate**

– – Active space – –

DUPUYTREN'S DISEASE

– Hand · 25 / 103 –

1 Basics

Epidemiology & risk factors

Who	**white ♂ of northern European (Scandinavian/Celtic) descent**, 40–60 y; ♂ earlier & more severe
Sex ratio	**2:1** (Orthobullets) · **10:1** (TO sources) – read the options
Genetics	**autosomal dominant, variable penetrance**; sporadic commoner
Digits	**ring > small > middle > index**; joints **MCP > PIP > DIP**
Bilateral	**~45%** but **rarely symmetrical**
TO EXCEPT	"**index & middle most common, asymmetrical**" = false
Risk factors	**DM, alcohol (cirrhosis), smoking**, antiepileptics/epilepsy, HIV, TB, COPD
NOT a risk factor (TO)	**occupation with vibrating tools/earth movers** (not confirmed)
Ectopic	**Garrod knuckle pads** (dorsal PIP, 40–50%) · **Ledderhose** (plantar, 10–30%) · **Peyronie** (penis, 2–8%)
Shoulder link (TO)	adhesive capsulitis associated with DM, thyroid, **Dupuytren**

Pathogenesis & diathesis (Garrod's pads)

Genes & molecular markers

Marker	Fact asked
Inheritance	**Autosomal dominant, variable penetrance** (sporadic commoner)
Cell	**Myofibroblast** – intracellular **α-actin**; linked by extracellular **fibronectin**
Collagen	**↑ type III : I**; nodules = **type III** (TO); residual cord = type I
Cytokines	**TGF-β1** (↓ myofibroblast apoptosis), β2, **bFGF, PDGF**, EGF, CTGF
Not involved	**CBFA-1** (osteoblast transcription factor)
CCH target	Lyses **I & III**; spares **IV** (vessel/nerve basement membrane)

Cell	**myofibroblast** (intracellular **actin** along long axis; linked by extracellular **fibronectin**)
Collagen	**↑ type III : type I** ratio; **nodules deposit type III** (TO)
Cytokines	**TGF-β1** (also blocks myofibroblast apoptosis), bFGF, PDGF, EGF, CTGF
NOT implicated	**CBFA-1** (osteoblast transcription factor)
Diathesis	**age <50, ♂ white, bilateral, family history, ectopic (Garrod, Ledderhose, Peyronie)** → aggressive & recurrent
Bad prognostic factors (TO)	**family history, ♂, DM, smoking, alcohol, bilateral, knuckle pads, plantar nodules**
NOT a bad prognostic factor (TO)	**epilepsy** (not associated per Geoghegan)

Garrod pads alone + mild contracture (MCP 5°) **observe**; **do not resect dorsal pads**



Garrod knuckle pads

Cord anatomy – spiral cord, neurovascular displacement & natatory ligament

Rule **bands = normal, cords = diseased**

Cord	From	Contracture	Note
Pretendinous	Pretendinous band	**MCP**	Palmar pits (superficial fibres)
Central	Distal pretendinous; inserts flexor sheath at PIP	**MCP** (± PIP)	Midline; **does not displace NV bundle**
Spiral	**Pretendinous band + spiral band + lateral digital sheet + Grayson lig**	**PIP** (TO)	**Displaces NVB central & superficial**
Lateral	Lateral digital sheet	PIP / DIP	–
Retrovascular	Dorsal to NVB; P1 → P3	**DIP** (± hyperextension with lateral cord)	Displaces nerve palmar & toward midline
Natatory	Natatory lig	**Web space**	–
Isolated digital	–	PIP	–

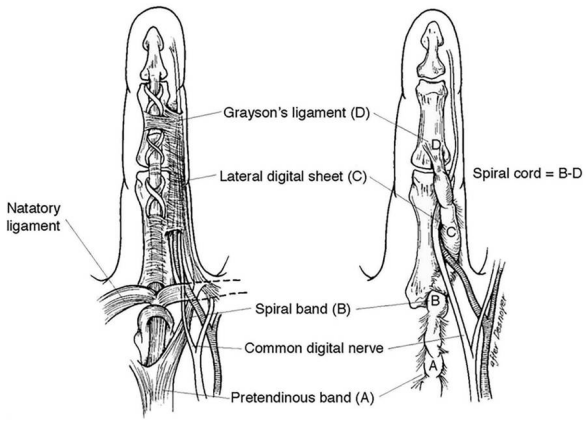
Spared (TO) **Cleland ligament** · **transverse lig of palmar aponeurosis**

Spiral cord position runs **dorsal to NVB proximally, volar distally**; cord **lateral & deep** to NVB → NVB pulled **central & superficial → at risk**

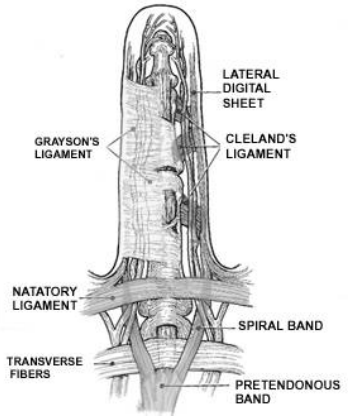
Spiral cord predictors **PIP flexion contracture (PPV 77%)** · **interdigital soft-tissue mass (71%)** · **nodule distal to MCP**

Ring finger radial digital nerve displaced **palmarly & ulnarly** (toward midline)

– – Active space – –



Spiral cord components



Digital fascia (McFarlane)

2 Mechanism

Luck stage	Histology
Proliferative	**Nodule**; hypercellular, large myofibroblasts, vascular, gap junctions; minimal matrix
Involutional	Myofibroblasts align along **tension lines**; ↑ type III collagen → contraction
Residual	**Acellular**; fibrocytes; dense collagen cord (type I predominant)

Surgery in proliferative stage ↑ stiffness & recurrence

3 Presentation

Signs (TO) palmar nodules · **pitting** · skin tethering · **tabletop test positive** (can't place hand flat) = surgical need

History **painless** nodule at **distal palmar crease** → cord → fixed flexion MCP/PIP

Exam palmar pits, skin blanching on extension, **Hueston tabletop test** + (can't lay palm flat)

Nodule beyond MCP clue to **spiral cord** (nerve midline & superficial)

– – Active space – –



Hueston tabletop test

4 Diagnosis & Classification

Diagnosis clinical

Key radiology findings

X-ray not needed; only to exclude arthritis/bony cause of stiffness

Ultrasound not needed; shows **thickened palmar fascia + nodules**

Differential	Distinguishing clue
Locked trigger finger	**Painful, tender A1**
Pulley rupture	Bowstringing, trauma
Ulnar claw	MCP **hyperextension** + IP flexion, ulnar sensory loss
Volkmann	Post-compartment syndrome; wrist flexion, thumb adduction, **MCP extension**, IP flexion

5 Management

Dupuytren

- Nodule only / no function loss / painful nodule → **observe (± steroid for painful nodule)**
- Palpable cord, MCP-predominant, wants to avoid surgery → **CCH or needle aponeurotomy (equal at 1 y)**
- MCP >30° or ANY PIP contracture (>15°), recurrence, wants lowest recurrence → **partial fasciectomy**
  - recurrent / diathesis / skin involved → **dermofasciectomy + FTSG**

– – Active space – –

Operative criteria

Treatment	Indication
Observation	Nodules only, no functional loss
Steroid injection	**Painful nodule** (✗ Wrong options: surgery for painful nodule)
CCH	Adult with **palpable cord**, MCP-predominant; max 2 cords/visit
PNA	Mild MCP contracture, **medically unfit / elderly**
Partial fasciectomy	**MCP ≥30°** · **any PIP (>15°)** · +ve tabletop · wants lowest recurrence
Total fasciectomy	Rare: diffuse, multiple joints, **recurrence**
Dermofasciectomy + FTSG	**Recurrent / diathesis** / skin involved
Arthrodesis / amputation	Salvage — severe small-finger PIP contracture

Surgical indication — **MCP ≥30°** · **PIP any (>15°)** · positive tabletop

NOT an indication (TO) — **painful nodule**

NOT a treatment (TO) — **Hyalase injection** · **high-dose radiotherapy** (only **low dose <30 Gy** for early disease)

Steroid injection — painful nodule; recurrence 50%; fat atrophy, depigmentation, tendon rupture

Collagenase & needle fasciotomy – mechanism, complications & outcomes

CCH (Xiaflex) — metalloprotease from *Clostridium histolyticum*; lyses **type I & III**

Why NV safe — **low activity vs type IV** (basement membrane of vessels/nerves)

Technique — 0.58 mg per cord; **max 2 cords/joints per visit**; manipulate 24–72 h; up to 3 injections/cord at 4-wk intervals

Manipulation at day 1 vs day 7 — **equivalent correction** (pain, skin tears same)

Meals & Hentz protocol (TO) — 4-step manipulation after CCH: flex PIP + extend MCP → flex MCP + extend PIP → extend both → press residual cord; max 3 attempts; stop if skin tears; night splint up to 4 mo

CCH-only side effects — **pruritic rash, axillary lymphadenopathy**

CCH major (1%) — **flexor tendon rupture**, CRPS, pulley rupture

Both CCH & PNA — **skin tears** (from manipulation; ↑ with age & degree of correction)

CCH vs PNA — **outcomes equal at 1 y** (90% full MCP extension); CCH costlier ($5,952 vs $1,540) & not fewer treatments

Reintervention	PNA	CCH
2 y	24%	**41%**
5 y	**61%**	55%

Both better at MCP than PIP; **PIP recurrence more severe**

PNA — office, 22–25 G needle; elderly/comorbid; risk **nerve/tendon injury**; recurrence up to 58%

Fasciectomy – indications, technique & complications

– – Active space – –

Partial (selective) fasciectomy	**gold standard**; removes diseased tissue only; **lowest recurrence**; best for **severe PIP**
Incisions	Brunner zig-zag, V-Y, **sequential Z-plasty → allows skin closure** ✗ Wrong options: recurrence
McCash open palm	transverse DPC wound left open → **↓ haematoma, ↓ stiffness**; slower healing
Total/radical fasciectomy	rarely; diffuse/recurrent; high complications, no ↓ recurrence
Dermofasciectomy + FTSG	recurrent/diathesis; **recurrence rare under graft**; grafts take even over NVB
Dermofasciectomy vs fasciectomy	**no proven difference** in ROM/recurrence/satisfaction
Skoog (TO)	selective fasciectomy of **pretendinous fibres only**
Luck (TO)	**subcutaneous fasciotomy** (3 puncture wounds; elderly/unfit)
Staged	mini-ex-fix across PIP 6 wk → fasciectomy
Amputation	severe small finger PIP contracture

Do CTR on separate occasion (TO)

Correction principles (TO)	**MCP contractures correct more easily than PIP** · **PIP < 30° fares better** · persistent PIP tightness after cord excision = **joint contracture → capsulotomy** · protect digital nerves near spiral band · PNF: higher recurrence, for elderly / low demand · dermofasciectomy + graft for recurrent aggressive disease

6 Complications

Complication	Rate / key fact
Wound (dehiscence, necrosis, skin tear)	**23% — commonest after fasciectomy**
Scar pain	17%
Infection	**Fasciectomy 7% · PNA 1% · CCH 0%**
CRPS	5.8%; ♀
Flare reaction	3.5%; ↓ by delaying splint to 1st visit
NV injury	**2–3% primary · 20% recurrent**; **spiral cord**; PNA
Haematoma	2%; ↓ with McCash
CCH skin tear	12%

Recurrence	Rate
CCH	~50% at 5 y (35% in some stems)
PNA	~50% at 3 y
Fasciectomy	**3.5–20%** at 4–5 y (long-term up to 50% TO)

Recurrence risk factors	**PIP disease**, small finger, **diathesis**, PNA/CCH > fasciectomy

– – Active space – –

NOT a recurrence risk factor (TO)	**MCP joint contracture**
PIP secondary structures	volar plate, accessory collaterals, flexor sheath → may need release

X vs Y

	CCH	Needle aponeurotomy	Partial fasciectomy
Best for	MCP, palpable cord	MCP, elderly/unfit	**PIP, severe, recurrence-averse**
Recurrence	~50% 5 y	~50% 3 y	**Lowest**
Unique complication	**Rash, axillary nodes**, tendon rupture	Nerve/tendon injury	**Wound complications 23%**
Infection	**0%**	1%	7%
Cost	**Highest**	Lowest	–

Cord	Joint
Central / pretendinous	**MCP**
Spiral	**PIP**
Retrovascular	**DIP**
Natatory	**Web space**

How it's asked

"Persistent PIP flexion after cord excision — next step?"	→ **PIP capsulotomy (joint contracture)** (TO)
"Cause of PIP contracture?"	→ **spiral cord** (TO)
"Natatory involvement?"	→ **web space contracture**
"Structure spared?"	→ **Cleland ligament** (TO)
"Spiral cord relative to NVB?"	→ **lateral & deep** (NVB pushed central & superficial)
"Collagen in nodules?"	→ **type III** (TO)
"CCH spares NV because?"	→ **type IV**
"CCH degrades?"	→ **I & III**
"Not implicated in pathogenesis?"	→ **CBFA-1**
"Not a bad prognostic factor?"	→ **epilepsy** (TO)
"Not a recurrence risk factor?"	→ **MCP contracture** (TO)
"Not a risk factor?"	→ **vibrating tools occupation** (TO)

"Not true"	→ **painful nodules = indication for surgery** (TO) · **high-dose radiotherapy** (TO) · **index/middle most common** (TO)
"Not a treatment"	→ **Hyalase** (TO)
"Meals & Hentz protocol"	→ Dupuytren (CCH manipulation)
"Skoog & Luck"	→ Dupuytren (TO)
"Garrod pads + 5° MCP"	→ **observe**
"CCH manipulation delayed 1 wk"	→ **equivalent correction**
"CCH vs PNA at 1 y"	→ **equivalent**
"Only after CCH"	→ **rash + axillary lymphadenopathy**
"Recurrence after PNA, MCP 35° + PIP 40°, lowest recurrence – commonest complication?"	→ **wound dehiscence/skin necrosis**
"Finding favouring fasciectomy over CCH"	→ **degree of PIP contracture**
"Z-plasty helps with"	→ **skin closure**
"Dermofasciectomy better for?"	→ **none – no proven difference**

– – Active space – –

EPIDERMAL INCLUSION CYST

– Hand · 26 / 103 –

1 Basics

Nature	painless, **benign, slow-growing** soft-tissue tumour; months–years after trauma
Rank	**3rd most common hand tumour**
Demographics	♂ **>** ♀; **3rd–4th decade**
Site	**distal phalanx / fingertip**

2 Mechanism

Pathogenesis	**penetrating injury** drives **keratinising epithelium** into subcutis or bone → **epithelium-lined cyst filled with keratin**

3 Presentation

Symptoms	**painless fingertip mass**; erythematous painful lesions rare
Exam	flesh/yellow/white; **well-circumscribed, firm, slightly mobile**; superficial, tethered to skin
vs ganglion	**firmer**, **does not transilluminate**
Large lesion near IP joint	↓ ROM
Digital nerve compression	↓ 2-point discrimination



Fingertip epidermal inclusion cyst

4 Diagnosis & Classification

Key radiology findings

Views	AP, lateral, oblique of digit/hand

X-ray	soft-tissue mass; **lytic distal phalanx lesion** if erodes bone → mimics malignancy/infection
Biopsy	before excision **if lytic distal phalanx lesion** (exclude neoplasm/infection)



Intraosseous epidermal inclusion cyst, distal phalanx

Histology

Gross	thick **white keratinous** material
Micro	**keratin-filled cyst lined by epithelial cells**
Differentials	tophaceous gout, foreign-body granuloma, sebaceous cyst, **pyogenic granuloma**, GCT tendon sheath, ganglion, enchondroma, glomus

5 Management

Observation	**not recommended**
Marginal excision	**remove entire capsule → curative**, low recurrence
Bone erosion	curettage + bone graft; amputation rarely (advanced destruction)
Indications	diagnosis, pain, loss of function, cosmesis
Malignant transformation	**never reported**

6 Complications

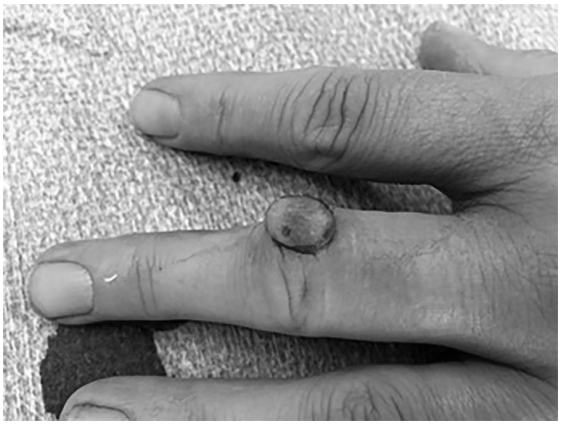
List	wound problems, infection, digital neurapraxia, recurrence (**low even with bone involvement**)

– – Active space – –

X vs Y

	Epidermal inclusion cyst	Pyogenic granuloma	Glomus tumour	GCT tendon sheath	Enchondroma
Growth	**Slow**, after penetrating injury	**Rapid**, after repeated trauma	Rare, benign	Benign nodular	Intramedullary
Look	Flesh/white, firm, tethered	**"Beefy red", bleeds on bumping**, ≤2 cm	Subungual	Nodule on tendon sheath	Bone lesion
Histology	Keratin, epithelial lining	**Collarette around lobules of endothelial cells**	Glomus body	–	Chondrogenic
Treatment	Marginal excision	**Excision + silver nitrate cautery** (office); 85% resolve, mean 1.6 Rx	–	–	–

Pyogenic granuloma benign **vascular** lesion; commonest **head & neck**; slight ♂; recurrence least with surgical excision



Pyogenic granuloma of ring finger

How it's asked

"Rapid, bleeding vascular finger mass after repeated trauma, excise + silver nitrate?"	→ **pyogenic granuloma** ✗ Wrong options: epidermal inclusion cyst · GCT · glomus · enchondroma
"Fingertip mass after penetrating injury, lytic distal phalanx?"	→ **epidermal inclusion cyst → biopsy, then marginal excision**
"Histology?"	→ **keratin-filled cyst with epithelial lining**

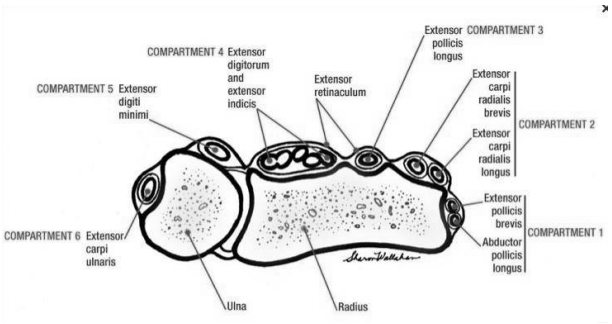
EXTENSOR TENDON COMPARTMENTS

– Hand · 27 / 103 –

1 Basics

Dorsal compartment contents

Dorsal wrist compartment	Contents
1	**APL, EPB**
2	**ECRL, ECRB**
3	**EPL** only (✗ Wrong options: ECRL)
4	**EDC, EIP, PIN** (floor)
5	**EDM**
6	**ECU**



Six dorsal wrist compartments

EIP + EDM (TO)	independent extensors lying **ULNAR to the EDC slip** of the same finger · EPL turns round **Lister's tubercle**
Related syndromes (TO)	**intersection** = 1st compartment crosses 2nd in distal forearm (pain, crepitus) · **drummer's wrist** = EPL attrition at Lister's · **Vaughan-Jackson** = ulnar → radial attritional ruptures (RA, prominent ulnar head)
PIN	**only nerve** in the dorsal compartments; consistently at **base (floor) of 4th**
Septa	1/2 radial surface thickest; **3/4 septum thinnest**; **5th compartment lowest resistance to failure** (Iwamoto)
Extensor retinaculum	check rein against **bowstringing → repair** at surgery (Palmer)
PIN ± AIN block	PIN alone: **1 cm ulnar to proximal Lister's tubercle**; PIN + AIN single injection: 1 cm ulnar + 3 cm proximal to Lister's (Grutter, 100% accurate)

– – Active space – –

Number of compartments (hand)

Hand compartments (TO) **10**

Hand compartment (TO)	Number
Dorsal interossei	4
Palmar (volar) interossei	3
Thenar + hypothenar	2
Adductor pollicis	1
Total	**10**

11th (TO) some authors add **mid-palmar** compartment

How it's asked

"Compartment incorrectly paired with contents?"	→ **3rd: EPL + ECRL** (ECRL is in 2nd)
"Only nerve in dorsal compartments?"	→ **PIN, floor of 4th**
"How many compartments in the hand?"	→ **10** (TO)

EXTENSOR TENDON INJURIES

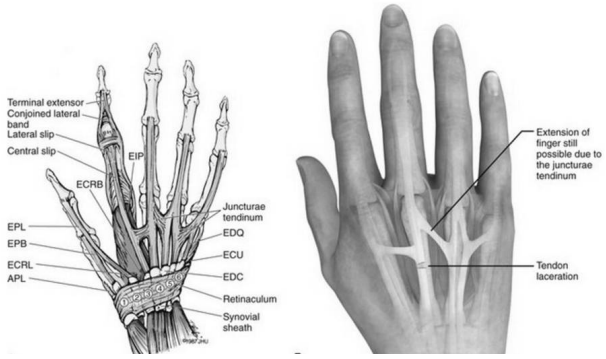
– Hand · 28 / 103 –

1 Basics

Epidemiology & anatomy

Commonest digit	**long finger**; commonest zone **VI**
Extensor mechanism	EDC → **central slip** (PIP extension) + **2 lateral bands** (DIP extension)
PIP flexion	central slip tensed, lateral bands sublux volar + slacken → DIP lax
Lateral band restraints (TO)	**triangular ligament** prevents volar subluxation; retinacular ligaments limit dorsal
Juncturae tendinum	link adjacent EDC → partial/temporary extension despite zone V cut (masks injury)
Retraction (TO)	extensor tendons **flat distally** · retract less when juncturae + sagittal bands intact → cut end found at wound
Extensor hood (TO)	over P1; receives **interossei + lumbricals** volarly → MCP flexion + IP extension · vincula supply from volar side

Retinacular ligament (TO)	Function	If damaged
ORL (Landsmeer) – P1 + flexor sheath → lateral bands / terminal tendon	Links PIP + DIP: **PIP extension tightens it → helps DIP extension**	Tight → contributes to **boutonnière**
TRL – flexor sheath at PIP → lateral band edges	**Stops lateral bands shifting dorsally**	**Lax / torn** (laceration, RA synovitis) → **swan neck** · tight + torn triangular → boutonnière
Triangular – dorsal, between lateral bands over P2	Holds bands together; prevents volar subluxation	**Attenuated → boutonnière** · contracture → swan neck



Juncturae tendinum

– – Active space – –

2 Mechanism

Zone	Mechanism
I	**Forced flexion of extended DIP**
II	Dorsal laceration / crush
III	Hyperflexion, blunt/penetrating trauma, **volar PIP dislocation**, rotation of flexed PIP (condyle buttonholes through triangular lig.) (TO)
V	**Fight bite**; **sagittal band rupture ("flea flicker")** = forced extension of flexed digit, long finger

3 Presentation

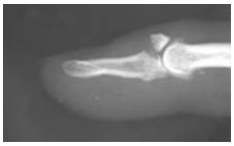
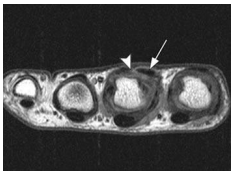
Zone I can't extend DIP (mallet)

Zone III / boutonnière (TO) **PIP flexed, DIP hyperextended, can't flex DIP**; may appear **10–14 d** after injury

Zone V extensor lag; **radial sagittal band** (stronger) rupture → tendon subluxes **ulnar**, MCP held flexed, no active extension

4 Diagnosis & Classification

Zones of extensor tendon injury

Zone (finger / thumb)	Level & key fact	Image
I	DIP / thumb IP (EPL) → **mallet finger**	
II	Middle phalanx / thumb P1 (EPL)	
III	**PIP (central slip)** / thumb MCP → **boutonnière**	
IV	Proximal phalanx / thumb MC; **adhesions common**	
V	**MCP** / thumb CMC → **fight bite, sagittal band**	
VI	Metacarpal; **commonest**; nerve/vessel injury likely	
VII	Wrist → **repair retinaculum** (bowstringing); immobilise **wrist 40° ext + MCP 20° flex, 3–4 wk**; adhesions common	
VIII	Distal forearm; penetrating; nerve injury; immobilise **elbow flexion + wrist extension**	

Central slip tests

Test	Method	Positive
Elson (most reliable, before deformity)	**PIP flexed 90° over table**, extend PIP against resistance	**DIP rigid** (normal = DIP **supple**)
Modified Elson (TO)	Both digits PIP fully flexed, middle phalanges back-to-back	Injured DIP extends better
Boyes (TO)	Extend PIP, actively flex DIP	**Can't flex DIP** – **late** sign

Tightness tests

	Intrinsic tightness	Extrinsic tightness
Cause	**Crush → interossei/lumbrical fibrosis**	**Extensor adhesions** (e.g. MC fracture)
PIP passive flexion ↓ when	**MCP extended** (Bunnell)	**MCP flexed**
Improves with	MCP flexion	**MCP extension**
Treatment	Interosseous tenotomy	**Tenolysis / fractional extensor lengthening** ± MC osteotomy

Key radiology findings

X-ray AP + lateral digit → exclude **bony mallet** / avulsion

MRI axial T1 sagittal band rupture

5 Management

Extensor tendon injury

- Laceration <50% + can extend against resistance → **PROTECTED EARLY MOTION**
- Laceration >50% (any zone) → **TENDON REPAIR**
- Closed injury → **SPLINT the joint of the zone**
  - Zone I mallet → **DIP extension splint 6 wk**
  - Zone III central slip → **PIP extension splint, DIP free**
  - Zone V sagittal band → **MCP extension splint 4-6 wk**

– – Active space – –

Operative criteria

Situation	Treatment
Laceration >50% width	Repair
Bony mallet + P3 volar subluxation	CRPP through DIP / extension-block pinning; **ORIF if >50% articular surface**
Fight bite to MCP	**Immediate I&D**, loose/delayed closure; cover **Eikenella**
Chronic / irreparable	Reconstruction (2-stage)
Chronic EPL rupture	**EIP → EPL transfer**

Mallet (zone I) splinting

DIP extension splint — acute (<12 wk), nondisplaced bony mallet, chronic (>12 wk) if **supple + congruent**; **full-time 6 wk** + part-time 4–6 wk; avoid hyperextension (skin necrosis); keep PIP free; noncompliance common

Central slip injury – Elson test & PIP extension splinting

Acute closed central slip — **PIP extension splint, DIP free to move (active DIP flexion)**; **6 wk** full-time + 4–6 wk part-time (Orthobullets, OB) · **5 wk** (OB) · **4–6 wk round the clock + nights** (TO)

Why DIP motion — prevents **oblique retinacular ligament contracture** / adhesions

✗ Wrong options: **Distractors** – full-time DIP splint (mallet), MCP extension splint (sagittal band), **dorsal blocking splint** (dorsal PIP dislocation), immobilising DIP

Surgery (TO) — only complete rupture/transection/bony avulsion after failed conservative trial (Doyle, Snow, Aiche)

Chronic boutonnière >6 wk (TO) — stiff/arthritic/RA → **PIP arthrodesis**; supple → lateral band relocation, **terminal tendon tenotomy (Fowler/Dolphin, not central slip)**, modified Littler

Central slip reconstruction — graft, extensor turndown, lateral band mobilisation, TRL, FDS slip

Pseudo-boutonnière (TO) — PIP flexion contracture **without DIP hyperextension**; after **dorsal PIP dislocation / volar plate injury**; can't passively extend PIP

EPL rupture – EIP transfer

Drummer's palsy (TO) — **EPL tendon**; chronic tendinitis from repetitive palmar flexion + ulnar deviation; rubs on **Lister's (dorsal) tubercle** → attenuation/rupture

Drummer's treatment (TO) — conservative / activity modification; rupture → **EI to EPL transfer**

EPL rupture after DRF — **nondisplaced/minimally displaced DRF**; ~**3 mo**; **nutritional compromise + attrition** → **EIP → EPL transfer** or graft
✗ Wrong options: primary repair · EPB · observation

ECU tendinitis/subluxation

ECU (extensor carpi ulnaris) — tennis forehand; **dorsoulnar pain** on **supination + wrist flexion + ulnar deviation**; X-ray normal

Subluxation — ECU retinaculum ruptures → tendon leaves sheath **in supination**, returns in pronation

Treatment	immobilisation; surgery if refractory

Rheumatoid extensor ruptures (Vaughan-Jackson, EDM)

Commonest RA tendon rupture (TO)	**EDM / EDQ** (extensor); **FPL** (flexor)
Vaughan-Jackson (TO)	**EDQ first → EDC small → ring → middle → index**; **ulnar → radial**; final stage of **caput ulnae** (dorsal ulnar head subluxation)
Mechanism (TO)	attrition on raw dorsal ulnar head + invasive tenosynovitis + ischaemia
EPL in RA (TO)	ruptures at **Lister's tubercle** (tortuous course)
Mannerfelt (TO)	**FPL** rupture in carpal tunnel over **scaphoid / radial osteophyte**
Clinical (TO)	painless "drop finger"; no active MCP extension; passive extension holds (tenodesis) → distinguish from MCP dislocation, **radial nerve palsy**
Bone work first (TO)	**Darrach** (low-demand) · **Sauvé-Kapandji** (younger) + dorsal synovectomy; wrist stabilisation if severe radial deviation

Ruptured tendons (TO)	Reconstruction
1 (EDQ)	Side-to-side to **ring EDC**
2 (EDQ + EDC small)	Side-to-side to **middle EDC**
3 (small, ring, middle)	**EIP transfer**
4 (small → index)	**FDS ring** or **ECRL** transfer

Post-op (TO)	dynamic extension splint 4–6 wk

Repair technique & reconstruction

Technique by level (TO)	distal (flat) tendons: **horizontal mattress** (most used) or **figure-of-8** · proximal (rounder): **modified Bunnell / modified Kessler** · **4-0 non-absorbable** (nylon / Prolene) · bury knots · check tension by **tenodesis effect**
Splint zones V–VIII (TO)	EPL repair: thumb full extension · fingers IP straight, MP neutral, wrist mild extension (counteract stronger flexors)
Zone V fight bite (TO)	dirty wound → debride, explore, repair sagittal band, cover
Strength	**number of strands crossing repair** > grasping loops; 4–6 strands → early active motion; epitendinous optional
Weakest	**post-op day 6–12**; fails at **knots**
Reconstruction	2-stage: **silicone rod → graft at 3–4 mo**; grafts: **palmaris longus** (commonest; absent 15%), plantaris (absent 19%, longer), toe extensor
Tenolysis	adhesions with ↓ flexion; wait **>3 mo** + full passive ROM; not with procedures needing immobilisation

Extensor laceration rehabilitation – relative motion splint

SAM (short-arc motion)	after **zone III repair**; ↑ arc, ↑ DIP motion, ↓ therapy time; **4 mm excursion**
Relative motion (yoke) splint	injured MCP in **relative hyperextension**; **zones IV–VII**
Yoke benefits	early active ROM, ↓ adhesions, ↓ ADL interference, adjacent digits free, **high compliance**, earlier RTW; **NOT "better at limiting motion"**

– – Active space – –



Yoke splint

6 Complications

Complication	Key facts
Adhesions	↓ flexion; **zones IV & VII**, older pts → **tenolysis at 3–6 mo**
Rupture	**5%**; days **7–10**; revision or reconstruction
Swan neck	Prolonged DIP flexion, dorsal lateral bands, PIP hyperextension → **Fowler central slip tenotomy**, SORL reconstruction
Boutonnière	Central slip + volar lateral bands → splint/serial cast → **terminal extensor tenotomy**, PIP volar plate release

X vs Y

	Boutonnière	Pseudo-boutonnière	Swan neck
PIP	Flexed	**Flexion contracture**	**Hyperextended**
DIP	**Hyperextended**	Normal / passive motion	Flexed
Cause	**Central slip** (volar PIP disloc.)	**Volar plate** (dorsal PIP disloc.)	Lateral bands dorsal
Tenotomy	**Terminal tendon**	–	**Central slip (Fowler)**

How it's asked

"Drummer's palsy involves?"	→ **EPL tendon** (TO)
"Thumb won't extend 3 mo after nondisplaced DRF – next?"	→ **EIP → EPL transfer**
"Tennis forehand, dorsoulnar pain on supination/flexion/ulnar deviation?"	→ **ECU tendon**
"DIP hyperextended, can't flex DIP, PIP flexed – injury?"	→ **EDC central slip** (TO)
"Exam finding in traumatic boutonnière?"	→ **PIP flexed, resisted extension → DIP rigid (Elson)**
"Central slip nonop – which active motion?"	→ **DIP flexion**
"Positive Elson, initial plan?"	→ **PIP splint 6 wk, DIP free**

"Relative motion splint benefits EXCEPT?"	→ **better at limiting digit motion**
"↓ PIP flexion with MCP flexed, improves with MCP extension — Dx & Rx?"	→ **extrinsic tightness → tenolysis / fractional lengthening**
"Most common tendon rupture in RA hand?"	→ **EDM** (TO)
"Lax / torn transverse retinacular ligament causes?"	→ **swan neck (lateral bands sublux dorsally)** (TO)
"Extensor repair technique for flat distal tendon?"	→ **horizontal mattress / figure-of-8, 4-0 non-absorbable** (TO)
"Vaughan-Jackson first tendon & direction?"	→ **EDQ, ulnar → radial** (TO)

– – Active space – –

EXTREMITY FLAP RECONSTRUCTION

– Hand · 29 / 103 –

1 Basics

Definitions & indications

Flap	tissue unit transferred **with its own vascular supply**
Pedicle	vascular portion; usually 1 artery + ≥1 veins
Indication	exposed **bone, tendon, cartilage, implant** ✗ Wrong options: STSG/FTSG on exposed tendon/bone/prosthesis (no viable bed, poor gliding)

Plastic surgery principles (TO)

Father of modern plastic surgery (TO)	**Sir Harold Delf Gillies** (Ten Commandments; Gillies lip flap) · **father of plastic surgery = Sushruta** (forehead flap)
Gillies commandments (TO)	make a plan · have a style · return normal to normal position · **do not throw away a living thing** (spare parts) · do not bear false witness against the defect · treat primary defect before secondary · **always provide a lifeboat** (plan B) · do not do today what can be put off till tomorrow (electrical burns) · no routine · do not covet thy neighbour's plastic unit
Paré principles (1564, TO)	remove superfluous · restore displaced · separate joined · join separated · supply defects of nature
NOT a principle (TO)	**"protect the surgical area"** (✓ replace like with like, ✓ always have an alternative plan)
Exam FAQs (TO)	facelift consent: **haematoma** first (commonest, 6–24 h) · nasal fracture: **Walsham** (lateral wall) / **Asch** (septum) forceps

Reconstructive ladder

Rung (simple → complex)
Primary closure
Secondary closure / secondary intention (± NPWT)
Skin graft
Local flap
Regional / distant flap
Free tissue transfer

Rule	choose the **lowest rung** that covers the defect (local before distant or free)

Reconstructive ELEVATOR (TO) has replaced the ladder → go **straight to a free flap** if it is the best option, even for a small defect · **dermal substitutes** (AlloDerm, Biobrane: acellular dermal matrix) let a wound granulate + take a graft

Tissue needed (TO)	Flap examples
Skin only	Groin, **ALT**, anteromedial thigh, radial forearm
Muscle (cover / functional)	**Latissimus dorsi, gracilis**, vastus lateralis, sartorius, gastrocnemius
Nerve	Sural graft, **vascularised ulnar nerve graft** (plexus)
Bone	**Fibula (commonest)**, iliac crest, scapula · first choice for IIIB+ bone gaps

4 Diagnosis & Classification

Blood supply classification

Type	Supply	Indication
Axial pattern	Single named AV pedicle	Length:width **>2:1**
Random pattern	Microcirculation, no named pedicle	Length:width **<2:1** (Orthobullets) · **~1:1 lower limb, up to 3:1 face** (TO) – read the options · V-Y, rotation, bipedicle
Axial examples (TO)	Named vessel along axis	Nasolabial (angular a.) · forehead (supratrochlear a.) · groin (SCIA); survives only a little beyond vessel end
Venous	Veins as arterial inflow + outflow	

Tissue type classification

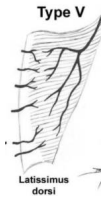
Type	Note
Cutaneous	Skin + subcut
Fascial	Fascia only (temporoparietal)
Muscle	Needs **skin graft** or musculocutaneous (composite); **no motor nerve → atrophy to 50%**
Bone	**Free fibula (peroneal a.) → diaphysis · free iliac crest (deep circumflex iliac) → metaphysis**
Composite	e.g. radial forearm (fasciocutaneous)

Mobilisation classification

Type	Use
Local	Tissue **adjacent** to defect
Distal random	Non-contiguous site; surrounding tissue inadequate; ratio <2:1
Distal axial	Surrounding tissue inadequate; ratio >2:1
Free	Local/distant tissue insufficient

– – Active space – –

Mathes & Nahai muscle vascular classification (TO)

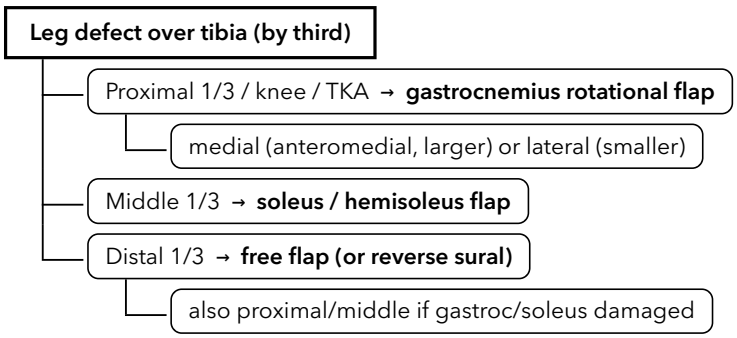
Type	Supply	Examples	
I	**Single** major pedicle	**Gastrocnemius**, TFL, rectus femoris	
II	**One dominant + minor** pedicles	**Soleus**, vastus lateralis, **gracilis**, peroneus brevis/ longus	
III	**Two dominant** pedicles (different regional arteries)	**Rectus abdominis**, gluteus maximus, serratus anterior, semimembranosus	
IV	**Segmental**, multiple similar-calibre pedicles	**Sartorius**, tibialis anterior, EDL	
V	**Single dominant + multiple secondary segmental**	**Latissimus dorsi**, pectoralis major	

Fasciocutaneous classification (TO)	Types
Cormack & Lamberty	**A** multiple perforators at one end · **B** single large perforator · **C** multiple small segmental perforators · **D** osteomyofasciocutaneous
Mathes & Nahai (pedicle source)	**A** direct cutaneous (groin, SIEA) · **B** septocutaneous (scapular/parascapular, lateral arm) · **C** musculocutaneous (median forehead, ALT)
Nakajima	A–F by course of vessel (know the name)

Special flap term (TO)	Meaning
Propeller	Island on a **single perforator**, axial rotation **up to 180°**; length = perforator→defect + defect; skeletonise perforator, fully island (avoid kinking); e.g. lateral malleolus
Supercharging	Extra vessel from an **UNRELATED distant** source
Turbocharging	Extra pedicle **from the SAME flap** (e.g. both DIEP pedicles)
Standard delay	Partial incision/elevation → full elevation after **10–14 days**; pedicled flaps (groin), not free
Strategic delay	Divide minor pedicles → survive on main pedicle; elevate after 2 wk
Prelamination	Build layers into flap at donor site first (cartilage under radial forearm for nose)
Prefabrication	Give tissue a **new dominant pedicle** → raise after **~6 wk**
Venous flaps (Thatte)	I unipedicled · II bipedicled · III arteriovenous

Hallock compound flaps (TO) solitary (one source) · combined: **conjoined** (territories joined physically, each own supply) vs **chimeric** (independent flaps linked to one source vessel) · classic chimeric = **subscapular system** (circumflex scapular → scapular/parascapular paddles + lateral border bone; thoracodorsal → LD + angular branch to scapular tip)

5 Management



Gastrocnemius flap – sural artery supply; knee & tibia coverage

Pedicle **medial / lateral sural arteries** (branches of **popliteal**); single source 85%; **Mathes I**

✗ Gastroc distractors genicular arteries · anterior/posterior tibial · peroneal · saphenous

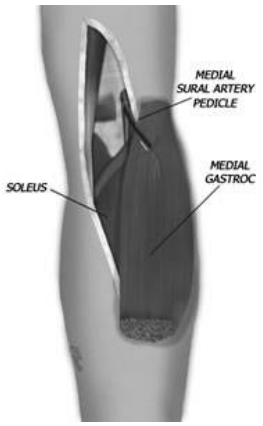
Medial head **medial + midline** defects, proximal 1/3 tibia; **preferred for anteromedial TKA wounds** (larger, more mobile)

Lateral head **lateral** defects; smaller, less excursion; lateral sural artery

Reach defects **up to 15 cm** from knee joint; fascial pie-crusting or proximal release (island) → **+5 cm**

TKA wound necrosis / PJI with exposed prosthesis **gastrocnemius flap first**
✗ Wrong options: latissimus/gracilis free (higher on ladder, donor morbidity) · STSG/FTSG

– – Active space – –



Medial gastrocnemius flap

Soleus supply	PT (medial) + peroneal (proximal) + popliteal trunk branches (Orthobullets) · **peroneal proximally, PT distally** (OB) · **PT medially, peroneal distal/lateral** (OB) – read the options ✗ Wrong options: anterior tibial perforators
Free flap	distal 1/3 tibia; **gracilis = most common free muscle donor** (**medial femoral circumflex a.**, anterior division obturator n.)
ALT free flap	**lateral femoral circumflex a.** ✗ Wrong options: popliteal

Upper-extremity flaps – lateral arm, radial forearm, latissimus & groin flaps (pedicles)

Flap	Pedicle	Key facts
Lateral arm	**Posterior radial collateral a.** (branch of **profunda brachii**), 3–5 septocutaneous perforators in lateral IM septum	Max width 8 cm, pedicle 8 cm; supine; primary donor closure; **posterior antebrachial cutaneous n. sacrificed** (forearm hypoaesthesia); protect **radial n.**; osteo/sensate/fascial variants
Radial forearm (rotational)	Radial a.	Local flap for dorsal hand/forearm when **PIA disrupted** by dorsal wound
Posterior interosseous	PIA (septum EDM–ECU)	not if dorsal wound disrupted septum
Latissimus dorsi myocutaneous	**Thoracodorsal a.** (branch of **subscapular**)	**Large defects around elbow**; Mathes V (TO)
Groin (axial, distant)	**Superficial circumflex iliac a.**	Mainstay UE coverage (thumb amputation); **LFCN at risk** (pierces iliac fascia medial to ASIS)

Radial collateral a.	runs with radial n. in spiral groove → pierces lateral IM septum → anterior + **posterior** radial collateral (posterior to septum)
✗ Lateral arm distractors	brachial, recurrent interosseous, posterior ulnar recurrent, middle collateral, radial recurrent, anterior radial collateral
Dorsal hand wound (MC + radius fx, exposed tendon/ bone)	**radial forearm rotational flap** (local) > groin (distant) > free lateral arm (free)

– – Active space – –



Groin flap

Big 5 free flaps (TO)

Flap (TO)	Pedicle	Size / parts	Key points
ALT	**Descending branch of lateral circumflex femoral a.** (septo- + musculocutaneous perforators)	Up to ~25 × 30 cm ± vastus lateralis	Versatile, low morbidity, bulky → suprafascial thin harvest · line **ASIS → knee**, main perforator at **junction of proximal ⅔ / distal ⅓**, rectus femoris–VL septum
Radial forearm ("Chinese")	Radial a. + venae, **always include cephalic vein**	Thin, pliable ± radius	Non-dominant arm; donor needs graft · stops 2 cm above styloid · **Allen: < 6 s safe, 7–11 s equivocal, ≥ 12 s do not harvest**
Latissimus dorsi	**Thoracodorsal a.** (subscapular, 3rd part axillary); thoracodorsal n. (C6–8)	Large; Mathes V	Free for lower-limb cover; **pedicled for elbow flexion** (BPI), back, breast · **TDAP** spares muscle · pivot at axilla
Gracilis	**Medial circumflex femoral a.** (type II) · anterior branch obturator n.	Muscle	Cover + **free functional transfer** (BPI elbow flexion, Volkmann) · pedicle **between adductor longus and gracilis** · keep incision medial
Free fibula	**Peroneal a.** (nutrient a. at mid-shaft; endosteal + periosteal)	**22–26 cm** bone; skin paddle ~12 × 20 cm	Peroneal from PT distal to popliteal fossa, between TP + FHL · **fibular head supplied by anterior tibial a.** → second anastomosis if head taken

Fibula contraindications (TO) PVD · heavy smoking · absent anterior tibial vessels · **peronea magna (0.2%) = absolute** · venous insufficiency · CTA not routine (pulses / Doppler)

Free flap monitoring (TO) **clinical = gold standard**: warm, light pink, not pale/congested, **slow bright red ooze on pricking** · adjuncts: handheld / laser Doppler, implantable devices, photoplethysmography, fluorescein, TcO_2

Groin flap (TO) SCIA; ~10–12 cm long, ≥ 8 cm wide; hidden scar · hand "in the pocket" **2–3 wk** until division (hygiene) · **scapular / parascapular** on circumflex scapular vessels (parascapular = ellipse along inferolateral border)

– – Active space – –

Bone flaps – free fibula (peroneal artery)

Free fibula — **peroneal a.** (1.5–4 mm) + 2 venae; pedicle 4–6 cm; ± skin paddle; keep distal **5–6 cm** (syndesmosis); defects **>6 cm** (trauma, infection, congenital, AVN) → 10 cm radius defect

Bone flap	Pedicle	Use
Free fibula	**Peroneal**	Long diaphyseal defects
Free iliac crest	Deep circumflex iliac	Metaphyseal
Medial femoral condyle	**Longitudinal br. descending geniculate**	Scaphoid nonunion
Medial femoral trochlea	Transverse br. descending geniculate	Scaphoid proximal pole nonunion
Distal radius vascularised graft	**1,2-intercompartmental supraretinacular a.** (radial a.)	Scaphoid AVN/nonunion
Lateral arm osteo	PRCA	Small bone only
Scapula (TO)	Circumflex scapular (or angular br. thoracodorsal)	Chimeric reconstructions
Radius (TO)	Radial a. (with radial forearm flap)	–
Calvarial (TO)	Superficial temporal / occipital a.	–
MFC osteo-periosteal (TO)	**Articular branch of descending genicular a.**	When an intra-articular surface is needed

Vascularised bone nonunion — up to **32%** without additional bone graft

Moberg flap limitations in the thumb

Moberg — volar advancement; medium defects **1.8–3 cm**; >1.5 cm → ↑ dehiscence, **parrot-beak nail**, ↓ padding → modifications (V-Y, Z-plasty, Burrow triangles, **O'Brien island + graft**)

3.5 cm pulp defect — **Moberg contraindicated**: needs **IPJ flexion >45° → IP stiffness**

Large thumb pulp options — **Foucher / Holevich FDMA flap** · **Littler neurovascular island** · **free great toe pulp**

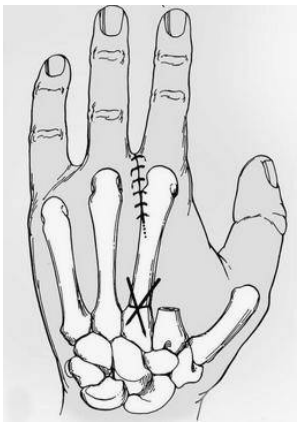
Ray transposition after digit amputation

Problem — central ray loss → objects fall through gap + **scissoring**

Ray amputated	Gap closure
Middle	**Index metacarpal transposition** (straight or step-cut; K-wire or plate)
Ring	**Little (small) metacarpal transposition**
Benign, ligaments intact	Suture deep transverse intermetacarpal ligaments / remove MC base fully

(✗ Wrong options: **Traps** – ring→long transposition leaves new gap) · DTML suture impossible if resected with **tumour** · 2nd toe transfer = multiple digit / thumb loss · iliac crest graft = thumb reconstruction

– – Active space – –



Index metacarpal transposition

Negative pressure wound therapy settings

Most efficient pressure	**125 mmHg** → 4× blood flow; granulation ↑63% continuous, **↑103% intermittent**; <50 or >250 → ↓ granulation
NPWT ≠ coverage	temporary; no infection reduction vs soft dressing (**WOLFF 2018**)

Timing

Free transfer	**within 7 days** of severe UE trauma → ↓ complications (Orthobullets) · coverage **within 7 days of definitive fixation** (OB) – orthoplastic team
Fasciocutaneous vs muscle flaps	equivalent bone healing, salvage, infection

6 Complications

Flap failure as infection predictor

Strongest predictor of wound infection	**flap failure**; 2nd = **time from definitive fixation to flap coverage** ✗ Wrong options: age · temporary ex-fix, time from debridement weaker

Failure	Treatment
Arterial insufficiency	**Immediate return to OR**
Venous congestion	Loosen dressings, remove sutures → OR if unrelieved

Donor site	cosmesis, graft pain, **seroma → aspirate** (excise if encapsulated)

How it's asked

"Type IIIB proximal 1/3 tibia – medial gastroc dominant supply?"	→ **sural artery** ✗ Wrong options: genicular · AT · PT · saphenous
"Sural artery supplies which flap?"	→ **gastrocnemius** ✗ Wrong options: soleus
"Lateral defect over revision TKA, exposed deep tissue?"	→ **gastrocnemius flap** ✗ Wrong options: lat dorsi/gracilis free · skin graft

– – Active space – –

"Anteromedial TKA wound – label for dominant supply?"	→ **medial sural artery** ✗ Wrong options: popliteal · geniculates
"Correct flap–artery pair?"	→ **lateral gastroc – lateral sural** ✗ Wrong options: hemisoleus–AT · medial gastroc–PT/peroneal · ALT–popliteal
"Latissimus dorsi vascular pattern?"	→ **single dominant + secondary segmental (Mathes V)** (TO)
"Lateral arm flap perforators arise from?"	→ **posterior radial collateral artery**
"Dorsal hand wound, exposed tendon/bone, dorsal forearm wound – coverage?"	→ **radial forearm rotational flap** ✗ Wrong options: STSG · PIA flap · groin · free lateral arm
"Groin flap artery & nerve at risk?"	→ **superficial circumflex iliac a.; LFCN**
"Latissimus flap for posterior elbow defect – artery?"	→ **thoracodorsal** ✗ Wrong options: subscapular · circumflex scapular
"10 cm radius defect – anastomosis artery?"	→ **peroneal** (free fibula)
"3.5 cm thumb pulp – contraindicated flap?"	→ **Moberg** (IP stiffness)
"Middle ray resected with DTML – prevent scissoring?"	→ **index metacarpal transposition** ✗ Wrong options: ring transposition · ligament suture
"Most efficient NPWT pressure?"	→ **125 mmHg**
"Father of modern plastic surgery?"	→ **Sir Harold Gillies** (TO)
"Supercharging vs turbocharging?"	→ **unrelated distant vessel vs vessel from the same flap** (TO)
"Allen test refill time contraindicating radial forearm flap?"	→ **≥ 12 s** (TO)
"Absolute contraindication to free fibula?"	→ **peronea magna** (TO)
"Best method of free flap monitoring?"	→ **clinical (colour, warmth, bright red bleeding on pricking)** (TO)
"Strongest predictor of infection after flap for open tibia?"	→ **flap failure**

FELON

– Hand · 30 / 103 –

1 Basics

Epidemiology – paronychia

Most common hand infection (TO)	**acute paronychia** ✗ Wrong options: felon, whitlow, subungual
Felon share	**15–20%** of hand infections
Felon site	**thumb > index** (TO)
Felon (TO)	infection of **subcutaneous tissue of distal pulp**
Paronychia (TO)	infection of nail fold (eponychium); ***S. aureus***; poor nail hygiene; risk: DM, steroids, chemo, immunocompromise, manual labourers, farmers

Perionychium term (TO)	Meaning
Perionychium	All tissue surrounding the nail
Nail bed	**Germinal matrix** (most nail growth; **lunula** = visible part) + **sterile matrix** (smooth gliding surface)
Hyponychium	Keratinised tissue distal to nail bed → **barrier to infection**
Paronychium	Skin fold each side of nail
Eponychium	Thin fold sealing proximal nail fold over plate

Eponychia (TO)	infection of one nail fold + eponychium · **runaround infection** = pus from one fold to the other via eponychium
Paronychia work-up (TO)	**screen for diabetes** · X-ray for osteomyelitis
Early paronychia, no pus point (TO)	**1st-generation cephalosporin** + warm **povidone-iodine soaks (1:10 in warm saline, 10–15 min twice daily)** · surgery: "follow the infection" (1–2 incisions, nail plate removal, volar exposure if spread)

Paronychia stage (TO)	Treatment
No pus point	**Antibiotics**
Abscess one side of nail	**I&D (single incision)**
Pus to opposite side + under nail	**Second incision + remove proximal third of nail**

– – Active space – –



Incisions for acute paronychia

Anatomy & pathophysiology

Pulp	closed sac; **fibrous vertical septae** periosteum → epidermis (stabilise pinch/grasp); eccrine sweat glands
Apical infection (TO)	infection limited to **skin of pulp tip** (vs felon = pulp abscess)
Pulp = "second eye" (TO)	rich sensory supply; septa give firm non-slip grip
Blood supply	digital arteries give **nutrient branch to epiphysis before entering pulp** → epiphysis spared → **children rarely get epiphyseal necrosis** (TO)
Pathophysiology	↑ pressure in micro-compartments → **pulp "compartment syndrome"** → diaphysis ischaemia > skin → **diaphyseal necrosis/sequestration** before spontaneous skin decompression

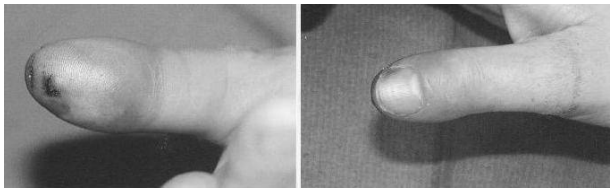
2 Mechanism

Penetrating	glucose needlestick, splinters
Local spread	from paronychia
No injury	**50%** (contamination via eccrine glands)

Organism	Setting
S. aureus	**Most common**
Gram −ve	Immunosuppressed
Eikenella corrodens	**Diabetic nail-biters**

3 Presentation

Symptoms	severe **throbbing pain**; tense, tender pulp
Swelling limit	**not proximal to DIP crease** unless sheath/joint involved



Felon

4 Diagnosis & Classification

Diagnosis	**clinical**; X-ray only if trauma (fracture/FB); no MRI, no labs
Differentials	**herpetic whitlow**, paronychia, glomus tumour, mucous cyst, psoriasis, **tophaceous gout**
Herpetic whitlow (TO)	**HSV-1**; distinction from felon mainly by **history**; **24–72 h burning prodrome** → erythema → **clear vesicles** coalescing around nail folds; turbid not purulent; **pulp NOT tense**; persists ~2 wk, resolves over next 1 wk

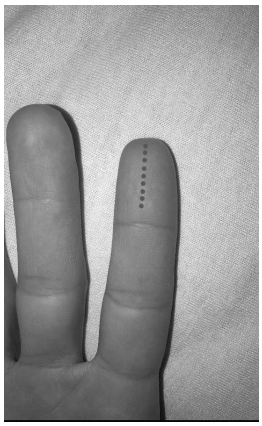
5 Management

Early felon (TO)	oral antibiotics + warm povidone-iodine soaks + immobilisation · **pus point → surgery**

- **Felon**
  - Early, no drainable abscess → **ORAL ABX + observation**
  - Abscess (most cases) → **BEDSIDE I&D + IV ABX, heal by SECONDARY INTENTION**
    - superficial / FB / visible drainage → **VOLAR LONGITUDINAL (midvolar)**
    - deep, no FB, no drainage → **MID-LATERAL (non-pressure side)**

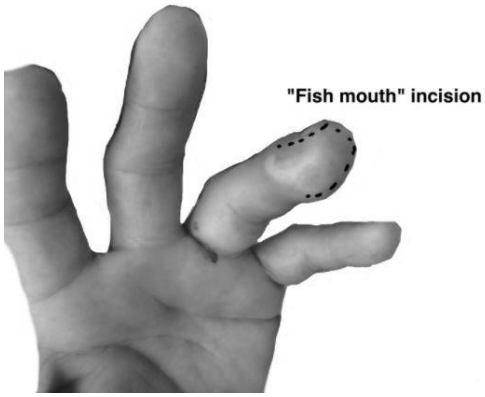
Incision & drainage technique

Placement	over **point of maximal tenderness**; **distal to DIP crease** (avoid contracture & sheath extension)
Mid-lateral side	**ulnar side digits 2–4**; **radial side thumb & small**
Avoid	**fish-mouth** (unstable pulp, **vascular compromise**) · double longitudinal / transverse (digital n./a. injury) · **proximal probing** (seeds flexor sheath)
Debridement	**break up septa**; avoid sheath & DIP joint; Gram stain + culture **before Abx**; gauze wick
Closure	**secondary intention**; soaks from 24–48 h; anti-*S. aureus* cover ✗ Wrong options: primary or delayed primary



Volar midline incision

– – Active space – –



Fish-mouth incision — avoid

Diabetic hand infection – renal failure & amputation

Diabetic triad	poor wound healing + neuropathy + vascular disease → ↑ infection, multiple procedures
↑ Amputation rate	**renal failure** · deep **polymicrobial / Gram −ve** infection ✗ Wrong options: insulin dependence, retinopathy, neuropathy, Gram +ve

6 Complications

Complications	**fingertip compartment syndrome** · **flexor tenosynovitis** · **osteomyelitis / diaphyseal sequestration** · **DIP septic arthritis** · digital tip necrosis

X vs Y

	Felon	Herpetic whitlow (TO)	Paronychia
Site	Distal pulp	Pulp / nail folds	Nail fold
Cause	*S. aureus*	**HSV-1**	*S. aureus*
Pulp	**Tense**	**Not tense**	–
Lesion	Abscess	Clear vesicles, turbid fluid	Pus at fold
Treat	**I&D**	No I&D (self-limiting ~3 wk)	Abx ± I&D

How it's asked

"Most common hand infection"	→ **acute paronychia** (TO)
"Midvolar pulp abscess, failed oral cephalosporin — next step?"	→ **I&D, midline volar incision, secondary closure** ✗ Wrong options: fish-mouth · IV Abx alone · colchicine
"Diabetic hand infection — ↑ amputation with"	→ **renal failure**
"Felon in diabetic nail-biter"	→ ***Eikenella***
"Mid-lateral incision side, index finger"	→ **ulnar**

"Pus tracking from one nail fold to the other via eponychium?"	→ **runaround infection** (TO)	– – Active space – –
"Children with felon rarely get epiphyseal necrosis — why?"	→ **epiphyseal artery arises proximal to the closed pulp space** (TO)	

– – Active space – –

FINGERTIP AMPUTATIONS & FINGER FLAPS

– Hand · 31 / 103 –

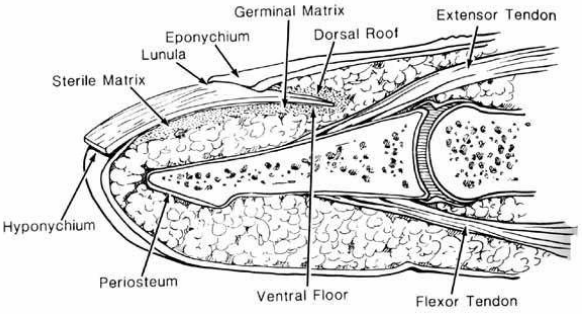
1 Basics

Goals

Goals	**sensate** tip · **durable** tip · **bone support for nail growth**; improper Rx → **stiffness**
Reconstructive ladder	primary closure → secondary intention → STSG → FTSG → random local flap → axial local → island local → distant random → distant axial → free tissue

Fingertip anatomy

Structure	Definition
Eponychium	Dorsal skin proximal to nail (covers nail fold)
Paronychium	Lateral nail folds
Hyponychium	Keratin plug under distal free nail edge
Lunula	White proximal nail; **divides sterile from germinal matrix**
Sterile matrix	Distal; **nail adheres** here
Germinal matrix	Proximal; **90% of nail growth**



Fingertip anatomy

Digital arterial arches

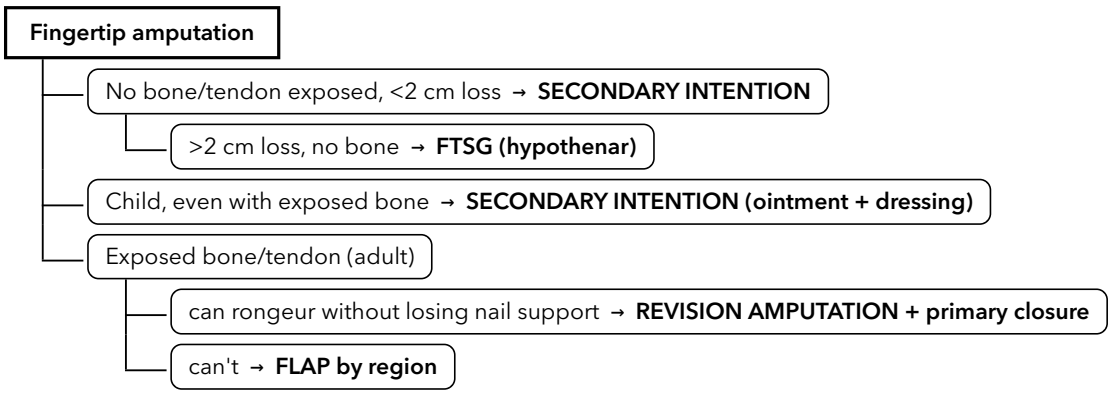
Proper digital aa.	radial + ulnar communicate via **proximal, middle, distal transverse arches**
Middle transverse arch	at **C3 pulley**, deep to flexors; supplies digital aa. between PIP & DIP → **required for retrograde digital artery flap**

3 Presentation

Mechanism	avulsion, laceration, crush
Exam	**laceration geometry** (transverse / dorsal oblique / volar oblique) guides Rx; **exposed bone?**; flexor/extensor involvement

X-ray AP + lateral for **bony involvement**

5 Management



Not replantation not indicated for distal fingertip amputation

Fingertip amputation – healing by secondary intention (children, small defects)

Indications adult/child, **no bone or tendon exposed, <2 cm skin loss**; **children with exposed bone**

6-y-old, exposed P3 → no change: ointment + dressing
✗ Wrong options: homodigital island · thenar · volar advancement · V-Y

6-y-old, lawn mower, 6 mm distal to lunula, contaminated, no bone, <1 cm → I&D + soft dressing + review within 1 wk
✗ Wrong options: replant · DIP revision · Moberg · Z-plasty

Adult 1 cm defect, no bone, sterile matrix intact → local wound care alone
✗ Wrong options: primary closure · revision · V-Y · digital island

Child >1 cm soft-tissue loss controversial: V-Y vs dressings

Technique irrigation + soft dressing → **day 7–10 water–peroxide soaks** → heals **3–5 wk** (Orthobullets) · ~4 wk (Neustein) · semiocclusive 2–8 wk (Germann)

Revision amputation outcomes better **static 2PD & arc of motion** than local flaps; commonest complaint **cold intolerance**

FTSG & revision amputation

FTSG **hypothenar** donor, closed primarily; bolster (cotton ball) 7 d → ROM after 7 d

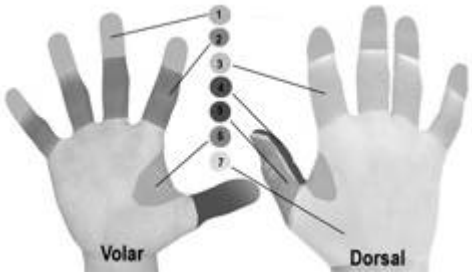
Avoid STSG **contractile, tender, less durable**

No skin graft on bare tendon without paratenon → adhesions

Revision amputation steps **ablate remaining nail matrix** (avoid nail remnants/cysts) · tendon insertions lost → **disarticulate DIP** · **transect digital nerves & tendons proximally** (traction neurectomy) · volar skin over bone to dorsal

– – Active space – –

Flap selection by region



Flap options by region

Region	Flap(s)
Tip – **transverse / dorsal oblique**	**V-Y advancement**, digital island artery
Tip – **volar oblique**	**Cross-finger (>30 y)** · **thenar (<30 y)** · digital island artery
Nail bed sterile matrix + eponychial loss	**Reverse cross-finger**
Volar proximal finger	Cross-finger (>30 y), axial flag (from long finger)
Dorsal proximal finger & MCP	Reverse cross-finger, axial flag
Volar thumb	**Moberg <2 cm** · **FDMA >2 cm** · **neurovascular island up to 4 cm**
Dorsal thumb	**FDMA (Kite)**
1st web space	**Z-plasty 60°**, **posterior interosseous fasciocutaneous** (if >75%)
Dorsal hand	**Groin flap** (superficial circumflex iliac a.)

Advancement flaps – V-Y & volar (Moberg) advancement

V-Y — **transverse / dorsal oblique** tip (more dorsal than volar loss); can't advance over phalanx to cover eponychium

Transverse ring tip, exposed bone → shorten P3 + nail bed removal + V-Y (nail bed removal prevents **hook nail**)
✗ Wrong options: PIP amputation · cross-finger · groin · thenar

Moberg (volar advancement) — **volar thumb pulp ≤2 cm**

Island volar advancement (thumb) — divide all but NV bundles → covers up to **3.5 cm** · up to 4 cm by elevating whole volar thumb skin; IP crease → nail bed
✗ Wrong options: <2.5 cm options

Thumb defect size source note — neurovascular island **up to 4 cm** (Orthobullets) · island volar advancement **2.5–3.5 cm** (OB4508)

Flap	Thenar	Cross-finger
Indication	**Volar oblique, index/long, <30 y**	Volar oblique / volar proximal, **>30 y**
Pros	**Cosmesis**, more fat/bulk, colour match, primary donor closure	**Less stiffness**
Cons	**PIP flexion contracture** (finger sutured to palm 2–3 wk), **donor tenderness**, limited size	–
Contraindications	**RA, Dupuytren's, age >30, DJD**	–

Thenar positioning flex **MCP & DIP** maximally to ↓ PIP flexion

– – Active space – –

Cross-finger & reverse cross-finger flaps

Cross-finger full-thickness; **volar** loss distal to PIP with **intact tenosynovium**; hinge adjacent to defect

IVDU ring finger volar defect after abscess I&D (exposed tendon) → cross-finger flap

Reverse cross-finger (RXFF) hinge **opposite** defect; dermis level (below follicles, above veins); **subcutaneous (adipofascial) flap turned over + FTSG**; donor paratenon left; divide **2–3 wk**; flap **5 mm wider** than defect; no graft over germinal matrix

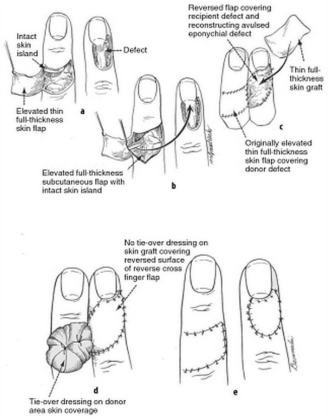
RXFF indications **eponychial fold loss + exposed extensor**, **sterile matrix loss + bare P3**, tendon without paratenon, dorsal PIP (boutonnière/burn), total nail-complex avulsion, burns

RXFF donor **dorsum of middle/proximal phalanx of adjacent finger**
✗ Wrong options: distal phalanx (too little tissue, endangers donor germinal matrix) · over DIP/PIP (skin too thin)

Blood supply RXFF dorsal cutaneous branches of proper digital a.

Index total nail complex avulsion alternative cross-thumb-to-index flap

Other dorsal options (wrong here) toe nail bed graft (nail bed only, not eponychium) · volar cross-finger (rare, distal thumb) · heterodigital reversed NV island (extensive defects, alternative to microsurgery)



Reverse cross-finger flap

First dorsal metacarpal artery (Kite) flap for thumb defects

Indication **dorsal thumb** with exposed tendon/bone (no paratenon/tenosynovium); also volar thumb >2 cm, web space

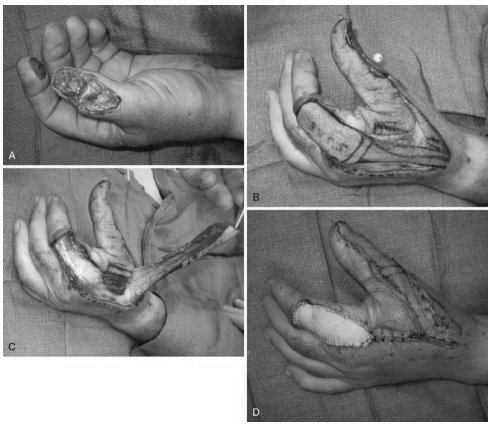
Vessel **FDMA, branch of radial a. at snuffbox**; **antegrade**, axial pattern; over index MC; ± 2nd DMA for large defects

Flap island from dorsoproximal index (over MCP/P1) + radial nerve branches + 1st DI fascia; pedicle **up to 7 cm**; covers **up to 3 × 5 cm**; donor FTSG; sensate via dorsal digital nerve branches

2 × 4 cm dorsal thumb, exposed tendon & bone → FDMA
✗ Wrong options: cross-finger · Moberg · FTSG · V-Y

Dorsal thumb avulsion after EPL repair/transfer → Kite flap
✗ Wrong options: VAC · wet-to-dry

– – Active space – –



Kite flap

Flap	Vessel / flow
FDMA (Kite)	**First dorsal MC a., antegrade**
Dorsoulnar/radial collateral (thumb)	Retrograde, distal thumb
Posterior interosseous	PIA, antegrade or retrograde; web space, dorsal hand, elbow
Homodigital / heterodigital NV island	Digital a., antegrade or reverse
Retrograde digital artery	**Needs middle transverse arch**
Axial flag	Dorsal MC aa.
Groin	Superficial circumflex iliac a.

Littler neurovascular island flap (TO)

Flap (TO) pulp skin from **ulnar side of middle** or **radial side of ring** finger (~2.5 × 1 cm) on its digital NV bundle → dissected to palm → **tunnelled to thumb**

Use (TO) **sensate cover for distal thumb / thumb pulp** (slicing, bike-chain injuries) · donor closed primarily (graft distally)

Complications (TO) commonest **venous congestion** (tight tunnel) · patient must **relearn localisation (cortical reorientation)**

Blast index unsalvageable beyond PIP (TO) **amputate at PIP level** + cover (keep functional length)

Homodigital retrograde island flap

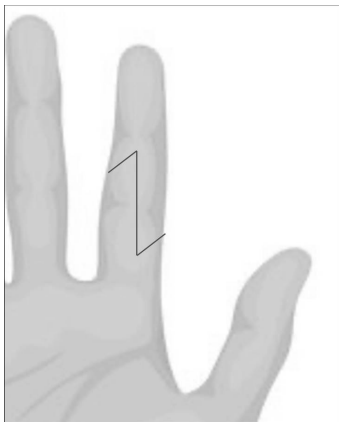
Retrograde flow digital artery flap pulp, dorsal & lateral defects **P3 → PIP**; requires **patent digital a. + middle transverse anastomotic arch**
✗ Wrong options: axial flag · groin · PIA · RXFF

Digital island artery **best axial pattern flap**; straight, dorsal oblique or volar oblique tip

Z-plasty principles

Z-plasty angle	Gain in length
30°	**25%**
45°	**50%**
60°	**75%**

Use	scar contracture revision (1st web burn, volar finger scar), prophylaxis; **60° = standard** ✗ Wrong options: 25° limbs · STSG · Moberg · V-Y

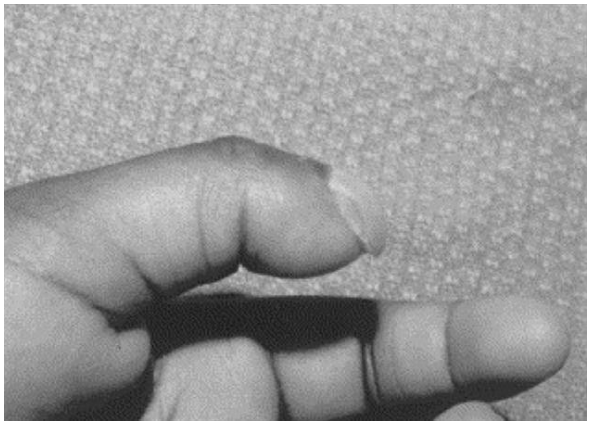


60° Z-plasty

6 Complications

Nail bed injury – germinal matrix

Split nail	**germinal matrix** scar → excise scar + matrix graft (adjacent finger / 2nd toe) ✗ Wrong options: eponychium · hyponychium · lunula · sterile matrix
Nail inclusion cyst	incomplete germinal matrix ablation at revision · or eponychial fold not splinted open → scars to sterile matrix
Nail bed repair	remove plate, repair (chromic), splint eponychial fold (nail/foil)
Hook nail	**tight tip closure / pulp overtightened** + **insufficient bony support** → prevent by nail bed removal when shortening



Hook nail deformity

Complications – digital neuroma

Symptomatic neuroma	<10%; **highest risk: avulsion mechanism** and **index finger** (nerve density)
Not risk factors	single digit, proximal level, ED vs theatre, small finger
Flap failure	inadequate arterial flow (**vasospasm → anastomotic thrombosis**) or venous outflow

– – Active space – –

Thenar flap — **recipient PIP flexion contracture**
✗ Wrong options: flap failure · colour mismatch · motor branch injury · recipient sensitivity (donor tenderness)

X vs Y

	Cross-finger	Reverse cross-finger
Defect	**Volar**	**Dorsal** (eponychium, nail bed, bare extensor)
Hinge	Adjacent to defect	**Opposite** defect
Tissue	Full-thickness skin + fat	Subcutaneous turnover + FTSG

	Moberg	FDMA (Kite)	NV island
Thumb defect	Volar **<2 cm**	**Dorsal**, volar >2 cm	Volar up to 4 cm

How it's asked

"Child, fingertip amputation, exposed bone?"	→ **ointment + dressing (no flap)**
"Child, contaminated tip, no bone, <1 cm?"	→ **I&D, soft dressing, review in 1 wk**
"Adult 1 cm defect, no bone, sterile matrix intact?"	→ **local wound care**
"Transverse tip, exposed bone?"	→ **shorten P3 + nail bed removal + V-Y**
"Moberg flap suits?"	→ **volar thumb defect**
"Largest defect for island volar advancement?"	→ **2.5–3.5 cm**
"Distal thumb amputation needing sensate cover?"	→ **Littler neurovascular island flap** (TO)
"Thenar flap candidate?"	→ **28 y, volar oblique index defect** ✗ Wrong options: 66 y · RA · dorsal · thumb
"Volar finger defect with exposed tendon, 29 y IVDU?"	→ **cross-finger flap**
"Commonest complication of thenar flap?"	→ **recipient PIP flexion contracture**
"Nail + eponychium avulsed, bare extensor?"	→ **reverse cross-finger flap** ✗ Wrong options: FTSG · toe nail bed · volar cross-finger
"RXFF for index sterile matrix loss — donor?"	→ **dorsum of middle phalanx of middle finger**
"Dorsal thumb defect with exposed tendon/bone?"	→ **FDMA (Kite) flap**
"Blood supply of Kite flap?"	→ **antegrade, first dorsal metacarpal artery**
"Flap needing middle transverse anastomotic arch?"	→ **retrograde flow digital artery flap**
"75% lengthening of web scar?"	→ **two-flap Z-plasty, 60°**
"Split nail after revision amputation?"	→ **germinal matrix**
"Neuroma after revision amputation — biggest risk?"	→ **avulsion mechanism**

FLEXOR CARPI RADIALIS TENDINITIS

– Hand · 32 / 103 –

1 Basics

Incidence rare; <1 / 100,000

Risk **repetitive wrist flexion**: **golf, racquet sports**, manual labour

Primary stenosing tenosynovitis in fibro-osseous tunnel

Secondary scaphoid fracture/cyst, DRF, **STT arthritis**, thumb CMC arthritis

Anatomy – FCR tunnel & insertion

Muscle **bipennate**

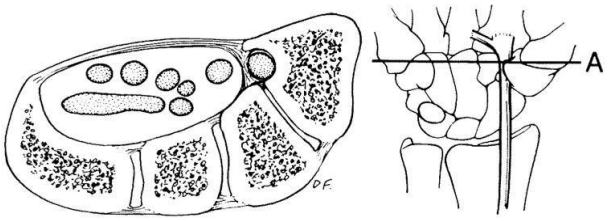
Sheath origin → trapezium; no fibrous sheath distal to trapezium

Tunnel entry **proximal border of trapezium**

Tunnel wall	Structure
Radial	Body of trapezium
Palmar	Trapezial crest, TCL
Ulnar	Retinacular septum (separates from carpal tunnel)
Dorsal	Septum reflection on trapezium

Zone	Tendon fills	Prone to
Within tunnel	**90%** (on rough trapezium)	Constriction, tendinitis, **attrition, rupture**
Proximal to tunnel	50–65%	Irritation from **osteophytes**

Insertion 1–2 mm slip → trapezial crest; **80% → 2nd MC**; **20% → 3rd MC**



FCR tunnel

3 Presentation

Pain **volar radial wrist**; tender along FCR at distal wrist crease

Provocative **resisted wrist flexion** · resisted radial deviation

No numbness/tingling

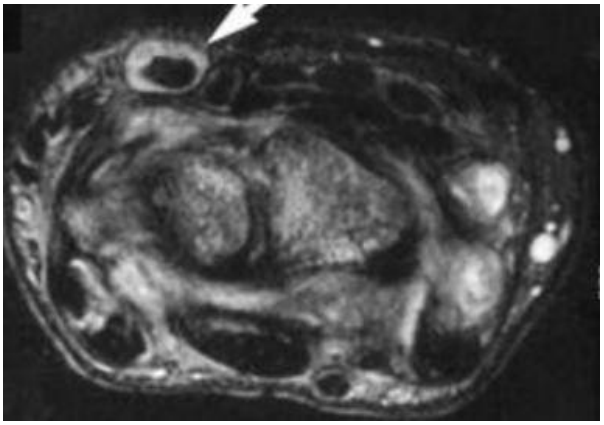
4 Diagnosis & Classification

Diagnosis clinical ± **MRI**

– – Active space – –

Key radiology findings

X-ray	normal in primary; secondary: healed scaphoid/DRF, STT/CMC arthritis, exostosis
MRI (T2)	**↑ fluid/signal around FCR sheath**; ganglion, scaphoid cyst



T2 MRI fluid in FCR sheath

Diagnostic injection	LA along FCR sheath relieves pain
Differentials	thumb CMC OA, scaphoid cyst, ganglion, De Quervain's, CTS

5 Management

FCR tendinitis

- First line → **REST / PT / ACTIVITY MODIFICATION, splint, NSAIDs**
  - fails (e.g. 6 wk) → **FCR SHEATH STEROID INJECTION (not into tendon)**
- Recalcitrant → **SURGICAL RELEASE of FCR sheath**

FCR sheath injection

Technique	near, **not into tendon** (rupture risk)
Outcome	effective in **primary**; fails in **secondary** (osteophytes)

Surgical release

Approach	volar longitudinal, proximal to wrist crease → proximal thenar eminence; reflect thenar muscles radially
Technique	open sheath distal forearm → trapezial crest; in tunnel **incise along ulnar margin**; mobilise tendon from groove (release trapezial insertion)
Nerves at risk	**palmar cutaneous br. median** · **lateral antebrachial cutaneous** · **superficial radial**

6 Complications

Disease	**FCR attrition/rupture**
Surgery	cutaneous nerve injury · deep palmar arch · **FPL** (lies superficial to FCR) · **FCR laceration within tunnel**
Poor prognosis	overuse, **workers' comp**, **no response to injection**, long duration

How it's asked

"Golfer, volar-radial wrist pain, T2 fluid round FCR, failed 6 wk PT — next?"	→ **FCR sheath injection** ✗ Wrong options: more PT · CTR · SRN decompression · FCU debridement
"Provocative test"	→ **resisted wrist flexion**
"FCR main insertion"	→ **2nd metacarpal base**

– – Active space – –

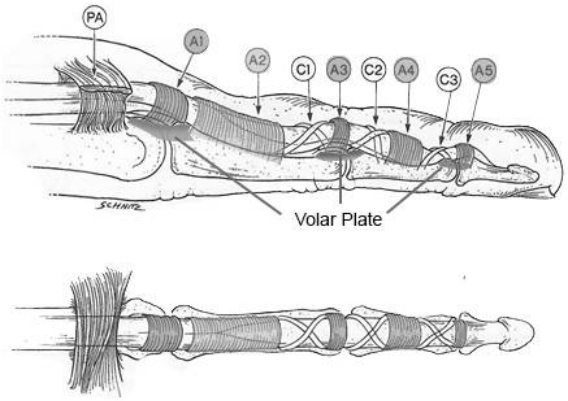
FLEXOR PULLEY SYSTEM

– Hand · 33 / 103 –

1 Basics

Pulley anatomy – annular/cruciform sequence, A2 & A4, thumb oblique pulley

Finger sequence	**A1, A2, C1, A3, C2, A4, C3** (prox → distal); 5 annular + 3 cruciate
Joint pulleys	**A1, A3, A5** → arise from **palmar plates** of MCP, PIP, DIP
Bone pulleys	**A2** = periosteum **proximal half of proximal phalanx** · **A4** = **midportion of middle phalanx**
Most important	**A2 + A4** → prevent **bowstringing**; minimum to preserve/reconstruct in flexor tendon surgery
Ideal	a pulley **proximal and distal to each joint**; preserve/reconstruct **≥3 pulleys**
Cruciate (C1–C3)	thin, collapsible; prevent sheath collapse/expansion; let annular pulleys **approximate in flexion**
Trigger finger	**A1** most commonly involved



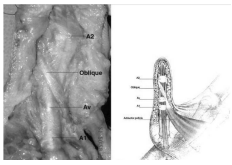
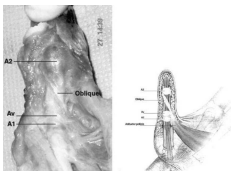
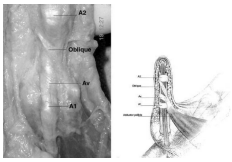
Finger pulleys

Thumb pulleys

Thumb sequence	**A1 – Av – oblique – A2** (2 annular + Av + oblique)
Oblique pulley	**most important in thumb**; analogue of finger C1 position; full FPL excursion, prevents bowstringing
Oblique origin	proximal half of proximal phalanx (Orthobullets) · ulnar side of PP base → radial side of DP base, 4 mm wide (Bayat, OB)
A1 (thumb)	at MCP volar plate; ~6 mm; **radial digital nerve closest (2.7 mm)** · ulnar 5.4 mm
A2 (thumb)	5–10 mm (Orthobullets) · 8 mm (Bayat); contributes **least to arc of motion**
Bowstringing rule	if **A2 intact**, cutting **A1 OR oblique** alone → no bowstringing; cutting **both A1 + oblique → bowstringing**
Restore kinematics (A2 intact)	repair **oblique OR A1**

Bayat caveat once **A1 + Av** both cut, oblique can no longer prevent bowstringing

– – Active space – –

Av pulley type	Image	Feature
I		Transverse, parallel to A1, **gap** to A1
II		**No gap** to A1
III		**Triangular/oblique**, fibres converge radially

2 Mechanism

Pulley rupture – climbers, grading & sequelae

Mechanism **extending force on flexed (crimped) digit**: **rock climbers**, Brazilian Jiu-Jitsu, baseball, lifting crimped; or laceration

Climbers 75% get UE injury; **30–50%** of these = closed pulley injury

Thumb pulley rupture rare; e.g. after **steroid injection** for FPL tenosynovitis → pain, clicking, ↓ ROM

Sequential ruptures needed for bowstringing; isolated rupture → early functional rehab

4 Diagnosis & Classification

Key radiology findings

Closed injury **MRI or US** to diagnose

MRI sign **volar subluxation of flexor tendon (FPL) away from phalanx** (normally apposed)

Clinical visible **bowstringing**; FPL intact (bowstringing needs tendon fixed at both ends → not FPL avulsion)

– – Active space – –



Thumb FPL bowstringing after oblique pulley rupture

Schöffl classification

Grade	Injury	Treatment
I	**Pulley strain**	Nonop
II	**Complete rupture A4** or **partial A2/A3**	Nonop, **taping**
III	**Complete rupture A2 or A3**	Nonop
IV	**Multiple ruptures** (A2/A3, A2/A3/A4) or single A2/A3 + **lumbrical/ collateral lig trauma**	**Surgical reconstruction**

Outcome (Schöffl) 92% nonop returned to previous climbing level

5 Management

Pulley injury

- Grade I–III → **NONOP (taping, functional rehab)**
- Grade IV → **PULLEY RECONSTRUCTION (not repair)**
  - A2 (proximal phalanx) → **3 LOOPS around bone, DEEP to extensor**
  - A4 (middle phalanx) → **2 LOOPS around bone, SUPERFICIAL to extensor**

Pulley reconstruction

Goal prevent bowstringing → **flexion contracture** & ↓ ROM; preserve/ reconstruct ≥3 pulleys, **A2 important**; A4 may be sacrificed acutely (unclear)

Thumb reconstruct **oblique pulley alone** = enough (tendon graft)

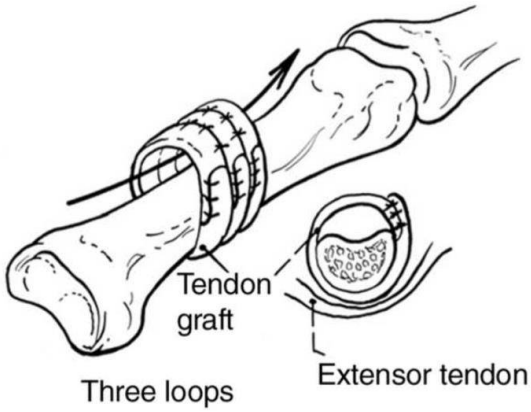
First step **excise all scar dorsal to flexor tendon**

Graft — **extensor retinaculum** (synovialised → **least gliding resistance**) · excised tendon · PL/plantaris · FDS · allograft

– – Active space – –

Technique	Examples	Note
Around-the-bone (encircling)	Bunnell single loop · **Okutsu triple loop**	**Biomechanically strongest** (triple loop)
Non-encircling	Kleinert ever-present rim · Karev belt loop · Lister extensor retinaculum · Doyle-Blythe PL through volar plate · Pulvertaft weave	Weaker

Hemostasis — prevents adhesions; check excursion after reconstruction



Okutsu triple loop

6 Complications

Untreated grade IV — **bowstringing → painful flexion contracture**

Encircling reconstruction — **phalangeal fracture** (most worrisome) · stiffness · persistent bowstringing (inadequate tension, **scar left dorsal to tendon**)

X vs Y

Effect	Cause
Bowstringing → flexion contracture	Neglected multi-pulley (A2–A4) injury
Quadriga	FDP **repaired short** → flexor lag of adjacent digits
Lumbrical plus	FDP injury **distal to lumbrical** → paradoxical IP extension on MCP flexion

How it's asked

"Annular pulleys arising from palmar plates"	→ **A1, A3, A5**
"Proximal → distal sequence"	→ **A1, A2, C1, A3, C2, A4, C3**
"Two most important pulleys"	→ **A2 and A4**
"Thumb, A2 intact – repair which to restore kinematics?"	→ **oblique or A1**

– – Active space – –

"Thumb clicking after FPL steroid injection, bowstringing on MRI"	→ **pulley rupture** ✗ Wrong options: FPL avulsion, RCL, gout
"Grade II pulley injury"	→ **complete A4 rupture** ✗ Wrong options: A2/A3 complete = III; multiple = IV
"Laceration through A2–A4, tendons intact, not reconstructed – sequel?"	→ **flexion contracture** ✗ Wrong options: quadriga, lumbrical plus
"Strongest reconstruction"	→ **Okutsu triple loop**
"Graft with least gliding resistance"	→ **extensor retinaculum**

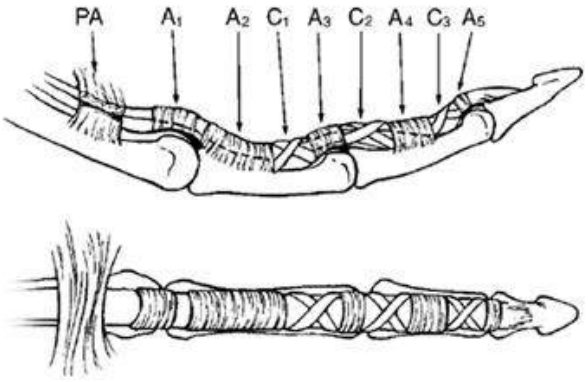
FLEXOR TENDON INJURIES

– Hand · 34 / 103 –

1 Basics

Anatomy & tendon nutrition

FDP	flexes **DIP**; common belly; **index/long = AIN**, **ring/small = ulnar**
FDS	flexes **PIP**; independent bellies; median n.; **absent to small finger in 25%**
FPL	AIN; **most radial** structure in carpal tunnel
FCR	inserts base 2nd MC; **median nerve lies immediately ulnar to FCR** at DRUJ level (deep FCR laceration → median nerve injury)
FCU	pisiform, hook of hamate, base 5th MC; ulnar n.
Camper chiasm	level of **proximal phalanx**; FDP passes through split FDS · **distal to the chiasma the FDP is SUPERFICIAL** → landmark in P1-level injuries (TO)
Lumbricals (TO)	arise **radial side of each FDP** → dorsal extensor expansion → MCP flexion + IP extension · **intrinsic-plus position = MCP 90° flexion, IPs straight**
Finger pulleys	**A2 & A4 (from periosteum) = most important vs bowstringing**; **A1, A3, A5 from volar plate** (MCP, PIP, DIP); C1–C3 collapsible
Thumb pulleys	**A1, Av, A2 + oblique**; **oblique (+ A1) most important vs bowstringing**; A2 contributes least
Pulley venting	**25% of A2** and **100% of A4** can be incised with little deficit
Nutrition in zone II (sheathed)	**synovial diffusion = primary** ✗ Wrong options: vincula
Chiasma rotation (TO)	FDS slips **rotate 180°** round FDP → restore rotation at repair; **never sacrifice both FDS slips** · **FDP vincula longa arises from FDS** · cruciate pulleys collapse in flexion → protect annular pulleys
Outside sheath	direct perfusion: vincula, bony insertion, reflected sheath vessels, longitudinal palmar vessels



Pulley system of the finger

2 Mechanism

Cause	volar laceration ± digital NV injury; closed avulsion (jersey finger)

– – Active space – –

Healing pathways **intrinsic** (tenocytes) · **extrinsic** (sheath/inflammatory cells → **adhesions**)

Intrinsic healing (TO) tenocytes of tendon + sheath · supple, vascular → **preserves gliding** = healing to promote → **repair the sheath too** (TO) · sheath repair no clinical difference (Orthobullets) – read the options

Phase	Days	Histology	Strength
Inflammatory	0–5	Cellular proliferation	None
Fibroblastic	5–28	Disorganised collagen	Increasing
Remodelling	>28	Linear collagen	Tolerates AROM

Weakest repair **post-op day 6–12** (OB) · **day 6–20** (TO) – read the options; fails at **knots**; **gap >3 mm** → failure

Early motion ↓ adhesions, ↑ excursion, better healing biology (esp zone II)

3 Presentation

Diagnosis – disrupted tenodesis cascade

Resting cascade involved digit lies **extended** out of cascade

Tenodesis passive **wrist extension → digits flex**; **DIP stays extended = FDP cut**; PIP stays extended = FDS+FDP

Extensor injury loss of digit extension with passive **wrist flexion**

Isolated tests FDP: block PIP, flex DIP · FDS: hold other fingers extended, flex PIP

Partial tear clue (TO) **painful active flexion → partial injury** · full active flexion does **not** exclude partial tear · straight finger alone ≠ complete tear · NV: static 2PD; delayed capillary refill → arterial injury

Key radiology findings

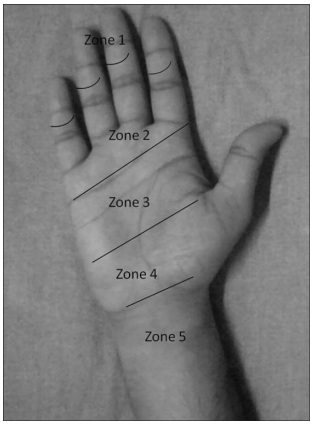
X-ray associated fracture; **bony FDP avulsion** (Leddy III large fragment caught at **A4**)

Ultrasound suspected (partial) laceration, retracted stump level

4 Diagnosis & Classification

Flexor zones (no man's land)

Zone	Boundaries	Key fact
I	Distal to FDS insertion	**Jersey finger**; FDP only
II	FDS insertion → **A1 / distal palmar crease**	**"No man's land" (Bunnell 1918)** (TO); **FDS + FDP in same sheath**; adhesions
III	A1 → distal edge TCL (palm)	**NV injury** → worse; good results otherwise; ± A1 release · **lumbrical origins → find + repair** (TO)
IV	**Carpal tunnel**	Adhesions; repair TCL **lengthened** if bowstringing
V	Proximal to carpal tunnel (forearm)	NV injury · deep cut → all tendons + median + ulnar nn. + both arteries = **"spaghetti wrist"** (TO)
Thumb T I–III	TI distal to IP · TII IP→MCP · TIII thenar	↑ re-rupture; early motion does not improve long-term; **avoid TIII incision (recurrent motor br.)**



Flexor zones

Jersey finger – FDP avulsion

Mechanism	forced DIP extension during grip; **ring finger** most common
Big bony fragment	**Leddy type III**, caught at **A4** (stays distal)
Jersey vs mallet (TO)	tendon **retracts → splinting does not work** → explore + reinsert (**pull-out button 6–8 wk**, K-wire or mini anchors via zig-zag)
Stem	can't flex DIP after grabbing jersey/sliding headfirst → **profundus insertion rupture**

5 Management

Flexor tendon laceration

- <60% width, no triggering → **trim frayed edges + early protected ROM (no repair)**
- 60–75% or ≥50% with triggering → **EPITENDINOUS repair of cut edge only**
- Complete → **core (4–6 strand) + running epitendinous, within 2–3 wk, early motion**
  - child / non-compliant → **4-strand + cast 3–4 wk**

– – Active space – –

Operative criteria

Situation	Criterion → action
Partial laceration	**<60%**, no trigger → no repair · **60–75%** or ≥50% + trigger → epitendinous · **>75%** → repair
Complete laceration	Core (**4–6 strands** for early active motion) + running epitendinous
Timing	**Within 3 wk** (2 wk ideal); FDP avulsion past ~3 wk → retracted; >6 wk pulleys collapse
Re-rupture	Scar **<1 cm** → resect + repair · **>1 cm** → graft (1-stage if sheath intact, 2-stage if collapsed)
Reconstruction	**Supple skin, sensate digit, vascularity, full passive ROM**
Single-stage graft	Pristine sheath + full ROM
Tenolysis	**>3 mo**, stable soft tissue, **full passive ROM**, active ≪ passive
Immobilise (✗ Wrong options: early motion)	**Children**, non-compliant

Partial lacerations – trimming vs epitendinous repair

<60% (e.g. 40–50%), no trigger — **no repair**; trim edges + **early protected ROM** (repaired partial tendons can be weaker)

Up to 75% partial — **epitendinous suture of cut edge alone** prevents gap >2 mm; no core (adds bulk), no circumferential needed

Complete laceration — always **core + epitendinous**

Repair biomechanics – core strands, suture purchase, FDS slip & rehab protocol strength

Strength ∝ number of core strands crossing repair (linear) – strongest single factor
✗ Wrong options: suture size · sheath repair

4–6 strands — needed for **early active motion**

Purchase — **10 mm** from cut edge (> 5 mm) (OB) · **5–7 mm** (TO) – read the options

Repair thresholds (TO, Mittal) — **> 60% → repair** · < 50% → conservative / epitendinous · **no primary repair after 6 wk** → reconstruction · best < 2 wk, acceptable ≤ 3 wk

Handling (TO) — **non-toothed forceps on endotendon**, not epitendon · stabilise with **25G needle** · **6-0 epitendinous** running (+20% strength, gliding) · named: Tajima (2-strand), Strickland, Savage, Becker

Zone I FDP (TO) — distal stump **≥ 1 cm → primary repair** · **< 1 cm → advancement / bone pull-out** (avoid > 1 cm advancement → quadriga)

Zone V repair order (TO) — **FDP + FPL → FDS → artery → nerve → superficial tendons (FCR, FCU, PL)** · zones III–IV: tendons first, then NV

Technique (TO) — cylindrical tendon → **core + peripheral (epitendinous)**, not simple/mattress · freshen ends with **No. 11 blade** · **modified Kessler-Mason** 2- or 4-strand **3-0 nylon/Prolene** core + **5-0 nylon running** · **knots buried between tendon ends** · non-absorbable monofilament (no braided / absorbable) · other named: Bunnell, cruciate, **Savage**, **Tang**, modified Becker

Other — larger calibre ↑ strength; **locking loops ↓ gap**; **dorsal core** stronger

Epitendinous	**simple running** (↓ gliding resistance; not Silfverskiöld); +20% strength; ↓ gap; ↓ bulk
Sheath repair	no clinical difference
Zone II FDS	repair it (↑ strength, outcomes); **repairing 1 slip → better gliding** than both
Timing	**within 3 wk** (2 wk ideal); retraction later
Incisions	cross creases transversely/obliquely (Brunner); **never longitudinal**
Excursion target (TO)	**minimum 3–5 mm, optimal 6–9 mm** · splint off at **6–8 wk** · complex repairs passive-only first 4 wk
Dorsal blocking splint (TO)	**4–6 wk**, **MCP 80–90° flexion, IPs straight** · supervised early active-assisted motion (esp. zone II)

Protocol splint (TO)	Position
Duran	Static dorsal block: wrist slight flexion, **MCP ~45°**, IPs near extension; isolated DIP → PIP → combined passive
Kleinert	Dorsal splint, **wrist ~30° flexion, MCP 30–40°**, **rubber bands nail → wrist** (palmar pulley); bands off at night, fingers strapped straight
Strickland (place & hold)	**Hinged** wrist splint, wrist 20° flexion, MCP 50°; tenodesis passive flexion → **hold ~5 s**
Mayo synergistic	Hinged; active wrist + finger motion

Rehab protocol	Force / excursion	How
Duran	Low / low	**Patient-assisted** passive flexion, active extension
Kleinert	Low / low	**Rubber-band dynamic** passive flexion, active extension
Synergistic (Mayo)	Low / **high**	**Active wrist motion** → greatest excursion
Early active (place & hold)	Moderate / high	Needs **4–6 strands**
Immobilisation	–	**Children, non-compliant**: wrist & MCP flexed, IPs extended

2-strand repair	OK for Duran/Kleinert/synergistic; **not for early active motion**

Paediatric flexor tendon injuries

Child (e.g. 4 y) zone II	**4-strand core + cast 3–4 wk** (wrist 35° flex, MCP 60–70°, IP extended; above elbow)
Differences from adults	**poor therapy cooperation**; delayed presentation commoner (may need staged); Botox may aid rehab

Wide-awake (WALANT) repair

Anaesthesia	**lidocaine + adrenaline** tumescent; **no tourniquet, no sedation**
Adrenaline in digit	**safe** (up to 1:1000); necrosis was **procaine**; antidote **phentolamine** (1 mg in 5 mL)
Doses	7 mg/kg lido; 1% + 1:100,000 if <50 mL; 0.5% + 1:200,000 if 50–100 mL; 0.25% + 1:400,000 if 100–200 mL; P1/P2 2 mL, P3 1 mL, palm 10–15 mL

– – Active space – –

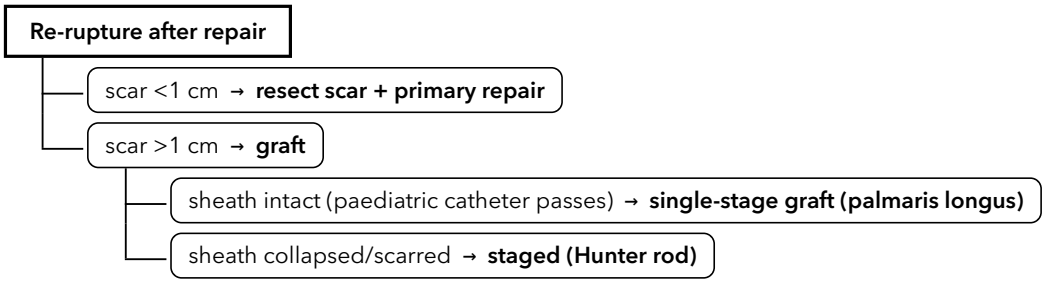
Primary goal	**active testing of repair** by awake patient (gaps, gliding through A2/A4)
Benefits	debulk on the spot, **vent A2 / divide A4 → ↓ late tenolysis**; facilitates early active motion
Post-op	immobilise 3 days → **half-fist 45/45/45** active midrange
WALANT mix (TO)	**lignocaine 2% + adrenaline + sodium bicarbonate** (bupivacaine for longer cases)

Other hand anaesthesia (TO)	Key points
Bier block	Esmarch exsanguination · **double cuff at SBP + 100 mmHg (or 2.5 × SBP)** · **0.5% plain lidocaine ~1 mL/kg** (TO – verify: usual max 3 mg/kg ≈ 0.6 mL/kg of 0.5%) via distal IV · wait 6–8 min → release **distal** cuff, keep proximal · **never release early** (hypotension, convulsions, arrhythmia, death) · **bupivacaine contraindicated**
Axillary block	Arm abducted + ER; inject above + below axillary artery · **does not block intercostobrachial n.**
Supraclavicular block	Whole upper limb; US-guided

✗ EXCEPT traps	"adrenaline contraindicated" · "brachial block + sedation" · "forearm tourniquet" · "surgeon passively flexes" · "doesn't reduce tenolysis"

Late reconstruction – two-stage grafting, FDS transfer & failed repair

Re-rupture (TO)	**re-repair if < 3 wk**; **graft if > 3 wk** · tenolysis for adhesions at **4–6 months** (TO)
Reconstruction ladder (TO)	do nothing → tenolysis → tendon transfer → primary graft → **two-stage graft (scarred bed)**
Re-rupture	15–25%



Advance FDP >1 cm	avoid → **quadrigia** & contracture
Reconstruction prerequisites	**supple skin, sensate digit, adequate vascularity, full passive ROM of joints**
Two-stage indications (TO)	delayed presentation · tendon loss · repeated adhesions · **collapsed sheath** · stage 1: rod sutured to **FDP stump** + **reconstruct A2/A4** with graft · stage 2 at **3–4 months** · junction by **Pulvertaft weave** (not Kessler)
Hunter (two-stage)	stage 1 silicone rod → pseudosheath; stage 2 at **≥3 mo** graft through **mesothelium-lined pseudosheath** (Pulvertaft proximal)
Paneva-Holevich	stage 1 rod + FDS–FDP loop in palm; stage 2 FDS reflected distally; **intrasynovial graft**, one tenorrhaphy healing at a time
Graft choice	**palmaris longus** (commonest; absent 15%) · plantaris (absent 19%, longer) · toe extensors · EIP · FDS
Pulleys	reconstruct one **proximal & distal to each joint**; do pulleys first

Tenolysis after reconstruction	**>50%**
Chronic FDP avulsion (4 mo), full passive ROM, palm mass	**two-stage grafting** (pulleys collapse >6 wk; no direct repair after ~3 wk)
Chronic FPL laceration (4 mo), full passive IP	**FDS ring transfer to FPL** (single stage) · alternative IP fusion
Active silicone rod	fatigues → not permanent

Attritional ruptures (rheumatoid, prominent plate)

Mannerfelt (TO)	**FPL rupture** in carpal tunnel from **scaphoid (STT) spur**; **most common flexor rupture in RA**
Treatment	**palmaris longus graft** (✗ Wrong options: primary repair (attritional)) or FDS transfer; debride osteophyte; IP fusion alternative (✗ Wrong options: MCP fusion)
Vaughan-Jackson (TO)	extensor ruptures **ulnar → radial** (EDM/EDC small first) from **distal ulna** (caput ulnae)
After volar DRF plate	**FPL attrition** (plate **distal to watershed** / prominent) → **remove plate + FPL reconstruction** (graft or FDS4)

6 Complications

Complication	Key fact
Adhesions	**Most common**; zone II; tenolysis at **3–6 mo** if passive ≫ active ROM
Re-rupture	15–25%; days 6–12 weakest
Joint contracture	Up to 17%
Quadrigia	FDP over-advanced/tethered → adjacent fingers can't fully flex
Lumbrical plus	FDP lax/long (graft too long, FDP cut distal to lumbrical) → paradoxical PIP extension on flexing
Swan-neck, trigger finger	– · **triggering** from over-tight pulley repair or nodular repair (TO) · contracture from prolonged flexed splinting + delayed therapy (TO)

Tenolysis requirements	soft tissues stable (>3 mo), **full passive ROM**; preserve A2/A4

X vs Y

	Quadrigia	Lumbrical plus
FDP	**Too short / tethered**	**Too long / lax**
Sign	Adjacent fingers lack full flexion	IP extension on attempted flexion

	Mannerfelt	Vaughan-Jackson
Tendon	**FPL** (± FDP index)	**EDM, EDC small → radial**
Bony cause	**Scaphoid / STT spur**	**Distal ulna**

– – Active space – –

Finger	Most important pulleys
Fingers	**A2, A4**
Thumb	**Oblique** (+ A1)

How it's asked

"No man's land?"	→ **zone II** (TO)
"Superficial tendon distal to Camper chiasma at P1 level?"	→ **FDP** (TO)
"Bier block — local anaesthetic contraindicated?"	→ **bupivacaine** (TO)
"Zone II, 40–50% laceration, no triggering"	→ **trim + early ROM, no repair**
"75% laceration with triggering, least suture"	→ **epitendinous of cut edge only**
"Managed by epitendinous alone"	→ **~70% partial FDP**
"Greatest effect on strength"	→ **number of core strands**
"Best load to failure + gliding"	→ **10 mm purchase + simple running epitendinous**
"2-strand repair — NOT strong enough for"	→ **early active motion**
"Zone II true statement"	→ **repair of 1 FDS slip improves gliding**
"Zone II nutrition"	→ **synovial diffusion**
"Median nerve immediately ulnar to"	→ **FCR**
"DIP stays extended with passive wrist extension"	→ **FDP**
"4-y-old zone II"	→ **4-strand + cast 4 wk**
"WALANT true"	→ **lido + adrenaline safe in digit · primary goal → active testing of repair**
"Re-rupture, 1.8 cm scar, sheath intact"	→ **excise + palmaris longus graft**
"Two-stage reconstruction true"	→ **supple skin, sensate, vascular digit**; tenolysis >50%
"Jersey finger 4 mo, palm mass"	→ **two-stage graft**
"Chronic FPL 4 mo"	→ **FDS transfer**
"RA, can't flex thumb IP"	→ **Mannerfelt (FPL) → PL graft**
"Pop 12 mo after volar plate, can't flex thumb IP"	→ **remove plate + FPL reconstruction**
"Baseball slide, big fragment on X-ray"	→ **FDP insertion (Leddy III)**

FROSTBITE

– Hand · 35 / 103 –

1 Basics

Epidemiology & risk factors

Demographics ♂:♀ **10:1**; **30–50 y**; homeless, outdoor workers

Distribution upper limb 20% · **lower limb 47%** · both 30% · head/face 3%

Most important host risk **alcohol abuse** (63% of admissions); homelessness

Other host mental illness, PVD, neuropathy, DM, malnutrition, chronic illness, prior cold injury, **African descent** (Caucasians better cold-induced vasodilatation)

Smoking ↓ NO + ↑ fibrinogen/platelets → **thrombosis**

Environmental factor	Threshold
Temperature	Risk **low > −10°C**; **high < −25°C**
Wind at −18°C	Freeze in **1 h at 10 mph** · **10 min at 40 mph**
Altitude	>17,000 ft
Contact	Water, ice, **metal** (↑ severity)

Spectrum of cold injury

Injury	Key features
Frostnip	Mildest; superficial skin only, **no dermal damage**; reversible
Chilblain (pernio)	**Nonfreezing, dry**; burning, pruritus; resolves 2 wk; chronic vasculitis in young/middle-aged ♀
Trench (immersion) foot	**Nonfreezing, wet**, <10°C; military
Frostbite	Freezing → tissue necrosis → amputation
Hypothermia	Core temp ↓; can be fatal

2 Mechanism

Limb phase	Events
I Cooling/ freezing	Vasoconstriction → **"hunting response"** (cyclic AV shunting q10 min); **intracellular ice** → membrane rupture; **extracellular ice** → sludging + **intracellular dehydration**
II Rewarming	Crystals melt; **endothelium permeable** → blisters/oedema; **avoid refreezing**
III Progressive injury	Inflammation, thrombosis, necrosis; **↓ PGE2**; **↑ PGF2α + thromboxane B2**
IV Resolution	Healing ± sequelae or gangrene

– – Active space – –

Numbers sensory nerve dysfunction **−10°C**; extracellular ice **−2 to −15°C**

Hypothermia CBT <35°C; shivering stops at **30–32°C** → rigidity, mimics death → **"no one is dead until warm and dead"**

3 Presentation

Hypothermia grades mild **32–35°C** · moderate **28–32°C** · severe **<28°C**; brady, AF/VF, resp acidosis, coma

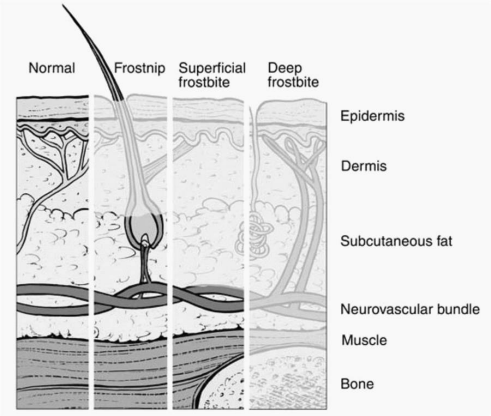
Blisters form **6–24 h after rewarming**; degree only defined **after rewarming**

Severe hand injury signs no Doppler signal, **absent cap refill**, dark purple digits, **haemorrhagic blisters** → bone scan / angiography

4 Diagnosis & Classification

Classification

Degree	Findings	Newer class / prognosis
1st	**Central white area + surrounding erythema**	Superficial – good
2nd	**Clear/cloudy blisters** within 24 h	Superficial – good
3rd	**Haemorrhagic blisters** (painless) / black eschar	**Deep – poor**
4th	Tissue necrosis	**Deep – poor**



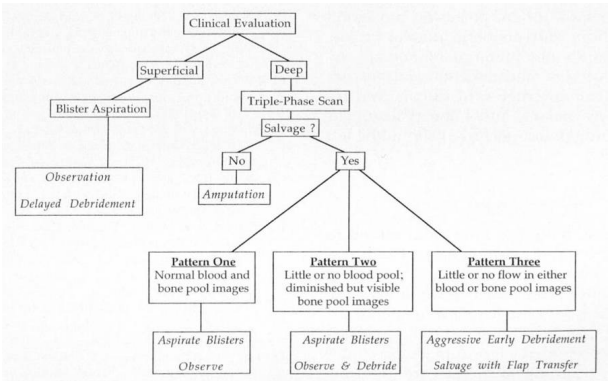
Frostbite degrees by tissue depth

Key radiology findings

Tc-99m bone scan assess severity: **1st at day 2** (absent uptake = poor prognosis, not necessarily necrosis); **2nd at day 5**

Day-5 blood/bone pool	Action
Normal	Expectant
Diminished	Observe ± early debridement
Absent	**Early debridement / amputation**; tPA candidate

MRI T2 ↑ signal in necrotic muscle; not the initial test



Bone scan algorithm

5 Management

Frostbite

- Field → **pad/splint, prevent refreezing; rewarm only if warmth can be maintained**
- ED → **warm IV fluids, tetanus, NSAIDs, RAPID REWARM 40-42°C water bath**
  - <24 h, no flow on bone scan, 2nd/3rd degree → **IV tPA**
- Necrosis → wait for demarcation (1-3 mo) → **LATE DEBRIDEMENT / AMPUTATION**

Prevention – insulated footwear

Most reliable prevention — **thermal insulation footwear** / reduce shoe heat loss (toes: little muscle → little heat)

✗ Wrong options: **Distractors** – hard-soled shoes (comfortable shoes ✓), industrial toe-caps, compression stockings, high-protein diet, warm drinks, triple socks, carbo-loading

Rewarming

Water bath — **40–42°C**; **30 min** (Orthobullets) · **15–30 min** (Golant); mild antibacterial

End point — skin **pliable and red-purple**

Avoid 50°C → thermal burn · lower temp → ↓ survival · **avoid freeze-thaw cycles**

Hypothermia rewarming — mild: external (passive/active; active → **"afterdrop"**) · moderate/severe: **internal core** (warm O_2/IV, lavage, bypass, CAVR ~1°C/15 min); avoid alcohol/sedatives

Acute treatment – blister care & thrombolysis

Blister	Management
Clear (superficial)	**Debride + topical aloe vera** (↓ PGF2α, TXB2)
Haemorrhagic (deep)	**Drain/aspirate, leave skin intact** → prevents dermal desiccation
Intact	Leave, dry gauze

Liquid nitrogen finger, haemorrhagic blister — **drain, leave roof intact**
✗ Wrong options: HBO · full debridement + flap · MRI · wet-to-dry

– – Active space – –

IV tPA (tissue plasminogen activator)	**within 24 h → ↓ digital amputation**; indication: **no flow on bone scan**, **2nd/3rd degree** ✗ Wrong options: superficial
Unconscious in snow, cold blue digits, no cap refill, 2PD >15 mm	**tPA** ✗ Wrong options: diltiazem · amlodipine [Raynaud's] · collagenase [Dupuytren] · diclofenac

tPA contraindications	
General	**Alcoholic** (head injury bleed), active bleeding, ICH/surgery <3 mo, **concurrent trauma**, major surgery <14 d, aneurysm/AVM, bleeding diathesis, pregnancy, labile HTN
Cold-related	**>24 h cold exposure**, **warm ischaemia >6 h**, **multiple freeze-thaw**

Other adjuncts	LMW dextran, anticoagulants; **HBO anecdotal**; IV Abx only if infected
Rehab	whirlpool hydrotherapy, PT/OT

Delayed debridement/amputation after demarcation

Timing	**after demarcation, 1–3 months → "frostbite in January, amputate in July"** ✗ Wrong options: days · weeks
Before demarcation, no sepsis	**moist dressings + observation** (like burns; much may be superficial)
Immediate surgery only	**escharotomy** (circumferential constriction) · **fasciotomy** (compartment syndrome)
Sympathectomy	↓ pain duration + time to demarcation; **no ↓ extent of necrosis**



Frostbite hand at 12 h

6 Complications

Adult	Detail / treatment
Persistent pain	**50%** (intolerable 15%)
Cold intolerance	–
Vasospasm	**Raynaud's**, colour change, **hyperhidrosis** → CCB, vasodilators, β-blockers, late sympathectomy
Neuropathy	↓ NCV → decompression (CTR)
Frostbite arthropathy	Subchondral bone loss, osteopenia, contractures **DIP > PIP** → arthroplasty

Child	Detail
Premature physeal closure	**1–2 y after exposure** (chondrocyte injury)
Late (>10 y age)	Joint laxity, angular deformity, **short digits**, excess skin, DJD → physeal arrest, osteotomy, arthrodesis

Severity ↑ **alcohol intoxication**, skin contact with metal/ice, high wind chill

X vs Y

	Chilblain	Trench foot	Frostnip	Frostbite
Freezing	No	No	Yes	Yes
Wet/dry	**Dry**	**Wet**	–	–
Tissue loss	No	±	No (reversible)	Yes

How it's asked

"Most reliable way to reduce toe frostbite?"	→ **thermal insulation footwear**
"Most important measure hiking in extreme cold?"	→ **reduce thermal heat loss from shoes**
"Liquid nitrogen, haemorrhagic blister – next?"	→ **drain, leave overlying skin intact**
"Found in snow bank, absent cap refill – medication?"	→ **tPA** (within 24 h)
"When to debride/amputate?"	→ **after demarcation, 1–3 months**
"Rewarming bath temperature?"	→ **40–42°C** ✗ Wrong options: 50°C
"Stranded days, gangrenous toes, no sepsis – next morning?"	→ **moist dressings + observation** ✗ Wrong options: amputation
"Growth problem after childhood frostbite?"	→ **premature physeal closure 1–2 y later**

– – Active space – –

GANGLION CYSTS

– Hand · 36 / 103 –

1 Basics

Epidemiology & sites

Commonest **most common hand/wrist mass (60–70%)**
✗ Wrong options: 40%

Age 20s–40s (TO); trauma history ≥10% (TO)

Lower limb 15–20% of all ganglia; ⅓ of these around **knee**

Site	%	Origin
Dorsal carpal	**70%** (Orthobullets) · 60–70% (TO)	**Scapholunate ligament / SL articulation** (most common stalk); over 3rd/4th compartments, between EDC & EPL
Volar carpal	20% (Orthobullets) · 13–20% (TO)	Radiocarpal or **STT** (Orthobullets, OB) · **radioscaphoid–SL interval > scaphotrapezial > metacarpotrapezial** (TO, in that order)
Volar retinacular (flexor sheath)	10%	Herniated tendon-sheath fluid; A1/A2 pulley
Dorsal DIP (mucous cyst)	–	DIP OA, **Heberden's nodes**

Children **volar > dorsal (1.2:1)**; higher ♀:♂

Pathology & origin

Causes trauma, **mucoid degeneration**, **synovial herniation**

Contents **mucin** from tendon sheath or joint; clear yellow viscous gel

Lining **no true epithelial lining**; **no true synovial lining**, fibrous layer (Orthobullets, OB4424) · "**synovial cells with mucin accumulation**" (OB8790 answer) – read the options
✗ Wrong options: epithelial

Histology **mucin-filled space with occasional spindled fibroblasts**

Associations (volar) median/ulnar nerve compression; hand ischaemia (vascular occlusion)

3 Presentation

Symptoms usually asymptomatic; cosmesis; may wax & wane

Exam **firm, well circumscribed**, **transilluminates**; fixed to deep tissue, **not fixed to skin**

Volar wrist **Allen's test** (radial artery adjacent)



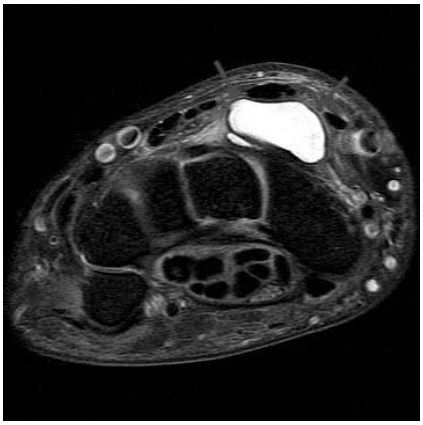
Dorsal wrist ganglion

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray	**Normal**
MRI	Not routine; well-marginated, **homogeneous fluid signal: T1 low, T2 high**; diagnostic if uncertainty remains
Ultrasound	**Homogeneous anechoic**, well-defined; **differentiates cyst from aneurysm**/vascular malformation; guides aspiration away from artery
Biopsy	Not routine; specimen always sent post-op

Most definitive confirmation (OB) **needle aspiration + cytology** (excludes myxoid neoplasm)
✗ Wrong options: observe 1 y · gadolinium MRI · X-ray · manual rupture



Ganglion MRI: homogeneous T2-bright mass

– – Active space – –

Histology vs MRI of look-alikes

Lesion	Histology	MRI
Ganglion	**Mucin + spindled fibroblasts**	**T1 low / T2 high**
GCT tendon sheath	**Uniform stromal + giant cells** / histiocytes with giant cells	**Low T1 + low T2**; recurs
Spindle-cell lipoma	Mature fat + spindle cells	High T1 + T2
Nerve sheath tumour	Fibroblasts + Schwann, mast cells, lymphocytes	T1 hypo / T2 hyper
Pyogenic granuloma	Lobular vascular proliferation + inflammation	T1 iso to muscle / T2 high
Abscess	PMNs	–

5 Management

Ganglion

- Child → **observation (± splint)**
  (76–80% resolve within 1 y)
- Adult, asymptomatic → **observation (1st line)**
- Adult dorsal, wants Rx → **aspiration (2nd line; avoid volar – radial artery)**
- Severe sx / NV compression / recurrence → **marginal excision of stalk + capsule cuff**

Recurrence rates

Treatment	Recurrence
Closed rupture	High (home remedy)
Aspiration	**50%** (Orthobullets, OB4424) · **40–60%** (OB211501); repeat aspiration success only **30–50%** (TO)
Observation	Recurrence 50–60% (OB)
Open excision	**15–20%** (volar higher than dorsal)
Arthroscopic excision	**Lower** than open (OB)
Intralesional excision	High

TO adjuncts steroid after aspiration / hyaluronidase before aspiration → **do not improve success**

Surgical excision

Gold standard (TO) surgical excision of **cyst + pedicle + cuff of joint capsule**; recurrence from missed tortuous capsular ducts

Margin **marginal excision**
✗ Wrong options: wide · intralesional

Arthroscopic (TO)	viewing **6R**, working **3-4** (through ganglion); excise root; if stalk not found → resect wide dorsal capsule
Mucous cyst	excise cyst + **resect DIP osteophyte**

Paediatric ganglia – observation & spontaneous resolution

Resolution	**76% within 1 y** (Orthobullets, OB217981) · **79%** (OB2880) · nearly 80% (OB211160) → **observe** ✗ Wrong options: aspiration · biopsy · MRI · excision
Aspiration in children	recurrence 13–85%, worse if off tendon; excision recurrence 6–35%
Volar palm base of index, 3 y	tendon-sheath ganglion → **reassure, will resolve**

Special locations – retinacular cyst, mucous cyst, Guyon's canal, proximal tibiofibular joint

Retinacular cyst	**2–8 mm, pea-sized**, from **A1/A2 pulley**, at **MCP flexion crease**; **does not move with tendon**; aspiration may work; excise with small portion of sheath/pulley
Mucous cyst	DIP ganglion in **OA** (dorsal osteophyte), **nail deformity** → excise + debride osteophyte ✗ Wrong options: inclusion cyst · synovial sarcoma · metastasis · felon



Mucous cyst with nail groove

Guyon's canal ganglion	ulnar to hook of hamate → ulnar motor palsy (**Froment**, **adductor pollicis sharp waves**, no numbness) → **excise**; ulnar **nerve dorsal + ulnar to artery**; DDx hypothenar hammer (jack-hammer)
Proximal tibiofibular ganglion, recurrent	**PTFJ arthrodesis = lowest recurrence** ✗ Wrong options: interposition arthroplasty · phenol / liquid nitrogen (ABC adjuvants) · aspiration

6 Complications

Aspiration	infection (rare), NV injury
Excision	infection, **radial artery injury (most common NV)**, **SL ligament injury**, stiffness, recurrence

– – Active space – –

How it's asked

"Dorsal wrist ganglion stalk origin?"	→ **scapholunate ligament** (TO, OB) ✗ Wrong options: STT · radioscaphoid–SL interval · CMC
"True of this dorsal wrist mass?"	→ **lacks a true epithelial lining** ✗ Wrong options: aspiration low recurrence · STT origin · wide excision
"Best way to confirm dorsal wrist ganglion?"	→ **needle aspiration + cytology**
"Volar wrist ganglion — resection & histology?"	→ **marginal excision; synovial cells with mucin** ✗ Wrong options: histiocytes + giant cells = GCT
"Histology of post-sprain dorsal wrist mass?"	→ **mucin-filled space with spindled fibroblasts**
"10-year-old, transilluminating dorsal wrist mass?"	→ **observation**
"Counsel parents of child with volar finger-base cyst?"	→ **will likely resolve with observation**
"True of paediatric ganglion?"	→ **>75% resolve within 1 y**
"True of volar wrist ganglion management?"	→ **open excision ~20% recurrence** ✗ Wrong options: aspiration 33% · 40% of masses · open better than arthroscopic
"DIP mass + nail deformity + dorsal osteophyte?"	→ **mucous cyst**
"Ganglion in Guyon's canal — nerve position?"	→ **ulnar nerve ulnar and dorsal to ulnar artery**
"Inconsistent with retinacular cyst?"	→ **moves with the tendon**
"Recurrent proximal tibiofibular ganglion — lowest recurrence?"	→ **PTFJ arthrodesis**

GIANT CELL TUMOR OF TENDON SHEATH

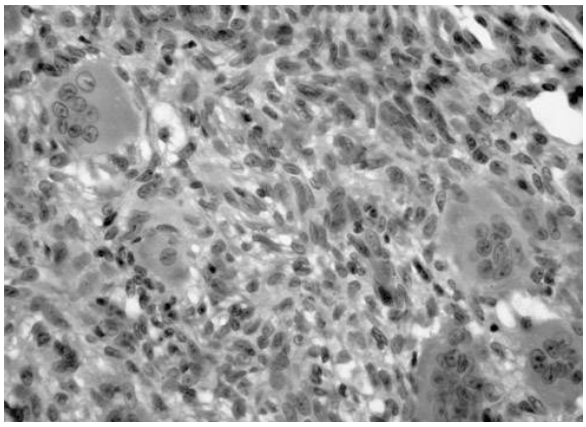
– Hand · 37 / 103 –

1 Basics

Synonyms	pigmented villonodular tumour of tendon sheath · **tenosynovial giant cell tumour (TGCT)**
Nature	**benign** nodular tumour from synovial lining of tendon sheath; idiopathic; **no metastasis, no malignant transformation reported**
Epidemiology	**2nd commonest hand soft-tissue tumour after ganglion** · **commonest SOLID soft-tissue mass** in hand · 3rd–5th decade
Site	**palmar radial 3 digits near DIPJ**; also foot/ankle tendon sheaths (dorsum, web space)
GCTTS vs PVNS	**histologically identical**; GCTTS almost always **extra-articular** · PVNS diffuse, intra-articular, larger joints

Genes & lab markers

Histology	**multinucleated giant cells**, sheets of **polygonal/histiocyte-like mononuclear cells**, **hemosiderin-laden** & **lipid/foam** macrophages, bland fibrous/spindled stroma, lymphocytes
Hemosiderin	present, typically **less than PVNS**
Gross	**lobulated, multicoloured: yellow-brown**, red, grey
CSF1R	CSF1R inhibitors for **diffuse-type TGCT** when surgery too morbid ✗ Wrong options: routine after excision



GCTTS histology: giant cells and hemosiderin

3 Presentation

Symptoms	slowly enlarging mass (months–years); pain with activity / **shoe wear** (foot); mechanical block, ↓ joint flexion
Exam	**firm nodular mass, does not transilluminate** (transilluminates = ganglion); may be fixed to bone

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

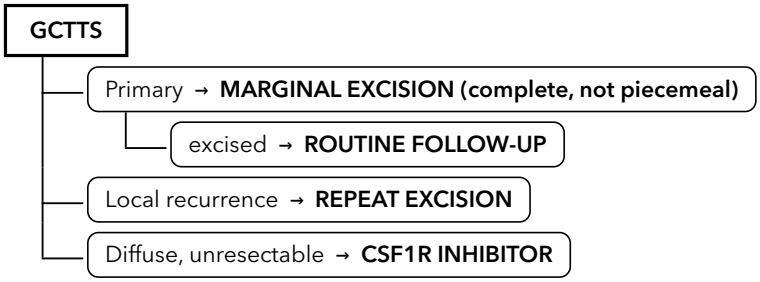
X-ray	soft-tissue swelling; **pressure erosion of bone** in up to **5%**
US	relation to tendon; homogeneously **hypoechoic**; **internal vascularity**
MRI	**modality of choice**; **↓ signal on T1 AND T2** (hemosiderin; Orthobullets, OB) · OB213042/OB219922: T1 iso/hypo, T2 may be **hyperintense**, enhances – read the options



MRI: low-signal GCTTS

Differential	ganglion (cystic, transilluminates), PVNS, desmoid, fibroma/fibrosarcoma, glomangioma

5 Management



Not indicated	radical/wide excision, ray resection, chemotherapy, routine radiation (lower recurrence reported but wound complications/fibrosis)

Local recurrence after excision

Commonest complication of incomplete excision	**recurrence** ✗ Wrong options: metastasis · malignant degeneration · pathological fracture · infection

Recurrence rate **5–50%** (Orthobullets) · 4–44% (Lanzinger) · 27% at mean 27 mo (Reilly)

– – Active space – –

Recurrence risk factors
Incomplete/piecemeal excision
Poorly encapsulated / diffuse, satellite lesions
Intra-articular extension, deep to volar plate
Flexor/extensor tendon involvement
Intraosseous involvement / osseous erosion
Adjacent **degenerative joint disease**
Distal location (DIP finger / IP thumb)
Not: age, sex, size, volar/dorsal, histology

X vs Y

	GCTTS	Ganglion	Epithelioid sarcoma	Gouty tophus	Synovial sarcoma
Key	Solid, yellow-brown, giant cells + hemosiderin	**Transilluminates**, cystic	**Commonest hand sarcoma**; epithelioid nodules, central necrosis/ ulceration	**White chalky**, negatively birefringent crystals	**Commonest foot sarcoma**; mineralised 30%, heterogeneous MRI

How it's asked

"Painful firm foot mass with shoe wear, GCTTS histology – true?"	→ **marginal resection** ✗ Wrong options: like GCT of bone · transilluminates · metastasis · radical excision
"Enlarging hand mass, multicoloured lobulated, giant cells?"	→ **GCTTS** ✗ Wrong options: ganglion · synovial sarcoma
"Foot web-space mass, hypointense T1 & T2, brown specimen?"	→ **GCTTS** ✗ Wrong options: melanoma · neuroma
"Finger mass eroding phalanx cortex, yellow-brown, hemosiderin?"	→ **GCTTS** ✗ Wrong options: epithelioid sarcoma · tophus
"Commonest complication of incomplete excision?"	→ **recurrence**
"After marginal excision – next step?"	→ **routine follow-up** ✗ Wrong options: wide re-excision · radiation · chemo
"Intraosseous, poorly encapsulated – counsel?"	→ **↑ local recurrence** ✗ Wrong options: CSF1R inhibitor · radiation

– – Active space – –

"Histologically identical to?" → **PVNS**

GYMNAST'S WRIST (DISTAL RADIAL PHYSEAL STRESS SYNDROME)

– Hand · 38 / 103 –

1 Basics

Epidemiology

Incidence	up to **25% of non-elite gymnasts**; radiographic physeal stress changes in **up to 85%** of gymnasts studied
Dose-response	training intensity correlates with ulna–radius length discrepancy
Load	distal radius bears **80% of axial load** on the extended wrist

2 Mechanism

Pathophysiology	wrist used as weight-bearing joint → supraphysiological repetitive load → **physeal inflammation** → microtrauma → **premature closure of distal radial physis** → **secondary ulnar overgrowth**
Associations	**distal ulnar overgrowth**, **positive ulnar variance**
Commonest injuring skills	front/back handsprings, saltos/flips

Distal radial physeal stress – growth arrest & Madelung deformity

Growth arrest risk in gymnasts	**distal radius** > scaphoid, distal humerus, clavicle, shoulder
Upper-limb weight bearers	stress fractures occur **distal to the elbow**
Physeal bar	forms in **ulnar aspect of distal radial physis** → relative ulnar overgrowth → ulnocarpal abutment → **Madelung deformity** (depends on growth remaining)

(✗ Wrong options: **Distractors** – distal radial *overgrowth*) · compression pseudofractures (Looser zones = osteomalacia/rickets) · radiocapitellar instability (nursemaid's) · ulnar longitudinal deficiency (club hand)
✗ Wrong options: – radius is short

3 Presentation

Symptoms	chronic wrist pain, usually **radial-sided**
Exam	**tenderness over distal radius**, ± swelling, ↓ flexion/extension from pain

4 Diagnosis & Classification

Diagnosis	clinical + radiographs

– – Active space – –

Key radiology findings

Modality	Finding
X-ray (AP + lateral)	**Widened distal radial physis with ill-defined, sclerotic borders**; chronic → **positive ulnar variance**
MRI (chronic / non-responders)	**Paraphyseal oedema**, early physeal bridging, radial bone bruise
CT	**Physeal bar** (ulnar side of distal radial physis)



Widened distal radial physis



Positive ulnar variance after physeal arrest

5 Management

- **Gymnast's wrist**
  - First line → **NSAIDs + REST + IMMOBILISATION 3–6 WK**
  - Premature physeal closure
    - Small bar → **PHYSEAL BAR RESECTION**
    - Large closure (~50% physis) → **ULNAR EPIPHYSIODESIS ± SHORTENING ± RADIAL OSTEOTOMY**

Prevention	proper technique, wrist guards/braces, ↓ repetitions during growth spurts, shoulder flexibility + thoracic mobility, core stabilisation
Return to play	pain resolved, normal wrist function, completed progressive rehab, proper technique
Prognosis	good with early treatment

6 Complications

Premature distal radial physeal closure	ulnar overgrowth → ulnocarpal abutment → **Madelung deformity**, DRUJ instability ✗ Wrong options: PRUJ

How it's asked

"Gymnast — which UE stress fracture most likely causes growth arrest?"	→ **distal radius**
"Physeal bar on CT, lost to follow-up — at risk of?"	→ **Madelung deformity** ✗ Wrong options: distal radial overgrowth · Looser zones
"X-ray finding?"	→ **widened physis, ill-defined sclerotic borders**
"First-line treatment?"	→ **rest, NSAIDs, immobilisation 3–6 wk**

– – Active space – –

HIGH-PRESSURE INJECTION INJURIES

– Hand · 39 / 103 –

1 Basics

Epidemiology

Incidence	rare; **1 in 600** hand traumas
Typical patient	**young male laborer**; paint, grease, solvents, diesel
Commonest digit	**non-dominant index finger** (>50%; 84% non-dominant hand in Bean series)

Pathophysiology

Force	**3,000–10,000 PSI**, up to 400 MPH
Skin breach	**100 PSI** breaks skin ✗ Wrong options: 80 PSI
4 components	initial injury → **chemical irritation** → inflammation → **secondary infection**
Spread	along planes of least resistance = **neurovascular bundles** ✗ Wrong options: flexor sheath
Vascular occlusion	local soft-tissue necrosis, thrombosis, ischaemia → compartment-syndrome-like picture

2 Mechanism

Injected substance & amputation risk

Substance	Destructiveness
Water, air	Least; non-toxic → only ones allowing nonop trial
Latex / water-based paint	Low
Grease, chlorofluorocarbon	Intermediate; aggressive debridement restores reasonable function
Oil-based paint	Highly inflammatory → chronic inflammation → amputation despite early surgery
Organic solvents (paint thinner, diesel, jet fuel, gasoline, oil)	**Highest amputation risk, up to 50%**

Most soft-tissue damage long term	**oil-based paint**
Highest amputation risk	**paint thinner / organic solvents** (some authors: primary amputation)
Temperature	**high temperature → worse** ✗ Wrong options: room temperature
Organic solvents & infection	may ↓ infection (do not support bacterial growth)

3 Presentation

History	document **time since injury** (average ~9 h to physician) + **material injected**
Entry wound	**benign-looking**, minimal/no pain; only **1 in 4** present to ED initially
Entry wound size	does not correlate with severity (>3 cm wound not prognostic)
Later	pain + paresthesias; finger **bloated, oedematous, tense, pale, cold**



Benign-looking entry wound, index finger

4 Diagnosis & Classification

Diagnosis	**clinical**: history + inspection

Key radiology findings

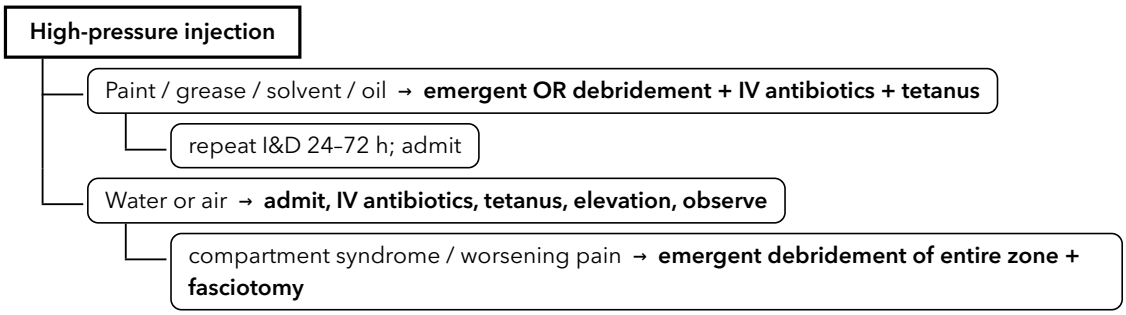
Radiographs	show spread of **radio-opaque paint/dyes**; often unremarkable



Radiograph: radio-opaque paint tracking

– – Active space – –

5 Management



Management – urgent debridement, antibiotics & decompression

First line (even 3 days late)	**immediate debridement in the OR** ✗ Wrong options: bedside I&D with digital block · IV then oral abx · amputation · clinic referral
Paint thinner, swollen finger	**surgical debridement & lavage** ✗ Wrong options: observation · corticosteroids · neutralising agent · antibiotics alone
Neutralising agents	only for **hydrofluoric acid** or chemo extravasation
Steroids & infection	do not affect amputation rate
Water injection + ↓ 2PD, swelling, erythema into palm, can't move	**emergent operative decompression** ✗ Wrong options: outpatient because water is non-toxic · because perfused · admit + IV abx only
Water injection observed → escalating pain	compartment syndrome → **immediate debridement of entire zone of injury** ✗ Wrong options: entry site only · cultures · cold compress · amputation
Antibiotics	**give before surgery**, don't delay; broad-spectrum **gram + and gram −**
Analgesia	**no digital block** and **no ice** (↓ perfusion)
Nonop care	tetanus, parenteral abx, elevation, early mobilisation, compartment monitoring; indicated only for **air, water, chicken vaccine**
Nonop outcome	**50%** ultimately need surgery; delay → ↑ reoperation & complications
Approach	**Bruner** palmar or **midaxial** incisions → full exposure of zone
Technique	wide debridement, **decompress compartments**, open tendon sheaths, remove all material, saline irrigation, **repeat I&D 24–72 h**
Post-op	**admit** for observation, possible second debridement/fasciotomy

Prognostic factors

Worst prognostic variable (MCQ) **delay in surgical treatment**

Factor	Key fact
Time to treatment	Debride **<6 h** → ↓ amputation & complications; **>10 h** → ↑ amputation
Force of injection	↑ pressure → worse
Volume injected	↑ volume → worse
Composition	Solvents / oil-based paints worst
Temperature	Hot → worse
Location	Tendon sheath involvement, proximal spread → worse



Debridement of injection injury

6 Complications

Amputation **48%** of finger injuries; ~50% with oil-based paint / organic solvents; Hogan series 30%

Infection **42% positive cultures, mostly polymicrobial**; delay ↑ risk; necrotic tissue = culture medium

Long term residual pain, stiffness, ↓ sensation, difficulty returning to work



Oil injection injury

– – Active space – –

How it's asked

"Most soft-tissue damage long term?"	→ **oil-based paint** ✗ Wrong options: water, latex, grease, CFC
"Highest amputation risk?"	→ **paint thinner** (organic solvent)
"Worst prognostic variable?"	→ **delay in surgery** ✗ Wrong options: wound >3 cm · grease · room temp
"Presents 3 days later, first-line?"	→ **immediate OR debridement**
"Paint thinner, swollen finger — next step?"	→ **surgical debridement & lavage**
"Water gun, ↓ 2PD, can't move finger?"	→ **emergent operative decompression**
"Observed water injection, now severe pain?"	→ **debride entire zone of injury**
"True statement?"	→ **antibiotics before surgery** ✗ Wrong options: 80 PSI · digital block · tracks flexor sheath · no timing difference
"Pressure that breaks skin?"	→ **100 PSI**
"Which may be managed nonop?"	→ **water, air**

HOOK OF HAMATE FRACTURE

– Hand · 40 / 103 –

1 Basics

Incidence **2–4% of carpal fractures**; ♂:♀ 2:1; often missed (mean **4 wk** to diagnosis)

Sports **golf, baseball, hockey**, racquet sports; **non-dominant hand** in batters

Anatomy

Articulations 4th & 5th MC, capitate, triquetrum

Flexor retinaculum hook = palpable attachment

Deep motor branch ulnar n. lies **under the hook**

Blood supply radial + ulnar foramina at base; **ulnar vessel absent 29%** → high nonunion

Guyon's canal border	Structure
Roof	Superficial palmar carpal ligament
Floor	Deep flexor retinaculum, hypothenar muscles
Ulnar	Pisiform, pisohamate lig.
Radial	**Hook of hamate**

Not a border **capitate**

Ulnar tunnel zones 1 (motor + sensory) & 2 (motor): **ganglion, hook of hamate fracture**; 3 (sensory): **ulnar artery thrombosis**

2 Mechanism

Direct blow to volar proximal palm grounding golf club, **checking a bat**, fouling a pitch, racket/oar strike; shear from ring/small flexors; FOOSH

Associated **bipartite hamate** (smooth cortices = normal variant, not fracture) · ring/small **flexor tendonitis/rupture** · **ulnar neuropathy in Guyon** (often motor) · other carpal fracture

3 Presentation

Symptoms **ulnar wrist pain** (commonest), hypothenar pain, pain with tight grip; ± ring/small paraesthesia

Exam **hook tenderness (80% sensitive)**; ↓ ulnar deviation, ↓ grip; Tinel at Guyon; intrinsic weakness

Hook of hamate pull test ulnar deviation + resisted DIP flexion ring/small → pain (70% sens)

Ulnar nerve clue **dorsal ulnar sensation normal** → lesion at wrist
✗ Wrong options: cubital tunnel

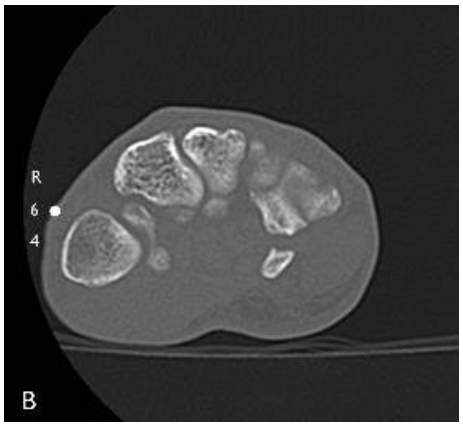
4 Diagnosis & Classification

Key radiology findings

Modality	Fact
PA/lateral	10% sensitivity; PA: **absent "eye sign"** (cortical ring)
Carpal tunnel view	**Best/most sensitive radiograph**, **first test** (40%)
Semi-supinated oblique	Adjunct
CT	Confirms if X-rays negative (92%; missed if cuts >1 mm) – **next step**
MRI	**Most accurate** (100%) with high suspicion
Bone scan	Confirms with CT (OB) · not typically indicated (OB) – read the options
✗ Wrong option: clenched fist	SL ligament
✗ Wrong option: MRA	Hypothenar hammer



Carpal tunnel view

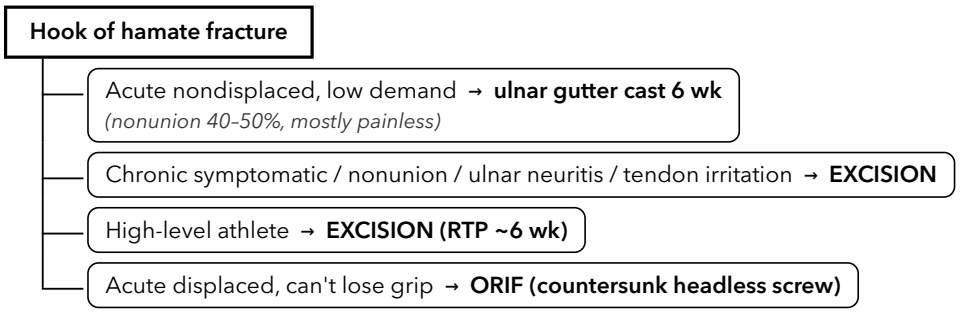


Axial CT hook fracture

Milch classification

Type	Description
I	**Hook** fracture (commonest): I-I avulsion · I-II middle · I-III base
II	Body fracture: IIA coronal · IIB transverse

5 Management



Treatment – fragment excision & return to play

Excision	**surgical treatment of choice**; fastest return; **RTP 6 wk** (some 3 mo)
Indications	chronic symptomatic nonunion, **ulnar neuritis**, flexor tendon irritation/ grating, high-level athlete

✗ Wrong options: **Traps** – Guyon release alone, carpal tunnel release, pisiform excision, cast for chronic nonunion
✗ Wrong options: tendon

Technique	incision along **ulnar border of ring finger**; release Guyon; **subperiosteal** removal (protect deep motor br.)
Excision cost	small finger FDP strength ↓ **10–15%**; FDP shifts 5 mm ulnar
ORIF	~100% union, theoretically better grip; screws **countersunk** (deep motor br.); neurolysis if neuritis

6 Complications

Complication	Fact
Nonunion	**Commonest**, ~50%; motion + poor supply
Ulnar neuritis	20% → excision
Closed rupture ring/small flexors	**Hook nonunion** (fulcrum); rare
Weak grip	Excision of large fragment
Recurrence	Pro baseball; prevent with extraperiosteal excision

Other closed ruptures volar plate DRF → **FPL** · closed DRF → **EPL** · scaphoid spur → radial flexors

How it's asked

"Baseball player, hypothenar pain + ulnar paraesthesia – view?"	→ **carpal tunnel view**
"Chronic golfer wrist pain, initial test"	→ **carpal tunnel view** ✗ Wrong options: bone scan, EMG, MRA
"Mechanic, interossei weak, dorsal sensation normal, X-ray"	→ **hamate nonunion**
"Not a border of Guyon's canal"	→ **capitate**

– – Active space – –

"Rower, oar into palm, carpal tunnel view"	→ **hook of hamate fracture**
"Pro baseball, 1 yr hypothenar pain, CT fracture — Rx"	→ **hook excision**
"Best literature outcome / RTP"	→ **surgical excision**
"Ulnar nerve sx + grating fingers 6 mo after FOOSH"	→ **excision of hook**
"Return to play after excision"	→ **6 wk**
"True of CT finding"	→ **excision → excellent RTP** ✗ Wrong options: normal variant · APB weakness · high union nonop
"Closed small-finger flexor rupture from"	→ **hook of hamate nonunion**

HUMAN BITE

– Hand · 41 / 103 –

1 Basics

Epidemiology

Frequency ~2% of bite wounds; **3rd after dog & cat**; ♂

Location **dorsal 3rd/4th MCP** = **"fight bite"**

Pathoanatomy

Mechanism clenched fist → tooth → **MCP capsule penetrated** → fist opens → bacteria **trapped under extensor tendon/capsule**

Also direct bite (child biting child)

Associated **extensor tendon laceration** (missed – proximal retraction)

Microbiology – polymicrobial & Eikenella

Most common isolate **polymicrobial** (~4 isolates/culture; saliva >600 species)

Commonest single organisms **α-haemolytic strep (S. viridans)** + **S. aureus** (S. aureus = overall most common pathogen)

Eikenella corrodens **facultative anaerobic gram-neg rod**; **characteristic of human bites** (rare in dog/cat); 7–29% (Orthobullets) · 17–30% / ~⅓ (OB)

Eikenella resistance **aminoglycosides, 1st-gen cephalosporins, clindamycin**

Bite	Infection rate	Signature organism
Dog	5–15%	Pasteurella canis, Capnocytophaga
Cat	**30–50%**	Pasteurella multocida
Human	High	**Eikenella corrodens**

3 Presentation

History punch to mouth, often **concealed** → treat as fight bite until proven otherwise; delayed presentation

Exam small **transverse, irregular** dorsal MCP wound; erythema, warmth, ± pus; pain on passive MCP ROM; **check extensor function**; volar side & NV usually spared

– – Active space – –



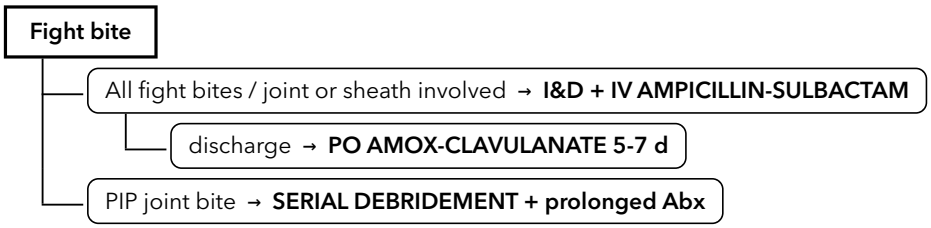
Fight bite over 3rd MCP

4 Diagnosis & Classification

Key radiology findings

X-ray	**tooth fragment (FB)** + fracture (MC head)
Culture	not in ED (contaminated); **deep aerobic + anaerobic in OR**

5 Management



I&D	debride wound + **joint capsule**; **leave open**; Gram stain + culture
MCP approach	**tendon-splitting** (Shewring)
IV Abx	cover **Staph, Strep, gram-neg → ampicillin/sulbactam**; alternatives: cephalosporin, gentamicin + penicillinase-resistant penicillin, **penicillin + metronidazole**
Outcome	MCP injuries satisfactory/good; **PIP injuries → 42% poor**

X vs Y

	Human bite	**Dog / cat bite**
Characteristic organism	**Eikenella corrodens**	**Pasteurella**
Also	S. viridans, S. aureus, anaerobes	Capnocytophaga (dog), Corynebacterium, Moraxella
Typical site	Dorsal MCP (fist)	Any (cat → joints/sheath)
Oral Abx	Amox-clav	Amox-clav

How it's asked

"Fight bite — most common culture result?"	→ **polymicrobial** ✗ Wrong options: Eikenella alone · Pasteurella
"Organism more frequent in human than animal bites?"	→ **Eikenella corrodens** ✗ Wrong options: Capnocytophaga, P. canis, S. aureus
"IV antibiotic for fight bite?"	→ **ampicillin-sulbactam**
"Eikenella resistant to?"	→ **clindamycin, 1st-gen cephalosporins, aminoglycosides**

– – Active space – –

HYPOTHENAR HAMMER SYNDROME

– Hand · 42 / 103 –

1 Basics

Epidemiology & anatomy

Definition	post-traumatic digital ischaemia from **ulnar artery thrombosis / aneurysm at Guyon's canal**
Demographics	rare (<1/100,000); ♂:♀ **9:1**; 40s–50s
Location	unilateral, dominant **ring ± small** finger; less often index/middle; **thumb spared**
Risk	**vibrating tools** (carpenters, mechanics, jackhammer, construction); baseball catchers, mountain biking, golf, volleyball, karate, MMA
Anatomy	distal 2 cm: ulnar artery **directly anterior to hook of hamate**, covered only by palmaris brevis, fat, skin; divides at Guyon's exit → deep branch + superficial palmar arch
Ulnar artery	**most common site of true and false aneurysms** in the hand; aneurysm = permanent dilatation **≥50%** of normal

Pathology – true aneurysm vs thrombosis & recurrence

Mechanism	single or **repetitive blunt impact** on hypothenar eminence → **hook of hamate acts as an anvil** → thrombosis or aneurysm → **distal embolisation** → ulcers, gangrene
Hook of hamate fracture	may precipitate ulnar artery aneurysm (catcher)

	True aneurysm (HHS)	False / pseudoaneurysm
Cause	**Blunt** (repetitive) trauma	**Penetrating** trauma
Wall	**All arterial layers**, true endothelial lining	No endothelial lining
Shape	**Gradual uniform dilatation**	Haematoma organises → recanalises → false lumen

3 Presentation

Symptoms	pain hypothenar eminence + **ring finger**, **cold sensitivity**, paraesthesia ulnar 2 digits (± ulnar nerve compression)
Signs	**hypothenar callus**, tenderness, blanching, mottling, cyanosis, pallor, **ulnar fingertip ulcers**, **splinter haemorrhages**, **pulsatile mass** if aneurysm
Allen's test	**positive with occlusion**; **negative with aneurysm** (patent); may show incomplete palmar arch

4 Diagnosis & Classification

Key radiology findings

First-line	**Doppler US** + **digital brachial index (DBI)** ✗ Wrong options: MRA first
Angiography (CTA / MRA / conventional)	**mandatory for diagnosis** (Orthobullets); contrast arteriography = **gold standard** (Hui-Chou)
Angiogram signs	**"corkscrew" tortuous ulnar artery**; **occlusion or aneurysm at hook of hamate**
Carpal tunnel view	hook of hamate fracture



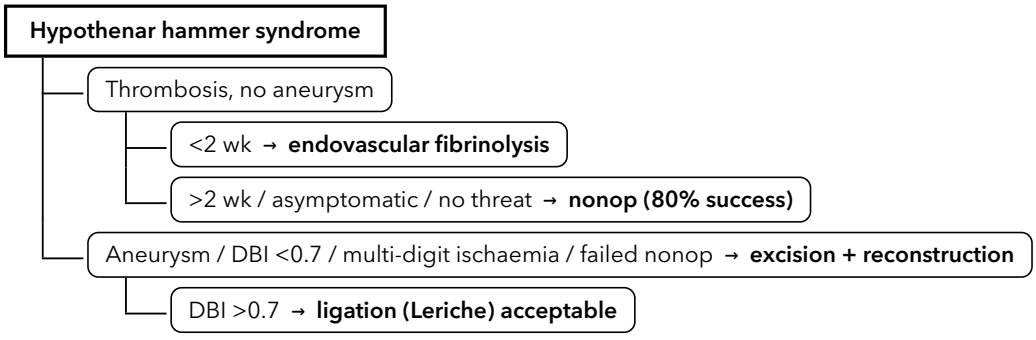
Angiogram: ulnar artery aneurysm at the hook of hamate

Digital brachial index criteria

DBI	Meaning / action
0.85–1.25	Normal
>0.7	Minimal symptoms, viable digit → **no surgery** / ligation acceptable
<0.7	**Reconstruction**
<0.5	Severe symptoms, **digital gangrene**

Differential **Raynaud's** involves the **thumb** (HHS spares it)

5 Management



Nonop smoking cessation, avoid trauma, vascular consult; **80% success**

– – Active space – –

Treatment – resection & arterial reconstruction

Operative criteria	**DBI <0.7** · **thrombosis with aneurysm** · ischaemia multiple digits · recurrent symptoms after nonop
Aneurysm	**always surgical**: **excision of segment + reconstruction ± vein graft** → prevents emboli, removes painful mass, relieves ulnar nerve, removes vasospasm trigger
Graft	**reversed interpositional vein graft** (saphenous, forearm, foot dorsum) or arterial (thoracodorsal, SIEA, descending LFCA); gap usually too large for direct repair
Not for aneurysm	fibrinolysis (acute thrombosis without aneurysm only) · **systemic thrombolysis** (never) · **stenting** (stenosis, not thrombosis) · warfarin · CCB / abstinence alone
Ligation	**contraindicated with incomplete palmar arch**
Ligation vs repair vs vein graft	**no outcome difference** (Demetri) ✗ Wrong options: "ligation superior"
Outcomes	vein graft patency **77% at 2 y** (Chloros); **78% occluded at ~10 y** yet patients satisfied (Endress)

6 Complications

Most common post-op complaint	**persistent cold intolerance** ✗ Wrong options: digital ischaemia

Vein-graft failure	Time	Cause
Acute	**<1 mo**	**Thrombosis** (patency 98% → <88% in 1st month)
Subacute	**1–18 mo**	**Intimal hyperplasia** = vascular smooth muscle migration + proliferation at anastomosis (30–50% fail)

Not failure causes at 6 mo	graft resorption, recurrent aneurysm, hamate nonunion compression, thrombosis (acute only)

X vs Y

	HHS	Raynaud's
Side	Unilateral, dominant	Bilateral
Digits	Ring ± small, **thumb spared**	Includes **thumb**
Cause	Blunt trauma, hook of hamate anvil	Vasospasm
Angio	Corkscrew, occlusion/aneurysm at hamate	–

How it's asked

"Jackhammer worker, hypothenar callus, pulsatile mass – true statement?"	**→ gradual dilatation of all arterial layers (true aneurysm)**
"DBI that necessitates reconstruction?"	**→ <0.7** ✗ Wrong options: 0.85

"First-line test?"	→ **Doppler US + DBI** ✗ Wrong options: MRA
"Most common complication after surgery?"	→ **persistent cold intolerance**
"Pathophysiology of blunt-trauma aneurysm?"	→ **true endothelial lining, uniform dilatation** ✗ Wrong options: recanalised false lumen
"Vein graft failed at 6 mo — why?"	→ **intimal hyperplasia (smooth muscle migration/proliferation)** ✗ Wrong options: thrombosis = acute
"HHS with aneurysm and ischaemia — next step?"	→ **excision + reconstruction ± vein graft** ✗ Wrong options: fibrinolysis · warfarin · CCB
"Aneurysm thrombosed, ulcer, DBI 0.4, incomplete arch?"	→ **ulnar artery reconstruction** ✗ Wrong options: ligation · stenting · thrombolysis
"Acute thrombosis <2 wk without aneurysm?"	→ **endovascular fibrinolysis**

– – Active space – –

INTERSECTION SYNDROME

– Hand · 43 / 103 –

1 Basics

Epidemiology & anatomy

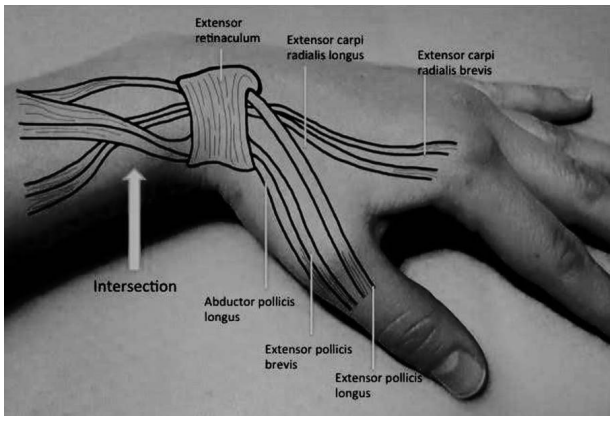
Definition inflammation where **1st compartment (APL, EPB) muscle bellies cross over 2nd compartment (ECRL, ECRB) tendons**; aka crossover tendinitis; = stenosing tenosynovitis of **2nd compartment**

Site dorsoradial forearm **~5 cm (4–6) proximal to wrist**

Demographics rare; ♂ = ♀ (Orthobullets) · **more in men, dominant hand, younger, shorter symptoms vs DQT** (Sato)

Risk **rowing**, canoeing, skiing, horse riding, weight lifting, racket sports

Compartment	Contents	Eponymous condition
1	APL, EPB	**De Quervain's**
2	**ECRL, ECRB**	**Intersection syndrome**
3	EPL	–
4	EIP, EDC	–
5	EDM	Vaughan-Jackson
6	ECU	Snapping ECU



Intersection of 1st compartment over ECRL/ECRB

Tendon	Insertion
ECRL	**Base of 2nd MC**
EPB	Dorsal base thumb proximal phalanx
EPL	Dorsal base thumb distal phalanx
APL	Radial base 1st MC
EIP	Extensor hood index MCP

Intersection type (TO)	Site	Imaging
Proximal (classic)	APL + EPB cross ECRL + ECRB in distal forearm	Request wrist **with distal forearm**
Distal	**EPL crosses ECRL + ECRB beyond Lister's tubercle**	Seen on routine wrist MRI

2 Mechanism

Mechanism **repetitive wrist extension** → friction at crossover → tenosynovitis

3 Presentation

Symptoms dorsal forearm/radial wrist pain, worse with pronation–supination

Signs tenderness dorsoradial forearm ~5 cm proximal; **crepitus / squeak / popping** with **resisted wrist extension + thumb extension**

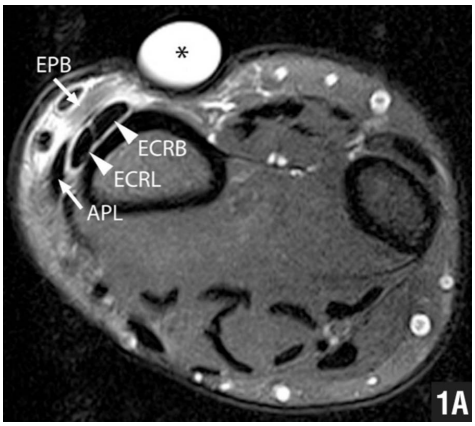
4 Diagnosis & Classification

Diagnosis **clinical**

Key radiology findings

X-ray not required (unremarkable)

MRI only if unclear; fluid-sensitive (STIR, FS PD, T2) → **peritendinous oedema around 1st + 2nd compartments** (most characteristic); tendinosis, muscle oedema, loss of comma shape



MRI: peritendinous fluid at the intersection

Differentials De Quervain's, Wartenberg's

5 Management

Intersection syndrome

- First line → **rest + wrist splint + steroid into 2nd compartment**
- Recalcitrant → **surgical release + debridement of 2nd compartment**

– – Active space – –

Second dorsal compartment (ECRL/ECRB) – injection & release

Injection into **2nd dorsal compartment (ECRL + ECRB)**
✗ Wrong options: 1st = DQT

Surgery rare; release **2nd compartment ~6 cm proximal to radial styloid** (Orthobullets) · **5–6 cm proximal to wrist** (OB) → decompresses **ECRL (inserts base 2nd MC)**

X vs Y

	Intersection syndrome	De Quervain's
Compartment	**2nd (ECRL, ECRB)**	**1st (APL, EPB)**
Site	**~5 cm proximal** to wrist, dorsoradial forearm	Radial styloid
Patient	Younger, men, dominant hand, rowers	♀ >40, **pregnant / lactating / peripartum**
Sign	Crepitus, resisted wrist + thumb extension	Finkelstein
Injection	2nd compartment	1st compartment

How it's asked

"EPL crossing ECRL/ECRB distal to Lister's tubercle?"	→ **distal intersection syndrome** (TO)
"Rower, tender 5 cm proximal to wrist, inject which compartment?"	→ **ECRL + ECRB** ✗ Wrong options: APL/EPB · EPL
"Compartment + structures affected?"	→ **compartment 2; ECRB + ECRL tendons** ✗ Wrong options: muscle bellies · compartment 1
"Release decompresses tendon inserting on?"	→ **base of 2nd MC (ECRL)** ✗ Wrong options: thumb P1 base = EPB · P2 base = EPL · 1st MC = APL · index hood = EIP
"Peripartum woman with radial wrist pain?"	→ **De Quervain's** ✗ Wrong options: intersection

INTRINSIC MINUS HAND (CLAW HAND)

– Hand · 44 / 103 –

1 Basics

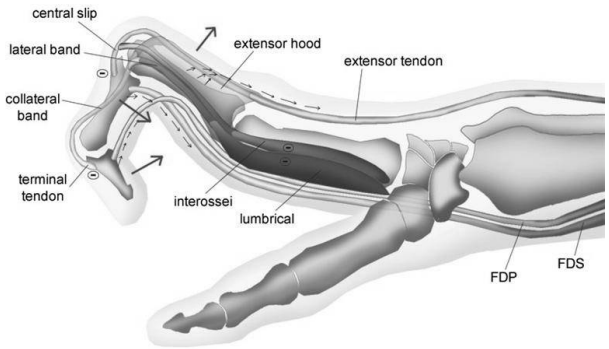
Etiology

Definition	**MCP hyperextension + PIP/DIP flexion** from **strong extrinsics vs deficient intrinsics**
Ulnar nerve palsy	cubital tunnel, ulnar tunnel
Median nerve palsy	Volkmann's ischaemic contracture, **leprosy**, crush hand not splinted intrinsic-plus
Others	**Charcot-Marie-Tooth** (HMSN), **compartment syndrome of hand**

2 Mechanism

Mechanism – strong extrinsics & deficient intrinsics

Loss of intrinsics	lose MCP flexion + IP extension
Strong EDC	unopposed **MCP extension**; not a significant PIP extensor (central slip force mostly from interossei)
Strong FDP + FDS	unopposed **PIP & DIP flexion**
Crush hand splinted in MCP extension	↑ extrinsic flexor tension → **PIP flexion** (claw); splint **intrinsic-plus**
Hand compartment syndrome	most consistent sign: **tense swollen hand in intrinsic-minus position**; Rx fasciotomy; full function unlikely after ischaemic contracture



Pathoanatomy

– – Active space – –

Deformity	Pathoanatomy
Claw / intrinsic minus	**Strong extrinsics + deficient intrinsics**
Intrinsic plus	**Spastic intrinsics + weak extrinsics** → MCP flexion, IP extension
Swan neck	**Volar plate laxity** + dorsal lateral band tethering → PIP hyperext, DIP flexion
Lumbrical plus	**FDP laceration distal to lumbrical origin** → paradoxical IP extension on flexing
Quadriga	**Adhered FDP** (one finger) → flexion lag adjacent digits

3 Presentation

Exam MCP hyperextension + IP flexion; ↓ grasp, grip, pinch

Ulnar palsy worst in **ring & little** (ulnar lumbricals); index/middle less (median lumbricals)



Claw hand

Ulnar nerve palsy – claw, ulnar paradox & Bouvier test

Low ulnar (wrist) cut (TO) **clawing ring + little** (lumbricals 3–4), **can't abduct/adduct** (interossei; can't abduct middle finger), **can't pinch** (adductor pollicis), thumb hyperextension

Ulnar supply in hand (TO) deep br.: ADM, FDM, ODM, **adductor pollicis**, **deep head FPB**, **all interossei** (PAD/DAB), **medial 2 lumbricals**; superficial br.: palmaris brevis

Lumbricals (TO) arise from FDP → dual supply like FDP

Ulnar paradox (TO) **higher lesion → less clawing**; due to **FDP** (ulnar half) paralysis at/above elbow; distal lesion spares FDP → more claw
✗ Wrong options: FDS, FCU, FCR

Sign rule (TO) every deformity is produced by the **muscle that still works, overacting** for the paralysed one · claw of **all 4 fingers = combined ulnar + median** (Duchenne sign)

Other ulnar sign (TO)	Finding	Cause
André-Thomas	Wrist flexes when trying to extend IPs	Tenodesis compensating for lumbricals
Bunnell O sign	Can't make round "O" thumb–index	Adductor pollicis + deep FPB lost; FPL takes over
Earle-Vlastou	Can't cross middle over index	Interossei
Egawa (table-top)	Can't abduct/adduct middle finger with MCP flexed	Interossei
Card test	Card pulled easily from between index + middle	Interossei
Pitres-Testut	Hand flat, can't abduct middle; can't make cone	Palmar + dorsal interossei
Mumenthaler	No hypothenar dimple on little abduction	Hypothenar + palmaris brevis
Sunderland	Can't oppose little finger to thumb	FDM, ODM, ADM
Inverted pyramid	Hollow triangle central palm (chronic)	Transverse head adductor pollicis wasting
Confrontation	Abducted little fingers opposed → palsied one pushed over	Interossei

Examine whole nerve length (TO) multiple sites possible (cubital tunnel + Guyon's + brachial plexus schwannoma) · recovery after repair: **radial > median > ulnar** (slowest)

Deformities (TO) **intrinsic plus = safe splinting position** (collaterals relaxed) · swan neck + boutonnière can develop in long-standing ulnar / combined palsy

4 Diagnosis & Classification

Bouvier test

Bouvier (TO) guides **claw correction**; block MCP hyperextension (~10° flexion) → watch active PIP extension

Result	Meaning	Surgery
Positive (PIP extends; assisted angle 0)	**Simple claw**; extensor mechanism intact	Any procedure **preventing MCP hyperextension → static** (MCP capsulodesis, pulley advancement)
Negative (PIP stays flexed)	**Complex claw**	Needs extensor augmentation / flexor or joint correction

– – Active space – –

Negative Bouvier subtype (TO)	Cause	Clue
Passive positive	**Stretched central slip** (long-standing)	–
Passive negative, tenodesis positive	**Long flexor shortening**	PIP extends with **wrist flexed**
Passive negative, tenodesis negative	**Stiff PIP** (volar plate / collateral contracture)	No extension even with wrist flexion

5 Management

Claw hand (progressive, ↓ QoL)

- Bouvier positive (simple) → **STATIC: capsulodesis / pulley advancement / tenodesis**
- Bouvier negative (complex) → **active tendon transfer + extensor/flexor/joint correction**

Goal **prevent MCP hyperextension**

Static procedures correct MCP hyperextension only (TO) no independent intrinsic action

Static procedure (TO)	Technique
Zancolli capsulodesis	**A1 cut**, U-flap of MP volar plate advanced proximally → MP ~30° flexion
Omer capsulodesis	Volar plate flap advanced to tighten capsule
Parkes tenodesis	Grafts from TCL, **volar to deep transverse MC ligament** → radial lateral bands
Fowler / Riordan tenodesis	Fowler: grafts to extensor retinaculum · Riordan: distally based strips of ECRL + ECU
Zancolli lasso	Low ulnar: ring + small FDS divided distal to A1, **looped round A1**, sutured to itself

– – Active space – –

Dynamic procedure (TO)	Donor → insertion	Notes
Modified Stiles-Bunnell (Littler)	**Ring FDS** → 2 slips → lumbrical canals → radial lateral bands ring + little	**Isolated LOW ulnar only** (FDP must work) · suture wrist neutral, MP 45–55°, IP extended · splint intrinsic plus 3–4 wk
Middle finger FDS	–	**High ulnar palsy** (ring FDS not expendable in high / combined) · risk swan neck, donor PIP contracture
Brand EE4T	**ECRB** + 4-tailed graft, **extensor route** through intermetacarpal spaces → lateral bands	Median function uncertain
Brand EF4T	**ECRL** + graft via **carpal tunnel (flexor route)**	Same insertions
Fritschi PF4T	**Palmaris longus**	Weak donor
Palande	Ring FDS in **5 slips** (or ECRL ± PL) → interossei + hypothenar	**Direct interossei activation** → abduction/ adduction

Lateral band insertion (TO) **ulnar side of index**, **radial side** of middle, ring, little

Other ulnar transfer (TO)	Procedure
Little finger abduction (Wartenberg)	**Fowler: ulnar half of EDM** volar to DTML → radial collateral ligament P1 base
Thumb adduction / key pinch	**Smith: ECRB + graft** via 2nd intermetacarpal space → adductor tubercle · Littler / Brand: ring FDS
Index abduction	**Bunnell: EIP + graft → 1st DI** · Bruner: EPB · Hirayama: PL · Riordan: ring FDS · Neviaser: accessory APL
High ulnar (FDP ring/little + FCU lost)	**Side-to-side FDP middle → ring + little** or middle FDS → FDP ring/little · FCR → FCU (rare)

Combined palsy (TO)	Order
Low	Prevent 1st web contracture → index abduction (APL) → thumb adduction (Smith ECRB / ring FDS) → opponensplasty (EIP) → claw correction (EF4T) last
High	**Finger flexion first** (ECRL → FDP) → BR → FPL → EIP opponensplasty → tenodesis / capsulodesis → **FDMA flap** (radial-innervated) for thumb-tip sensation

Ulnar nerve transfers (TO) **AIN (PQ branch) → deep motor ulnar** at forearm · fascicle topography **sensory–motor–sensory** (motor central) · sensory: median 3rd web fascicle → ulnar sensory; dorsal cutaneous end-to-side to median · release Guyon's / carpal tunnel

How it's asked

"Crush hand splinted in MCP extension – deformity?"	→ **PIP flexion**
"Claw after crush/compartment syndrome – pathoanatomy?"	→ **strong extrinsics + deficient intrinsics** ✗ Wrong options: spastic intrinsics = intrinsic plus

– – Active space – –

"Ulnar paradox due to"	→ **FDP** (TO)
"Wrist cut: claw ring/little, can't pinch, can't abduct middle — nerve?"	→ **ulnar** (TO)
"Bouvier test helps correction of"	→ **claw deformity** (TO)
"EE4T stands for?"	→ **ECRB, Extensor route, 4 Tails (Brand)** (TO)
"Direct interossei activation transfer?"	→ **Palande** (TO)
"Stiles-Bunnell ring FDS transfer — contraindicated in?"	→ **high ulnar palsy** (FDP ring/little lost) (TO)

INTRINSIC PLUS HAND

– Hand · 45 / 103 –

1 Basics

Definition **MCP flexion + PIP & DIP extension** from **spastic/contracted intrinsics** vs weak extrinsics

Pathoanatomy tight interossei + lumbricals → MCP flexion, IP extension; **weak EDC** (no MCP extension balance); **weak FDS/FDP** (no IP flexion balance)

2 Mechanism

Trauma crush, vascular injury, **compartment syndrome** (e.g. IVDU overdose); oedema/haematoma trapped by interosseous fascia → fibrosis

Rheumatoid MCP dislocation + ulnar deviation → spastic intrinsics

Neurologic **TBI, CP, CVA, Parkinson's**

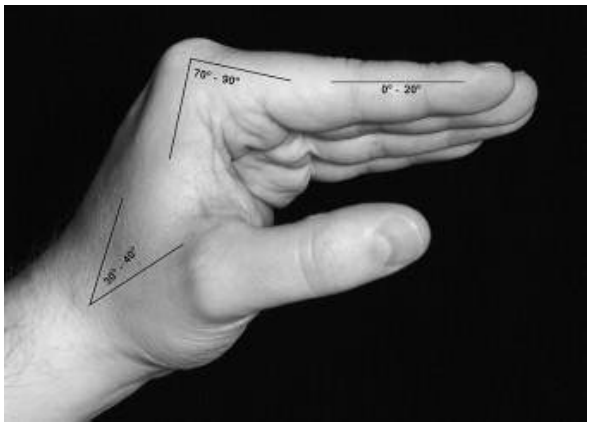
Other repetitive grip (carpenter), prolonged immobilisation, ischaemia, iatrogenic

3 Presentation

Intrinsic tightness (Bunnell test) & posture

Symptom difficulty **gripping large objects** (e.g. hammer)

Posture MCP flexion, PIP + DIP extension



Intrinsic plus posture

Bunnell (intrinsic tightness) test **positive = ↓ passive PIP flexion with MCP extended** vs MCP flexed → **intrinsic tightness**

Finding	Source of stiffness
PIP flexion **↓ with MCP extended**	**Intrinsic tightness**
PIP flexion **↓ with MCP flexed** (more with MCP extended)	**Extrinsic tightness** (extensor adhesions)
PIP motion unchanged by MCP position	**Joint capsule** contracture

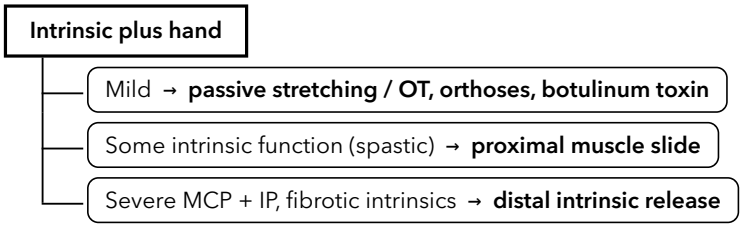
– – Active space – –

(✗ Wrong options: **Distractors** – **quadriga** (flexion lag adjacent to repaired FDP)) · **lumbrical plus** (paradoxical PIP extension on active flexion) · central slip rupture (boutonnière)

4 Diagnosis & Classification

Diagnosis	**clinical**; X-ray not required (normal)

5 Management



Proximal muscle slide	**subperiosteal elevation of interossei** → lengthens MTU
Distal intrinsic release	resect intrinsic tendon **distal to transverse fibres** (spares MCP flexion)
Release needed (Espiritu)	extensor hood index 59%, middle 65%, ring 26%, small 33%

X vs Y

	Intrinsic plus	Intrinsic minus (claw)
MCP	**Flexion**	**Hyperextension**
PIP / DIP	**Extension**	**Flexion**
Imbalance	Spastic intrinsics, weak extrinsics	Strong extrinsics, deficient intrinsics

How it's asked

"Passive PIP flexion possible with MCP flexed but not extended?"	→ **intrinsic tightness (positive Bunnell)** ✗ Wrong options: extrinsic tightness · quadriga · lumbrical plus
"Crush injury, same Bunnell finding – stiffness from?"	→ **intrinsic musculature** ✗ Wrong options: PIP/MCP capsule
"Posture of intrinsic plus hand after compartment syndrome?"	→ **MCPJ flexion with PIPJ & DIPJ extension** ✗ Wrong options: MCP extension + IP flexion = intrinsic minus

JERSEY FINGER

– Hand · 46 / 103 –

1 Basics

Definition	**FDP avulsion** from base of distal phalanx (**zone I**); aka rugby/rugger jersey finger
Ring finger	**75%**; ring tip 5 mm more prominent in grip (~90%) → ↑ pull-away force
FDP innervation	ulnar n. + AIN
Zone I	distal to FDS insertion
Blood supply	**vincula**: short (breve) near insertion · long (longum) from FDS chiasma region; palm retraction shears **both** → tendon necrosis
Quadriga basis (TO)	FDP of long, ring, little share **common muscle belly**

2 Mechanism

Forced DIP extension during max FDP contraction	grabbing a jersey/flag in a tackle

3 Presentation

Diagnosis, retraction level & urgency

Rest	finger in **slight extension** at DIP vs cascade
Key sign	**no active DIP flexion**; MCP/PIP normal
Palpable mass	retracted tendon in palm / distal palmar crease (Type I) – MRI can show level
Paradoxical PIP extension	on attempted flexion



Jersey finger: loss of DIP flexion in ring finger

- - Active space - -

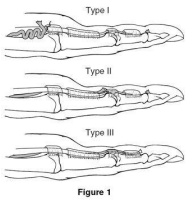
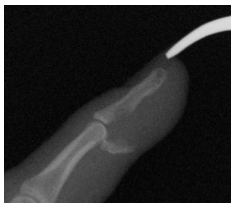
4 Diagnosis & Classification

Key radiology findings

X-ray may show **avulsion fragment** (level of fragment = level of retraction)

MRI level of proximal retraction

Modified Leddy & Packer classification

Type	Retraction / fragment	Vincula	Repair window	Image
I	**Into palm** (lumbrical origin, ≥A1); no fragment	**Both ruptured → most threatening** (TO)	**7–10 d**	
II	**PIP level** (A3 / Camper's chiasma), ± small fragment; **most common** (TO)	Short ruptured, **long intact**	Several weeks (Orthobullets) · **6 wk** (OB) · up to **3 mo** reported (TO)	
III	**Large fragment** held at **DIP / A4 pulley** (Orthobullets, TO) · held at **A2** in one TO stem – read the options	**Both intact →** best prognosis	Weeks; **ORIF** (screw, K-wire, pull-out wire)	
IV	**Double avulsion**: fragment + tendon off fragment → tendon usually into palm (Smith 1981)	As I/II	**ORIF fragment first, then tendon**	
V	Avulsion + **comminuted distal phalanx** (Va extra-, Vb intra-articular) (Al-Qattan 2001)	Minimal damage (TO)	Fix if possible, else tendon-to-bone ± DIP stabilisation	

Urgency depends on **degree of tendon retraction**

✗ Wrong options: presence or size of fragment · PIP ROM · athletic level

5 Management

Jersey finger (all need surgery)

- Acute (<3 wk; ideally <1 wk) → **tendon repair / reinsertion (button or anchor)**
  - bony (III/IV) → **ORIF fragment (IV: + tendon reinsertion)**
- Chronic (>3 mo) + full DIP PROM → **TWO-STAGE TENDON GRAFT**
- Chronic (>3 mo) + stiff DIP → **DIP ARTHRODESIS**

No fracture on X-ray (TO)	Type I/II → **immediate surgical repair** ✗ Wrong options: splint in extension/flexion · observe months
No nonop	extension block splint, buddy taping, early ROM → no DIP function

Acute repair technique – advancement limit & anchor placement

Fixation	direct repair or reinsertion with **dorsal (nail) pull-out button** or **suture anchor** (anchors replacing buttons: button irritation) (TO)
Advancement limit (TO)	**≤1 cm** (excursion balance of common FDP belly) → >1 cm = **quadriga** / DIP flexion contracture
2.5 mm advancement	**acceptable**
Anchor	**45° retrograde** (distal-volar → proximal-dorsal) in **proximal metaphysis** of distal phalanx ✗ Wrong options: perpendicular · dorsal cortex breach · antegrade (more gap) · articular surface
DIP pinning	optional, not required
No DIP arthrodesis acutely	repair instead
Do not delay	don't finish 8-week season before repair
Rehab	**Duran** (patient-assisted passive) or **Kleinert** (dynamic splint passive)
After repair	check **symmetric resting cascade** (TO: over-tension → quadriga)

Chronic injury – two-stage grafting

Chronic (>3 mo) + full PROM	**excise palmar mass + 2-stage grafting** (Hunter: silicone rod → pseudosheath → PL/plantaris graft)
Avoid single-stage graft	adhesions in fibro-osseous tunnel
Avoid active silicone rod	not effective
No direct repair late	too much tension, tendon won't reach
DIP arthrodesis	chronic + **stiff** DIP; young guitarist with supple finger → **grafting**
Outcome	reconstruction poorer than acute repair, stiffness

6 Complications

Quadriga	over-advancement (>1 cm) → adjacent fingers' flexion & grip ↓
DIP flexion contracture	over-advancement

– – Active space – –

X vs Y

Differential – central slip injury

	Jersey finger	Central slip / boutonnière (TO)
PIP	Flexes normally	**Flexed, can't extend**
DIP	**Can't flex**	**Hyperextended, can't flex**
Test	No active DIP flexion with PIP blocked	**Elson** (rigid DIP = +), modified Elson, Boyes (late)
Acute Rx	**Surgery**	**PIP extension splint 4–6 wk**, DIP free

Boutonnière pathology (TO) central slip rupture → lateral bands volar (transverse retinacular lig. contract) → PIP flexion + DIP hyperextension; may appear **10–14 d** after injury

Chronic boutonnière (TO) supple: **terminal tendon tenotomy (Fowler/Dolphin)**, lateral band relocation, Littler; stiff/RA/arthritic: **PIP arthrodesis**

Pseudo-boutonnière (TO) PIP flexion contracture **without** DIP hyperextension

Other diagnoses (wrong)	Finding
FDS avulsion	PIP flexion lost
Distal extensor (mallet)	DIP extension lost
Sagittal band	MCP extensor subluxation

How it's asked

"Football tackle, can't flex DIP, X-ray normal — Rx"	→ **immediate surgical repair** (TO)
"Most threatening Leddy-Packer type"	→ **Type I** (TO)
"Large fragment trapped at pulley, ring finger, 5 d"	→ **Type III**, vincula intact (TO)
"Ring finger after tackle, can't flex DIP — Dx"	→ **FDP avulsion**
"Rugby, 1 wk, distal palm tender, can't flex DIP"	→ **acute tendon repair**
"Urgency of fixation depends on"	→ **degree of tendon retraction**
"Max advancement to avoid quadriga"	→ **1 cm** (TO)
"Correct statement about repair"	→ **2.5 mm advancement acceptable**
"Suture anchor position"	→ **45° retrograde in proximal metaphysis**
"5 mo, palmar mass, full PROM"	→ **excise mass + 2-stage grafting**
"Guitarist, 6 mo, supple DIP"	→ **two-stage flexor reconstruction** ✗ Wrong options: arthrodesis
"DIP hyperextended can't flex, PIP flexed"	→ **central slip injury** (TO)

KIENBOCK'S DISEASE

– Hand · 47 / 103 –

1 Basics

Definition – lunate osteonecrosis

Definition — **avascular necrosis of the lunate** → progressive wrist pain, abnormal carpal motion

Osteochondrosis, crushing type (TO) — Kienböck = **lunate**
✗ Wrong options: scaphoid, radial head, capitulum

Osteochondrosis (TO)	Site
Köhler	Navicular
Kienböck	**Lunate**
Panner	Capitulum
Scheuermann	Vertebral ring epiphysis
Calvé	Vertebral central nucleus
Thiemann	Phalangeal epiphyses
Freiberg	Metatarsal heads
Iselin *(traction)*	5th MT base
Osgood-Schlatter *(traction)*	Tibial tuberosity
Sever *(traction)*	Calcaneal apophysis
Sinding-Larsen-Johansson *(traction)*	Inferior pole patella
OCD *(splitting)*	–

Typical patient — ♂ **20–40 y**, often **manual labourer** (TO); history of trauma; **dominant** hand

Lunate ossification — **2–4 y**

Aetiology & risk factors (ulnar negative variance, vascular pattern)

Multifactorial — repetitive microtrauma + altered biomechanics + ↑ intraosseous pressure + ulnar negative variance → **"all of the above"**

Ulnar negative variance — ↑ **radiolunate contact stress**; **most predominant risk factor** (TO); up to **78%** of cases (TO)

↓ Radial inclination — ↑ radiolunate pressure; risk ↑ as inclination ↓ → **not increased radial inclination** (TO)

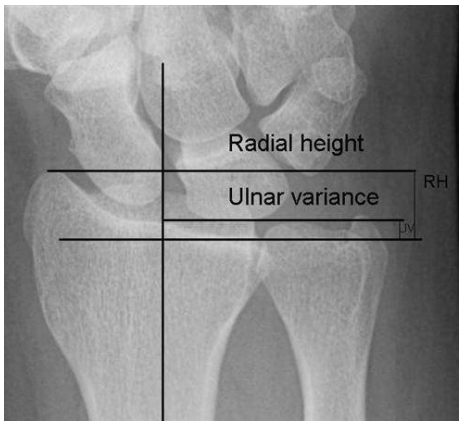
Lunate morphology (TO) — small lunate · **type I (triangular, no medial facet)** = weaker trabeculae → risk; type II/III square

Vascular (TO) — **fewer penetrating arteries** (esp. volar from palmar radiocarpal arch) → ↑ risk; sparse intraosseous collaterals

Venous congestion — ↑ **intraosseous pressure (>40 mmHg vs capitate)** · **wrist extension raises intraosseous pressure (~40 mmHg) → avoid extension** (TO)

– – Active space – –

Acute lunate fracture (TO)	**no correlation** with later Kienböck · MRI: transient **focal** oedema vs Kienböck diffuse, progressive
Medical conditions	scleroderma, sickle cell, SLE, corticosteroids
NOT a cause	**increased carpal height** (carpal height ↓ in Kienböck)
Ulnar variance associations	**negative** → Kienböck + ulnar impingement · **positive** → TFCC tear, ulnar abutment, SL dissociation, triquetral arthrosis
Ulnar variance axis (TO)	measure using the **radial shaft**
Measuring ulnar variance	PA, **shoulder 90° abduction, elbow 90°**, neutral rotation; **more positive in pronation**, negative in supination

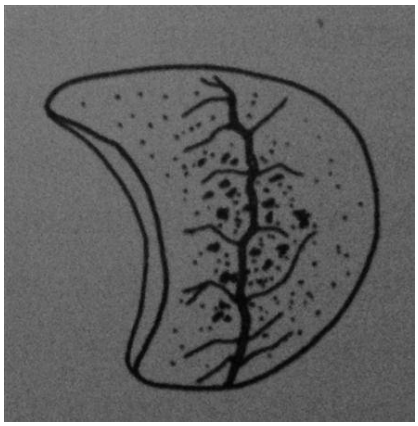


Ulnar variance

Anatomy – lunate blood supply

Pattern	Fact
Y	–
X	–
I	**31%** of Kienböck patients (Orthobullets/OB); **highest AVN risk** (also TO)

Single palmar supply	**25%** of lunates (TO) → ↑ osteonecrosis risk
Capitate	supply also poor → AVN possible



I-pattern lunate supply

3 Presentation

Symptoms — **dorsal wrist pain**, **activity-related** / worse with grip loading; insidious, usually no trauma event (TO)

Exam — ± swelling, **radiocarpal tenderness**, ↓ flexion–extension arc, **↓ grip strength**

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray (AP, lat, oblique)	Normal (stage I) → **sclerosis** (II) → **collapse** (III) → carpal OA (IV)
MRI	**Best for early disease / IOC** (TO); **↓ T1 signal**; rules out ulnar impaction
CT	**Most sensitive for staging / early collapse**; extent of necrosis, trabecular destruction, geometry
✗ Wrong option: bone scan	Non-specific
✗ Wrong option: US, angiography	Not used
✗ Wrong option: clenched-fist AP	SL interval

Stage IIIA → IIIB — **radioscaphoid angle >60°** (fixed scaphoid flexion) · **↓ carpal height ratio (CH / MC3 height)** = capitate migration → IIIB (TO)

Earliest X-ray sign (TO) — **↑ lunate density (sclerosis)**

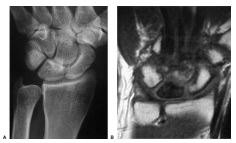
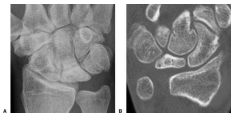
MRI T1 pattern (TO) — **uniform ↓ = Kienböck · focal ↓ → think ulnar impaction, intraosseous ganglion, enchondroma** · MRI = most sensitive (TO)



Kienböck MRI

– – Active space – –

Staging (Lichtman) & imaging

Stage	Findings	Treatment	Image
I	X-ray normal, **MRI changes**	**Immobilisation + NSAIDs** (cast (TO))	
II	**Lunate sclerosis**, normal shape	**Ulnar −ve → joint levelling** · **ulnar neutral/+ → radial wedge osteotomy or STT fusion** · core decompression · revascularisation	
IIIA	**Collapse**, no scaphoid rotation, carpal height preserved	**Same as II**	
IIIB	Collapse + **fixed scaphoid flexion**, ↓ carpal height	**PRC, STT fusion or SC fusion**	
IIIC (TO)	Collapse + **coronal fracture**	–	–
IV	**Adjacent intercarpal OA**	**Wrist fusion, PRC, limited intercarpal fusion** (± TWA)	

History (TO) first classification by **Stahl**, modified by Lichtman

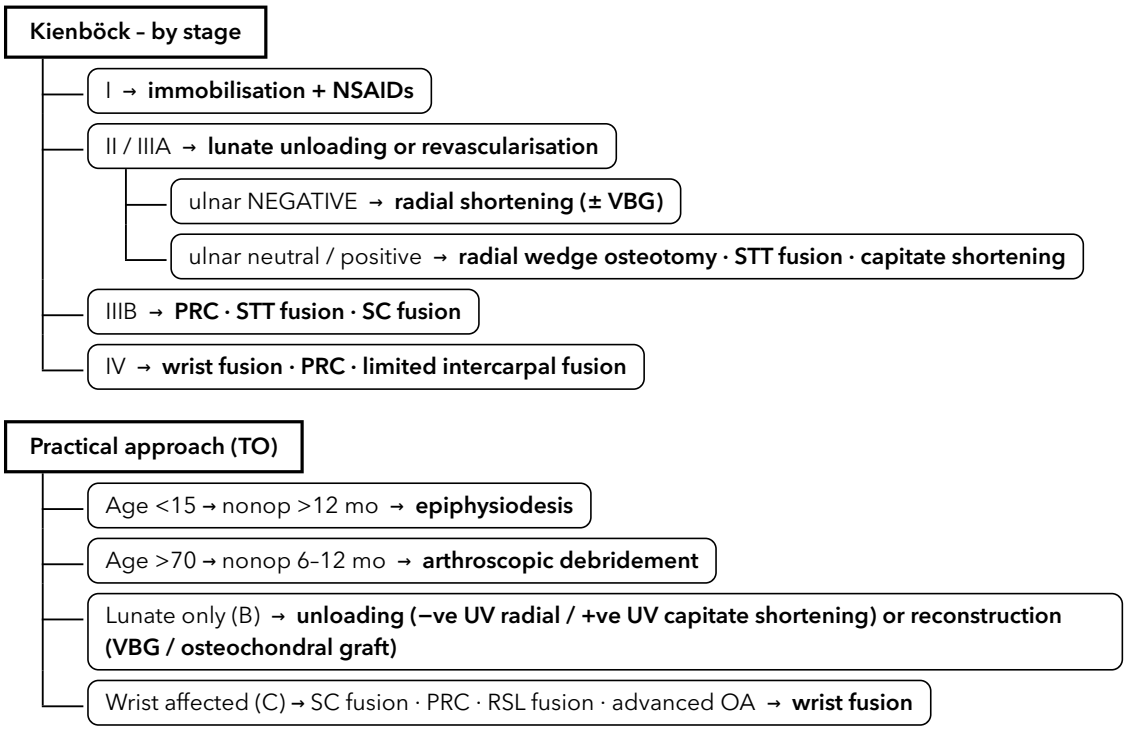
Lichtman X-ray + MRI based; low interobserver variability (TO)

Articular grading for advanced disease (TO, Bain–Begg)

Grade	Nonfunctional surfaces	Recommendation
0	None	Joint levelling · forage ± graft · VBG
1	Proximal lunate	RSL fusion · PRC · osteochondral graft
2a	Proximal lunate + lunate facet radius	**RSL fusion**
2b	Proximal + distal lunate	PRC · lunate replacement · capitate lengthening
3	3 surfaces (capitate preserved)	Hemiarthroplasty (SC fusion if radial column intact)
4	All 4	Total wrist fusion / TWA (SC fusion if radial column intact)

Synovectomy all patients (TO)

5 Management



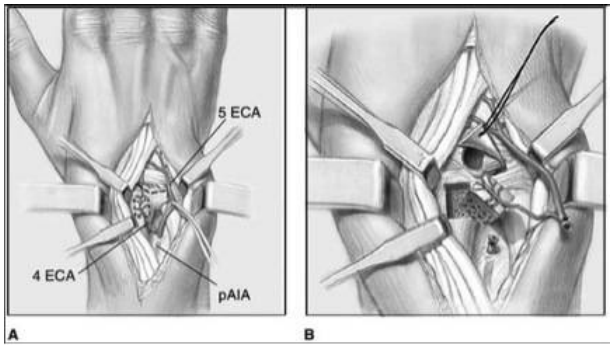
Conservative care (TO)	**no role after diagnosis** (TO) · stage I immobilisation + NSAIDs (Orthobullets) – read the options
Treatment categories (TO)	core decompression · joint levelling / unloading · revascularisation (distal radius / pronator quadratus VBG) · salvage (carpectomy, partial fusion)

Early stages – radial shortening, core decompression & vascularised bone graft

Stage II, ulnar −ve (TO)	**VBG + radial shortening** ✗ Wrong options: STT fusion · PRC · VBG + radial wedge
Radial shortening osteotomy	**treatment of choice** stage II ulnar minus; **extra-articular**; more evidence than ulnar lengthening
Radial shortening amount (TO)	**2–3 mm only** · **first 2 mm → 90% load reduction**; 3 mm no extra benefit · oblique cut → 2 mm wafer, transverse → 3 mm · **diaphyseal** osteotomy compresses better than metaphyseal
Ulnar lengthening (TO)	distal ⅓ ulna · complications **ulnocarpal impingement, nonunion**
Ulnar-positive variance, stage I–IIIA (TO)	radial **and** ulnar shortening **contraindicated → capitate shortening + capitohamate fusion** · do not release 4th extensor compartment · excise hamate tip
Radial closing wedge osteotomy (TO)	for abnormal radial inclination · shifts load **radiolunate → radioscaphoid** · osteotomy **proximal to DRUJ**
Failed immobilisation stage II	**→ radial shortening** ✗ Wrong options: continue cast · PRC · STT · fusion
DRUJ (Tolat type II)	remodelling after shortening **without arthritis** → not a contraindication
Ulnar shortening	✗ wrong option: **worsens** (↑ radiolunate force)
Distal radius core decompression	stage I–IIIA; **local vascular healing response**

– – Active space – –

Not Kienböck options	TFCC repair, SL reconstruction, lunate chondroplasty; lunate excision / implant **rarely used**
Limited intercarpal fusion	before total wrist arthrodesis
Partial fusions	STT, **capitate shortening ± capitohamate**, SC → stage II (neutral/+), IIIA/IIIB; must address **DISI**
VBG	stage I–IIIB; best in stage III; long-term data lacking
VBG options	pisiform · distal radius on **pronator quadratus** pedicle · dorsal metacarpal artery branches · **4+5 ECA**
VBG pedicles (TO)	**1,2-ICSRA short pedicle** · 2,3-ICSRA good arc · **4,5 = best** (large calibre, arc) · sources: 2nd / 3rd MC base, dorsal distal radius
4+5 ECA advantage	**largest arc of rotation** · large diameter · **ulnar location** (✗ Wrong options: radial) · **dorsal** approach (✗ Wrong options: volar) · antegrade flow not unique
After VBG	**2–3 mo immobilisation / unloading** (STT/SC pinning or ex-fix) required



4+5 ECA graft

Radiolunate contact stress

Procedure	↓ Radiolunate stress
STT fusion	3%
SC fusion	12%
Capitohamate fusion	0%
Ulnar lengthening 4 mm	**45%**
Radial shortening 4 mm	**45%**
Capitate shortening + CH fusion	**66%** (but ↑ radioscaphoid 26%) · **60%** (TO) – read the options

Advanced disease – STT fusion/pinning

Temporary STT pinning	**adolescent** with Kienböck + progressive pain → ↓ radiolunate, ↑ radioscaphoid load ✗ Wrong options: radial club hand, scaphoid fracture, hypoplastic thumb, gymnast's wrist
IIIB	PRC (also lunate excision + STT/SC fusion); STT superior to PRC in some studies
Salvage IIIB (TO)	**SC fusion better than STT fusion** (preferred) · PRC · lunate arthroplasty: **pyrocarbon preferred over silastic** · wrist fusion for advanced collapse
Stage IV (TO)	interposition arthroplasty or wrist fusion

IV wrist fusion (remove arthritic joint) ; TWA long-term data lacking



STT pinning

Paediatric Kienböck – conservative care

Skeletally immature, stage I **short arm cast + NSAIDs (4 wk)**
✗ Wrong options: core decompression + ulnar shortening · PRC · STT pinning/fusion

Nonop outcome (adult) majority progress → surgery

Other carpal osteonecrosis (capitate) – salvage

Carpal AVN frequency scaphoid (trauma), lunate (idiopathic) > **capitate** > hamate > trapezium > trapezoid > triquetrum > pisiform

Eponym	Bone
Kienböck	Lunate
Preiser	**Scaphoid**
Köhler	Navicular

Capitate AVN causes trauma, **retrograde palmar-dominant supply** (like scaphoid), hypermobility / **ligamentous laxity**, steroids, vibration, repetitive extension

Capitate AVN treatment cast first

Group	Treatment
Pre-collapse, no OA (young)	Decompression / **VBG** (4+5 ECA local; **free MFC flap**) → best preserves vascularity & anatomy
Collapse, no OA	Proximal pole excision + interposition · **pyrocarbon** resurfacing · graft · fusion
Collapse + OA	**Midcarpal fusion** (salvage), SCL / 4-corner fusion

Not for capitate AVN **PRC** (radius articulates with dysvascular capitate) · TWA · radial wedge osteotomy · curettage + non-vascularised graft · implant arthroplasty in teen

Carpal coalition **LT** commonest (1%, 60% bilateral; West African descent)

– – Active space – –

X vs Y

	Kienböck	Capitate AVN	Ulnar impaction
Bone	Lunate	Capitate proximal pole	Ulnar lunate / triquetrum
Variance	**Negative**	–	**Positive**
MRI	↓ T1 lunate	Proximal pole capitate cystic change	MRI rules it out (Orthobullets)
Key Rx	Radial shortening (ulnar −ve)	VBG / midcarpal fusion	Ulnar shortening

How it's asked

"NOT a risk factor"	→ **↑ radial inclination** (TO)
"All are etiologies EXCEPT"	→ **↑ carpal height**
"Cause of Kienböck"	→ **all of the above** (multifactorial)
"Negative ulnar variance on X-ray – at risk of"	→ **lunate collapse** ✗ Wrong options: TFCC tear, ulnar abutment, SL dissociation
"Kienböck involves"	→ **lunate** (TO)
"Most sensitive study to stage, sclerosis no collapse"	→ **CT** ✗ Wrong options: bone scan, US, angiography
"Collapsed lunate + flexed scaphoid, PRC advised"	→ **IIIB** (radioscaphoid >60°)
"MRI sclerosis + negative ulnar variance – ideal Rx"	→ **VBG + radial shortening** (TO)
"Procedure inciting local vascular healing"	→ **distal radius core decompression**
"Stage II failed immobilisation"	→ **radial shortening osteotomy**
"Accepted Rx after failed immobilisation"	→ **radius core decompression** ✗ Wrong options: ulnar shortening, TFCC repair
"T1 low lunate, ulnar minus, no OA"	→ **radial shortening** ✗ Wrong options: ulnar shortening
"4+5 ECA main advantage"	→ **large arc of rotation**
"Radial shortening for Kienböck – how much?"	→ **2–3 mm (first 2 mm gives 90% unloading)** (TO)
"Ulnar-positive Kienböck stage II – procedure?"	→ **capitate shortening + capitohamate fusion** (TO)
"Focal T1 low signal in lunate – think?"	→ **ulnar impaction / intraosseous ganglion / enchondroma** (TO)
"Temporary STT pinning indicated for"	→ **adolescent Kienböck**
"7-y-old stage I"	→ **short arm cast + NSAIDs 4 wk**

"Jackhammer, MRI proximal capitate change — diagnosis"	→ **capitate AVN** ✗ Wrong options: Preiser, Köhler
"Capitate AVN, best restores vascularity / native anatomy"	→ **vascularized bone graft**
"Capitate AVN pre-collapse, failed cast"	→ **4+5 ECA VBG** ✗ Wrong options: PRC · 4CF
"Capitate AVN salvage"	→ **midcarpal fusion** ✗ Wrong options: PRC, TWA
"Carpal bone ossifying at 2–4 y removed in PRC for osteonecrosis"	→ **lunate**

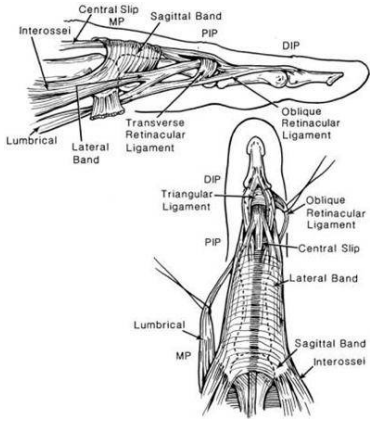
– – Active space – –

LIGAMENTS OF THE FINGERS

– Hand · 48 / 103 –

1 Basics

Retinacular ligaments



Oblique retinacular ligament

Oblique retinacular ligament (Landsmeer)	links PIP & DIP; **volar to PIP axis, dorsal to DIP axis**; prox phalanx (lateral volar) → terminal extensor
ORL biomechanics	PIP flexion → relaxes → DIP flexes; PIP extension → tightens → DIP extends
ORL tight	rest DIP extended + PIP flexed; **can't flex DIP with PIP extended**, can after PIP flexed
ORL pathology	contracture → **boutonnière**; **ORL reconstruction → swan neck** treatment
Transverse retinacular ligament	flexor sheath at PIP → lateral bands; **attenuation → swan neck** (dorsal bands); contracture + triangular lig. attenuation → **boutonnière**
Triangular ligament	dorsal, distal to PIP; prevents lateral band subluxation; contracture → swan neck

Tightness test	MCP extended	MCP flexed
Intrinsic tightness	↓ PIP flexion	↑ PIP flexion
Extrinsic (extensor) tightness	↑ PIP flexion	↓ PIP flexion
ORL tightness	DIP flexion only after PIP flexed	–

Digital cutaneous ligaments

	Cleland's ("C" = ceiling)	Grayson's ("G" = ground)
Relation to digital nerve	**Dorsal**	**Volar**
Dupuytren's	**Not involved**	Involved

– – Active space – –



Cleland's and Grayson's ligaments

Expansion hood & MCP ligaments

Central slip base of middle phalanx → **PIP extension**

Lateral band distal phalanx → **DIP extension**; lumbricals, EI, interossei insert

Lumbrical **volar** to deep transverse MC ligament · **interossei dorsal** to it

MCP cam "snoopy head" MC head → collaterals **loose in extension, tight in flexion**

MCP collateral part	Shape/position	Tight in	Test
Proper	Cord, dorsal; MC posterior tubercle → P1 base	**30° flexion**	Stress in 30° flexion
Accessory	Fan, volar; MC head → volar plate & DTML	**Extension**	Stress in extension

RCL vs UCL RCL more horizontal

Deep transverse MC ligament joins 2nd–5th MC heads at volar plates; prevents splaying

Natatory ligament most superficial; resists abduction; web space

Sagittal bands volar plate → extensor; centralise extensor over MCP

Volar plate prevents hyperextension; **checkrein ligaments** attach to P1; loose in flexion, tight in extension; thumb sesamoids

5 Management

PIP contracture release sequence

Volar PIP flexion contracture release (test extension after each step)

- 1 → **retract flexors, release CHECKREIN ligaments**
- 2 → **ACCESSORY COLLATERAL + VOLAR PLATE**
- 3 → **PROPER COLLATERAL off proximal phalanx**
- no active extension → **extensor tenolysis**

After checkreins **accessory collateral ligament + volar plate**
✗ Wrong options: proper collateral · dorsal capsule · extensor

– – Active space – –

How it's asked

"Released after checkrein ligaments in PIP contracture?"	**→ accessory collateral ligament + volar plate**
"Ligament dorsal to digital nerve, spared in Dupuytren's?"	**→ Cleland's**
"MCP proper collateral tight in?"	**→ 30° flexion**
"Transverse retinacular ligament attenuation?"	**→ swan neck**
"Lumbrical relation to DTML?"	**→ volar**

LUMBRICAL PLUS FINGER

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1 Basics

Anatomy – lumbricals

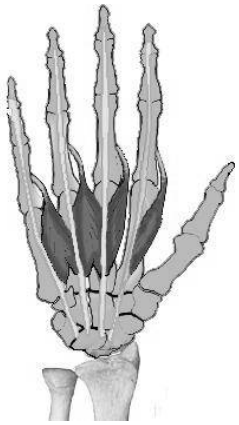
Lumbrical	Pennation	Nerve	Origin
1st	**Unipennate**	**Median**	Radial side FDP2
2nd	**Unipennate**	**Median**	Radial side FDP3
3rd	**Bipennate**	**Ulnar**	FDP3 & FDP4
4th	**Bipennate**	**Ulnar**	FDP4 & FDP5

Insertion all on **radial** side of extensor expansion (lateral bands)

Commonest digit **middle finger** (2nd lumbrical)

Why middle FDP 3/4/5 share a **common muscle belly**; distal FDP3 cut → FDP shifts **ulnarly** (pull of bipennate 3rd lumbrical) → tightens 2nd lumbrical → amplifies effect

Why not index **independent FDP belly**; can fist with FDS2 alone → avoids paradoxical extension



Lumbrical origins and innervation

2 Mechanism

Cause – amputation at DIP level

Definition **paradoxical IP (PIP + DIP) extension** on attempted finger flexion

– – Active space – –

Pathoanatomy lumbrical arises from FDP → FDP divided distal to lumbrical origin → FDP contraction pulls lumbrical → pulls **lateral bands** → IP extension

Cause	Note
FDP disruption distal to lumbrical origin	**Most common**: transection or **avulsion** (jersey finger)
DIP amputation	Distal to central slip insertion → PIP extends on flexion
Amputation through **middle phalanx shaft**	
"Too long" tendon graft	Excess flexor graft length

✗ DIP amputation distractors central slip rupture, swan neck, boutonniere, **quadrigia** (FDP advanced >1 cm → adjacent digit flexion weakness, common muscle belly)



Pathoanatomy of lumbrical plus

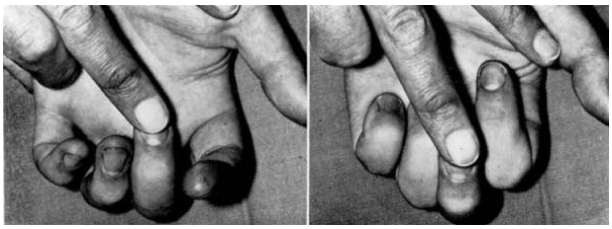
3 Presentation

History volar digital laceration (FDP transection) or axial traction on flexed digit (FDP avulsion)

Symptom one digit **sticks out / catches on clothes** when gripping or making a fist

Exam paradoxical IP extension with grip ("**fingers extend while holding a beer can**")

Diagnosis clinical: one digit's IPs extend when fisting all fingers



Lumbrical plus finger on grip

5 Management

Lumbrical plus finger

- Minimal symptoms → **observe**
- FDP laceration, repairable → **tenodesis FDP to terminal tendon / reinsert to distal phalanx**
- FDP retracted / segmental loss → **lumbrical release at base of flexor sheath (palm)**

Tenodesis / reinsertion for FDP lacerations; **do not suture flexor–extensor mechanisms over bone**

Lumbrical release **transect in palm at base of flexor sheath** → abolishes paradoxical PIP extension

No acute routine release	phenomenon not consistent enough
Contraindication	**do not divide lumbricals 1 & 2 with ulnar nerve palsy** (interossei also lost → no IP extensors left)

X vs Y

	Lumbrical plus	Quadrigia
Cause	FDP too loose / cut distal to lumbrical, long graft, DIP amputation	FDP **too tight** (advanced >1 cm)
Sign	Paradoxical IP extension on fisting	Adjacent fingers can't fully flex, weak grip
Fix	Lumbrical release in palm	Release tethered FDP

How it's asked

"Complication of DIP-level amputation?"	→ **lumbrical plus finger** ✗ Wrong options: quadrigia, swan neck, boutonniere, central slip rupture
"Paradoxical PIP extension on making a fist?"	→ **lumbrical plus**
"Where to release the lumbrical?"	→ **palm, base of flexor sheath**
"Most common finger?"	→ **middle (2nd lumbrical)**
"Don't release lumbricals 1 & 2 when?"	→ **ulnar nerve palsy**

– – Active space – –

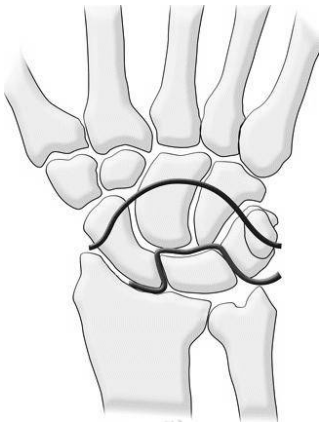
LUNATE DISLOCATION (PERILUNATE DISSOCIATION)

1 Basics

Epidemiology & categories

Incidence	rare, high energy; **missed in ~25%** at first presentation (Herzberg 25%, Dunn 27.5%)
Carpal instability class (TO)	**CIC** (complex) = **perilunate dislocations & fracture-dislocations** → intrinsic (SL, LT = dissociative) + extrinsic (midcarpal capsule = non-dissociative) ± fractures ✗ Wrong options: pure LT dissociation · Colles dorsal tilt · RA ulnar translocation
Perilunate dislocation	**lunate stays in fossa**, carpus dislocates (usually dorsal)
Lunate dislocation	lunate forced volar (or dorsal), carpus aligned → lunate usually **into carpal tunnel**
Perilunate types	transscaphoid-perilunate · perilunate · transradial-styloid · transscaphoid-transcapitate-perilunate
Most common variant	**dorsal trans-scaphoid perilunate fracture-dislocation** (OB, TO)

Arc	Injury
Lesser arc	**Purely ligamentous** (rapid force)
Greater arc	Ligament + fractures of radius, ulna or carpus (slower force: transradial styloid, transscaphoid, transcapitate)



Greater vs lesser arc

Ligament anatomy

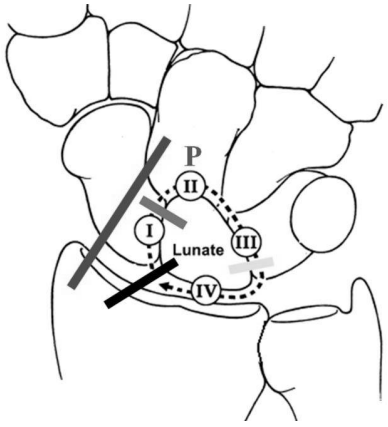
SLIL / LTIL	interosseous; major proximal-row stabilisers; SL prevents **DISI**, LT tear → **VISI**
Space of Poirier	weak volar area **between RSC and long radiolunate** (OB) · between radiocapitate and LRL (TO); over capitolunate joint ✗ Wrong options: between dorsal intercarpal and radiotriquetral

Short radiolunate (SRL)	**remains intact**, tethers lunate to radius ✗ Wrong options: first ligament injured
Dorsal intercarpal	typically remains intact
Ligament of Testut (RSL)	neurovascular conduit, little structural role

2 Mechanism

Mechanism – Mayfield progression

Position	**wrist extension + ulnar deviation + intercarpal supination**; axial load, FOOSH, fall from height
Ligament order	**SL → capitolunate → LT → dorsal radiocarpal** → lunate rotates + dislocates ✗ Wrong options: SL → LT → CL · DRC first
First ligament injured	**scapholunate** ✗ Wrong options: SRL
Bony sequence (Mayfield)	**scaphoid extension → scaphoid failure (space of Poirier opens) → distal row dissociation → triquetral hyperextension → LT failure → dorsal dislocation of carpus**
Reduction (Mayfield)	intercarpal **pronation, radial deviation, palmar flexion**



Progressive perilunar instability: clockwise ligament disruption

3 Presentation

Symptoms	acute swelling, pain, deformity
Acute CTS	**~25%** (Orthobullets) · **25–50%** (OB) · **25–45%** (TO) · 24–45% (Stanbury) · 50% (Dunn) – read the options; highest in **Mayfield IV**
Vascular exam	rarely abnormal; DRUJ injury uncommon

4 Diagnosis & Classification

Key radiology findings

Views **PA + lateral**; lateral = **most diagnostic** (TO)

View	Sign
PA	**Break in Gilula's arcs** (3 arcs), lunate–capitate **overlap**, **↓ carpal height**, crowded carpus, SL diastasis
PA	**"Piece-of-pie" sign** = triangular lunate (palmar rotation)
Lateral	**Loss of collinearity radius–lunate–capitate**; SL angle >70°
Lateral	Perilunate: lunate in fossa, capitate dorsal · Lunate: **"spilled teacup"** lunate volar, capitate drops into fossa
CT	Fracture lines (trans-scaphoid/capitate), planning (TO)
MRI	**Not required** / doesn't aid diagnosis (OB); ligament extent (TO)

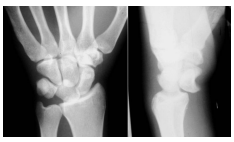
Radial styloid fracture avulsion of radiocarpal ligaments

Subacute lesser arc case (TO) **overlapping carpal bones** · **scaphoid ring sign + SL gap** · **lunate rotated 90°** · capitate dorsal to lunate → ORIF + ligament reconstruction (dorsal approach)



Lateral: spilled teacup

Mayfield classification

Stage	Orthobullets / OB	TO wording	Image
I	**SL dissociation**	SL dissociation / **scaphoid fracture** / rotation	
II	+ **lunocapitate** disruption	Lunocapitate dislocation through **space of Poirier** / perilunate dislocation	
III	+ **LT** disruption = "perilunate"	LT disruption / triquetrum dislocation	
IV	**Lunate dislocated** from fossa (usually volar) → **median n. compression**; DRC fails	Lunate extruded volarly into carpal tunnel	

Stage I (TO) **"not always a scaphoid fracture"** ✓
✗ Wrong options: LT injury · lunate extruded · RSC detachment

Stage II (TO) capitate–lunate joint through **space of Poirier**

Stage IV ligament **dorsal radiocarpal** fails (Orthobullets/OB) · long radiolunate shown disrupted (OB219974 illustration) – read the options

5 Management

Perilunate / lunate dislocation

- Acute → **emergent closed reduction + splint (neuro exam before & after)**
  - then ORIF + ligament repair ± CTR (all <8 wk)
    (worsening / persistent ACTS → emergent CTR + ORIF)
- Chronic >8 wk → **proximal row carpectomy**
- Chronic + degenerative change / failed PRC → **total wrist arthrodesis**

Acute management – urgent reduction, neurological exam & missed diagnosis

Next step **urgent closed reduction + splint**
✗ Wrong options: cast · MR arthrogram · bone scan

Before reduction **neurological exam** (repeat after)
✗ Wrong options: vascular · Allen · MRI · DRUJ

Technique finger traps, elbow 90°, **5–10 lb for 15 min → wrist extension + traction → flexion with dorsal pressure on carpus** (volar pressure on lunate for lunate dislocation); sugar-tong splint

– – Active space – –

Benefit of emergent reduction	↓ **median nerve** damage, ↓ **cartilage** damage, ↓ AVN (TO)
Median nerve (Budoff)	usually static, non-progressive → no emergent release; **delayed onset / progressive CTS → emergent CTR + ORIF**
Closed treatment definitive	**never**; universally poor, redislocation

Definitive ORIF ± carpal tunnel release

Timing	all acute <8 wk; surgery can wait **~4 d** for swelling (Budoff) · standard **within 7–10 d** (TO); **delay >3 wk** worsens outcome (TO)
Lunate volar + median symptoms	**ORIF via dorsal + volar approaches + extended CTR** ✗ Wrong options: CR + cast · CRPP · ORIF without CTR · PRC acute

Approach	Gives	Limits
Dorsal (Lister's)	Proximal row + midcarpal exposure, SL/LT repair	not for CTR
Volar (extended CT)	**CTR**, close space of Poirier rent, volar ligaments	–
Combined	Easier reduction, distal scaphoid, CTR	Swelling, devascularisation, ↓ flexion/grip

Case-based approach choice (TO)	trans-scaphoid trans-styloid dorsal perilunate → ORIF scaphoid + perilunate fixation · **volar approach if palmar comminution** or **volar perilunate dislocation** · dorsal perilunate without fracture **not reducible closed → open reduction + SL anchor repair** · combined trans-styloid + SL gap + DISI → ORIF + K-wires across SL, LT, scaphocapitate
Steps	fix fractures (scaphoid headless screw) · **SL repair with suture anchors** · protect with K-wires (SL, LT, scaphocapitate) or cerclage · LT repair optional
Post-op	thumb spica → cast **≥6 wk** (Orthobullets); **K-wires out 8–12 wk** (TO)

Chronic/late injuries – proximal row carpectomy & RSC ligament

Chronic (>8 wk)	**PRC**; arthrodesis only if PRC fails / degenerative ✗ Wrong options: ORIF: cartilage loss + contracture · more splint/cast
PRC stability	capitate head in lunate fossa + **preserve RSC** → prevents **ulnar translocation** ✗ Wrong options: SRL · SL · RSL
Radial styloidectomy	**≤4 mm** to preserve RSC; **6 mm destabilises** (Nakamura)
PRC approach	dorsal (scaphoid, triquetrum) + volar (lunate + median decompression)

6 Complications

Post-operative lunate ischaemia – observation

Transient lunate ischaemia	radiodense lunate in **12.5%**, **1–4 mo** post-injury → **observation** (benign, self-limiting; no Kienböck progression)

Not Kienböck treatments STT pinning (adolescent), capitate shortening (Lichtman II–III), PRC (advanced), cast (stage I)

Outcome	Figure
Post-traumatic arthritis	**40–60%** (TO); **56% at 6 y** (Herzberg); well tolerated, does not correlate with symptoms
Wrist motion loss	**30–50%** (TO)
Grip	**70–80%** of contralateral (TO)
Persistent paraesthesia	17.5% (Dunn)
Other	Scaphoid nonunion/AVN (trans-scaphoid), lunate AVN, full function unlikely

X vs Y

	Perilunate dislocation	Lunate dislocation
Mayfield	III	**IV**
Lunate	**In fossa**	**Volar, in carpal tunnel**
Capitate	Dorsal to lunate	Back in line with radius
Lateral	Capitate off lunate	**Spilled teacup**
ACTS	Less	**Highest**

How it's asked

"True of Mayfield stage I?"	→ **not always a scaphoid fracture** (TO)
"CIC is best represented by?"	→ **perilunate dislocations and fracture-dislocations** (TO)
"Mayfield II disrupts?"	→ **capitolunate joint through space of Poirier** (TO)
"Sequence of perilunate injury (bony)?"	→ **scaphoid extension → scaphoid failure → distal row dissociation → LT failure → dorsal carpal dislocation**
"First ligament injured?"	→ **scapholunate** ✗ Wrong options: SRL · DRC
"Order of ligament failure?"	→ **SL → CL → LT → DRC**
"Perilunate + thumb/index paraesthesia – next step?"	→ **urgent closed reduction + splint**
"Before reduction obtain?"	→ **neurological exam**
"True statement?"	→ **missed in ~25%** ✗ Wrong options: ACTS rare · SRL first · MRI aids · Poirier dorsal
"Volar lunate + hand numbness – definitive?"	→ **ORIF dorsal + volar + extended CTR**

– – Active space – –

"Stage IV + median numbness — definitive?"	**→ ORIF + carpal tunnel decompression**
"PRC — preserve which ligament?"	**→ radioscaphocapitate**
"Perilunate missed 8 wk ('sprain'), limited ROM?"	**→ PRC** ✗ Wrong options: ORIF · arthrodesis
"Radiodense lunate 2 mo after ORIF?"	**→ observation**

LUNOTRIQUETRAL LIGAMENT INJURY & VISI

– Hand · 51 / 103 –

1 Basics

Epidemiology

Incidence LT injury **less common than SL** injury

LT ligament – volar fibres strongest

Shape **C-shaped intrinsic** ligament: dorsal + palmar transverse parts + weak proximal membranous (fibrocartilaginous) part

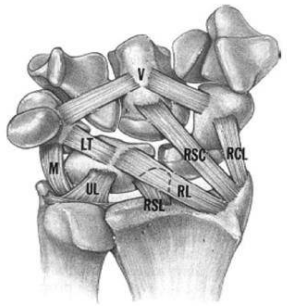
Ligament	Strongest part	Yield strength (Berger)
LT	**Volar** (thickest; transmits triquetral extension moment)	Volar **301 N** vs dorsal 121 N
SL	**Dorsal**	Dorsal **260 N** vs volar 118 N

Dorsal LT most important **rotational constraint** (but weaker)

Dorsal radiocarpal (radiotriquetral) ligament extrinsic; **secondary restraint to VISI**; loss → lunate flexes easily

Volar long & short radiolunate extrinsic; torn in advanced injury

Short radiolunate stays attached to lunate in perilunate dislocation; not involved in LT tear



Volar ligaments of the carpus

2 Mechanism

Mechanism wrist **hyperextension** or extension + **radial deviation**

Kinematics LT torn → scaphoid flexes lunate unopposed; triquetrum extends → **lunate flexes with scaphoid**

VISI **LT ligament + dorsal radiotriquetral + volar radiolunate** injury; a **CID** pattern

– – Active space – –

VISI in normal wrists may occur with **ligamentous laxity**
✗ Wrong options: DISI (always pathological)

Rocking lunate

Rocking lunate (TO) **tear of both SL + LT ligaments**
✗ Wrong options: SL alone · LT alone · + capitolunate

Lunate balance (TO) lunate held between scaphoid (flexion) and triquetrum (extension) via SL & LT; lose one → dominated by the other

(TO)	SL disruption	LT disruption
Scaphoid	Flexes	–
Lunate	Follows triquetrum → **extends**	Loses triquetral pull → **flexes**
SL angle (normal <60°)	↑	↓

Chronic ulnar impaction **ulnar positive variance** → long ulna impacts triquetrum → degenerative LT tear ± **TFCC** tear

3 Presentation

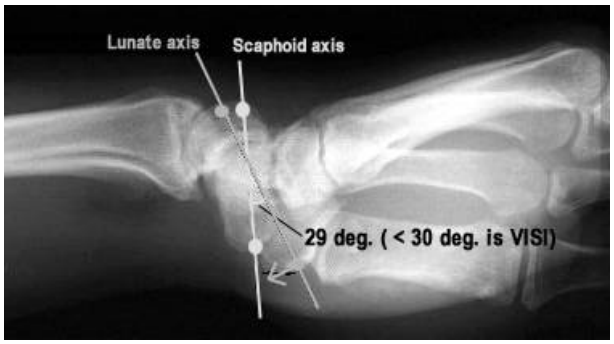
Symptoms **ulnar-sided wrist pain**, worse with **pronation + ulnar deviation** (power grip); painful snap with ulnar deviation, pronation, axial load

Test	Technique	Positive
LT shuck / ballottement	Hold lunate, shuck triquetrum dorsal–palmar	Pain, crepitus, laxity
Kleinman shear	Stabilise radiolunate, load triquetrum AP (forearm neutral)	Pain or **clunk**
LT compression	Triquetrum displaced ulnarly during radioulnar deviation	Pain

4 Diagnosis & Classification

Key radiology findings

View	Finding
Lateral	**Volar-flexed lunate → SL angle <30°** (normal **47°**) = **VISI**
Lateral	**Capitolunate angle >15°** (zigzag; normally co-linear)
AP	**Break in Gilula's arc**; proximal translation of triquetrum ± LT overlap; often **no LT widening** (unlike SL)
AP clenched fist	Incongruent LT articulation
Arthroscopy	Diagnostic when X-rays normal



VISI: SL angle <30° on lateral

5 Management

LT ligament injury

- Initial / acute stable → **observe / cast (better if acute)**
- Acute instability → **CRPP (multiple K-wires) + ligament repair ± dorsal capsulodesis**
- Chronic instability → **LT fusion**
  - due to ulnar positive variance → **arthroscopic LT debridement + ulnar shortening**

Cast	more successful when acute; only **37%** operated (Atkinson & Watson)
Acute	BLB autograft reconstruction & LT fusion **out of favour** acutely
LT fusion	chronic; complication **nonunion**

X vs Y

	SL injury	**LT injury**
Frequency	Commoner	Rarer
Pain	Dorsoradial	**Ulnar**
Instability	**DISI** (always pathological)	**VISI** (may be laxity)
SL angle	**>60°** (↑)	**<30°** (↓)
AP X-ray	SL widening	Gilula break, no widening
Stronger part	**Dorsal**	**Volar**
Test	Watson scaphoid shift	Shuck / Kleinman shear

How it's asked

"Strongest portion of the injured ligament (VISI, positive ballottement)?"	→ **volar fibres of LT** ✗ Wrong options: dorsal LT, dorsal SL
"Rocking lunate results from?"	→ **SL + LT tear** (TO)
"SL angle in VISI?"	→ **<30°** (normal 47°)
"LT disruption effect on SL angle?"	→ **decreases** (TO)

– – Active space – –

"Chronic LT tear with ulnar positive variance?"	→ **arthroscopic debridement + ulnar shortening**
"Chronic LT instability?"	→ **LT fusion** (risk nonunion)

MACRODACTYLY (LOCAL GIGANTISM)

– Hand · 52 / 103 –

1 Basics

Epidemiology

Definition	**nonhereditary congenital digit enlargement**; all tissues (skin, subcut fat, **nerve, joint, bone**) (TO)
Incidence	very rare; no known risk factors
Laterality	**90% unilateral** (Orthobullets + TO)
Multiple digits	**70%** involve >1 digit; majority >1 digit (TO)
Commonest digit	**index** > long > thumb > ring > small; hand or foot
Distribution	follows **nerve territory**; **median nerve** commonest

Genes & lab markers

Genetics	**somatic (post-zygotic) gain-of-function PIK3CA mutation**, mTOR pathway (TO) · "no genetic correlation known" (Orthobullets) – read the options

Associations & growth pattern (lipofibromatous hamartoma)

Lipofibromatous hamartoma of median nerve	seen in **macrodactyly** (TO); **adult homolog** ✗ Wrong options: syndactyly, polydactyly, idiopathic CTS
Median nerve in carpal tunnel	enlarged in majority → **significant CTS** (TO)
Most common associations (TO)	**neurofibromatosis**, hemangiomatosis, **congenital lipofibromatosis**
Other associations	**Proteus**, **Bannayan-Riley-Ruvalcaba**, **Maffucci**, **Ollier**, **Milroy** (Orthobullets + TO)
Foot pattern (TO)	overgrowth **greatest plantarly and distally** → convex sole, dorsiflexed toes ✗ Wrong options: dorsal
Bone involvement (TO)	usually limited to **phalanges**

3 Presentation

Symptoms	asymmetry at birth or over time, pain, ↓ use, cosmesis; foot: pain, **shoe fitting** (TO)
Exam	thick **fibrofatty** tissue; ↓ ROM from soft-tissue constraint

– – Active space – –



Macrodactyly of multiple digits

4 Diagnosis & Classification

Key radiology findings

Radiographs biplanar hand views → **enlarged phalanges** ± joint malalignment, angled phalanges

CT / MRI not typically needed; **angiography** only for surgical planning



Radiograph: enlarged phalanges

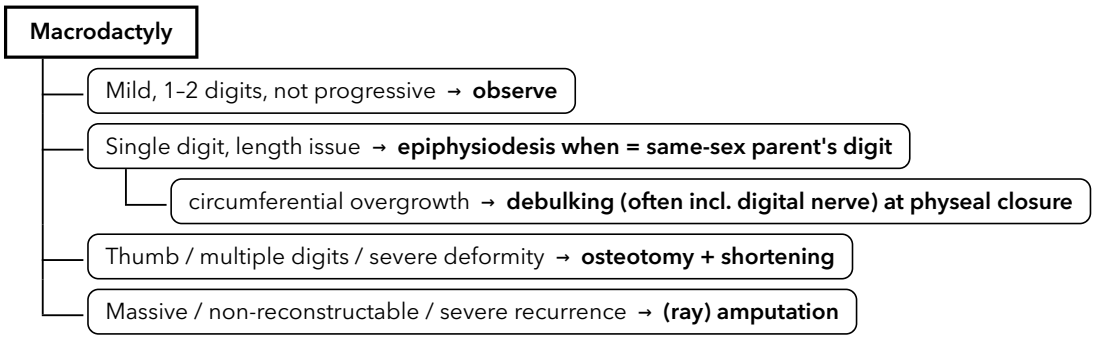
Functional classification

Type	Feature	Prognosis
Static	Present at birth; grows **linearly** with other digits	Asymmetry **does not worsen**
Progressive	Less obvious at birth; **disproportionate growth**	Asymmetry **worsens**

Flatt (TO)	Type
I	Gigantism + **lipofibromatosis**
II	Gigantism + **neurofibromatosis**
III	Gigantism + **digital hyperostosis**
IV	Gigantism + **hemihypertrophy**

Incidence ~0.2/1000 (TO) · thumb: **MP arthrodesis** if unstable · ray amputation last resort (TO)

5 Management



Epiphysiodesis

Timing	when digit reaches **adult length of same-sex parent's same digit** (father for a boy) (TO + OB) ✗ Wrong options: now · debulk now · ray amputation · osteotomy at maturity
Level (TO)	**proximal & middle phalanges** (± metacarpal) → stops **longitudinal** growth
Most common approach	epiphysiodesis for single digit
Debulking (TO)	for **circumferential** growth; removes enlarged digital nerve + skin/subcut; done **at time of physeal closure** unless massive
Angulation (TO)	osteotomy, can be at level of physeal closure → combining procedures ↓ number of operations
Avoid nerve resection	hypertrophic nerve resection fails to prevent overgrowth (Ishida)
Avoid compression wrapping	
Ray amputation (TO)	massive bone + soft-tissue enlargement; **procedure of choice for severe recurrence** after debulking/reduction syndactyly; "often required" = ✓ true (TO)
Foot options (TO)	amputation, ray amputation, epiphyseal ablation, osteotomies, nerve stripping, defatting
Post-op	soft-tissue care, early ROM

6 Complications

Complications	**digital stiffness**, chronic pain / oedema; recurrence

How it's asked

"10-y boy with macrodactyly – advice?"	→ **epiphysiodesis of PIP when finger = father's** (TO)
"Best intervention for proportional finger?"	→ **epiphysiodesis when finger reaches father's adult length** ✗ Wrong options: nerve resection
"All except – macrodactyly?"	→ **max overgrowth dorsal & distal** (false – it is plantar & distal) (TO)
"Lipofibromatous hamartoma of median nerve seen in?"	→ **macrodactyly** (TO)

– – Active space – –

"Commonest digit?"	→ **index**
"Severe recurrence after debulking?"	→ **ray amputation** (TO)

MADELUNG'S DEFORMITY

– Hand · 53 / 103 –

1 Basics

Epidemiology

Who	**adolescent females**; commonly **bilateral**; ♀ > ♂ (TO); common in **gymnasts**
History (TO)	described by Malgaigne, then **Otto Madelung 1878**

Aetiology – volar-ulnar physis & SHOX (Léri-Weill)

Cause	**impaired growth of the volar (palmar) & ulnar part of the distal radial physis** ✗ Wrong options: physeal mismatch radius/ulna · DRUJ fusion · nutritional · unrecognised trauma
Physis asked	**ulnar volar physis of distal radius** ✗ Wrong options: radial dorsal · ulnar dorsal · radial volar · distal ulna
Vickers ligament	thick **radiolunate** band (volar), tethers lunate to distal radius **proximal to the physis** → restricts ulno-palmar growth; 0.5–1.0 cm diameter; ± anomalous palmar radiotriquetral ligament
Result	**↑ radial inclination** (coronal) + **↑ volar tilt** (sagittal) + **ulnocarpal impaction**; **dorsal subluxation of distal ulna** (TO)

Genes & lab markers

Item	Fact
Inheritance	**Autosomal dominant** (Orthobullets, TO) · SHOX on X & Y → **sex-linked dominant** (OB) – read the options
Gene	**SHOX** (short stature homeobox), **pseudoautosomal region / tip of sex chromosomes** (TO)
Léri-Weill dyschondrosteosis	SHOX mutation → **mesomelic dwarfism + bilateral Madelung**
Other associations (TO)	**Turner syndrome**, achondroplasia
Madelung-like (TO)	**Multiple hereditary exostoses**, **Ollier disease**

SHOX gene distractors (TO)	radial club hand → **Sonic hedgehog** · CPT → **NF1** · Blount → no named gene
Other dysplasias (OB)	cleidocranial → **CBFA-1** (AD) · pseudoachondroplasia → **COMP** chr 19 (AD) · Ellis-van Creveld → **EVC** (AR) · achondroplasia → **FGFR3** (AD)

3 Presentation

Course	**asymptomatic until late childhood/adolescence**; deformity **painless**, worsens with growth; advanced → pain limits function (TO)

– – Active space – –

Symptoms	**ulnar impaction**, **median nerve irritation**
Exam	**ulnar + volar displacement of hand**, prominent distal ulna, **restricted forearm rotation**, family history +

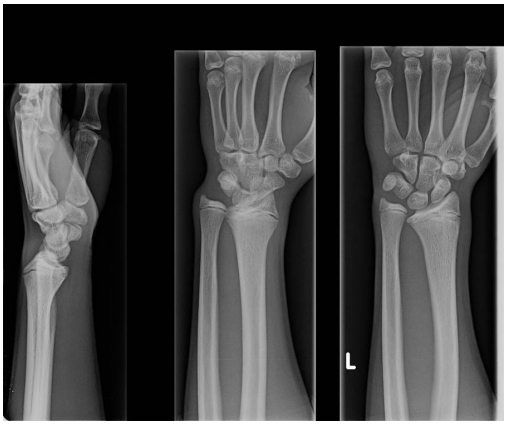


Clinical Madelung deformity

4 Diagnosis & Classification

Key radiology findings

Modality	Finding
X-ray	Undergrowth of **volar-ulnar corner of radius**; **↑ radial inclination**, **↑ volar tilt**; **lunate uncovered**; carpus **wedge-shaped with lunate at apex**, subluxed ulnar + palmar (TO)
X-ray (TO)	↑ radius–ulna width; **positive ulnar variance**; **↓ carpal angle**; triangular distal radial epiphysis; carpus migrates proximally
X-ray	± proximal synostosis
MRI	**Thick Vickers ligament** distal radius → lunate



Madelung radiograph: increased radial inclination and volar tilt



T1 MRI: thick Vickers ligament

Vender & Watson classification (TO)

Type	Example
Post-traumatic	Physeal injury (e.g. gymnast's wrist)
Dysplastic	**Dyschondrosteosis**, diaphyseal achalasia (as printed in TO)
Genetic	**Turner syndrome**
Idiopathic	–

5 Management

Madelung's deformity

- Asymptomatic → **OBSERVATION (stop repetitive wrist loading if painful)**
- Skeletally immature, pain / ↓ ROM → **PHYSIOLYSIS + VICKERS LIGAMENT RELEASE**
  - marked deformity → **+ DOME OSTEOTOMY ± ULNAR SHORTENING**
- Skeletally mature → **RADIAL OSTEOTOMY ± ULNAR PROCEDURE**
  - painful DRUJ instability → **DRUJ ARTHROPLASTY (controversial)**

Operative options

Procedure	Indication / note
Physiolysis + Vickers release (volar approach; bar resection + fat graft)	Pain / ↓ ROM in immature; **83% radiographic improvement**, no ROM loss (OB); prophylactic release in mild cases – efficacy unknown
Radial corrective (dome) osteotomy ± ulnar shortening	Pain, ↓ ROM, cosmesis, function; dome corrects coronal + sagittal; severe → staged distraction ex-fix
Adolescents (TO)	**Physiolysis + ligament resection + dome osteotomy** preferred
Mature (TO)	Ligament resection + dome osteotomy · closing wedge + ulnar shortening (**Milch**) · opening wedge · + **Darrach** · + **Sauvé-Kapandji**
DRUJ arthroplasty	Painful DRUJ instability + ↓ pronosupination; controversial

– – Active space – –

Acute/recent cases (TO) dorsiflexion in **full arm plaster 4 wk**

6 Complications

List incomplete physiolysis / premature arrest · violation of radiocarpal/ulnocarpal joint · under-correction · recurrence · osteotomy nonunion · **continued ulnar impaction if radial osteotomy alone**

X vs Y

	Madelung	Reverse Madelung
Ulnar variance	**Positive**	Negative
Physeal deficit	Volar-ulnar distal radius	–
Carpal adaptation	Adaptive **VISI** (↑ volar tilt)	–

How it's asked

"Inheritance pattern?"	→ **autosomal dominant** (TO)
"SHOX mutation → which deformity?"	→ **Madelung** (TO) ✗ Wrong options: radial club hand · Blount · CPT
"SHOX sex-linked mutation → which dysplasia?"	→ **Léri-Weill dyschondrosteosis**
"Cause of Madelung?"	→ **impaired growth of volar-ulnar distal radial physis**
"Which physis is disrupted?"	→ **ulnar volar physis of distal radius**
"Tethering structure?"	→ **Vickers ligament** (radiolunate)
"Radial osteotomy alone – complication?"	→ **continued ulnar impaction**

– – Active space – –

MALLET FINGER

– Hand · 54 / 103 –

1 Basics

Definition	disruption of **terminal extensor tendon distal to DIP** (tendinous or bony avulsion); "baseball / drop finger"; extensor **zone 1** (Kleinert & Verdan)
Demographics	young–middle-aged ♂, older ♀; work/sport (ball sports)
Digits	**long (commonest) > ring > small**, dominant hand (Orthobullets, TO) · "ulnar three fingers" (TO)

2 Mechanism

Commonest	**sudden forced flexion** (impaction) of extended fingertip → DIP hyperflexion avulses tendon
Bony mallet	**axial load** on extended finger → P3 abuts P2 head
Less common	dorsal sharp/crush laceration
Untreated	terminal tendon retracts → central slip overpulls P2 → **swan neck** (PIP hyperextension + DIP flexion) (TO)

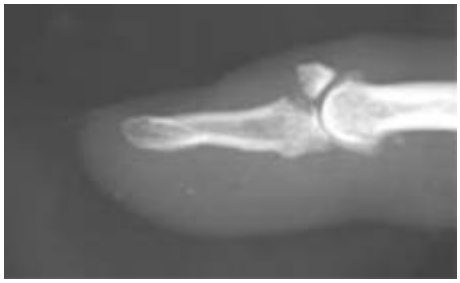
3 Presentation

Exam	painful swollen DIP; fingertip rests **~45° flexion**; **no active DIP extension** (TO)

4 Diagnosis & Classification

Key radiology findings

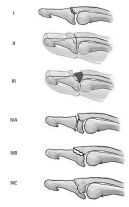
X-ray	AP + lateral finger: **bony avulsion at dorsal P3 base** or normal bones (tendinous); **assess volar subluxation of P3** (key decision)
US	not needed; loss of tendon motion, tears, fluid at insertion



Bony mallet on lateral X-ray

– – Active space – –

Doyle classification

Type	Description	
I	Closed ± small dorsal avulsion fracture	
II	Open (laceration)	
III	Open, deep abrasion with skin + tendon loss	
IV	Mallet fracture: **A** physeal (child) · **B** 20–50% articular · **C** >50% articular	

Wehbe & Schneider classification (TO)

Type		Subtype	Articular surface
1	**No DIP subluxation**	1	**<1/3**
2	**DIP subluxation**	2	**1/3–2/3**
3	Epiphyseal/physeal	3	**>2/3**

Most used Wehbe & Schneider (TO)

5 Management

Mallet finger

- Tendinous / bony, P3 NOT subluxed → **FULL-TIME DIP EXTENSION SPLINT (6–8 wk)**
  - chronic, supple, congruent → **still SPLINT TRIAL**
- Volar subluxation of P3 (even after splinting) → **CRPP (extension block pinning) / ORIF**
- Painful stiff arthritic DIP → **DIP ARTHRODESIS**
  - swan neck → **Fowler central slip tenotomy / lateral band or FDS tenodesis**

Nonoperative – continuous DIP extension splinting (bony & tendinous, delayed, residual lag)

Treatment of choice (TO) **extension splinting of DIP only**; **no PIP immobilisation**; DIP in extension / slight hyperextension
✗ Wrong options: PIP extension block · buddy strap

Partial articular avulsion, no subluxation (TO) **splint distal phalanx in extension**
✗ Wrong options: ORIF

Displaced fragment but joint reduced still **full-time DIP extension splint** (surgery ↑ complications)

Regimen **24 h/day** (✗ Wrong options: 12 h), **6–8 wk** (✗ Wrong options: 2–4 wk); PIP free; then night splint

Splint duration	Source
6–8 wk full-time, then **4 wk nocturnal** (controversial)	Orthobullets
6 wk full-time, then **night 2–6 wk**	TO
Bony 6 wk · tendinous ≥8 wk	OB218821
Bony 6–8 wk · tendinous **10–12 wk**	OB219900

Splint details **volar** splint fewer complications than dorsal; avoid hyperextension (skin necrosis); swan-neck posture → add **dorsal blocking PIP splint**; stack splint; flexion exercises from 6 wk; extra night splinting no proven benefit

Delayed presentation splint trial still works: **chronic >12 wk** supple, congruent, non-arthritic (Orthobullets) · **>4 wk** without fixed deformity or fragment >30% (TO) · 1 month (OB) → **chronic >6 wk ≠ surgery** (TO)

Compliance key to success

Outcome op = nonop for satisfaction & lag; ~80% satisfactory

Residual lag <10° up to **40%** after op or nonop → **no functional deficit** (TO, OB)
✗ Wrong options: dexterity loss · arthritis · swan neck/boutonnière

Operative indications – volar subluxation & pinning; swan-neck risk

Indication	Source
Volar subluxation of P3 = absolute	Orthobullets, OB
>50% articular surface · >2 mm articular gap/step-off · open injury (relative)	Orthobullets, TO
All open injuries + large fragment >30% with DIP subluxation	TO
Chronic failing splint trial	TO, OB
Chronic >12 wk without contracture, segmental loss → **tendon reconstruction** (~50% complications)	Orthobullets
Painful stiff arthritic DIP → **arthrodesis**	All

Fragment threshold conflict **>50%** (Orthobullets) · **>30% with subluxation** (TO) – read the options

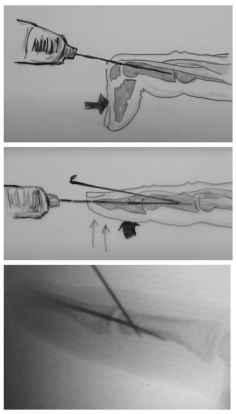
Bony mallet not fixed at earliest (TO) – splint first; re-X-ray in splint before deciding

Subluxed after closed reduction + splint → reduction and pinning
✗ Wrong options: repeat splint · PIP+DIP splint · fusion in young

~50% articular + volar subluxation (athlete) → CRPP
✗ Wrong options: play in cast · fusion · fragment excision + dermatotenodesis · 12 wk splint

Extension block pinning sequence (TO, Ishiguro) **DIP flexion → K-wire through extensor tendon into middle phalanx (dorsal block) → DIP extension (reduces fragment) → K-wire across DIP**

– – Active space – –

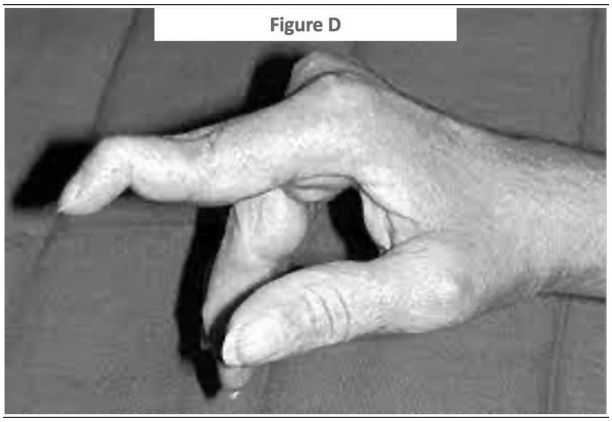


Extension block pinning

Pinning hierarchy (TO)	simple K-wire < **dorsal extension block (Ishiguro)**
ORIF options	dorsal midline; pins, dorsal blocking pin, screw, plate, suture-button
Tendon repair	direct repair + trans-articular pin; tenodermodesis; spiral ORL reconstruction
Swan neck correction (TO)	**Fowler central slip tenotomy**, **lateral band tenodesis**, FDS tenodesis; mild may correct with DIP treatment alone

6 Complications

Complication	Fact
Extensor lag	<10° in 40%, no deficit; **1 mm terminal tendon lengthening → 25° lag**
↓ DIP flexion	**1 mm shortening**; adhesions
Swan neck	Terminal retraction → central slip overpull + **volar plate & transverse retinacular lig. attenuation** → dorsal lateral bands → PIP hyperextension; **triangular ligament contracture maintains** (TO)
Skin (maceration, ulcer, nail)	**~70%** with splinting; dehiscence/infection after surgery



Swan neck after untreated mallet

X vs Y

	Mallet → swan neck	Boutonnière	Lumbrical plus
Lesion	Terminal tendon	**Central slip** + triangular lig. attenuation, volar lateral bands	FDP distal to lumbrical origin
PIP	Hyperextension	Flexion	Paradoxical extension
DIP	Flexion	Hyperextension	–

How it's asked

"True regarding mallet finger?"	→ **extension splinting of DIP only** (TO) ✗ Wrong options: PIP + DIP · fix bony mallet early · >6 wk = surgery
"Partial articular avulsion, no subluxation?"	→ **splint distal phalanx in extension** (TO) ✗ Wrong options: buddy splint · ORIF · PIP extension block
"Sequence of extension block pinning?"	→ **DIP flexion → wire through tendon into P2 → DIP extension → wire across DIP** (TO)
"Non-displaced bony mallet, 2 days?"	→ **DIP extension splint 6–8 wk**
"15 y, displaced fragment, joint reduced?"	→ **full-time DIP extension splint**
"Tendinous mallet regimen?"	→ **24 h/day DIP splint for 6–8 wk** ✗ Wrong options: 12 h · 2–4 wk · PIP
"7° lag after 8 wk splint – outcome?"	→ **no functional effect**
"Bony mallet, 1 month, no subluxation?"	→ **trial of extension splint 6 wk** ✗ Wrong options: pinning · screw · fusion
"Subluxed P3 after closed reduction + splint?"	→ **reduction and pinning**
"Untreated mallet risks which deformity?"	→ **swan neck** ✗ Wrong options: boutonnière · lumbrical plus · Dupuytren · scissoring
"~50% fragment + volar subluxation, athlete?"	→ **CRPP**

– – Active space – –

MCP DISLOCATIONS

– Hand · 55 / 103 –

1 Basics

Epidemiology

Incidence	rare, **<1/100,000/y**; less common than IPJ dislocations
Direction	**dorsal** most common
Digit	**index** > **thumb**
Associated	base P1 / MC head fractures in up to **50%**

Anatomy

Osteology	condyloid; **cam-shaped MC head**; shallow concave P1 base; sesamoids in volar plate (thumb: FPB, APB)
Proper collateral	**primary stabiliser**; dorsal MC head → volar P1 base; **tight in flexion**
Accessory collateral	volar to proper → volar plate; **tight in extension**
Volar plate	thick fibrocartilage distally, membranous proximally; **A1 pulley attaches to it**; attaches to **deep** transverse MC ligament ✗ Wrong options: superficial
Dorsal capsule	weak, thin, loose
Sagittal bands	keep extensor central during MCP flexion ✗ Wrong options: interposed

2 Mechanism

Dorsal	FOOSH → **hyperextension → volar plate avulsed from MC neck/head** ✗ Wrong options: hyperflexion · rotation
Volar	hyperextension or hyperflexion

3 Presentation

	Finding
Simple dorsal	**P1 hyperextended** on MC head, **PIP flexed**
Complex dorsal	**Bayonet** P1 (dorsal to MC shaft), **skin dimpling proximal palmar crease**, palpable MC head in palm, digit in extension deviated toward central digits, can't flex MCP
Volar	**Extensor lag**, dorsal skin depression proximal to P1 base

4 Diagnosis & Classification

Key radiology findings

Views	AP, **lateral (best view)**, oblique
MCP volar plate (TO)	attached to **P1, not the metacarpal** → displaced dorsally · collaterals taut in flexion · **tendon noose round MC neck** blocks reduction · complex: P1 **not markedly hyperextended**
Complex signs	**joint space widening** (volar plate interposed) · **sesamoid in joint = diagnostic of complex**



Lateral X-ray: dorsal MCP dislocation

Classification

Direction	Mechanism	
Dorsal (most common)	Hyperextension	
Volar	Hyperextension / hyperflexion	

Complexity	Definition	Treatment
Simple (subluxation)	**No interposition**; volar plate draped over MC head; P1 base touches MC head	**Closed reduction**
Complex (complete)	**Volar plate ± sesamoid interposed** dorsal to MC head	**Open reduction**

5 Management

MCP dislocation

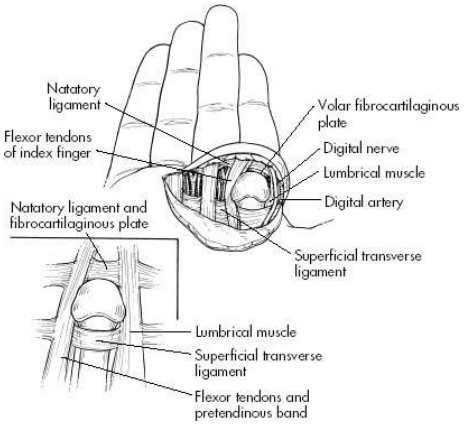
- Simple → **CLOSED REDUCTION**
  - fails / complex / delayed → **OPEN REDUCTION**
- Complex (sesamoid in joint, widening, dimpling) → **OPEN REDUCTION (dorsal approach, split volar plate)**

Complex dorsal MCP dislocation – volar plate interposition & reduction manoeuvre

Commonest block	**volar plate** ✗ Wrong options: flexor tendon · extensor · lumbrical · collaterals · 1st DI · natatory lig.
Cadaver (Afifi)	only **volar plate division** allows reduction; flexors move ulnar, lumbricals radial, radial digital nerve superficial
Dorsal reduction manoeuvre	**wrist flexion** (relax intrinsic & extrinsic flexors) + **direct dorsal-to-volar (distal & volar) pressure on P1 base** (same for thumb, index, small)
Avoid longitudinal traction	pulls volar plate into joint → **converts simple → irreducible complex** ✗ Wrong options: wrist extension tightens noose
Traction source conflict	avoid traction (Orthobullets, OB) · "gentle" traction with dorsal pressure (OB219826) – read the options
Volar reduction	pressure on volar P1 with MCP flexed → immobilise **30° flexion 2 wk** → AROM in dorsal blocking splint
After dorsal reduction	early ROM + dorsal blocking splint

Noose (dorsal complex)

Digit	Radial side	Ulnar side
Index	**Lumbrical**	**Flexor tendons**
Thumb	Intrinsics	**FPL**
Small	Flexors + lumbrical	ADM + FDM



Index noose: lumbrical and flexor tendon

Volar dislocation blocks **dorsal capsule**, avulsed volar plate / collateral ligament, **juncturae tendinum (4th–5th EDC) in small finger**, **1st DI in thumb**

Open approach	Pros	Cons
Dorsal (midline, split extensor/ capsule; thumb EPL–EPB interval; **split volar plate longitudinally**)	**↓ NV injury**, easy **MC head fracture fixation**, 14 min shorter, no 2nd approach	–
Volar (oblique, skin only; release **A1**)	Better volar plate access	**NV bundle at risk** (radial digital n. tented), hard osteochondral fixation; 42% needed 2nd dorsal approach

CMC fracture-dislocations – fixation

Mechanism axial load (punch), hyperflexion; ~1% of hand/wrist injuries; often **5th ± 4th**, dorsal **hamate fracture**

Scenario	Treatment
All 5 CMC fracture-dislocations	**Closed vs open reduction + internal fixation** (✗ Wrong options: cast · splint)
Open contaminated	External fixation
Acute 4th CMC fracture-dislocation	**CRPP** (✗ Wrong options: IM fixation · fusion · splint)
Subacute >2 wk, still dislocated	**Open reduction + fixation** (scar blocks closed reduction)
Acute (<3 d) isolated ulnar CMC dislocation	Closed reduction + splint can work (Zhang, Storken)
Severely comminuted MC base	CMC arthrodesis
Chronic painful	Fusion / arthroplasty

4th CMC fracture-dislocation **contraindicates 5th CMC suspension arthroplasty** (ECU slip)

6 Complications

Stiffness injury trauma, prolonged immobilisation

Post-traumatic arthritis / osteonecrosis repeated closed attempts, prolonged dislocation, traumatic open reduction

Premature physeal closure rare

X vs Y

	Simple dorsal	Complex dorsal
Posture	P1 hyperextended, PIP flexed	Bayonet, P1 parallel dorsal
Palm	–	**Skin dimpling**
X-ray	Congruent contact	**Sesamoid in joint**, widening
Treatment	Wrist flexion + dorsal pressure	**Open (dorsal), split volar plate**

– – Active space – –

How it's asked

"Irreducible dorsal index MCP — interposed structure?"	→ **volar plate**
"Mechanism + cause of failed closed reduction?"	→ **hyperextension → volar plate entrapped dorsal to MC head** ✗ Wrong options: flexor–lumbrical buttonhole
"Simple dorsal MCP subluxation — next?"	→ **wrist flexion + distal/volar pressure on P1** ✗ Wrong options: axial traction · surgery first
"Reduction manoeuvre for 5th MCP / thumb MCP dorsal dislocation?"	→ **dorsal pressure on P1 with wrist flexed** ✗ Wrong options: wrist extension · longitudinal traction
"True of interposed structure?"	→ **A1 pulley attaches to it** ✗ Wrong options: prevents bowstringing (A2/A4) · superficial TML · midline tracking (sagittal band)
"Cross-section — which structure blocks?"	→ **volar plate** ✗ Wrong options: DTML, sagittal band, proper collateral, dorsal capsule
"Fracture-dislocation of all 5 CMC joints?"	→ **closed vs open reduction + internal fixation**
"4th/5th CMC dislocation seen at 3 wk?"	→ **open reduction + fixation**
"Acute 4th CMC fracture-dislocation?"	→ **CRPP**

MCP JOINT ARTHRITIS

– Hand · 56 / 103 –

1 Basics

Epidemiology

Frequency in hand **DIP > thumb CMC > PIP > MCP** (MCP least common)

Cause mostly **inflammatory** (RA); primary OA rare (**6.1% ♂, 1.4%** ♀; OB: 1–5%); post-traumatic rare

Sex OA & post-traumatic → **men**, younger · inflammatory → **women**; 4th–5th decade

Risk intra-articular fracture, dislocation, rheumatic disease

Trend ↓ incidence with **biologics**

Anatomy

Joint diarthrodial; MCP = **77% of total finger flexion arc**; ADLs need ~35° (50%)

Volar plate ↑ congruence, prevents hyperextension

Deep transverse MC ligament links **2nd–5th MC heads** at volar plate

Collateral component	Tight in
Accessory	**Extension**
Proper	**30° flexion**

Thumb MCP 2 sesamoids; **adductor pollicis → ulnar sesamoid**; **FPB → radial sesamoid**; most thumb motion at **CMC**

Genes & lab markers

Disease	Mediator	Lab / aspirate
RA	**Neutrophil** proteases, ROS; synovial hypertrophy	**RF, anti-CCP**
Gout	MSU → **↑ IL-1β**	**Needle-shaped, strongly negatively birefringent** (yellow parallel)
CPPD	Ca pyrophosphate in synovium	**Rhomboid, weakly positively birefringent**
SLE	**Interferon-inducible genes ↑**	**ANA**
Haemochromatosis	Iron → ↑ IL-1R	**Transferrin saturation, ferritin**

3 Presentation

OA pain, deformity · **inflammatory** – waxing/waning, other joints · **erythema** → gout/CPPD · **periarticular tophi** → gout

– – Active space – –

RA fingers — **ulnar deviation** (also secondary to **radial wrist deviation**) · synovitis attenuates dorsal capsule → **volar translation of proximal phalanx** · volar capsule laxity → **swan neck**

RA thumb — EPL displaced ulnar & volar → no active MCP extension · MCP flexion + IP hyperextension → **boutonnière** · **90/90 deformity** · CMC adduction → secondary MCP hyperextension · **gamekeeper's deformity** (UCL attrition) → **first web contracture**



Thumb boutonnière: MCP flexion, IP hyperextension

4 Diagnosis & Classification

Key radiology findings

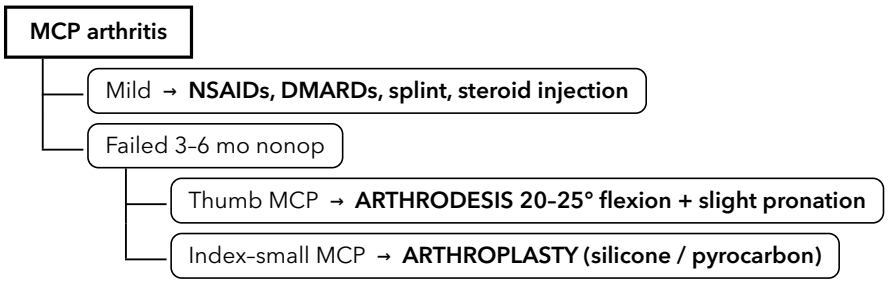
Views — AP, lateral, oblique hand

Disease	X-ray sign
OA	Narrowing, osteophytes, subchondral sclerosis, cysts
RA	**Bone loss dorsal base of proximal phalanx**; volar subluxation, ulnar drift
Gout	**Punched-out "rat-bite" erosions, overhanging edges**, sclerotic margins
SLE	**Jaccoud's** (deforming **non-erosive**) · erosive = **"rhupus"**
Haemochromatosis	**Hook-like osteophytes radial 2nd & 3rd MC heads**; symmetrical narrowing
CPPD	**2nd–5th MCP**, radial hook osteophytes

Larsen radiographic grading (RA)

Grade	Finding
0	No change
I	Slight; periarticular swelling
II	Erosions + definite narrowing
III	Medium destruction; joint space poorly defined
IV	Severe destruction, collapse
V	Mutilating; articular surfaces gone

5 Management



Nonop injections **temporary**; intrinsic-plus MCP blocking splint + dexterity exercises ↑ ROM; topical NSAID if GI/renal compromise

Perioperative DMARDs **continue methotrexate & sulfasalazine**; cyclosporine/tacrolimus if severe; **stop biologics** (operate at end of dosing cycle)

Thumb MCP arthrodesis position

Gold standard **arthrodesis** (silastic arthroplasty = option); well tolerated as motion at CMC & IP

Position **20–25° flexion + slight pronation → key pinch** (OB, Orthobullets) · "15° pronation" (Orthobullets technique)

Advanced TMC disease ↑ flexion to **20–40°** offloads TMC

Fixation	Key numbers
K-wire	**83–100% fusion**; ↓ pinch/dexterity; **16.7% complications** (nonunion, **EPL rupture**)
Tension band	**95% success**; ↓ infection/nonunion vs K-wire
Plate	Union **11.8 wk**; positioned 10–15°
IM screw	Mean 18°, union **10 wk**, 96% pain relief; MC head fracture, cut-out

Immobilise 4–6 wk / until union

Finger MCP arthrodesis

Indication index in **high-demand / young with instability**; otherwise **arthroplasty preferred** (motion)

Finger	Fusion flexion
Index	**20–25°**
Long	**30°**
Ring	**35°**
Small	**40°**

Index union 84.6% at 11.6 wk

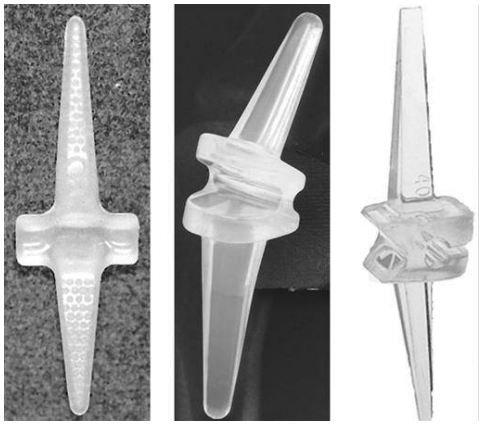
– – Active space – –

Rheumatoid MCP arthroplasty

Indication — **severe finger MCP arthritis / fixed deformity** (ulnar drift, volar subluxation) → **MCP arthroplasty** = least extensor lag, best ulnar drift correction
✗ Wrong options: tenosynovectomy · soft-tissue realignment alone · resection arthroplasty

Soft tissue — release ulnar collateral, tension radial; repair sagittal band, **centralise extensor**; intrinsic release/transfer (swan neck)

Results — arc **27–60°**; extensor lag **9–22°**; ~12° loss by 63 mo; pinch ↑ (index radialised)



Swanson silicone implants

Implant contraindications (pyrocarbon vs silicone)

	Silicone	Pyrocarbon	Metal-plastic SRA
Design	**One-piece linked hinge**, spacer; fibrous capsule; 30° preflexed	**Unlinked, non-constrained** surface replacement; no ingrowth	Porous Ti MC + cemented PE
Stability from	**Implant itself**	**Collateral ligaments + volar plate**	Soft tissues
Best for	**Erosion / deformity (RA)**	**Milder, no deformity; OA**	–
Survival	63–97%	RA 81% (10 y), 70% (16 y); OA > inflammatory	RA 95% / 85% (10/20 y); non-inflammatory 76% (10 y), satisfaction 72%

Pyrocarbon contraindication — **deficient collateral ligaments**

Not contraindications — deficient radial sagittal band, radial lateral band, MC head subchondral bone, intermetacarpal ligaments

Thumb MCP arthroplasty — RA with fixed deformity or limited IP/CMC motion; silastic survival **83% at 5.8 y**; bone change 4%, fracture 2%

6 Complications

Nonunion (thumb fusion) — **9%**; risk **tension band**, smoking, DM → revision + distal radius graft

Infection	**0.1–1%**; ***S. aureus***; risk DMARDs, RA, DM → superficial 2 wk Abx · deep I&D + removal
Instability	RA: **SRA 31%, pyrocarbon 19%**; **pyrocarbon highest acute dislocation**; revision 71% success → **revise to silicone**
Pyrocarbon	↑ failure & recurrent deformity in **inflammatory** arthritis

– – Active space – –

How it's asked

"RA, fixed MCP ulnar drift – least extensor lag at 1 y?"	→ **MCP arthroplasty** ✗ Wrong options: tenosynovectomy, EIP→EDQ, resection
"Thumb MCP fusion position"	→ **20–25° flexion**
"Thumb MCP fusion – flexion & rotation"	→ **20° flexion + slight pronation** ✗ Wrong options: supination · 40°
"Pyrocarbon vs silicone contraindication – true?"	→ **deficient collaterals contraindicate pyrocarbon**
"Thumb vs fingers"	→ **thumb = arthrodesis; index–small = arthroplasty**

– – Active space – –

METACARPAL FRACTURES

– Hand · 57 / 103 –

1 Basics

Epidemiology

Incidence	**40% of all hand injuries**; most common hand injury
Who	**men 10–29 y**
Site	**neck = commonest site**; **5th MC = commonest bone**

Anatomy

Arch	palmar concave; **1st, 4th, 5th = mobile borders**; **2nd + 3rd = stiff central pillar** (index most firmly fixed)
CMC mobility	2nd/3rd CMC stiff → **less tolerance of angulation** in index/long
Transverse (deep) intermetacarpal ligament	**primary restraint to shortening** of isolated 3rd/4th shaft fractures; stays intact in shaft fracture ✗ Wrong options: rotation · MCP/CMC stability · fracture prevention

Tendon	Insertion	Deforming role
ECRL / ECRB	Base MC II / III	–
ECU (PIN)	**Base MC V**	**Pulls 5th MC base dorsal + proximal**
APL	Base MC I + trapezium	Bennett/Rolando
Opponens pollicis / digiti minimi	MC I / MC V	–

Interossei	3 palmar + 4 dorsal arise from MC shafts

2 Mechanism

Mechanism	direct blow / punch; rotational + axial load; high energy → **multiple fractures**
Associations	open wounds (tendon, NV injury), **compartment syndrome** (crush, multiple fractures)

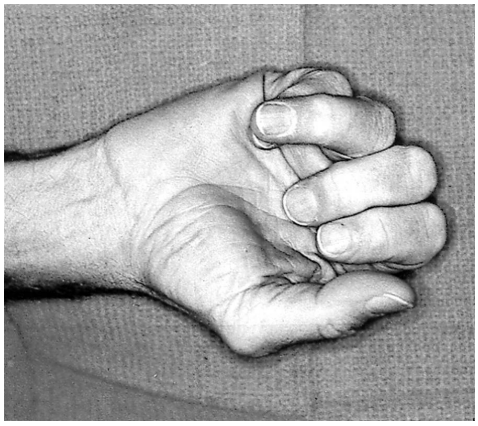
Boxer's fracture

Definition (TO)	**transverse or short oblique fracture of the 5th MC neck** (less often 4th) ✗ Wrong options: head · shaft
Facts (TO)	**most common MC fracture**; punching with clenched fist; **apex dorsal angulation**; closed reduction + cast or K-wire

3 Presentation

Fight bite over MCP	**open until proven otherwise**; dorsal wounds over MC fractures almost always open

Malrotation line up fingernails in partial + full flexion, compare sides → **scissoring/ overlap**; best assessed after block



Assessing malrotation in flexion

Shortening compare knuckles with other hand

Base deformity suspect **CMC dislocation**

4 Diagnosis & Classification

Key radiology findings

View	Best for
PA + lateral	All
ER oblique (supinated)	**4th/5th CMC** fracture-dislocation
IR oblique (pronated)	**2nd/3rd CMC**
Brewerton	**MC head** fractures
Roberts	**Thumb CMC**
CT	Inconclusive CMC injury, multiple CMC dislocations, complex head fractures; **after reduction** of 5th CMC injury

CMC fracture-dislocation clues MC shortening on AP; dorsal base subluxation / dorsal hamate fragments on lateral

Acceptable angulation & shortening; extensor lag

Digit	Shaft angulation (Orthobullets/ OB)	Shaft angulation (TO)	Neck angulation	Shortening
Index & long	**10–20°**	**>10°** → fix	**10–15°**	2–5 mm
Ring	**30°**	**>25°** → fix	**30–40°**	2–5 mm
Little	**40°**	**>45°** → fix	**50–70°**	2–5 mm

Conflict ring 30° / little 40° (Orthobullets, OB) · **ring 25° / little 45°** (TO) – read the options

– – Active space – –

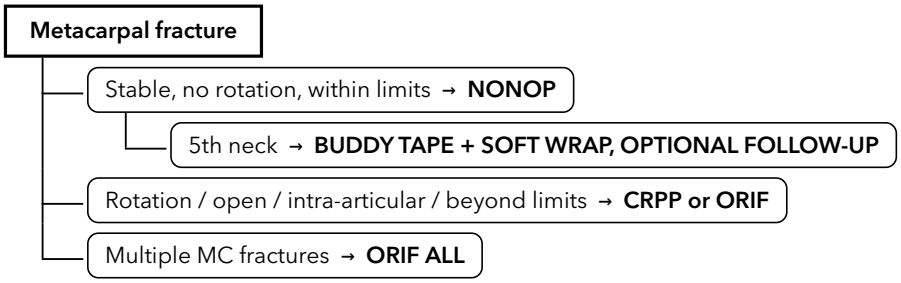
Shortening	**2–5 mm acceptable** (Orthobullets, OB) · fix if **>5–6 mm** (TO)
Shortening effect	**7° extensor lag per 2 mm** (linear, not digit-dependent) + **8% ↓ power per 2 mm** (↓ interosseous force; intrinsics 40–90% of grip)
Shortening >2–5 mm	**MCP extension lag + ↓ grip** (loss of extensor tension)
Apex dorsal angulation (index/long)	**pseudoclawing** (MCP hyperextension, PIP extension lag) because stiff 2nd/3rd CMC cannot compensate; reduce if >10°
Rotation	overlap, no extensor lag, no pseudoclaw

Operative criteria

Indication	Threshold
Malrotation	**Any scissoring** / >10° (TO) – **absolute**, least tolerated
Angulation	Beyond table limits
Shortening	>5–6 mm (TO)
Open fracture	Any
Intra-articular	Any (head: **no articular step acceptable**)
Multiple MC fractures	Any – loss of adjacent support
Associated CMC dislocation, extensor lag (TO)	Any

Not indications	comminution alone, manual-labour occupation, ring sagittal angulation 15° (TO)

5 Management



Nonoperative care – buddy taping & early return

Cast	**MCP 70–90° flexion**, PIP free, **4 wk**
Jahss reduction	MCP 90° flexed, dorsal push through proximal phalanx, stabilise shaft
5th neck, no rotation (<50–70°)	**buddy taping 4–5 + soft wrap**, **optional follow-up** → better satisfaction, **earliest RTW** ✗ Wrong options: ulnar gutter splint/cast · CRPP
Athletes, minimally displaced spiral ring MC	ORIF: **no faster union, but faster return to play**; no better grip/ROM

Rotational deformity – scissoring as fixation indication

Strongest fixation indication — **scissoring of digits with flexion**
✗ Wrong options: 20° 4th neck angulation (30° OK) · 4 mm shortening · comminution · occupation

Untreated multiple MC shaft fractures — rotational malunion → **digits overlap in fist** → corrective osteotomy

Not MC fracture sequelae — sagittal band rupture (can hold passive extension) · loss of key pinch / Wartenberg / ulnar numbness (ulnar neuropathy)

Operative fixation – multiple fractures, ORIF & intramedullary screws

Multiple shaft fractures — **ORIF**; **fix all three**
✗ Wrong options: casting → stiffness · ex-fix binds extensors · removable splint → shortening

Techniques

Fixation	Fact
CRPP	Antegrade from base or retrograde via collateral recess; remove pins **4 wk**
Lag screws	Long spiral/oblique; ≥2 screws
Dorsal plate	**Transverse**; cover with periosteum; extensors less adherent at MC level → low adhesion
IM headless screw	**Only axially stable** (transverse) patterns; corrects angulation/rotation; not for oblique/shortened/comminuted (**least amenable = long oblique axially unstable**); unstable if fracture length >2× shaft diameter
Head ORIF	Dorsal approach, split extensor or release/repair sagittal band; screws in collateral recess / headless / K-wires; hardware not proud
Severe head comminution	Ex-fix or **MCP arthroplasty**; late arthritis → fusion

Post-op — **early motion critical**; head fracture commonest complication = **stiffness**

Neck ORIF — difficult (little distal bone)

Fifth CMC fracture-dislocation – deforming forces & pinning

Incidence — **<1% of hand trauma**; often **missed**; 5th ± 4th; axial load on clenched fist

Deforming force — **ECU** (insertion base 5th MC) → **innervated by PIN**
✗ Wrong options: radial (ECRL only) · ulnar · AIN

Acute — **closed reduction + splint first → CT after reduction**
✗ Wrong options: CT first · immediate CRPP/ORIF

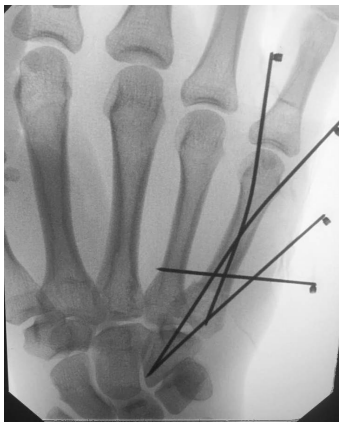
Reduction lost at 1 wk — **CRPP**
✗ Wrong options: repeat reduction · discontinue immobilisation · arthrodesis

Delayed / irreducible — **open reduction + pinning**; **comminuted intra-articular 5th base + displaced 4th shaft (>30°) → fix both**

Arthrodesis — chronic / painful post-traumatic arthritis only

Dorsal ulnar approach — **dorsal branch of ulnar nerve** (from **medial cord**), arises **~6 cm proximal to ulnar styloid** → retract (neuroma)

– – Active space – –



4th/5th CMC fracture-dislocation

6 Complications

List **stiffness** (commonest, head) · malunion (rotation → scissoring; angulation → pseudoclaw; shortening → extensor lag) · tendon irritation (plate) · arthritis

X vs Y

	Shortening	Apex dorsal angulation	Malrotation
Sign	**MCP extensor lag**, ↓ grip	**Pseudoclawing** (index/long)	**Scissoring / overlap**
Tolerance	2–5 mm	Digit-specific	**None**

	Shaft	Neck
Angulation tolerance	Less	**More**
Little finger	40° (45° TO)	**50–70°**
Ring	30° (25° TO)	**30–40°**

How it's asked

"Fixation indications EXCEPT?"	→ **15° sagittal angulation in ring finger** (TO)
"Extensor lag + ↓ grip after closed treatment – cause?"	→ **5 mm shortening**
"Effect of shortening?"	→ **7° extensor lag per 2 mm**
"Role of transverse intermetacarpal ligament?"	→ **limits shortening**
"Little finger neck – max acceptable angulation?"	→ **50–70°**
"Index/long healed with deformity – pseudoclawing cause?"	→ **30° apex dorsal angulation**
"Strongest indication to fix 4th neck?"	→ **scissoring with flexion**
"Multiple MC shafts untreated – complaint?"	→ **digits overlap in fist**

"Buddy taping + optional follow-up?"	→ **angulated 5th MC neck without rotation**
"ORIF vs nonop in athlete?"	→ **same union time, faster return to play**
"Fastest RTW for 5th neck <50°?"	→ **buddy taping 4th–5th**
"Multiple MC shaft fractures (polytrauma) – definitive?"	→ **ORIF**
"Three MC fractures from airbag?"	→ **fix all three**
"Least amenable to IM headless screw?"	→ **oblique, axially unstable, shortened shaft**
"Comminuted 5th base + angulated 4th, 2 wk old?"	→ **fix both**
"4th/5th CMC redislocated at 1 wk?"	→ **CRPP**
"Deforming muscle – nerve?"	→ **ECU – PIN**
"Dorsal ulnar approach nerve – from?"	→ **medial cord** (dorsal branch ulnar n.)
"Acute 5th base fracture-dislocation – next step?"	→ **closed reduction + splint** (then CT)
"Boxer's fracture is?"	→ **5th MC neck fracture** (TO)

– – Active space – –

NAIL BED INJURY

– Hand · 58 / 103 –

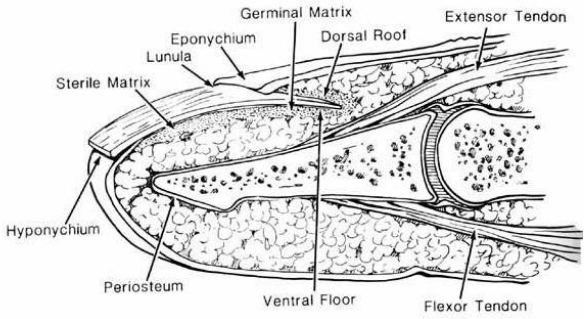
1 Basics

Epidemiology

Frequency	fingertip injuries = **most common hand injury in ED**; ♂:♀ **3:1**
Mechanism	crush between objects, closing door, saw, snowblower, hammer
Associated	**DIP fracture/dislocation**, **distal phalanx fracture**

Anatomy

Structure	Definition
Perionychium	Nail + nail bed + surrounding skin
Paronychium	Lateral nail folds
Hyponychium	Skin distal + palmar to nail
Eponychium	Dorsal nail fold (proximal)
Lunula	White proximal nail
Sterile matrix	**Distal to lunula**, adheres to nail
Germinal matrix	**Proximal to sterile**; **most nail production**; extensor insertion **1.2–1.4 mm proximal**



Fingertip anatomy

4 Diagnosis & Classification

Key radiology findings

X-ray	**AP, lateral, oblique** of finger → rule out **distal phalanx fracture**; child: **physeal separation (Seymour)**

Injury types

Type	Finding	Treatment
Subungual haematoma <50%	Crush; blood under nail	**Trephination** (sterile needle / electrocautery)
Subungual haematoma >50%		**Nail removal, I&D, nail bed repair**
Laceration	Nail intact + haematoma >50%	Nail removal + repair; tetanus + antibiotics
Avulsion, minimal matrix loss	Higher energy; ± DP fracture	Repair ± fracture fixation (**reduce fracture**); tetanus + antibiotics
Avulsion/crush, significant matrix loss		Repair + **STSG or nail matrix graft** (adjacent injured finger or **2nd toe**) ± fixation



Subungual haematoma

5 Management

Nail bed injury

- Subungual haematoma <50% → **trephination**
- Haematoma >50% / laceration → **nail removal + nail bed repair**
- Avulsion ± DP fracture → **repair + reduction ± fixation**
  - significant matrix loss → **matrix graft (toe / adjacent finger)**

Repair technique – tissue adhesive & Seymour fracture

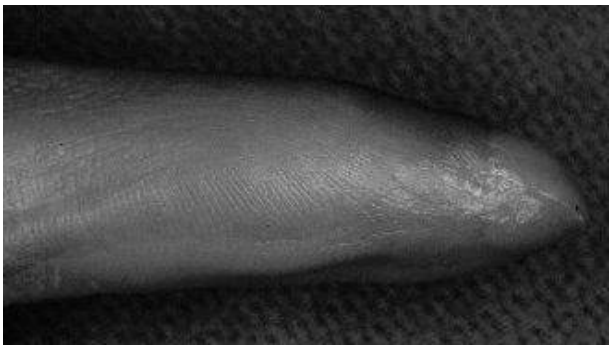
Nail	remove → **soak in Betadine** during repair
Suture	**6-0 or smaller absorbable**
Dermabond (2-octylcyanoacrylate)	RCT (Strauss, level 1): **only advantage = faster procedure**; no better cosmesis, function, pain, infection (all equal)
Splint eponychial fold	original nail, aluminium, or non-adherent gauze (prevents fold adhesion)
Seymour fracture	child, nail torn **transversely beneath eponychium** + **juxta-epiphyseal distal phalanx physeal separation** (nail bed trapped) → **I&D + physeal reduction + nail bed repair + immobilisation** ✗ Wrong options: splint alone · home Betadine soaks · I&D + splint without reduction · pinning first

– – Active space – –

Early repair superior to secondary correction (91% good–excellent in children); outcome depends on **germinal matrix** injury

6 Complications

Complication	Cause	Treatment
Hook nail	Matrix advanced for coverage **without bony support**	**Remove nail + trim matrix to level of bone**
Split nail	**Matrix scarring**	**Excise scar + replace matrix** (± graft)



Hook nail

How it's asked

"Dermabond vs suture for nail bed – advantage?"	→ **faster procedure** ✗ Wrong options: cosmesis · function · pain · infection
"Child, nail torn under eponychium + DP physeal fracture – next step?"	→ **I&D, reduction, nail bed repair, immobilisation** (Seymour)
"Subungual haematoma <50%, nail intact?"	→ **trephination**
"Hook nail treatment?"	→ **trim matrix to level of bone**
"Split nail cause?"	→ **matrix scar** → excise + matrix graft

NERVE CONDUCTION STUDIES

– Hand · 59 / 103 –

1 Basics

Uses

Components	**NCV + EMG** (= EDS)
Uses	localise **compression**/neuropathy; **UMN vs LMN**; root vs plexus vs peripheral; **severity & prognosis** (neurapraxia good; axonotmesis/neurotmesis poor); denervation/reinnervation; **worker's compensation**/secondary gain
Indications	CTS, cubital tunnel, cervical & lumbar radiculopathy, shoulder nerve dysfunction (winging)
Pretest probability	EDS most useful when nerve injury **likely** (severity, location, prognosis); low probability + nonspecific sx → inconclusive/misleading; adjunct after Hx + Ex
Fibres tested	**large myelinated** motor + sensory (position, vibration, some light touch)

4 Diagnosis & Classification

Nerve conduction velocity – measures



NCV technique

Stimulus	**0.1–0.2 ms square wave**; motor studies add a 2nd proximal stimulus (cancels NMJ delay)
NCV	**distance ÷ latency** (cathode → recording electrode); NCV ∝ 1/latency
Latency	stimulus onset → response onset (incl. NMJ transmission + muscle propagation)
NCV determinants	**myelin thickness**, internode distance, temperature, age

Age	NCV (% adult)
Newborn	**50%**
1 y	**75%**
5 y	**100%**

– – Active space – –

Amplitude	baseline → negative peak (mV) = **number of functioning axons/muscle fibres**
Duration	spread of CVs; ↑ in acute demyelination
Late responses (proximal lesions, e.g. GBS)	**F-wave**; **H-reflex** (Iα at knee → soleus, **S1 root**; tibial/sciatic neuropathy)

Demyelination – increased latency

Demyelination	**↑ latency** + **↓ conduction velocity**; amplitude normal/↓; evoked response absent/prolonged ✗ Wrong options: ↑ amplitude · normal evoked response · unchanged CV/latency
Velocity vs latency	velocity **less specific** than latency; **<52 m/s abnormal**
Clinical focal demyelination	↓ DTRs, ↓ vibration, intermittent pain/paraesthesia; **no motor strength/ sensibility loss**

Interpreting NCS in carpal tunnel syndrome

Abnormal for CTS (Orthobullets NCS note)	**distal sensory latency >3.2 ms · motor latency >4.3 ms**
Mild–moderate CTS	**↑ latency + ↓ NCV, normal amplitude** = focal demyelination, axon number intact
Severe/chronic CTS	**↓ amplitude** (SNAP/CMAP) = axonal loss from ischaemia → surgery less predictable
Prognosis	mild–moderate + temporary relief from steroid injection → **CTR predictably relieves**
EDS in CTS	not needed for diagnosis; only EDS grades mild → severe; ambiguous cases, outcome prediction
SNAP vs CMAP	sensory nerve action potential vs compound motor action potential

Findings on NCV

– – Active space – –

Condition	Latency	Velocity	Amplitude	Evoked response
Normal	N	UL **>45 m/s** · LL **>40 m/s**	N	N
Axonal	N	N	↓	Prolonged
Demyelinating	↑	↓	N/↓	Absent/prolonged
Anterior horn cell	N	N	↓	N or polyphasic, prolonged
Myopathy	N	N	↓	N
NMJ	N	N	↓	N
Neurapraxia – proximal / distal to lesion	Absent / **N**	Absent / **N**	Absent / **N**	Absent / **N**
Axonotmesis – proximal / distal	Absent / Absent	Absent / Absent	Absent / Absent	Absent / N
Neurotmesis – proximal / distal	Absent / Absent	Absent / Absent	Absent / Absent	Absent / Absent

Electromyography

Purpose root vs nerve vs muscle; acute vs chronic; reinnervation; nerve continuity

Insertional activity **↑ (>300–500 ms) = early denervation** (polymyositis, myotonia, myopathy); **↓ = prolonged denervation** (fibrosis)

Normal spontaneous **end plate potentials & spikes**

Fibrillations **single** muscle fibre; denervated fibre membrane oscillation; appear **3–5 wk** after lesion; also muscular dystrophy

Fasciculations **group** of fibres; **ALS**, spinal muscular atrophy, polio, syringomyelia; "**bag of worms**"

Other abnormal positive sharp waves, complex repetitive discharges, myokymic discharges

Condition	Insertional	Spontaneous	Minimal activity	Interference
Normal	N	Silent	Bi/triphasic	Complete
Axonal neuropathy	↑	**Fibs / PSW**	Bi/triphasic	Incomplete
Demyelinating neuropathy	N	**Silent**	Bi/triphasic	Incomplete
Anterior horn cell	↑	Fibs/**fasciculations**, PSW	**Large polyphasic**	Incomplete
Inflammatory myopathy	↑	Fibs, myotonia	**Small polyphasic**	Early
Noninflammatory myopathy	N	N	Small polyphasic	Early
NMJ disorder	N	N	Bi/triphasic (↓ amplitude/ duration)	Early/N
Neurapraxia	N	**Silent**	None	None
Axonotmesis / neurotmesis	↑	**Fibs / PSW**	None	None

– – Active space – –

X vs Y

	Demyelination	Axonal loss
Latency	↑	N
Velocity	↓	N
Amplitude	N (or ↓)	↓
EMG	Silent	Fibs, PSW
CTS grade	Mild–moderate	Severe
CTR outcome	Predictable	Less predictable

	Fibrillation	Fasciculation
Unit	**Single fibre**	**Group of fibres**
Classic	Denervation (3–5 wk)	ALS, anterior horn disease

How it's asked

"Demyelination on NCS?"	→ **increased latency** ✗ Wrong options: ↑ amplitude · unchanged velocity
"↑ latency, ↓ NCV, equal amplitude + transient injection relief – counsel?"	→ **mild–moderate CTS; surgery predictably relieves**
"Amplitude reflects?"	→ **number of functioning axons**
"NCV of 1-year-old?"	→ **75% of adult**
"Fasciculations seen in?"	→ **ALS / anterior horn disease**
"Fibrillations appear after?"	→ **3–5 weeks**
"H-reflex tests?"	→ **S1 root**

PERIPHERAL NERVE INJURY & REPAIR

– Hand · 60 / 103 –

1 Basics

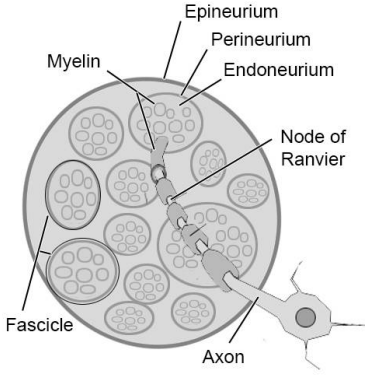
Epidemiology

Incidence	16.9 / 100,000 / yr; 2.6% of UE trauma, 1.2% of LE trauma; ~3% of UE/ hand injuries
Demographics	♂ **80%**, age 32–39
Risk factors	penetrating / high-energy trauma, crush, displaced fractures
Commonest UE nerve injury (OB)	**ulnar nerve**, working-age males

Nerve microanatomy (epineurium, perineurium, endoneurium)

Layer	Covers	Function / MCQ fact
Epineurium	Group of fascicles → whole nerve	**Cushions against external pressure**; collagen + elastin; focus of surgical repair
Perineurium	**Individual fascicle**	**Tensile strength + elasticity**; blood–nerve barrier; limits diffusion (protects from oedema)
Endoneurium	**Individual axon** (+ Schwann cell, myelin)	Forms **Schwann cell tube**; nourishes axon; **must be intact for full recovery**
Myelin	Axon	Schwann cells; ↑ conduction velocity; saltatory at **nodes of Ranvier**

Cross-section labels (OB)	A axon · B myelin · C epineurium · **D perineurium** · E endoneurium · F node of Ranvier
✗ distractors	fibronectin/laminin (ECM, directional branching), N-cadherin (adhesion, growing axons)
Neuron	cell body <10% of mass (metabolic centre); axon conducts; dendrites receive
Blood supply	extrinsic vessels in loose tissue + intrinsic plexus in epi-, peri-, endoneurium



Nerve cross-section with fascicles

– – Active space – –

Anatomical rules & nerve identity (TO)

Rule 1 (TO) every nerve **changes compartment at coracobrachialis insertion**

Rule 2 (TO) every nerve **enters the forearm between two heads of a muscle**

Nerve identity (TO) **ulnar = musician's nerve** (fine movements) · **median = labourer's nerve** (power grip) · radial = extensor nerve

Radial nerve (TO) **maximum motor loss, minimal sensory disability**

Fibre types

Fibre	Diameter	Myelin	Speed	Carries
A	10–20 µm	Heavy	Fast	Touch (Aβ)
Aδ	Thin	Myelinated	Medium-fast	**Pain + temperature** (TO) – sharp, first pain
B	<3 µm	Moderate	Medium	Autonomic
C	<1.3 µm	None	Slow	Pain – dull, delayed (TO)

Fibre (TO)	Function	Diameter (µm)	Velocity (m/s)
A alpha	Proprioception, motor	12–20	70–120
A beta	Pressure, touch	5–12	30–70
A gamma	Muscle spindle (motor)	3–6	15–30
A delta	Touch, cold, **fast pain**	2–5	12–30
B	Autonomic	< 3	3–15
C	Autonomic, **slow pain**	0.3–1.3	0.7–2.3

Mnemonic (TO) "Prime Minister Pressurized To Sign The Contract Fast"

Sensory receptors

Receptor	Adaptation	Senses	Test
Merkel	**Slow**	Pressure, texture, low-freq vibration	**Static 2PD**
Meissner	Rapid	Light touch	Moving 2PD
Pacinian	Rapid	**High-freq vibration**, rapid indentation (~1 mm ovoid)	Vibration
Ruffini	Slow	Skin stretch	

Sensory test (TO)	Receptor / meaning
Light moving touch (Ten test)	Patient grades 0–10 vs normal side · A-beta · **excellent screening**; limited in polyneuropathy / bilateral compression
Vibration threshold	**Fast-adapting** (Meissner, Pacinian) · high frequencies affected first
Semmes-Weinstein (cutaneous pressure threshold)	**Slow-adapting** (Merkel, Ruffini)
Static 2PD / moving 2PD	Slow / fast adapting · pins **perpendicular** to digit · **> 8 mm = non-functional** · useful after **nerve repair**; **not useful in mild chronic compression**

Quality vs quantity (TO) **thresholds = quality** of innervation (demyelination) · **2PD = quantity** (receptor density, axonal loss)

2 Mechanism

Stretch **8% elongation → ↓ microcirculation 50%**; **15% → ↓ 80%**; e.g. stingers, suprascapular n. in volleyball, CPN after TKA valgus correction

Compression **30 mmHg → paresthesia**, ↑ latency; **60 mmHg → complete conduction block**; endoneurial oedema → fibrosis → ↓ gliding

Chronic compression conduction change from **Schwann cell proliferation + apoptosis**; Schwann cells ↑ VEGF
✗ Wrong options: Wallerian degeneration · axon transection · cell body apoptosis

Compressive neuropathy (cubital tunnel, NCV 31 m/s) **neurapraxia of ischaemic origin**
✗ Wrong options: axonotmesis · endoneurial disruption

Laceration sharp better than crush; ends retract

Classic injury	Nerve
Anterior shoulder dislocation	Axillary
Holstein-Lewis (distal 1/3 humerus), Saturday night palsy, extension SCH	Radial
Distal humerus ORIF, flexion SCH, OR positioning	Ulnar
Extension SCH, prominent olecranon TBW pins (anterior ulnar cortex)	AIN
Posterior hip dislocation	Sciatic
TKA valgus correction	CPN
Percutaneous tibial plating, holes 11–13	Superficial peroneal

Regeneration physiology

Wallerian degeneration **distal** to injury (TO); axoplasm + myelin degraded by **phagocytes**; proximal reaction = primary/retrograde degeneration (TO)
✗ Wrong options: Schwann cells

Sequence distal Wallerian → Schwann cells proliferate, line basement membrane → **proximal budding after ~1 month → antegrade** growth **1 mm/day**
✗ Wrong options: distal → proximal · neurotrophic factors direct phagocytosis

Advancing Tinel ~**1 inch/month** (TO); **most reliable sign of recovery**

Guidance contact guidance (Schwann basal lamina), neurotropism, neurotrophic factors

Order of recovery **sympathetic → pain → temperature → touch → proprioception → motor**; loss in reverse; **motor first lost, last to return**

Myelin disrupted before axon

3 Presentation

Diagnosis clinical (injury + deficit); EMG/NCS mainstay; exploration confirms traumatic injury

– – Active space – –

High vs low palsy (TO) high = proximal to major motor branches → more paralysis but **less deformity** (antagonists lost) · low = fewer muscles, **more deformity** → **ulnar paradox**: high ulnar palsy claws less than low

Claw hand causes (TO) ulnar palsy · median palsy · combined palsy · **Volkmann ischaemic contracture** · **leprosy** · mechanism: intrinsic loss → EDC hyperextends MCP, FDS/FDP flex IP

Nerve	Autonomous sensory zone (TO)
Median	**Tip of index finger**
Ulnar	**Tip of little finger**
Radial	**First dorsal web space**

Radial palsy (Holstein-Lewis) last to recover **wrist extension**
✗ Wrong options: elbow extension: spared at this level

4 Diagnosis & Classification

Injury classification (Seddon / Sunderland)

Seddon	Sunderland	Damaged	Wallerian	Tinel advances	Recovery
Neurapraxia	**1**	Focal demyelination (ischaemia); axon intact	No	No (no motor march; simultaneous proximal + distal recovery)	**Complete**, days–6 wk
Axonotmesis	**2**	**Axon**; endo-, peri-, epineurium intact	Yes	Yes	**Full** (intact endoneurial tube)
Axonotmesis	**3**	Axon + **endoneurium**; peri-/epineurium intact	Yes	Yes	**Most variable**, incomplete
Axonotmesis	**4**	Axon + endo- + **perineurium**; **only epineurium** intact	Yes	No	**Neuroma-in-continuity**; poor without surgery
Neurotmesis	**5**	All layers (transection)	Yes	No	None without repair
–	**6 (Mackinnon)**	Mixed degrees in one nerve			Mixed; neuroma in continuity (TO)

Sunderland 2 = axon disrupted, endoneurial tube intact (TO); regeneration because **endoneurium** intact
✗ Wrong options: epineurium, perineurium, myelin, Schwann cell

Tinel progression **Sunderland II & III** (TO)

Neurotmesis range Sunderland **5** (Orthobullets classification) · **3rd–5th** (Orthobullets treatment section, OB214026) – read the options

Full recovery prevented by **endoneurial injury = neurotmesis** (structure A in figure)

	Sunderland	Seddon	Description of Injury	Recovery Period
Mildest	I	Neurapraxia	Conduction block, nerve is in-continuity, Wallerian degeneration does not take place	≤ 3 months
	II	Axonotmesis	Axon not continuous, nerve itself remains intact, axonal sprouting, Wallerian degeneration	1 inch per month
	III		During healing, excessive scarring of the endoneurium occurs that hinders axon regeneration	< 1 inch per month where it is slowed by the scar tissue; determined by degree of scarring and involved fascicles
	IV		Nerve is still in-continuity, scar build up blocks nerve regeneration	Surgical intervention required to re-establish nerve transduction by removing scar tissue and reconnecting nerve segments
Most Severe	V	Neurotmesis	Rupture of the nerve, it is no longer a continuous fiber	Recovery requires surgical intervention

Seddon / Sunderland table

Key radiology findings

High-resolution US dynamic, superficial nerves; localises entrapment, cut ends, neuromas

MRI / MR neurography brachial plexus; **better for muscle denervation**, deep structures, static resolution

Sensory receptors & electrodiagnostics

Study	Finding	Meaning
NCS	↑ latency, ↓ velocity, ↓ SNAP/CMAP	Large myelinated fibres; compression/demyelination
EMG	**Fibrillations, positive sharp waves**	**Acute denervation** (axonotmesis / neurotmesis)
EMG	Fasciculations, myokymia	Neurogenic
EMG	CRDs, myotonic discharges	Myopathy
Polyphasic MUPs		**Early reinnervation** (TO)
Chronic MUPs (large amplitude, long duration)		Established reinnervation (TO)
↓ SNAP & CMAP		Axonal loss (quantitative) (TO)
↓ / absent insertional activity		**Poor potential for surgical reinnervation** (TO)
↓ recruitment		Loss of innervated MEPs (TO)

Thresholds DSL **>3.2 ms**, DML **>4.3 ms** abnormal; velocity **<52 m/s** (Orthobullets) · **<50 m/s across elbow** (OB); amplitude ↓ 50% vs contralateral

SNAPs affected before CMAPs; EMG normal until late

Cubital tunnel NCS **prolonged distal sensory latency** + ↓ velocity
✗ Wrong options: fibrillations, PSW, fasciculations = EMG · ↑ velocity

Neurapraxia NCV slowing/conduction block, **no fibrillations**

EMG value often only objective evidence (workers' comp)

- - Active space - -

5 Management

Nerve deficit

- Closed injury / neurapraxia / Sunderland 1–2 → **observe + serial EMG (baseline 3–4 wk)**
  - no recovery by 3 mo → **explore**
- Sharp / iatrogenic / vascular injury / progressive → **early exploration + repair (7–14 d)**
- Gunshot with transection → **repair at 1–3 wk (zone of injury declares)**

Gap after resecting to healthy fascicles

- Tension-free → **primary epineurial (or fascicular) repair**
- Small sensory gap (≤20 mm) → **collagen conduit / processed allograft**
- >5 cm or motor/mixed → **autograft (sural; cable if size mismatch)**

Surgical repair

Repair timing (TO)	Window
Primary	**Within 1 week**
Delayed primary	**2–4 weeks**
Secondary	**> 1 month**

Gold standard technique (TO) — **epineural repair** · fascicular in select cases · **grouped fascicular**: median + ulnar in distal ⅓ forearm, sciatic in thigh — no proven benefit over epineural

Grafting (TO, median lectures) — gap **> 2.5 cm** → reversed autograft (**sural** commonest; saphenous, LABC) · **collagen conduit = only synthetic with good results** · no distal end → **direct muscular neurotisation** (less than normal function)

Graft threshold — gap **> 2–3 cm** (TO) · **> 5 cm** for autograft (Orthobullets) — read the options

Repair window — best **7–14 d**; unpredictable after **6 mo**; reinnervation by **18 mo** goal; time limit **18 mo**

Epineurial repair — resect proximal neuroma + distal glioma to **"mushrooming" fascicles**; align with epineurial vessels; **tension-free**; gain length by transposition/ neurolysis

Fascicular repair — perineurial sutures; outcome **= epineurial**; ↑ scarring; matching by topography, **stimulation (proximal: sensory in awake pt; distal: motor before Wallerian)**, histology

Direct muscular neurotization — stump into muscle; ↓ neuroma; **TMR, RPNI** for amputees

Timing of repair after ballistic injury

GSW with transected radial nerve — definitive repair **1–3 wk** after injury
✗ Wrong options: at initial debridement · at fracture fixation · after 3 mo · wait for neuroma

GSW without transection / closed humerus fracture observe **3 mo** (mostly neurapraxia); GSW brachial plexus → observe

Repair options – conduit, allograft & autograft by gap length

Option	Gap	Key facts
Collagen conduit	**≤20 mm** (Orthobullets, OB) · <3 cm (OB4644) · tubular wrap <1 cm (OB219234)	**Type I collagen** (= **ligaments**; not cartilage/NP II, basement membrane IV, vessels III); relies on **fibrin clot**; nutrient + neurotrophic access; **8 mm digital gap → as good as autograft** (✗ Wrong options: PGA, GTMC, silicone, end-to-end under tension)
Decellular allograft	Sensory **<3 cm**; effective up to **7 cm**	**Scaffold (microtubules, laminin) for host Schwann cell migration**; no living cells, no immunosuppression; not yet for motor/mixed
Non-processed allograft		Needs immunosuppression – rarely used
Vein conduit	Digital	not >5 cm, not superior
Autograft	**>5 cm (50 mm)**	**Gold standard**; graft **≥10% longer** than gap; sural (palm level), MABC, LABC, terminal PIN; **cable** for mismatch; donor morbidity

Quality of recovery drops with gaps **>5 mm**

Segmental defect, motor recovery **nerve autograft** best
✗ Wrong options: allograft, conduits, vein



Nerve graft reconstruction

Prognosis – nerve, level & order of recovery

Favourable **younger age (most important)**, **distal level (2nd)**, sharp / stretch injuries

Prognostic factors (TO) age · level · time to repair · **associated vascular injury** · sharp vs crush

Unfavourable older age, proximal level, crush/blast, delay, **ulnar nerve**, concomitant bone/soft-tissue injury, graft length

Best motor recovery after graft (<6 mo) **radial**
✗ Wrong options: median, ulnar, tibial, peroneal

Ulnar poor outcomes (worse than tibial, radial, femoral, MC)

✗ Wrong options: **Prognostic factor** – **type of autograft** (sural vs saphenous); ✓ time to repair, level, length, rehab

– – Active space – –

High median laceration + 6 cm sural cable graft – poorest recovery

thumb opposition (intrinsics; endplates survive ~**12 mo**)
✗ Wrong options: wrist flexion, finger flexion, FPL, sensation

Critical period of delay (TO)

Nerve	Critical period
Radial	**15 mo**
PIN	9 mo
Ulnar high / low	9 / 15 mo
Median high / low	9 / 12 mo
Sciatic	12–15 mo
CPN	12 mo

Nerve transfers – AIN to ulnar motor (SETS)

Indication proximal injury: deliver axons before **motor endplate** loss; priority shoulder abduction/ER, elbow flexion, hand

Goal	Transfer
Shoulder abduction + ER	**Spinal accessory → suprascapular**
Shoulder abduction	**Medial triceps branch (radial) → axillary** (Leechavengvong)
Elbow flexion	**Oberlin**: FCU fascicle (ulnar) → biceps branch (MC); double Oberlin adds FCR/FDS fascicle (median) → brachialis
Radial (TO)	**Median → radial**: spare **FDS / FCR fascicles → PIN + ECRB branch** · **preserve PT branch + AIN** · intra-op stimulator / wide-awake
Radial sensory (TO)	Median / ulnar web-space fascicles → radial sensory · **LABC → superficial radial n.** (↓ painful neuroma, not full sensation)
Median motor (TO)	**Brachialis branch (MC) → median n.** (arm fascicles: lateral sensory, medial motor) · **ECRB branch (radial) → AIN**
Median sensory (TO)	**Dorsal ulnar sensory fascicles → median digital territory**
FDS / ECRB (TO)	**Median nerve** runs close to nerve to ECRB → basis of **ECRB-to-FDS and FDS-to-ECRB** transfers
Ulnar intrinsics	**Terminal AIN (from PQ) → deep motor ulnar, supercharged end-to-side (SETS)**

SETS purpose **maintain finger abduction/adduction** (interossei) while repaired ulnar nerve regrows; **motor only**
✗ Wrong options: sensation

Proximal forearm ulnar laceration **ulnar repair + end-to-side AIN → deep motor branch**
✗ Wrong options: end-to-end: requires dividing deep branch · repair alone

SETS contraindication **absent preop CMAPs** (endplates gone)
✗ Wrong options: scarce fibs, intrinsic atrophy = indication, ↓ SNAP, present CMAP

Martin-Gruber 15%; some intrinsic function spared in high ulnar injury

Success ≥ M3 (many M4); easier rehab than tendon transfer; nerve transfer within **6 mo**, tendon transfer any time
✗ Wrong options: cortical re-education

Tendon transfers for irreparable palsy (radial, ulnar claw, peroneal)

Tendon transfer = salvage, not primary treatment (TO) nerve recovery unlikely · late presentation · failed nerve repair · needs supple joints, **adequate sensation**, expendable donor

Principles supple joints first; synergistic; similar power (cross-sectional area); straight line of pull; one tendon one function; **lose one MRC grade**; better <30 y, distal

Smith excursion rule	cm
Wrist flexors / extensors	**3**
EDC, FPL, EPL	**5**
FDS, FDP	**7**

Palsy	Function	Transfer
Radial	Wrist extension	**PT → ECRB**
Radial	Thumb extension	**PL (or FDS IV) → EPL**
Radial	Finger MCP extension	**FCU → EDC (Jones)** · FCR → EDC (Brand) · FDS → EDC (Boyes)
Median	Opposition	**EIP → APB (Burkhalter)** · **FDS ring (Riordan)** · **PL → APB (Camitz)** (TO)
Median	Thumb IP flexion	**BR → FPL**
Median	Index flexion	Side-to-side FDP
Ulnar	Simple claw (**+ Bouvier**)	**Zancolli lasso** (FDS around A1), MCP volar plate capsulodesis, A1/A2 release
Ulnar	**Complex claw (– Bouvier)**	**FDS → lateral bands** (or FCR + grafts)
Ulnar	Ring/small flexion	Side-to-side FDP to median FDP
Shoulder ER		Lower trapezius → infraspinatus
Elbow flexion		Steindler flexor-pronator advancement; pec major, lat, triceps
CPN foot drop	Dorsiflexion	**Tibialis posterior → dorsum of foot**

Camitz in long-standing CTS (TO) **PL → APB** via same incision
✗ Wrong options: Burkhalter, Riordan, Jones

Bouvier test block MCP hyperextension → active IP extension? **Yes = simple claw** (static procedure) · **No = complex claw** (dynamic FDS → lateral bands)

Radial palsy >12 mo, no conduction **PT → ECRB, FCU → EDC, PL → EPL**
✗ Wrong options: neurolysis, grafting, AIN → ECRB / FCR → PIN nerve transfers

Radial palsy natural history 85–90% recover in 2–3 mo; EMG at 3 mo

– – Active space – –

CPN palsy after THA, 6 mo, brace intolerant, supple	**tibialis posterior transfer** ✗ Wrong options: repeat decompression · arthrodesis: fixed deformity/ arthritis · BKA

6 Complications

Neuroma after digital amputation

Neuroma	disorganised proximal regrowth; amputees up to **30%** (Orthobullets)
Digital amputation	**<10%** symptomatic (6.6–7.8%); **>½ need revision within 1 yr**; rate higher than after neurorrhaphy ✗ Wrong options: improves spontaneously · revision after 1 yr
Symptoms	sharp/burning pain, cold intolerance, pressure hypersensitivity
Treatment	gabapentin/antidepressants, phenol/botox, rehab → resection, **TMR**, **RPNI**
Grafting	donor neuroma; allograft rejection

Metabolic neuropathy – vitamin B12

B12 deficiency	**peripheral sensory neuropathy** (✓ check B12); **↓ DTR**, Babinski, fatigue ✗ Wrong options: ↑ DTR · inability to whistle = FSHD · hydrophobia = rabies

X vs Y

	Neurapraxia	Axonotmesis	Neurotmesis
Wallerian	No	Yes	Yes
EMG fibs/PSW	No	Yes	Yes
Advancing Tinel	No	Yes (II, III)	No
Recovery	Complete, simultaneous	Variable, proximal → distal	Needs repair

	Simple claw	Complex claw
Bouvier	Positive	Negative
Extensor mechanism	Intact	Stretched
Procedure	Static: capsulodesis, Zancolli lasso, pulley release	**Dynamic: FDS → lateral bands**

How it's asked

"Cushions nerve against external pressure?"	**→ epineurium**
"Cross-section labels A–F?"	**→ axon, myelin, epineurium, perineurium, endoneurium, node of Ranvier**
"Item D in nerve cross-section?"	**→ perineurium**

"Axon disrupted, endoneurial tube intact — Sunderland?"	→ **2** (TO)
"Wallerian degeneration is?"	→ **distal to severed nerve** (TO)
"Advancing Tinel in which Sunderland?"	→ **II and III** (TO)
"Sunderland 2 regenerates because intact?"	→ **endoneurium**
"Cubital tunnel, slowed NCV — classify?"	→ **neurapraxia, ischaemic origin**
"Regeneration after axonotmesis?"	→ **proximal budding → antegrade migration**
"Chronic compression conduction change from?"	→ **Schwann cell proliferation + apoptosis**
"Injury preventing full recovery?"	→ **endoneurium (A); neurotmesis**
"8 mm digital gap — as good as autograft?"	→ **collagen conduit**
"Segmental defect — best motor recovery?"	→ **nerve autograft**
"Conduit collagen = primary collagen of?"	→ **ligaments (type I)**
"Decellular allograft purpose?"	→ **scaffold for host Schwann cell migration**
"Gap requiring autograft over conduit?"	→ **50 mm**
"Critical period, radial nerve?"	→ **15 months** (TO)
"Best motor regeneration after graft?"	→ **radial nerve**
"NOT prognostic after repair?"	→ **type of autograft**
"Holstein-Lewis palsy — last to return?"	→ **wrist extension**
"High median 6 cm graft — poorest recovery?"	→ **thumb opposition**
"GSW, transected radial nerve — repair when?"	→ **1–3 weeks**
"Contraindication to SETS?"	→ **absent preop CMAPs**
"Purpose of AIN → ulnar SETS?"	→ **maintain finger abduction/ adduction**
"Proximal forearm ulnar laceration — best early hand function?"	→ **ulnar repair + end-to-side AIN transfer**
"Long-standing CTS, opponensplasty?"	→ **Camitz (PL → APB)** (TO)
"Claw, negative Bouvier?"	→ **FDS → lateral bands**
"Radial palsy 2 yr after humerus ORIF?"	→ **PT → ECRB, FCU → EDC, PL → EPL**
"Persistent foot drop after THA, brace intolerant?"	→ **tibialis posterior transfer**
"Autonomous sensory zone of radial nerve?"	→ **first dorsal web space** (TO)
"High ulnar palsy claws less than low palsy?"	→ **ulnar paradox** (TO)
"Semmes-Weinstein monofilaments test which receptors?"	→ **slowly adapting (Merkel, Ruffini)** (TO)
"2PD reflects quality or quantity of innervation?"	→ **quantity (axonal loss)** (TO)

– – Active space – –

"Fibres for pain + temperature?"	→ **A delta** (TO)
"Polyphasic MUPs?"	→ **early reinnervation** (TO)
"Slowly adapting, static 2PD?"	→ **Merkel**
"Cubital tunnel NCS finding?"	→ **prolonged distal sensory latency**
"B12 deficiency causes?"	→ **peripheral sensory neuropathy**
"Neuroma after digital amputation – counsel?"	→ **<10%; those affected need surgery before 1 yr**

PHALANX DISLOCATIONS

– Hand · 61 / 103 –

1 Basics

PIP joint **most commonly injured joint in hand**; **dorsal > volar**

DIP usually dorsal or lateral; volar rare; often **open** (tight envelope)

Mechanism direct blow; fracture-dislocations by **avulsion** or **impaction shear**

Anatomy

Stabiliser	PIP	DIP
Proper collateral	Proximal phalanx condyle → volar base P2	P2 condyle → volar base P3
Accessory collateral	→ volar plate	→ volar plate
Dorsal	**Central slip** (dorsal base P2)	Terminal extensor (dorsal base P3)
Volar tendon	FDS (volar shaft P2)	FDP (volar base P3)
Volar plate	**Checkrein ligaments** proximally; resists **hyperextension**	no checkreins; resists hyperextension

Collaterals resist radial/ulnar deviation

Ligament box (TO) proper collateral (**primary stabiliser**, inserts volar ⅓ of P2) + accessory collateral + volar plate → **at least 2 sides must fail for PIP dislocation** · PIP collateral fails **proximally**, volar plate avulses **distally** · volar plate taut in extension resists dorsal displacement

P1 head asymmetry (TO) slight condylar asymmetry → **~9° supination** during flexion

3 Presentation

Skin puckering soft-tissue interposition

Stability tests lateral stress in **full extension** (secondary stabilisers) and **30° flexion** (collaterals); hyperextension (volar plate); **Elson** (central slip); full active ROM = stable

Collateral grades I pain, no laxity · II laxity, firm endpoint · III no endpoint

Neuro traction neurapraxia of adjacent digital nerves

4 Diagnosis & Classification

Key radiology findings

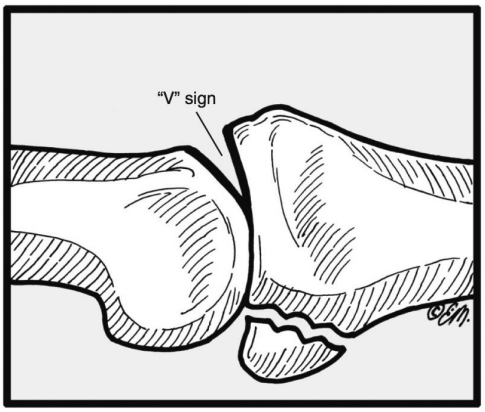
Views AP, lateral, oblique

Dorsal PIP types (TO) **I** hyperextension, surfaces still touching · **II** bayonet · **III** fracture-dislocation

– – Active space – –

Complex dorsal PIP (TO) volar plate interposed + soft-tissue **"tendon noose"** round P1 → open reduction

Dorsal "V sign" dorsal joint widening = **incomplete reduction** (TO) · = **subtle dorsal subluxation of P2** → unstable



V sign

PIP dislocation types

Type	Lesion	Reduction	After reduction
Dorsal simple	Hyperextension + axial load; collaterals torn, **volar plate** sheared ± small avulsion; P2 touches condyles	Volar force + flexion (often by patient)	Stable → **buddy tape 3–6 wk** · unstable → **extension block splint**
Dorsal complex	**Bayonet**; P2 off condyles; **volar plate blocks** traction	Hyperextend then palmar force	Open (dorsal approach between central slip & lateral band) if fails
Volar simple	**Central slip rupture**	Flexion + dorsal force at P2 base	**PIP full extension splint**, DIP + MCP free
Volar rotatory	**One collateral torn**, condyle **buttonholes between central slip & lateral band**	**MCP + PIP 90° flexion + traction** (relaxes lateral band)	Open via **dorsolateral** approach if fails
Lateral	One collateral + partial volar plate	Closed	Buddy tape; block = **lateral band**

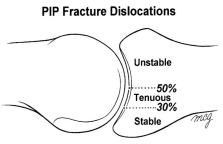
Blocks to reduction closed dorsal PIP: **volar plate** · open dorsal PIP: **FDP** · lateral: **lateral band** · closed dorsal DIP: **volar plate** (open: FDP) · volar DIP: **terminal tendon**

Volar PIP splint duration **6–8 wk** (Orthobullets) · **4–6 wk** (OB) · **6 wk** (OB) – read the options



Rotatory volar PIP dislocation

Hastings classification (PIP fracture-dislocation, % P2 articular surface)

Type	%	Stability	Image
I	**<30%**	Stable	
II	**30–50%**	Tenuous	
III	**>50%**	Unstable	

Commonest osseous injury with dorsal PIP fracture-dislocation — **middle phalanx volar (palmar) lip fracture**

Volar fracture-dislocation — hyperflexion → central slip → **dorsal lip** avulsion

Pilon — comminuted dorsal + volar lips (axial load)

5 Management

Dorsal PIP dislocation/fracture-dislocation – extension block splinting

Primary goal — **maintain joint reduction → congruent gliding arc**
✗ Wrong options: anatomic articular reconstruction · volar plate reattachment · immobilise to union

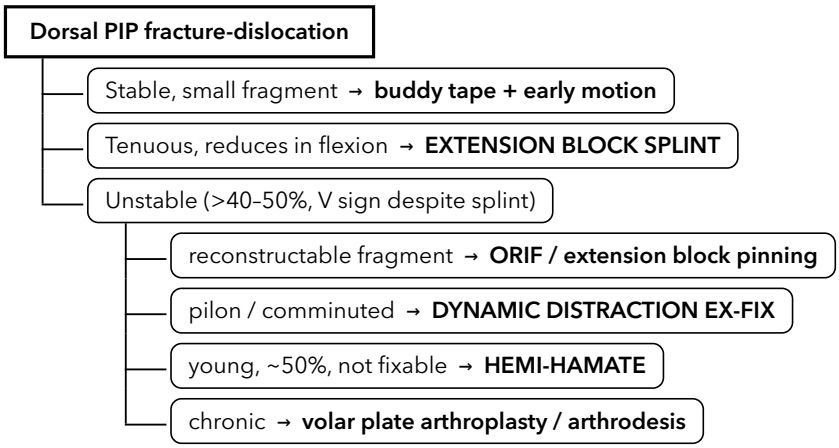
Best predictor of outcome — **congruent gliding joint** > early motion > anatomic surface

Avoid hinging — subluxation → poor outcome

Operative threshold	Source
<40% stable → closed + splint; **>40%** unstable → CRPP/ORIF	Orthobullets
<25% + reduces in flexion → EBS; **>25%** or incomplete reduction → surgery; **>50%** → open/VPA	OB
Stable <30% buddy · tenuous 30–50% EBS if reduced in <30° flexion · unstable >50% surgery	OB (Hastings)

– – Active space – –

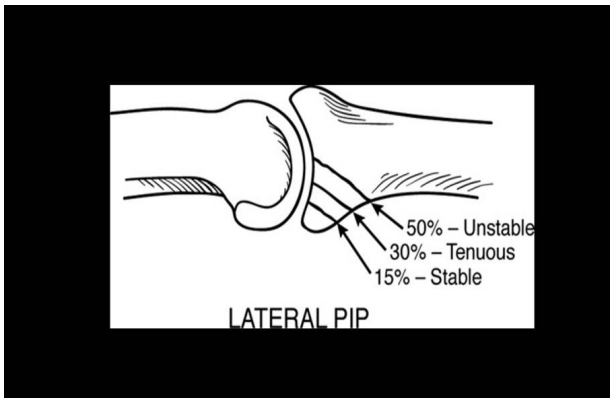
Stability threshold (TO)	**< 40% volar articular surface → usually stable** · **> 40% or comminution → unstable** (proper collateral detaches with volar fragment)
EBS indications (TO)	no incongruity · **stable in ≤ 30° flexion** · collaterals intact · start 30°, **extend 10°/wk** · avoid in **short swollen fingers** (→ extension block pinning)
Extension block pinning (TO)	K-wire **obliquely between central slip and lateral bands** → early motion
Dynamic traction (Suzuki frame, TO)	pilon / multifragment **30–50%** surface, reducible joint · **proximal–distal hook distance ≤ 2.5 cm** (avoid over-distraction)
Cerclage wiring (TO)	severe pilon with central impaction when ORIF impossible
Pilon (TO)	axial compression → dorsal + volar fragments separate, **central cartilage depression**, P2 base widening → dynamic traction or arthroplasty
Extension block splint (EBS)	stable reduced dorsal **fracture**-dislocation <40%; block **~40°**, reduce **10°/wk** over 3 wk, active flexion within splint
Redislocates only in full extension, <25% involvement	**EBS 3 wk**
Avoid cast 6 wk	stiffness



Unstable/chronic fracture-dislocations – volar plate arthroplasty, hemi-hamate & ORIF

Acute >50% volar rim, V sign at 30° flexion, young athlete	**ORIF** ✗ Wrong options: buddy · EBS · VPA in young acute
Volar plate arthroplasty	acute unstable or **chronic** dorsal FD (>40% comminuted volar surface, e.g. neglected 5 wk); excise fragments, trough, advance plate, dorsal sutures
VPA must release	**collateral ligaments** (else hinging)
VPA technique (TO)	transverse trough at P2 base · mobilise volar plate **4–6 mm distally** · release accessory collaterals · **central exit hole mandatory** (avoids tethering lateral bands) · flex DIP during suture passage
Volar exposure	resect **proximal C2, entire A3, distal C1** pulleys
Hemi-hamate	**non-vascularised osteochondral autograft with hyaline cartilage** from **hamate**; young, ~50% P2 base, chronic/irreducible ✗ Wrong options: MFC, iliac crest, fibula, distal radius

Hemi-hamate graft (TO)	from **dorsal distal hamate centred over 4th/5th CMC joints** · half dorsal-volar thickness + half radial-ulnar width · orientation restores volar buttress
Older sedentary chronic	**PIP arthrodesis**
Intra-articular proximal phalanx condylar fracture with step	**ORIF** ✗ Wrong options: buddy, splint, arthrodesis in young



Hemi-hamate reconstruction

Volar PIP dislocation – central slip injury & reduction/splinting

Most often injured (TO)	**central slip** ✗ Wrong options: sagittal bands, lumbrical, juncturae
Dorsal PIP (TO)	volar plate injured → early ROM after reduction
Volar PIP Rx	**closed reduction + full-time PIP extension splint**; DIP + MCP free (lateral band motion) ✗ Wrong options: EBS = dorsal
Untreated	**boutonnière** (3 wk later: can't extend PIP → central slip)
Open central slip repair	subacute/chronic developing boutonnière
Rotatory volar	no in-line traction (noose tightens) · no hyperextension · not always open

DIP dislocations & fracture-dislocations

Reduction	traction + dorsal pressure on P3 + flexion; I&D if open
Splint	dorsal: slight flexion 2–3 wk if unstable · volar: **extension 6 wk**
Irreducible closed dorsal DIP	**volar plate** entrapment → open reduction ✗ Wrong options: central slip, rotatory, collateral, MCP natatory/transverse ligament
Open reduction	after 2 failed attempts; FDP repair if attached to volar fragment
>40% unstable	VPA · comminuted without soft-tissue loss — arthrodesis · with NV/soft-tissue loss — amputation
Tuft fracture	no specific Rx

– – Active space – –

6 Complications

Complication	Seen in
Pseudoboutonnière (PIP flexion contracture)	Commoner after volar; usually resolves with therapy
Swan neck	Dorsal (volar plate injury)
Boutonnière	Volar (central slip)
Extensor lag	Volar
DIP stiffness	DIP
Mallet	Volar DIP dislocation

How it's asked

"Commonest osseous injury with dorsal PIP fracture-dislocation"	→ **P2 volar lip fracture**
"How many sides of the PIP ligament box must fail to dislocate?"	→ **two** (TO)
"Suzuki dynamic traction — maximum hook distance?"	→ **≤ 2.5 cm** (TO)
"Which X-ray best treated by dorsal extension block splint?"	→ **stable reduced dorsal PIP fracture-dislocation <40%**
"Stable dorsal FD, no hinging — next step"	→ **EBS + active ROM** ✗ Wrong options: cast 6 wk · VPA
"Redislocates in full extension, <25% involved"	→ **EBS 3 wk**
"Most important predictor of outcome"	→ **congruent gliding joint**
"Primary objective"	→ **maintain joint reduction**
"Intra-articular proximal phalanx fracture with step-off"	→ **ORIF**
"Neglected 5-wk dorsal FD, >40% comminuted volar surface"	→ **volar plate arthroplasty**
"Structure excised in VPA"	→ **collateral ligament**
"Chronic irreducible PIP FD, osteochondral autograft donor"	→ **hamate**
">50% volar rim, V sign despite 30° flexion splint"	→ **ORIF**
"Irreducible closed DIP dislocation — block"	→ **volar plate**
"Most often injured in volar PIP dislocation"	→ **central slip** (TO)
"Volar PIP dislocation — initial Rx"	→ **closed reduction + full-time extension splint**
"Boutonnière 3 wk after reduced volleyball injury — structure"	→ **central slip**
"Rotatory volar PIP — reduction manoeuvre"	→ **MCP + PIP 90° flexion + traction**
"Post-reduction volar PIP splinting"	→ **PIP full extension, DIP + MCP free**

PHALANX FRACTURES

– Hand · 62 / 103 –

1 Basics

Epidemiology

Frequency	commonest skeletal injuries; **10% of all fractures**
Sex	♂:♀ **2:1**
Location	**distal > middle > proximal phalanx**; **small finger** commonest (38% of hand fractures)
Mechanism by age	10–29 y sports · 40–69 y machinery · >70 y falls
Association	distal phalanx # → **nail bed injury**

Anatomy

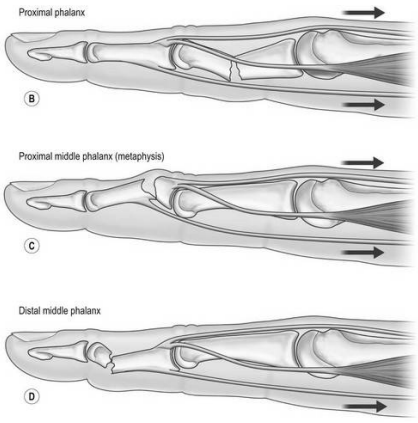
Parts	P3: tuft, shaft, base · P1/P2: head, neck, shaft, base
Insertions	terminal extensor → dorsal base P3 · **FDP → volar base P3** · central slip → dorsal base P2 · **FDS → volar shaft P2**
FDP origin	**anteromedial proximal ulna** (upper ¾ ulna, olecranon/coronoid, IOM) ✗ Wrong options: epicondyles; lumbricals arise from FDP tendons
IP joint	hinge; dynamic stability from pinch/grip compression; passive from **proper + accessory collaterals**
Digital NV	dominant artery on side **closer to midline**; nerve **volar** to artery

Deforming forces – interossei & central slip (apex volar)

Fracture	Proximal fragment	Distal fragment	Apex
P1 shaft (transverse)	**Flexed by interossei**	**Extended by central slip**	**Volar**
P2 distal to FDS insertion	Flexed by FDS	–	**Volar**
P2 proximal to FDS insertion / P2 base	Extended by central slip	Flexed by FDS	**Dorsal**
P2 with long intact FDS	–	–	No angulation
P2 neck	–	–	Volar
Seymour (P3 epiphysis)	Terminal tendon on proximal epiphysis	FDP on distal fragment	Dorsal (**mallet look**)

Proximal fragment flexor	**interossei** (Orthobullets, most OB) · "lumbrical insertions" (OB217355) – read the options
P1 base	transverse # in **proximal 6–9 mm** more stable (collaterals, capsule, intrinsics)
Reduction manoeuvre	**flex MCP** → extensor mechanism acts as tension band (**intrinsic-plus**)

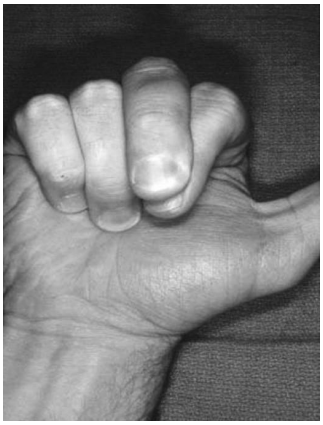
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Deforming forces in phalanx fractures

3 Presentation

Exam	swelling, deformity (angular, **rotation**, shortening), **open wounds**
Scissoring	rotational component; assess via **tenodesis**
NV	2-point discrimination; cap refill <2 s



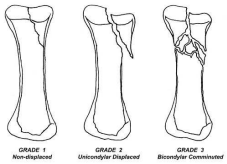
Scissoring = malrotation

4 Diagnosis & Classification

Key radiology findings

Views	**PA, lateral, oblique** of digit (orthogonal)
X-ray	P1 apex volar; P2 apex volar (distal to FDS) / dorsal (proximal to FDS)
CT	articular displacement, comminution

Condylar (head) fractures – P1 & P2

Type	Description	Image
I	**Stable**, undisplaced	
II	**Unstable unicondylar**	
III	**Unstable bicondylar / comminuted**	

Shaft & base patterns

Bone	Patterns
P1	Shaft: short oblique, **long oblique**, **spiral**, transverse · base: extra-articular, intra-articular lateral base
P2 shaft	Transverse, short oblique, long oblique, spiral
P2 base – volar	Hyperextension/axial → **volar plate avulsion**; unstable if **>40%** articular surface (Orthobullets)
P2 base – dorsal	Hyperflexion → **central slip avulsion**
P2 base – lateral	Collateral avulsion
P2 pilon	Complete articular (axial load; dorsal + volar cortex) → **unstable in all directions**
P3 tuft	**Crush**; stable (nail plate + pulp); **nail matrix/pulp laceration**
P3 shaft	Transverse, longitudinal
P3 base	**Usually unstable**: shear (axial, **>20%** surface) or avulsion (terminal tendon / FDP)

PIP fracture-dislocation stability (OB)

Volar lip (P2 base) involvement	Stability
<30%	Stable → closed reduction
30–50%	Tenuous; depends on post-reduction stability
>50%	**Unstable** → surgery
Needs **>30° PIP flexion** to hold	Unstable regardless of size

Conflict volar base unstable if **>40%** (Orthobullets) · **>30% / >50%** bands (OB) – read the options

– – Active space – –

5 Management

P1 / P2 extra-articular

- <10° angulation + <2 mm shortening + no rotation → buddy tape / splint 3 wk → **aggressive ROM**
- >10° · >2 mm · ANY rotation · open → **CRPP or ORIF**
- Unstable pattern (spiral, long oblique, comminuted) → **fixation**
  - needs early ROM (e.g. musician) → **ORIF (lag screws ± plate)**

Operative criteria

Bone	Nonoperative	Operative
P1 & P2	Extra-articular **<10° angulation, <2 mm shortening, no rotation**; undisplaced intra-articular	**>10° angulation**, **>2 mm shortening**, **any rotation**, displaced intra-articular, unstable/irreducible, **grossly contaminated open**
P3	Most: closed reduction ± splint (nail matrix may block reduction)	Displaced/irreducible shaft · **dorsal base >25%** · **displaced volar base with large FDP fragment** · nonunion · nail bed injury → remove nail, repair bed, replace nail (keeps eponychial fold)

EXCEPT trap **5° apex dorsal** angulation not an indication (atypical, <10°)

Operative indications & fixation methods

Pattern	Fixation
Transverse / short oblique P1	**Two 0.045-in K-wires longitudinally through flexed MCP (metacarpal head)** or PIP; single wire risks rotational malunion
Very proximal P1	**Eaton-Belsky** pinning through metacarpal head
Long oblique / spiral P1	**Open reduction + lag screws (1.3 mm > 1.5 mm), radial approach**; CRPP ≈ lag screws long term
Oblique extending to condyles	**Lag screw** fixation
Unicondylar / bicondylar head	Unstable → **ORIF**
Comminuted bicondylar / comminuted shaft	**Plate**, radial approach (↓ adhesions)
P2	Crossed K-wires, extension-block pinning, collateral recess pinning, mini-fragment, volar plate arthroplasty, **hemi-hamate (>50% articular)**
P3	Longitudinal/crossed K-wires, extension-block, lag screw
Intramedullary screw	Shaft #s; not too-proximal P1

Plate **allows early ROM** (main reason for ORIF) but **higher complications** (52% in Guerrero; hardware removal, tenolysis/capsulotomy 30%) than CRPP/lag screw; one RCT: plate via mid-lateral → ↑ TAM

Distal phalanx nonunion ORIF **high risk of symptomatic implant** → removal

Nonoperative care – immobilisation & early motion

– – Active space – –

Position **intrinsic-plus (MCP flexion)** → prevents MCP stiffness
✗ Wrong options: MCP + IP extension

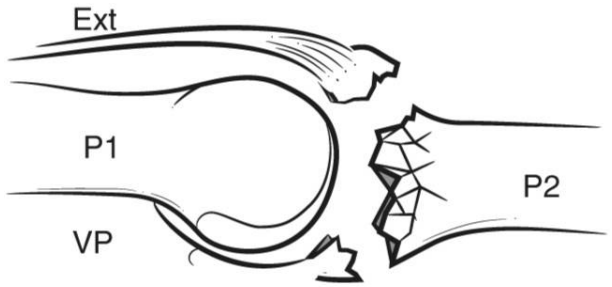
Duration **3 wk immobilisation → ROM** (Orthobullets, OB) · buddy-tape 4 wk (Meals, OB)
✗ Wrong options: immediate ROM without immobilisation → displaces

Nonop P1 in young <60% active motion maintained

Intra-articular fractures & avulsions (pilon, volar lip)

Injury	Treatment
Bony mallet	Extension splint 6–8 wk
DIP fracture-dislocation, small articular piece	Closed reduction ± splint
Undisplaced P3 #	None / Alumafoam for comfort
Unicondylar P1 head	**ORIF**
Dorsal PIP fracture-dislocation, congruent + stable 30–90°	**Extension-block splint**
Dorsal PIP # -dislocation, subluxed	Extension-block pinning / pin volar fragment / ORIF
Large volar articular piece	**ORIF** (buttress plate / screws)
Pilon P2 base	**Dynamic external fixator** (✗ Wrong options: static)
Unstable dorsal PIP subluxation, comminuted volar lip / malunion with arthrosis	**Hemi-hamate arthroplasty** (needs intact dorsal cortex)
Volar fracture-dislocation (dorsal lip #)	Extension splint, CRPP, transfixion, mini/ hook plate, screws

HHA not for pilon or dorsal lip # (dorsal cortex not intact)



Pilon fracture of P2 base

Paediatric phalangeal neck/condylar fractures

Malunion blocking subcondylar fossa (7 y, PIP 40° arc, stable) → **ostectomy** of subcondylar fossa
✗ Wrong options: observe: P1 physis proximal → no remodelling · osteotomy: AVN of small fragment · stretching/dynamic splint: bony block

– – Active space – –

6 Complications

Loss of motion

Most common complication (incl. after ORIF PIP fracture-dislocation) — risk: prolonged immobilisation, intra-articular #, extensive dissection → aggressive therapy → surgical release

Malunion & shortening – PIP extensor lag

Apex volar P1 malunion — effectively **shortens extensor → PIP extensor lag**
✗ Wrong options: MCP motion · PIP flexion · scissoring without rotation

Number	Effect
1 mm P1 shortening	**12° PIP extensor lag**
16° apex volar	**10° extensor lag**
<15° apex volar	Loss of extension only
>25° apex volar	Flexion + extension lost

Site → lag — **P1 shortening → PIP lag**; P2 distal to FDS → DIP lag; P2 proximal to FDS no PIP lag; MC neck no lag

Treatment — symptomatic → **corrective osteotomy at malunion** (preferred) · metacarpal osteotomy (limited) · soft-tissue release

Nonunion — **<2%**, atrophic, bone loss/NV compromise → resection, graft, plate; ray amputation/fusion

X vs Y

	P1 shaft	P2 distal to FDS	P2 proximal to FDS / base
Apex	**Volar**	Volar	**Dorsal**
Malunion lag	**PIP**	DIP	No PIP lag

	CRPP	Lag screws	Plate
Pattern	Transverse/short oblique; long oblique	**Long oblique / spiral**	Comminuted, bicondylar
Early ROM	Less	✓	✓
Complications	Low	Low	**Highest** (stiffness, hardware)

How it's asked

"Cause of apex palmar P1 deformity?" — **→ central slip on distal fragment + interossei on P1 base**
✗ Wrong options: swan neck · boutonnière · intrinsic fibrosis

"Interossei and central slip deform P1 how, respectively?" — **→ proximal flexion, distal extension**

"True of ORIF of distal phalanx nonunion?"	→ **high risk of symptomatic implant**
"Optimal fixation, long oblique P1?"	→ **open reduction + 1.3 mm lag screws, radial approach**
"Oblique P1 into condyles, can't move PIP/DIP?"	→ **reduction + lag screw**
"P1 operative indications EXCEPT?"	→ **5° apex dorsal angulation**
"Long oblique P1 with rotation — best outcome, fewest complications?"	→ **CRPP** ✗ Wrong options: plate: more complications
"Indication for surgery after reduced P1 fracture?"	→ **5° rotational deformity**
"Why plate instead of cast/CRPP in a violinist?"	→ **need for early ROM**
"Undisplaced transverse phalanx fracture — treatment?"	→ **immobilise with MCP flexed**
"Reduced P1 fracture, no rotation/shortening/ angulation?"	→ **short immobilisation, ROM at 3 wk**
"17° volar malunion — which movement lost?"	→ **PIP extension**
"Untreated angulated P1 fracture — sequela?"	→ **loss of PIP extension**
"Shortened healing causing PIP extensor lag — which bone?"	→ **proximal phalanx**
"Child, PIP motion blocked by subcondylar malunion?"	→ **ostectomy**
"Acute pattern best for ORIF?"	→ **unicondylar phalangeal head fracture**
"Dynamic external fixator — which fracture?"	→ **pilon fracture of P2 base**
"Origin of tendon avulsing P3 volar base?"	→ **anteromedial proximal ulna (FDP)**
"Commonest complication of ORIF of PIP fracture-dislocation?"	→ **loss of motion**
"Unstable dorsal PIP subluxation — hemi-hamate for which film?"	→ **dorsal fracture-dislocation with large volar lip fragment** ✗ Wrong options: pilon · dorsal lip · pure dislocation

– – Active space – –

PHYSICAL EXAM OF THE HAND

– Hand · 63 / 103 –

1 Basics

Digital nerve anatomy

Median nerve after TCL splits into smaller **lateral** + larger medial portion

Lateral portion **recurrent motor branch** + **3 proper palmar digital nerves**: radial thumb, ulnar thumb, **radial index → one common branch off median nerve**
✗ Wrong options: radial nerve origin for any of them

Thumb–index pattern (Jolley)	%
Type 1: radial digital thumb + common digital to 1st web (→ radial index + ulnar thumb)	**69**
Type 2: common digital to thumb (→ radial + ulnar thumb) + proper radial index	6
Type 3: **trifurcation** of 3 proper digital nerves	25

Nerve vs artery **digits: nerve VOLAR to artery** ("it hurts before it bleeds"); **palm: nerve DORSAL** (superficial arch volar)

Radial digital n. index (Lourie) at intersection of central index line with **proximal palmar crease**

Dorsal branches proper digital n. at base of P1 → dorsal branch → supplies dorsum of distal phalanx (joins superficial radial)

3 Presentation

Inspection & palpation

Finding	Meaning
Erythema / white / blue-purple / black spots	Cellulitis / arterial insufficiency / venous congestion / melanoma
Trophic change (hair, sweat)	Sympathetic derangement
Thenar atrophy	**Median** (CTS)
Interossei atrophy	**Ulnar** (cubital tunnel, cervical radiculopathy)
Subcutaneous atrophy	Post-steroid injection
Cascade sign	Flexed fingers converge on **scaphoid tubercle**; non-convergence → **malrotation**
Warm / cool	Infection / vascular

Normal range of motion

Joint	Arc
MCP	0–85°
PIP	0–110°
DIP	0–65°
Wrist	60° flexion · 60° extension · **50° radioulnar deviation arc**

Neurovascular exam

Nerve	Motor test
Radial	**Thumb IP extension** against resistance
Median – recurrent motor	**Palmar abduction** of thumb
Median – AIN	Thumb IP + index DIP flexion (**OK sign**)
Ulnar	**Cross fingers** / abduct against resistance

Vascular radial + ulnar pulse, **Allen's**, capillary refill; **sensation** – 2-point discrimination

4 Diagnosis & Classification

Special tests – wrist & DRUJ

Test	Pathology
Watson (scaphoid shift)	**SL instability** – thumb on distal pole, radial→ulnar deviation; dorsal pain/**clunk**
Reagan (LT ballottement)	**LT instability** – pisotriquetral unit vs lunate; laxity + pain
Kleinman shear	LT instability
Lichtman	**Midcarpal instability** – axial load, radial→ulnar deviation; **clunk in ulnar deviation**
TFCC grind / **ulnocarpal abutment**	TFCC tear / ulnar impingement – ulnar deviation + axial load; pain, pop/click
Fovea sign	TFCC or **ulnotriquetral ligament split tear**
ECU snap	ECU instability
Piano key	**DRUJ instability**

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Special tests – hand

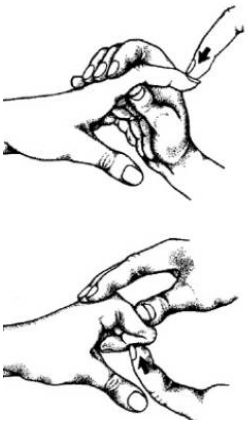
Test	Pathology / technique
Grind	**Thumb CMC** arthritis – axial load + rotate 1st MC; pain, crepitus, instability
Finkelstein	**De Quervain** – fist over thumb + ulnar deviation; pain 1st compartment
FDP test	Hold MCP + PIP extended → flex DIP
FDS test	Hold other fingers fully extended (blocks FDP) → tested finger flexes PIP
Gamekeeper's	**Thumb UCL** – radial stress in flexion + extension; **>30° laxity** or gross vs other side
Tinel (CTS)	Percuss distal palmar crease midline → median paraesthesia
Phalen (CTS)	Gravity wrist flexion, **max 2 min**
Froment (ulnar)	Paper key pinch → **thumb IP flexes**
Wartenberg (ulnar)	Fingers adducted + extended → **small finger drifts into abduction**
Jeanne (ulnar)	Key pinch → **thumb MCP hyperextension**



Froment sign

Intrinsic tightness test (Bunnell)

Technique	passively flex PIP with **MCP extended**, then with **MCP flexed**
Intrinsic tightness (positive)	↓ **PIP flexion with MCP extended** (lateral bands tensioned); PIP flexes easily with MCP flexed
Extrinsic tightness	↓ PIP flexion with **MCP flexed**; flexes easily with MCP extended



Bunnell intrinsic tightness test

5 Management

Functional position of the hand (TO)

Functional position (TO) **wrist 20° extension · MCP 70° flexion · fingers 10° flexion**
✗ Wrong options: wrist neutral / flexed · MCP 20°

Position of safe immobilisation (POSI) (TO)	Range
Wrist	**10–45° extension**
MCP	**60–90° flexion**
IP joints	Slight flexion → full extension
Thumb	**45° abduction**

X vs Y

	Intrinsic tightness	Extrinsic tightness
PIP flexion with MCP **extended**	↓	Easy
PIP flexion with MCP **flexed**	Easy	↓

How it's asked

"Positive intrinsic tightness test?"	→ **↓ PIP flexion with MCP extended**
"Numb both sides of thumb + radial index – nerve origin?"	→ **one common branch off median nerve** ✗ Wrong options: radial nerve
"Digital nerve relation to artery in the finger?"	→ **volar** (dorsal in palm)
"Functional position of wrist and hand?"	→ **wrist 20° extension, MCP 70° flexion, fingers 10° flexion** (TO)
"Clunk in ulnar deviation under axial load?"	→ **midcarpal instability (Lichtman)**
"Piano key sign?"	→ **DRUJ instability**
"Gamekeeper's positive threshold?"	→ **>30° laxity**

– – Active space – –

PIN COMPRESSION SYNDROME

– Hand · 64 / 103 –

1 Basics

Definition compressive neuropathy of **PIN** (deep motor radial n.) → extensor compartment weakness, **no sensory loss**

Incidence ~3/100,000/yr; manual labourers, ♂, bodybuilders

Causes repetitive pronosupination microtrauma · trauma (Monteggia, radial head) · **space-occupying (lipoma = commonest lesion, ganglion)** · **RA radiocapitellar synovitis** · iatrogenic

Anatomy – arcade of Frohse, innervated muscles & recovery order

Origin radial n. splits at **anterior radiocapitellar joint**, just proximal to supinator

Course between 2 heads of **supinator**, around **radial neck**, over APL origin → posterior interosseous membrane → **floor of 4th compartment** (dorsal wrist capsule sensation)

Cutaneous **none**

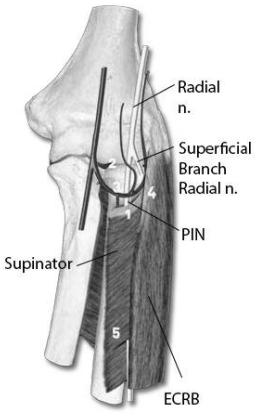
PIN innervates	Not PIN
Supinator, EDC, EDM, ECU, APL, EPB, EPL, EIP (± ECRB)	**ECRL, BR** (± ECRB) = radial n. proper; **APB** = median recurrent motor

EMG fibrillations **EPL / supinator / APL**
✗ Wrong options: any set with ECRL, BR, APB

Last to recover **EIP** (most distal)

5 sites of compression	Note
Fibrous bands anterior to radiocapitellar jt	Between brachialis & BR
Leash of Henry	Recurrent radial vessels at radial neck
ECRB edge	Medio-proximal
Arcade of Frohse	Proximal edge superficial supinator – **commonest spontaneous site**
Distal supinator edge	

Not lateral intermuscular septum compresses **radial nerve** proper, not PIN



Radial tunnel compression sites

3 Presentation

Symptoms	insidious; forearm/wrist pain (site-dependent; just distal to lateral epicondyle = arcade); **weak finger MCP, thumb, wrist extension**
Wrist	extends with **radial deviation** (ECRL intact, ECU lost); can't extend in neutral/ulnar deviation
Thumb	**can't extend thumb** (EPL)
Preserved	PIP extension (intrinsics, ulnar) · thumb flexion (AIN) · sensation
Resisted supination	↑ pain
Tenodesis	**normal** ✗ Wrong options: tendon rupture

4 Diagnosis & Classification

Diagnosis	clinical

Key radiology findings

X-ray / MRI	not routinely needed; **MRI** to locate a **mass** + plan resection
EMG	level of compression, rule out other neuropathy

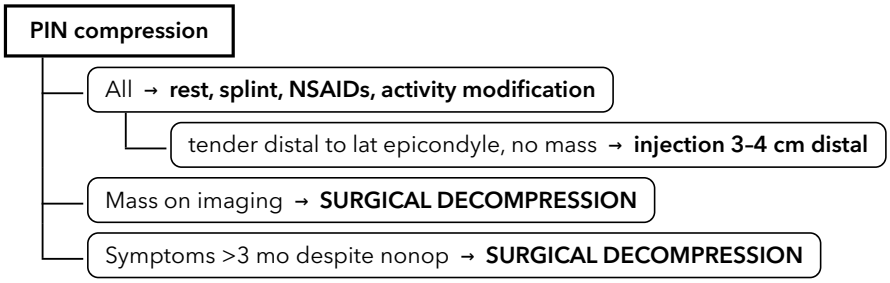
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Differential – radial tunnel syndrome & rheumatoid extensor rupture

Condition	Key finding
PIN syndrome	Pain distal to lateral epicondyle (mobile wad) + **extensor weakness** + **intact sensation**
Radial tunnel syndrome	Mobile wad pain, **no motor deficit**
RA extensor tendon rupture	**Abnormal tenodesis**
RA PIN palsy	**Peri-elbow synovitis** → normal tenodesis, can't initiate or hold extension
Sagittal band rupture	Can't initiate extension from fist but **can maintain** after passive extension
Wartenberg	Superficial radial n. → dorsal thumb numbness, normal motor
C7 radiculopathy	Sensory loss, triceps + wrist flexor weakness
Intersection syndrome	1st/2nd compartment crossing pain

Others brachial plexus, peripheral neuropathy

5 Management



Approach **anterolateral elbow** (commonest) or posterior

Release brachialis–BR bands → leash of Henry → ECRB edge → arcade of Frohse + distal supinator

Outcomes variable; spontaneous motor recovery **75–97%** (non-traumatic); improves up to **18 mo**

6 Complications

Neglected fibrosis of PIN muscles → **tendon transfers**

Chronic pain

How it's asked

"EMG fibrillations in PIN compression"	→ **EPL / supinator / APL**
"Commonest site of spontaneous PIN entrapment"	→ **arcade of Frohse** ✗ Wrong options: lateral intermuscular septum
"Lesion at arcade of Frohse – exam finding"	→ **can't extend thumb** ✗ Wrong options: can't flex thumb · PIP extension lost

"Last muscle to recover"	→ **EIP**
"RA, can't extend fingers, normal tenodesis, wrist radially deviates"	→ **peri-elbow synovitis compressing PIN**
"Exam corroborating PIN syndrome"	→ **pain distal to lateral epicondyle, extensor weakness, intact sensation**

– – Active space – –

– – Active space – –

PISIFORM FRACTURE

– Hand · 65 / 103 –

1 Basics

Epidemiology & anatomy

Incidence	**<1% of carpal fractures**; rare, often missed
Associated	**50% isolated**, 50% with other carpal or distal radius fractures
Pisiform	pea-shaped **sesamoid within FCU**; most **ulnar & palmar** bone of proximal row; stabilises ulnar column (prevents triquetral subluxation)

2 Mechanism

Mechanism	**direct impact / FOOSH**; or sudden **FCU contraction**

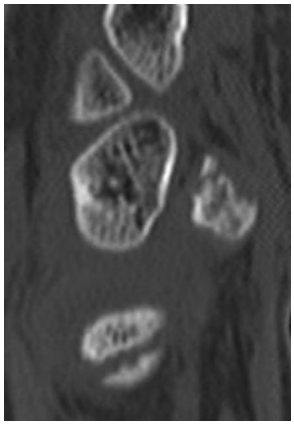
3 Presentation

Symptoms	**ulnar-sided wrist pain** after fall, ↓ grip
Exam	**hypothenar tenderness** + swelling; rule out other carpal / DR injuries

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray	PA, lateral, ER oblique, carpal tunnel; **best seen on ER oblique or carpal tunnel view** (routine views miss it)
CT	Delineates pattern, guides treatment; shows nonunion
MRI	Suspected fracture with negative X-rays → **marrow oedema**



Pisiform fracture on CT

Differential	hamate body fracture · **hook of hamate fracture** · FCU tendonitis

5 Management

Pisiform fracture

- Acute → **SHORT ARM CAST 30° FLEXION + ULNAR DEVIATION, 6–8 WK**
- Severely displaced symptomatic / painful nonunion → **PISIFORMECTOMY**

Symptomatic nonunion – pisiform excision

Cast position	**30° wrist flexion + ulnar deviation**, 6–8 wk (relaxes FCU)
Painful nonunion / pisotriquetral incongruity / arthritis	**pisiform excision** ✗ Wrong options: ORIF pisiform · hook of hamate excision · more splinting · SL repair · LT fusion
Outcome	reliable pain relief; **no loss of wrist flexion strength**, grip or motion (75% complete relief)

6 Complications

List	malunion · **nonunion** · chronic ulnar wrist pain · ↓ grip

X vs Y

	Pisiform fracture	Hook of hamate fracture
Best view	ER oblique / carpal tunnel	Carpal tunnel / CT
Symptomatic nonunion	**Pisiform excision**	Hook excision
Bone	Sesamoid in FCU	Radial border of Guyon's canal

How it's asked

"Ulnar wrist pain 6 mo after fall, pisiform nonunion on CT – treatment?"	→ **excision of pisiform** ✗ Wrong options: ORIF pisiform
"Hypothenar pain 9 mo, splint failed, displaced pisiform fracture?"	→ **pisiform excision**
"Best view?"	→ **ER oblique / carpal tunnel view**
"Cast position?"	→ **30° flexion + ulnar deviation**

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POLAND SYNDROME

– Hand · 66 / 103 –

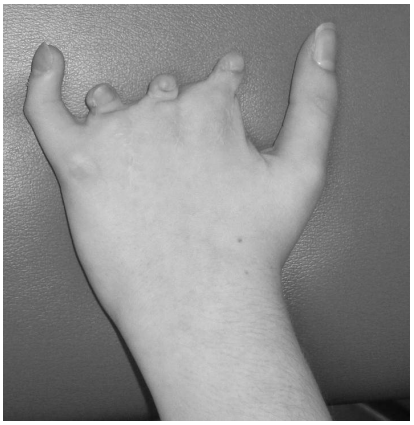
1 Basics

Definition & epidemiology

Triad	**unilateral chest wall hypoplasia (absent sternocostal head of pectoralis major)** + ipsilateral hand/forearm hypoplasia + **symbrachydactyly**
Incidence	**1 in 32,000** live births; ♂ **3×**; **right side ~75%**; sporadic (no family history)
Cause	**subclavian artery hypoplasia / disruption in utero**
Syndactyly link	Poland = **10% of syndactyly** cases

Associations – syndactyly & absent pectoralis major

Most common hand malformation — **symbrachydactyly** = short central digits (absent/short **middle phalanx**) + syndactyly ± nubbins



Symbrachydactyly

Syndactyly type — **simple complete** of short digits (Orthobullets) · usually **simple incomplete** (Ireland) – read the options; can involve all digits ± thumb

Syndactyly associated	Not associated
Poland (chest wall + limb hypoplasia)	**Tay-Sachs**
Apert (AD, craniosynostosis, complex syndactyly)	
Holt-Oram (heart–hand)	
Carpenter (acrocephaly)	

Not seen in Poland — **polydactyly**, **macrodactyly**, **lower-limb** anomalies (pes planus, accessory navicular)

3 Presentation

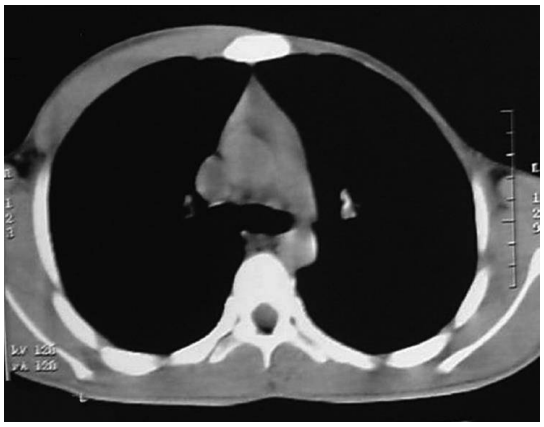
Chest — hypoplasia/absence pec major, pec minor, deltoid, serratus anterior, external oblique, latissimus; **breast/nipple aplasia**; hypoplastic ribs/scapula; **Sprengel**; scoliosis; **dextrocardia**; renal/GU anomalies

Hand / forearm	syndactyly, hypoplastic/absent MC or phalanges, absent flexors/extensors, carpal coalition, **radioulnar synostosis** (no forearm rotation), nail agenesis

4 Diagnosis & Classification

Key radiology findings

CT chest	**absent pectoralis major** / unilateral chest wall hypoplasia
X-ray hand	hypoplastic digits, short/absent middle phalanges, malformed MCs, absent carpal ossification



CT chest: absent pectoralis major

✗ CT distractors	**pectus excavatum** (Marfan), **pectus carinatum**, rib osteochondroma
Classification (Catena)	severity ↑ with more proximal upper-limb involvement

5 Management

Syndactyly release	most patients, **early (by ~1 y)**; some need revision releases or 3-finger hand
Technique	complete release → **skin graft** needed; **one side of a digit at a time** (vascular safety); local commissure flap + interdigitating **zigzag dorsal/palmar flaps**
Other	**first web release** (opposition), toe-to-hand transfer; chest wall reconstruction / muscle transfer near **sexual maturity**
Outcome	function improves but **hand stays hypoplastic**; deficits persist vs idiopathic syndactyly (Kramer)

6 Complications

After release	skin graft failure, excessive tension, poor flap planning, **digital artery injury**, **web creep**, nail deformity

– – Active space – –

X vs Y

	Poland	Apert
Syndactyly.....408	**Simple**, short central digits	**Complex**, bilateral
Other	Absent pec major, unilateral	**Craniosynostosis**, AD
Side	Unilateral (R)	Bilateral

How it's asked

"Syndactyly associated EXCEPT?"	→ **Tay-Sachs** ✗ Wrong options: Apert, Poland, Holt-Oram, Carpenter
"Hand malformation most associated with Poland?"	→ **symbrachydactyly** ✗ Wrong options: Apert complex syndactyly · constriction band · radial club hand · thumb polydactyly
"Absent pec major in 3-year-old – associated?"	→ **syndactyly** ✗ Wrong options: polydactyly · macrodactyly · foot anomalies
"Symbrachydactyly + forearm lacks rotation – which CT?"	→ **unilateral absent pec major / chest wall hypoplasia** ✗ Wrong options: pectus excavatum/ carinatum
"Symbrachydactyly X-ray – also found?"	→ **absence of pectoralis major** ✗ Wrong options: craniosynostosis · Lisch nodules · ash-leaf · leukocoria

POLYDACTYLY OF HAND

– Hand · 67 / 103 –

1 Basics

Definition extra digit; **most common congenital hand malformation**; diagnosis clinical

Form	Digit	Ethnicity
Preaxial	Thumb (radial)	**Caucasian** > African American
Postaxial	Small finger (ulnar)	**10× African American** (1.2% vs 0.09%)
Central	Index/long/ring	Often with **syndactyly**

Genes & associations

Preaxial usually **unilateral, sporadic**; ✓ M > F; **Type VII (triphalangeal) → Holt-Oram, Fanconi anaemia**, Blackfan-Diamond, imperforate anus, cleft palate, tibial defects

Preaxial spectrum triphalangeal thumb, preaxial polysyndactyly, **mirror hand** = AP-axis limb-bud error

Postaxial **autosomal dominant** (African American); sex-linked recessive modifier hypothesised

Postaxial in Caucasian complex genetics → work up for **chondroectodermal dysplasia / Ellis-van Creveld**

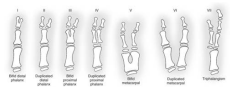
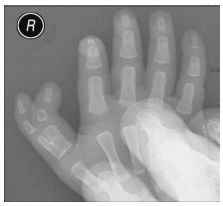
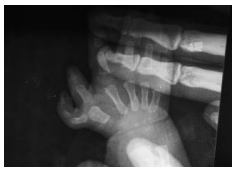
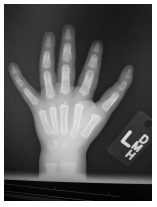
Pollex abductus abnormal **EPL–FPL connection**, ~20% hypoplastic/duplicated thumbs; abducted digit + **absent IP crease**

Look-alikes constriction band = pseudosyndactyly (post-formation injury) · syndactyly → **Apert, Poland** · macrodactyly → **NF, Klippel-Trenaunay-Weber**

- - Active space - -

4 Diagnosis & Classification

Wassel classification (preaxial)

Type	Level	Image
I	Bifid distal phalanx	
II	Duplicated distal phalanx (**2nd commonest, 15%**)	
III	Bifid proximal phalanx	
IV	**Duplicated proximal phalanx (commonest, 43%)**	
V	Bifid metacarpal	
VI	Duplicated metacarpal	
VII	**Triphalangia**	

Postaxial classification

Type	Description	Treatment
A	Well-formed digit	Formal reconstruction (type 2 combination): **keep radial digit**, reconstruct collaterals from ulnar remnant
B	Rudimentary skin tag	**Tie off in nursery / amputate <1 y**

5 Management

Preaxial (thumb) duplication – Wassel classification & reconstruction

Goals thumb **80% of contralateral size**; resect smaller (usually **radial**) thumb; preserve/reconstruct **UCL (ulnar collateral)** for pinch; one-stage

Procedure	Wassel type	Technique
Bilhaut-Cloquet (type 1 combination)	**I, II, III**	Remove central bone + nail, join halves; **~20% late deformity** (stiffness, angulation, growth arrest, nail)
Type 2 combination	**III, IV** (V–VI need intrinsic/collateral transfer)	Keep one skeleton + nail, augment with soft tissue, **ablate radial** digit
Type 3 combination (on-top plasty)	V, VI, VII	Segmental distal transfer
Triphalangeal (VII)	VII	Excise vs fuse extra phalanx

Equal-size type IV excise **radial** thumb → keep ulnar thumb's **MCP UCL**

Ulnar-side scar after resection **ulnar (valgus) MCP instability**, weak pinch
✗ Wrong options: dorsal/volar/radial instability · EPL loss

Principles (TO) use parts of both digits · repair collateral ligament · reinsert tendons · adequate web · cover articular surfaces

Triphalangeal thumb (TO) five-fingered hand, **delta phalanx**, unstable MCP → web creation, delta excision, **pollicisation**

Central polydactyly

Rx **early** osteotomy + ligament reconstruction → prevent angular growth deformity



Postaxial polydactyly

– – Active space – –

X vs Y

	Preaxial	Postaxial
Ethnicity	Caucasian	**African American**
Genetics	Sporadic (VII syndromic)	**AD**
Syndrome work-up	Triphalangeal → Holt-Oram, Fanconi	**Caucasian → Ellis-van Creveld**
Excise	**Radial** thumb	**Ulnar** digit

How it's asked

"Bilhaut-Cloquet indicated for which Wassel type?"	→ **Type II** (I–III) ✗ Wrong options: IV–VII
"Which image is preaxial?"	→ **duplicated thumb**
"Ulnar-side thumb resection scar, weak pinch — finding?"	→ **ulnar instability of thumb MCP** (UCL)
"Caucasian with this anomaly — work up for Ellis-van Creveld"	→ **postaxial polydactyly**
"More common in African Americans"	→ **postaxial polydactyly**
"Inheritance of postaxial polydactyly"	→ **autosomal dominant**
"Commonest Wassel type"	→ **IV**

PREISER'S DISEASE (SCAPHOID AVN)

– Hand · 68 / 103 –

1 Basics

Definition	**idiopathic AVN of scaphoid**; Preiser 1910; not fracture
Epidemiology	rare, **<1/100,000/y**; mean age **45 y**
Ranking	scaphoid = **2nd commonest carpal AVN** after lunate (Kienböck)

3 Presentation

Symptoms	chronic, progressive **dorsoradial wrist pain**, atraumatic

4 Diagnosis & Classification

Key radiology findings

X-ray	**sclerosis + fragmentation of proximal pole**, **no fracture**
MRI	complete vs partial involvement; STIR marrow oedema, cysts, partial collapse; contrast MRI classifications (Kalainov, Schmitt)



Preiser's: proximal pole sclerosis

Herbert & Lanzetta classification (1994)

Stage	X-ray
1	Normal X-ray, **+ bone scan**
2	↑ density proximal pole, generalised osteoporosis
3	**Fragmentation** proximal pole ± fracture
4	**Carpal collapse** ± OA

- - Active space - -

5 Management

Preiser's disease

- Young, Herbert 1–2, minimal sx → **NSAIDs + IMMOBILISATION (effective 20%)**
- Fails nonop, stage 1 → **ARTHROSCOPIC RESECTION / drilling**
- Stage 2 → **VASCULARISED BONE GRAFT / revascularisation**
- Stage 3 (collapse) → **PRC or scaphoid excision + 4-corner fusion (salvage)**

Other options microfracture drilling, allograft replacement

X vs Y

	Preiser's	Kienböck's	Trapezium AVN
Bone	**Scaphoid**	**Lunate**	Trapezium
Frequency	2nd commonest	Commonest	Exceedingly rare

How it's asked

"Atraumatic radial wrist pain, scaphoid sclerosis/ collapse on X-ray & MRI?"	→ **Preiser's disease** ✗ Wrong options: Kienböck's · acute SL injury · basilar thumb OA · trapezium AVN
"Herbert stage 1?"	→ **normal X-ray, positive bone scan**
"Salvage for collapse?"	→ **PRC or 4-corner fusion**

PRONATOR SYNDROME

– Hand · 69 / 103 –

1 Basics

Epidemiology

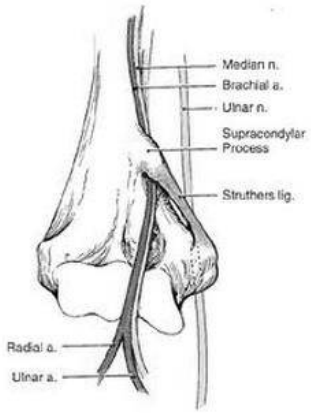
Definition	compressive neuropathy of **median nerve proper at the elbow / proximal forearm** (proximal median nerve entrapment, PMNE)
Incidence	**rare, <1/100,000/y**
Demographics	♀ > ♂; **5th decade**
Risk	**well-developed forearm muscles** (weightlifters, climbers, MMA / jiu-jitsu); repetitive pronosupination
Association	**medial epicondylitis**

Compression sites

Site	Note	Provocative test
Supracondylar process	Osseous remnant distal humerus, **1%**	Elbow X-ray
Ligament of Struthers	Supracondylar process tip → medial epicondyle (✗ Wrong options: arcade of Struthers = ulnar)	–
Lacertus fibrosus (bicipital aponeurosis)	Biceps tendon → deep volar forearm fascia	**Resisted elbow flexion + supination**
Between humeral & ulnar heads of pronator teres	**Most common site**	**Resisted pronation with elbow extended**
FDS aponeurotic arch	Rare; FDS = medial epicondyle → volar P2 digits 2–5	**Resisted FDS of middle finger**
Gantzer muscle (accessory FPL head)	Listed by Adler & Wolf (OB)	–
Palmaris profundus, FCR brevis (TO)	Rare	–

Distances (TO)	PT heads **3–7.5 cm** distal to humeral epicondylar line · FDS arch **~6.5 cm** distal
Branching (TO)	**PT first** (around elbow) → **ulnar-side** branches to PL, FCR, FDS (two) → **AIN = only branch from the RADIAL side** (FPL, FDP index + middle, PQ) → localises median lesions
Pronator teres attachments	humeral head **medial epicondyle** + ulnar head **coronoid** → **mid-lateral radial shaft**

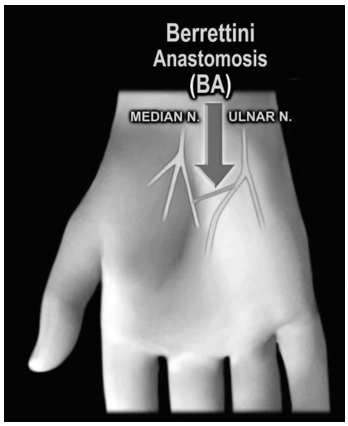
– – Active space – –



Compression sites of median nerve at elbow

Median–ulnar anastomoses (Berrettini)

Anastomosis (TO)	Site	Fibres / direction	Clue
Martin–Gruber	Forearm, distal to medial epicondyle	**Median/AIN → ulnar**, motor	Intrinsics work despite ulnar lesion
Marinacci	Forearm	**Ulnar → median** ("reverse MG"); **rarest**	Ulnar stimulation at elbow → **larger CMAP** than at wrist
Riche–Cannieu	**Palm**	**Recurrent median ↔ deep ulnar**; motor ± sensory	Normal thenar function despite advanced CTS; all-ulnar hand
Berrettini	**Hand**	**Common digital nerves** of ulnar + median (sensory)	**Most frequent** anomaly



Berrettini anastomosis between common digital nerves

3 Presentation

Paraesthesia	thumb, index, middle, radial ½ ring (as CTS), **worse with repetitive pronosupination**
Differentiators from CTS	**aching proximal volar forearm pain** · numbness of **palm/thenar eminence (palmar cutaneous branch**, arises 4–5 cm proximal to carpal tunnel) · **no night symptoms**
Tinel	**+ proximal volar forearm / medial antecubital fossa**; **– at wrist**; no Phalen/Durkan

Motor	weakness of median-innervated forearm + hand muscles (e.g. **FPL, FDP index, FCR**; trouble using a key)
Other	pain on palpation of lacertus fibrosus; **scratch collapse +**

4 Diagnosis & Classification

Key radiology findings

X-ray	**elbow films mandatory → supracondylar process**
EMG/NCV	usually **inconclusive/negative**; help exclude other sites or **double crush**
Differentials	AIN syndrome, CTS, pronator teres strain



Supracondylar process on elbow X-ray

5 Management

Pronator syndrome

- Mild–moderate → **rest + splint (avoid forearm rotation) + NSAIDs 3–6 mo**
- Failed 3–6 mo nonop → **decompress median nerve at elbow, ALL 5 sites**
  (≈80% relief; lacertus release possible under local anaesthesia)

Surgical release (TO)	lazy-S proximal forearm → release **lacertus fibrosus** (almost subcutaneous) → **step-lengthen humeral head of PT** → release tendinous attachment of **deep (ulnar) head** → **excise FDS arch**
After failed CTR + forearm signs	**decompress median nerve at elbow** ✗ Wrong options: revision CTR · AIN decompression: AIN purely motor · ulnar decompression · more night bracing

– – Active space – –

X vs Y

	Pronator syndrome	CTS	AIN syndrome
Sensory	Radial 3½ **+ palm/thenar eminence**	Radial 3½, not palm	**None**
Night sx	**No**	**Yes**	No
Tinel	**Proximal forearm**	Wrist	–
Pain	**Proximal volar forearm ache**	–	–
Worse with	**Pronosupination**	Wrist flexion	–
NCS	Often negative	Diagnostic	–

Nerve (OB distractors)	Supplies
Median proper	**Pronator syndrome** source
AIN	Radial FDP, FPL, PQ; **no sensory**
PIN	Forearm extensors; no sensory
Recurrent motor median	OP, APB, superficial FPB
Ulnar	FCU, FDP ring/small, adductor pollicis, deep FPB, interossei, hypothenar

How it's asked

"Failed CTR, no night sx, forearm pain, lacertus tender – next step?"	→ **decompress median nerve at elbow**
"Jiu-jitsu, numb radial digits worse with pronosupination, Tinel – at wrist – nerve?"	→ **median nerve proper**
"Pain with resisted pronation, elbow extended – structure's attachments?"	→ **medial epicondyle, coronoid, radial shaft (pronator teres)** ✗ Wrong options: FDS · lacertus · brachioradialis · TCL
"Pain with resisted elbow flexion + supination?"	→ **lacertus fibrosus**
"Pain with resisted middle-finger FDS?"	→ **FDS arch**
"Connection between common digital nerves of median and ulnar?"	→ **Berrettini** (TO)
"Reverse Martin–Gruber?"	→ **Marinacci** (TO)
"Mandatory imaging?"	→ **elbow X-ray (supracondylar process)**
"Only median branch in forearm arising from radial side?"	→ **AIN** (TO)

PYOGENIC FLEXOR TENOSYNOVITIS

– Hand · 70 / 103 –

1 Basics

Epidemiology

Incidence	**2.5–9.4%** of hand infections
Severity (TO)	**most serious hand infection** → sheath gliding destruction, tendon necrosis, osteomyelitis, amputation
Commonest digits (TO)	**ring, middle, index**
Risk factors	**diabetes**, **IVDU**, immunocompromise

Anatomy – sheaths & spaces

Digit	Sheath extent
Index, middle, ring	**DIP → just proximal to A1** (MC neck); **isolated** (✗ Wrong options: horseshoe)
Thumb (FPL)	IP → **radial bursa**
Little	DIP → **ulnar bursa**

Sheath anatomy (TO)	closed synovial tube **A1 → just beyond DIP** · parietal + visceral (epitenon) layers joined at both ends · synovial fluid nourishes tendon
Pathogenesis (TO)	↑ pressure in closed sheath → compresses **vincula** (2nd nutrition source) → tendon necrosis, stiffness, amputation
Bursal extent (TO)	radial + ulnar bursae reach **~2 cm proximal to TCL** · anatomically separate but usually continuous
Parona's space	**between pronator quadratus fascia and FDP conjoined tendon sheaths** → radial–ulnar bursa communication → **horseshoe abscess** (thumb ↔ little finger)



Flexor sheaths and bursae

– – Active space – –

Microbiology – MRSA in IV drug users

Organism	Setting
Staph aureus (40–75%)	**Most common** (gram + cocci; not gram + bacilli)
MRSA (29%)	**IV drug users**
Staph epidermidis, β-haemolytic strep, Pseudomonas	Skin flora
Mixed / gram-negative	Immunocompromised
Eikenella	**Human bite** / fight bite; IVDU licking needles
Pasteurella multocida	Animal bite

Haematogenous PFT (TO) mostly **gonococcal**

Distractors HSV → **herpetic whitlow** (vesicles, healthcare workers, **do not debride**, acyclovir) · Candida → **chronic paronychia** · E. coli → uncommon

2 Mechanism

Route **penetrating trauma** (commonest) · direct spread from felon, septic joint, deep space · rarely haematogenous

3 Presentation

Timing presents **24–48 h** after onset; palmar aspect of one digit

Kanavel signs & diagnosis

Kanavel sign	Note
Flexed (partially flexed) resting posture	Not hyperextended
Tenderness along entire flexor sheath	not extensor sheath (TO) · not localised to A1 (= trigger finger)
Pain on passive extension	**Most constant + first sign** (TO)
Fusiform swelling	4th sign **added later, not by Kanavel** (TO) · not discrete swelling

Most reliable Kanavel sign **tenderness along the flexor sheath** (TO – not localised to a joint/phalanx → excludes pyarthrosis, osteomyelitis) · pain on passive extension = most constant + first (TO) – read the options

Early PFT (TO) all signs may be absent; signs also track improvement

Bursal infection (TO) Kanavel signs **subtler** (auto-decompression) · tenderness: **radial bursa** at wrist crease–**thenar** junction · **ulnar bursa** at wrist crease–**hypothenar** junction

NOT a Kanavel sign **pain on passive flexion** (TO ×2, OB) · tenderness over extensor sheath (TO ×2)

✗ Wrong option inability to flex DIP → **FDP rupture**



Kanavel signs

4 Diagnosis & Classification

Diagnosis **clinical (Kanavel)**; labs may show normal WBC with mildly ↑ ESR/CRP

Key radiology findings

Modality	Fact
X-ray	Not required; exclude **foreign body** · osteomyelitis, pyarthrosis (TO)
Lateral X-ray (TO)	**Volar soft tissue at proximal phalanx > 7 mm thicker than dorsal** → predicted PFT (one study)
Labs (TO)	CBC, ESR, CRP to follow disease
MRI	**not needed for diagnosis**; cannot separate infective vs inflammatory; extent if equivocal

Differential felon, cellulitis, deep space infection, collar-button abscess, herpetic whitlow, septic arthritis, gout

Non-septic tenosynovitis (TO) gout, calcific tendinitis, RA, acute stenosing tenosynovitis → less acute, no excruciating pain → **aspirate sheath**; no pus → **NSAIDs**

5 Management

- **Suspected PFT**
  - Early (<24–48 h), no purulence → **ADMIT, IV ANTIBIOTICS, SPLINT, ELEVATE**
    - no improvement in 24 h → **I&D**
  - Late presentation / purulence / ischaemia → **URGENT I&D + CULTURE-SPECIFIC IV ANTIBIOTICS**
    - Early disease → CLOSED CATHETER IRRIGATION (A1 → **A5**)
    - Advanced / necrosis → **OPEN I&D (BRUNER / MIDLATERAL)**

Surgical drainage – indications, technique & stiffness risk

Best outcome predictor **early IV antibiotics** correlate most closely with good outcome

Non-operative window (TO)	**only within 24 h of onset** · dorsal block splint, elevation, IV antibiotics · **operate if no improvement in 12 h** (TO) · 24 h (Orthobullets) — read the options
Empirical (OB)	**vancomycin + piperacillin/tazobactam**
Surgery indicated by	**delayed presentation (>24–48 h)**, purulence, failed 24 h antibiotics ✗ Wrong options: immunocompromise · age · malignancy · all four Kanavel signs alone
Closed irrigation	catheter **proximal to A1** → out at **A5 / DIP**; thumb exits just distal to carpal tunnel; little finger with ulnar bursa → 2nd catheter A1 → wrist

Surgical option (TO)	Details
Limited exposure (Neviaser)	Incisions at **A5 distally + A1 proximally**; catheter → continuous irrigation every **1–2 h**; only very early cases
Mid-lateral + palmar incision over A1	Decompresses pulleys + digit; debride necrotic sheath; irrigate from palm until clear
Brunner zig-zag	Advanced infection, wider exposure; more scarring

Postoperative (TO)	wounds **left open**, dorsal slab, secondary intention · **IV antibiotics 7–10 days → oral to complete 4 weeks** · **active movement within 12–48 h**
Open vs closed	**equally effective**; open → trend to more complications/reoperation; closed largely replaced open ✗ Wrong options: two-incision superior
Incision error	**volar midline incision perpendicular to creases → flexion contracture →** Z-plasty revision (✓ Bruner zig-zag, ✓ midlateral)
A1 release	does not reduce contracture risk
Poor-outcome risk factors	**age >43 y**, **DM**, PVD, renal failure, **subcutaneous purulence**, **digital ischaemia**, **polymicrobial** infection

Spread – thenar space & Parona's space

Thumb PFT spreads first to	**thenar space** ✗ Wrong options: central · hypothenar · carpal tunnel (not a palmar space; posterior adductor = secondary)
Palmar spaces	thenar, central (midpalmar), hypothenar
Thumb + little finger + distal forearm swollen	**Parona's space** → horseshoe abscess

6 Complications

List	**stiffness** · tendon/pulley rupture · spread · soft-tissue loss · **osteomyelitis** · amputation

X vs Y

	PFT	Herpetic whitlow	Felon	Trigger finger
Site	Whole volar digit	Fingertip vesicles	Pulp abscess	A1 nodule
Posture	Flexed	Normal	Normal	Locking
Treatment	**I&D + IV abx**	**Acyclovir; do not debride**	Incise	Injection / A1 release

How it's asked

"Kanavel signs seen in?"	→ **tenosynovitis** (TO)
"Not a Kanavel sign?"	→ **pain on passive flexion** (TO)
"All true except?"	→ **tenderness over extensor sheath** (TO)
"Most constant / earliest sign?"	→ **pain on passive extension** (TO)
"IVDU index finger PFT – true statement?"	→ **early IV antibiotics correlate with good outcome** ✗ Wrong options: index → horseshoe · MRI confirms
"Purulence found at surgery – exam finding?"	→ **pain on passive extension**
"IVDU hand infection – commonest pathogen?"	→ **MRSA**
"Advanced PFT with ischaemia – next step?"	→ **open I&D**
"Contracture after I&D – technical error?"	→ **volar midline incision perpendicular to creases**
"Indication for urgent surgery?"	→ **delayed presentation**
"Risk of stiffness/amputation?"	→ **age >43**
"Thumb PFT spreads first to?"	→ **thenar space**
"Most reliable Kanavel sign?"	→ **tenderness along the flexor sheath** (TO)
"Haematogenous flexor tenosynovitis – organism?"	→ **gonococcus** (TO)
"Thumb PFT + swollen little finger + forearm?"	→ **Parona's space** (PQ fascia–FDP sheaths)

– – Active space – –

QUADRIGA EFFECT

– Hand · 71 / 103 –

1 Basics

Pathoanatomy

Definition	**active flexion lag of fingers adjacent** to a digit with injured/repaired **FDP**; injured finger reaches palm
Common belly	FDP **long, ring, little** share a common muscle belly + distal cross-connections (forearm, carpal tunnel sheaths) → excursion = shortest tendon (TO)
Index FDP	often **independent belly** → can be shortened without quadriga
FDS	**separate muscle bellies** → FDS repair/advancement **no quadriga**
Name	reins of 4 chariot horses: one tight rein → slack in the others
Zone	most quadriga injuries **zone I**

2 Mechanism

FDP over-advancement or tethering – quadriga

Causes	**FDP over-advancement/over-tensioning** at repair · **adhesions** (Neu: mostly after crush) · FDP **retraction** · **"over-the-top" FDP repair / flexor–extensor tenodesis after DIP disarticulation** · too-short tendon graft
Threshold	**>1 cm** advancement (Orthobullets; Malerich: 1 cm tolerable, **1.5 cm → ulnar 3 digits lose 40° flexion**) · OB7866 option states **>1.5 cm** → risk of quadriga – read the options

FDP avulsion reinserted (button) despite proximal migration → tight, adhesed, improperly tensioned FDP → **quadriga** (TO)



Quadriga: adjacent fingers cannot reach palm

3 Presentation

Symptoms	can't close fingers tightly (e.g. over cricket ball, TO); **weak grip**

Exam fist: **fingers adjacent to injured digit don't reach palm**; **full passive motion**; ↓ grip

Radiographs usually not required

5 Management

Quadriga

- Mild, QoL unaffected → **OBSERVATION**
- Severe, function-limiting → **RELEASE FDP of injured digit**
  (or tendon lengthening / tenolysis / release of intertendinous connections)

FDP release 85% resolution (Neu)

Prevention don't advance FDP **>1 cm**

X vs Y

	Quadriga	Lumbrical plus
Cause	FDP **too short/tethered** (over-advanced, tenodesed, adhesions)	FDP **lax/detached** (retracts after amputation at/ proximal to DIP, long graft)
Finding	Injured finger reaches palm; **adjacent fingers lag**	**Paradoxical PIP/IP extension** on attempted fist

Distractor	Why wrong
Elson test (DIP hyperextends with PIP 90°)	Central slip injury
Can't actively extend MCP but holds it once placed	Sagittal band rupture
Intrinsic tightness	↓ PIP flexion with MCP extended
AIN injury	Thumb IP + index DIP flexion lost
Ulnar nerve injury	Intrinsic minus / claw ulnar 2 digits
FDP repair rupture	Affected finger can't flex, not adjacent

How it's asked

"FDP avulsion reinserted despite proximal migration, can't grip cricket ball at 6 mo?"	→ **quadriga effect** (TO) ✗ Wrong options: lumbrical plus · swan neck · intrinsic tightness
"Ring FDP repaired, long & small can't flex fully – intraoperative cause?"	→ **overtensioning of FDP**
"Shortened 12 mm, adjacent fingers can't reach palm, full passive motion – factor?"	→ **common flexor muscle belly**
"Injured finger flexes fully, others don't – which injury?"	→ **FDP laceration with repair** ✗ Wrong options: FDS repair · nail bed · P1 fracture · dorsal MCP dislocation

– – Active space – –

"Shortening that causes it?"	**→ 2 cm** ✗ Wrong options: 0.5 cm
"FDP advanced to stump after DIP amputation; only ring tip reaches palm?"	**→ quadriga** ✗ Wrong options: lumbrical plus
"Flexor–extensor tenodesis at revision amputation → 3-mo finding?"	**→ can't make full fist, adjacent fingers ↓ ROM**
"True about FDP avulsion?"	**→ advancement >1.5 cm risks quadriga** ✗ Wrong options: timing irrelevant to retraction · chronic → tenodese FDP to FDS · repair method irrelevant
"Treatment of symptomatic quadriga?"	**→ release FDP of injured digit**

– – Active space – –

RADIAL CLUBHAND (RADIAL DEFICIENCY)

– Hand · 72 / 103 –

1 Basics

Epidemiology

Definition (TO)	longitudinal **failure of formation** along pre-axial (radial) border; hand **perpendicular** to forearm + **absent thumb**
Incidence	1 / 100,000 (Orthobullets) · 1 / 50,000–100,000 · 1 / 30,000–100,000 (OB)
Sex	♂ predilection (**3:2**)
Bilateral	**50–72%** (Orthobullets) · **38–58%** (OB)
Syndromic	**30–44%** associated with syndromes

Bilateral = usually syndromic · unilateral often idiopathic (TO) · mesenchymal defect: radius, thumb, thenar muscles, radial vessels, **radial half of median nerve**, radial forearm muscles (TO)

Aetiology – apical ectodermal ridge defect

Embryology	**defect of apical ectodermal ridge** ✗ Wrong options: teratogen · AD · FGF excess (OB)
Isolated RCH	**spontaneous mutation** (most common cause; vs cleft hand, Madelung, syndactyly, symphalangism = AD)
Teratogens (TO)	**thalidomide, valproic acid, phenobarbital** (conflicts with OB "not linked to specific teratogens")

Genes & lab markers

Condition	Gene / inheritance
Radial clubhand	**Sonic hedgehog** (TO, Orthobullets)
Holt-Oram	**AD**, **TBX5** chr **12q24.1** (TO: "TXBS", chr 12); >70% mutated
TAR	**AR**
Fanconi anaemia	**AR**; impaired DNA-damage response; **↑ chromosomal breakage**
Madelung (TO)	**SHOX** (pseudo-autosomal region of sex chromosomes); **Léri-Weill dyschondrosteosis**
Congenital pseudarthrosis tibia (TO)	**NF1**
Blount's (TO)	No named gene

Associated syndromes & work-up (TAR, Fanconi, Holt-Oram, VACTERL)

Mandatory screen	**CBC + renal US + echocardiogram**

– – Active space – –

NOT associated **osteogenesis imperfecta**

Syndrome	Key features	Test
TAR	**AR**; **thumb PRESENT**; bilateral absent radii; platelets <50/nL, **transient** (improves in 1st year); lower limbs involved; NICU	**CBC**; platelet transfusion
Fanconi	**AR**; **aplastic anaemia** at **6–9 y** (1st decade); **most common inherited aplastic anaemia**; **lethal**; ↑ malignancy (**AML**, solid tumours); café-au-lait, microcephaly, triangular face, short stature, abnormal kidneys, ↓ fertility (TO); **↑ MCV, ↓ platelets** (TO)	**CBC, smear, chromosomal breakage** (diepoxybutane / mitomycin C) → **bone marrow transplant**
Holt-Oram	**AD**, **TBX5**; **ASD/VSD** (75%); heart block/conduction disease; absent/triphalangeal thumb → phocomelia; **radioulnar synostosis**; abnormal carpal bone in **all**; clavicle/scapula abnormality (TO)	**Echocardiogram**
VACTERL	Vertebral, Anal, Cardiac, TE fistula, oEsophageal atresia, Renal, Limb (TO: + radial)	**Spine X-rays**, **bronchoscopy** (TEF)
VATER	Vertebral, Anal, TE fistula, oEsophageal atresia, Radial, Renal (TO)	–
Heart-hand II (Tabatznik)	AD; short thumb DP, **arrhythmias** (✗ Wrong options: malformation), no gene	–

Life-threatening AR condition **Fanconi** → chromosomal breakage before pancytopenia

3 Presentation

Exam hand perpendicular to forearm, **absent thumb**, short forearm; **examine elbow** (flexion, biceps)

Thumb predicts radius **thumb hypoplasia severity = greatest predictor of RLD severity** → Blauth V thumb → **Bayne-Klug IV (absent radius)** (Orthobullets) · **severity of radial deficiency does NOT correlate with thumb hypoplasia** (TO) – read the options

Subtle (TO) mild hypoplastic thumb, missing/fused carpals · **severe** – total radial absence, short humerus, hypoplastic glenoid



Radial clubhand

4 Diagnosis & Classification

Key radiology findings

X-ray radius partly/entirely absent, thumb often absent; ulna may be curved (TO)

Carpal ossification **capitate 1–2 mo, lunate 2–4 mo** → no carpal ossification = <12 mo old

TAR X-ray absent radius **with thumb**

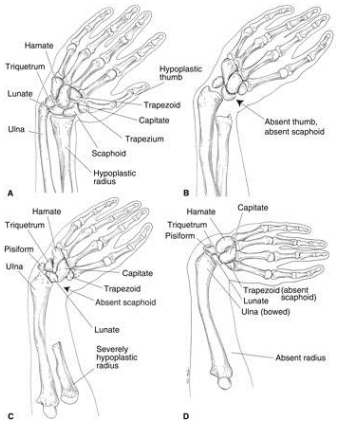


TAR: absent radius, thumb present

Bayne & Klug classification

Type	Orthobullets	TO	Modified (OB / TO table)
0	–	–	Thumb absent/hypoplastic, **normal radius**
I	Deficient distal radial epiphysis	Radius **shorter than ulna**, distal physis present	**>2 mm short**; ± synostosis / radial head dislocation
II	Deficient distal + proximal epiphyses	Small, thin; **no growth plate**	**Hypoplasia proximal + distal**
III	**Present proximally** (partial aplasia)	Small radial segment only	**Absent distal physis** + proximal hypoplasia
IV	**Completely absent** (total aplasia)	Completely absent, ulna curved	**Absence proximal + distal**

Type IV **most common** (and most severe); Heikel type 4 most common (TO)



Bayne & Klug

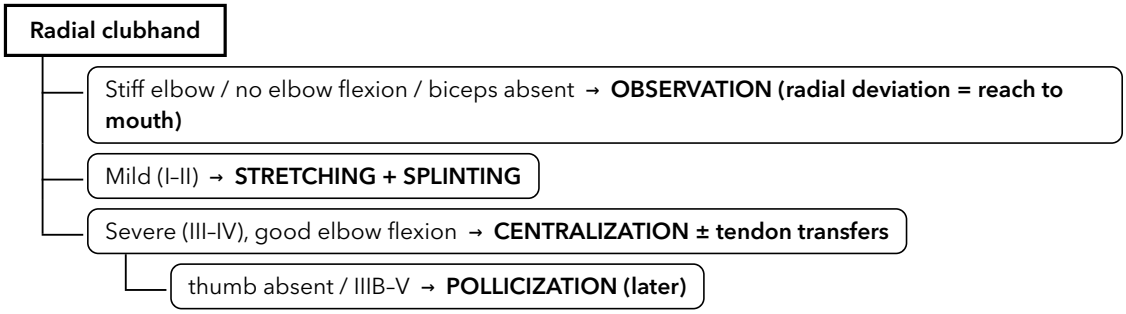
– – Active space – –

Blauth thumb hypoplasia (TO)

Type	Findings (TO)	Treatment (TO)
I	Minor generalised hypoplasia	**No surgery** / 1st web lengthening
II	**Absent thenar intrinsics**, narrow 1st web, **UCL insufficiency**	**Opponensplasty + UCL reconstruction**
IIIA	II + extrinsic abnormalities, **stable CMC**	**Opponensplasty + UCL reconstruction**
IIIB	**Unstable CMC** (Manske)	**Pollicization**
IIIC	Remnant MC head only (Buck-Gramcko)	Pollicization
IV	**Pouce flottant** (floating thumb)	**Pollicization**
V	**Aplasia**	**Pollicization**

Pollicization prerequisites (TO) **flexible, functional index finger** (side-to-side grip) · wrist oriented on forearm · **after 2 y**

5 Management



Infant (TO) stretch & splint from early infancy; **above-elbow splint / serial casting 0–6 mo** · stretching + night splint 6–18 mo · operate **2–3 y** (TO)

Avoid (TO) aggressive early surgery → **distal ulnar growth arrest**; address **fibrous anlage**

Treatment – centralisation indications, contraindications & thumb reconstruction

Centralization carpus on ulna → ↑ reach, wrist tendon balance, ± ↑ limb length

Timing **6–12 mo** (Orthobullets, TO) · not before **12 mo** for anaesthesia safety (OB) · 2–3 y (TO age guide)

Technique resect carpus, **shorten ECU**, ± ulnar angular osteotomy (**spare distal ulnar physis**); ± 2-stage with **distraction ex-fix**; then **tendon transfers**

Current approach (TO) **bilobed flap + anlage resection + ulnar osteotomy + temporary longitudinal wire** carpus → ulna

Others (TO) **radialization** (distraction, carpus on ulna, radial tendons transferred ulnarly), Ilizarov, **vascularised epiphyseal transfer** (MTPJ – OB)

Thumb reconstruction ~**18 mo** (Orthobullets) · pollicization **after 2 y** (TO)

Riordan casting (TO) first 6 mo · Ponseti-like · lift **2nd metacarpal**, ulnar-deviate with pressure over distal ulna · above-elbow casts

Distraction ex-fix (TO) after 6 mo · uniplanar · **2 metacarpal + 2 ulnar pins** · **1 mm/day**

Procedures (TO) — **Buck-Gramcko tendon transfers** (milder) · **centralisation (Sayre 1893)** – ulna into carpal notch, overcorrect · **radialisation** – ulna under radial carpus, ↓ recurrence · **ulnarisation (Paley)** – **FCU transfer** augments wrist extension · **centralisation = stability · radialisation = mobility** (TO)

Pollicisation (TO) — ideal **12–18 mo** (lecture A) · up to **3 yr** (brain plasticity, lecture B) · after 2 y (existing TO) – read the options · shorten + **derotate 160°** + extend · **EIP → EPL · EDC → APL · 1st dorsal interosseous → APB · 1st palmar interosseous → adductor pollicis · index MP joint becomes thumb CMC** · above-elbow splint 4–5 wk, K-wires out 4–6 wk

Adolescents (TO) — wrist flexion needed → **no wrist fusion**; otherwise **fusion + partial correction**

Operative criteria

Centralization	Factor
✓ Indication	**Good elbow flexion + biceps intact**, types III/IV
Contraindication	**Lack of elbow flexion** (<90°) → can't reach mouth
Contraindication	**Older child with good function**
Contraindication	Elbow extension contracture relying on radial deviation (Orthobullets)
Contraindication	Proximate terminal condition · severe soft-tissue contracture
Relative contraindication	**Age <12 mo** (OB)
Not contraindications	Cardiac defects, ↓ grip, **unstable carpus**, scoliosis, TEF, imperforate anus, hypoplastic thumb

X vs Y

	Radial clubhand	TAR	Madelung
Thumb	**Absent**	**Present**	Normal
Gene	Sonic hedgehog / spontaneous	AR	**SHOX**
Blood	Fanconi: pancytopenia	**Thrombocytopenia (transient)**	–

Phocomelia (TO) — intercalated deficiency, long bones absent, hand present ("seal limb") · **thalidomide**, **Roberts syndrome** · orthotics / prosthesis · **ulnar longitudinal deficiency = rarest**; stiff elbow treated first

How it's asked

"Radial clubhand – associated EXCEPT"	→ **osteogenesis imperfecta**
"Tests to order"	→ **CBC + renal US + echo**
"Rule out AR lethal condition"	→ **CBC, smear, chromosomal breakage (Fanconi)**
"Absent radius, thumb present"	→ **TAR**
"ASD/VSD, TBX5, AD family history"	→ **Holt-Oram**

– – Active space – –

"Bilateral RLD + radioulnar synostosis + syndactyly — add which test?"	→ **echocardiogram**
"Congenital UL anomaly most often from spontaneous mutation"	→ **radial clubhand**
"Aetiology of isolated type IV"	→ **apical ectodermal ridge defect**
"Pollicization treatment of choice"	→ **Blauth IIIB + flexible index finger** (TO)
"Relative contraindication to centralization in infant (no carpal ossification)"	→ **age** ✗ Wrong options: limited elbow extension, cardiac defect
"7-mo, good elbow flexion, has thumb — surgery?"	→ **centralization ± tendon transfers** ✗ Wrong options: pollicization · free fibula
"Contraindication to centralization"	→ **lack of elbow flexion**
"Blauth V thumb — radius?"	→ **absent proximal + distal (type IV)**
"SHOX mutation"	→ **Madelung** (TO)

RADIAL TUNNEL SYNDROME

– Hand · 73 / 103 –

1 Basics

Epidemiology

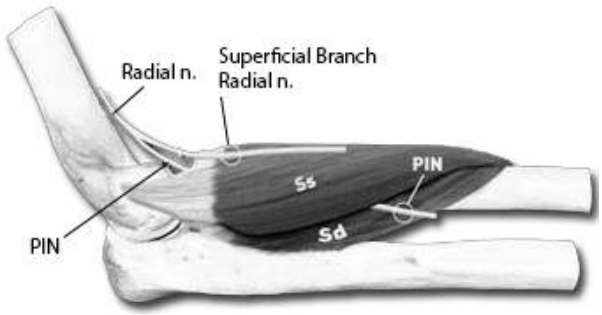
Definition	compressive neuropathy of **PIN** in the **proximal forearm (radial tunnel) → pain only**, no motor or sensory deficit
Incidence	rare, **~3/100,000/y**; ♂ > ♀
Lateral epicondylitis	hard to distinguish; **coexists in 5%**
Risk	constant prono-supination with 1 kg force, elbow 0–45° flexion

Anatomy – radial tunnel & PIN

Radial tunnel **~5 cm**; radiocapitellar joint → past proximal edge of supinator

Boundary	Structure
Lateral	**Brachioradialis, ECRL, ECRB** (mobile wad)
Medial	**Biceps tendon, brachialis**
Floor	Radiocapitellar capsule

Nerve origin	radial nerve ← **posterior cord** ✗ Wrong options: lateral/medial cord · anterior division
PIN course	between **2 heads of supinator** · contacts radial neck · over APL origin · along posterior interosseous membrane
PIN motor	**ECRB** (often radial n. proper), **EDC, EDM, ECU**, supinator, APL, EPB, EPL, EIP
PIN sensory	**dorsal wrist capsule** via terminal branch in **floor of 4th extensor compartment**; not cutaneous
Wrist extension in PIN palsy (TO)	**radial deviation** (ECRL intact, ECU lost)



PIN between supinator heads

– – Active space – –

2 Mechanism

Diagnosis & compression site (arcade of Frohse, posterior cord origin)

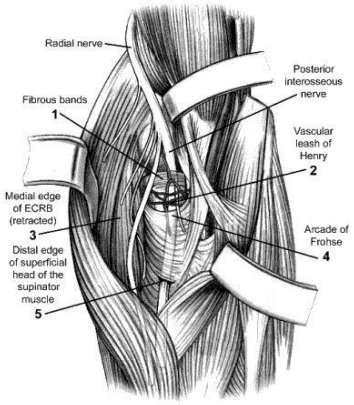
Compression site (proximal → distal)	Note
Fibrous bands **anterior to radiocapitellar joint** (between brachialis & BR)	
Leash of Henry (radial recurrent vessels fanning over PIN at radial neck)	
Medial (medio-proximal) edge of ECRB	
Arcade of Frohse = proximal aponeurotic edge of **superficial** supinator	**Most common**
Distal edge of supinator	

Arcade of Frohse naming (TO) compression here sometimes designated **PIN / supinator syndrome**

NOT PIN sites (TO) **ligament of Struthers** (supracondylar process → medial epicondyle, <1%, **median** n.) · **Gantzer muscle**

Tendinous supinator next to bone → does not compress PIN

Pressures radial tunnel normal **50 mmHg → 250 mmHg** with supinator stretch (forced wrist flexion + pronation)



Compression sites

3 Presentation

Pain **deep aching dorsoradial proximal forearm**, lateral elbow → wrist; ↑ rotation, lifting; ± night pain

Weakness **due to pain**, not denervation (subtle at most)

Tenderness mobile wad over supinator arch, **3–5 cm distal to lateral epicondyle** (~4 cm; TO: 5 cm)

Sensory **none**; 1st dorsal web changes → think more proximal

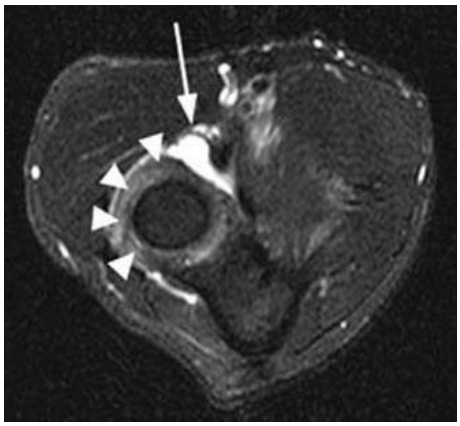
Provocative test	Finding
Resisted long (3rd) finger extension	Pain at radial tunnel
Resisted supination (elbow & wrist extended)	Pain
Passive pronation + wrist flexion	Pain (↑ pressure to 250 mmHg)
Radial tunnel injection	**PIN palsy + pain relief = diagnostic**

4 Diagnosis & Classification

Diagnosis	**clinical** (history + exam)
EMG/NCV	**inconclusive / normal**: pain carried by **unmyelinated group IV (C) + small myelinated IIA** fibres; large myelinated fibres normal
Diagnostic LA injection	into point of tenderness; avoid spread to lateral epicondyle

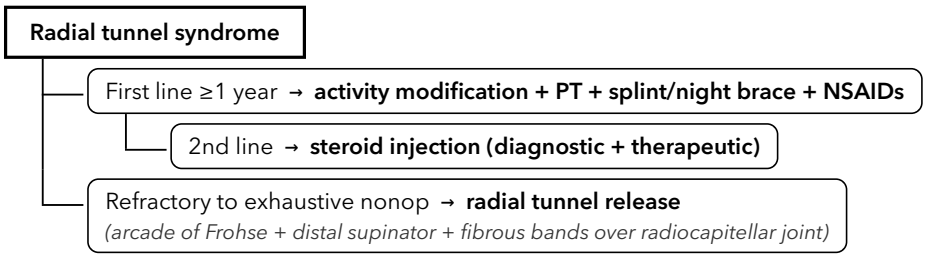
Key radiology findings

MRI	usually **negative**; may show **denervation oedema/atrophy** supinator/extensors · thick ECRB edge · prominent leash of Henry · PIN swelling
MRI other causes	tumour, ganglion, radiocapitellar synovitis, bicipital bursitis, radial head fracture/dislocation



Denervation oedema

5 Management



First-line treatment – therapy & activity modification

First line	**PT + activity modification** (± night bracing) **≥1 year** ✗ Wrong options: injection first · surgery
Avoid	prolonged elbow extension + forearm pronation + wrist flexion
Steroid injection	**2nd line**; **70% improve at 6 wk**, **60% pain-free at 2 y**; TO: if associated lateral epicondylitis

– – Active space – –

Radial tunnel release

Releases **arcade of Frohse**, **distal edge of supinator**, fibrous bands over radiocapitellar joint

Approach	Interval
Dorsal (Henry / Kaplan)	**ECRB–EDC**
Dorsal	**BR–ECRL**
Dorsal	**Transmuscular BR-splitting**
Anterior	**BR–biceps** → used for **mass on MRI**

Posterior approach preferred; safe access to supinator + distal edge

Outcomes – conflict **50–90% good/excellent, "disappointing"**, max recovery **9–18 mo** (Orthobullets) · **70–90%** success (Orthobullets techniques) · "excellent in most" (TO)

Lower success group	Success
Multiple entrapment neuropathies	60%
Concomitant lateral epicondylitis	40%
Workers' compensation	**30%**

X vs Y

	Radial tunnel	PIN syndrome	Wartenberg	Lateral epicondylitis
Pain	**Yes**	Yes	Yes	Yes
Motor	**No**	✓ **finger/thumb/ wrist ext weak**	No	No
Sensory	**No**	No	✓ **dorsoradial hand** (superficial radial)	No
Tenderness	**3–5 cm distal** to epicondyle	–	Distal forearm	**Directly over** epicondyle
EMG	Normal	Abnormal	–	Normal

Other mimics **intersection syndrome** (1st/2nd compartment, crepitus with wrist extension; ECRL/ECRB tenosynovitis) · **pronator syndrome** (numb radial 3 digits) · **AIN** (thumb IP flexion weak) · **C6–7 radiculopathy** (EDS may show denervation) · radiocapitellar arthritis

How it's asked

"PIN compressed at all EXCEPT"	→ **ligament of Struthers** (TO)
"Does NOT cause PIN entrapment"	→ **Gantzer muscle** (TO)
"Most common site of PIN compression"	→ **arcade of Frohse (free proximal aponeurotic supinator margin)** (TO)
"RTS – all true EXCEPT"	→ **weakness of MCP extension** (TO)

"RTS – NOT true"	→ **motor weakness** (TO)
"Pain 4 cm distal to lateral epicondyle, pain on resisted long finger extension"	→ **radial tunnel syndrome** ✗ Wrong options: lateral epicondylitis, intersection
"Arcade of Frohse compression – symptom?"	→ **dorsoradial forearm pain worse with passive pronation** ✗ Wrong options: dorsoradial numbness = Wartenberg · thumb IP weakness = AIN
"Nerve's origin in brachial plexus"	→ **posterior cord**
"Untreated, normal EMG – first line"	→ **PT + activity modification** ✗ Wrong options: steroid injection · release
"Diagnostic injection result"	→ **PIN palsy + pain relief**

– – Active space – –

RADIOCARPAL FRACTURE DISLOCATION

– Hand · 74 / 103 –

1 Basics

Epidemiology

Definition **total loss of contact between carpus and radius**

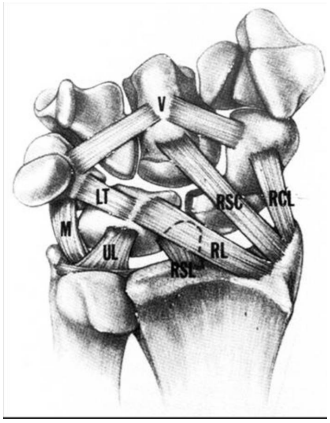
Incidence **0.2%** of wrist injuries; ♂ > ♀; high energy

Direction **dorsal > volar**

Anatomy – radiocarpal ligaments

Ligament	Attachment (radius)	Restrains
Short radiolunate	**Volar lunate facet** (volar-ulnar "critical corner")	**Volar translation** of carpus (primary)
Radioscaphocapitate (RSC)	**Radial styloid**	**Ulnar translation**
Ulnolunate, ulnotriquetral	Volar TFCC (→ base ulnar styloid)	–

Distal radius biconcave triangular surface; apex = radial styloid; base = sigmoid notch



Volar wrist ligaments

2 Mechanism

Force high-energy shear / **rotation on hyperextended, pronated wrist** (falls from height, MVC)

Associations open #; **radial styloid avulsion (RSC)**; **volar lunate facet # (short RL)**; ulnar styloid #; scaphoid/lunate/trapezium #; **median > ulnar nerve injury**; **irreducible DRUJ (soft-tissue interposition)**; SL/LT dissociation

3 Presentation

Exam deformity (dorsal/volar), swelling, **pale cold hand** (radial/ulnar artery occlusion common), numbness; note crepitus

4 Diagnosis & Classification

Key radiology findings

View	Look for
PA	Radiolunate alignment (normally **⅔ of lunate on radius**; complete disruption → **ulnar translation**), intercarpal widening, SL/LT dissociation, **broken Gilula arcs**
Lateral	Dorsal/volar dislocation; loss of lunate–radius colinearity
Teardrop view	**10° proximal tilt** on lateral → **lunate facet fracture**
CT	Comminution, articular depression
MRI	SL and LT integrity

Differentials perilunate/lunate dislocation, distal radius fracture



Radiocarpal dislocation, lateral

Fernandez type IV (TO) **avulsion** mechanism = radiocarpal fracture-dislocation → **always ORIF + ligament repair**

Moneim classification

Type	Description
1	Radiocarpal fracture-dislocation **without** intercarpal dissociation
2	**With** intercarpal dissociation

– – Active space – –

Dumontier classification

Group	Description	Image
1	**Purely ligamentous** or small cortical avulsion off radius	
2	**Large radial styloid fragment** (≥⅓ scaphoid fossa)	

5 Management

Radiocarpal fracture-dislocation

- Stable after closed reduction / unfit → **cast up to 6 wk**
- Irreducible / unstable after reduction → **ORIF + radiocarpal pinning + ligament repair**
  - still unstable → **ex-fix / radiolunate pin 4–6 wk or dorsal bridge plate**

Operative technique – three-column fixation

Column	Fixation
Radial (styloid)	K-wire / compression screw / plate; anchor repair if small/ligamentous
Intermediate (lunate facet)	**Screw or tension-band wire**; **repair short RL + RSC**
Ulnar (controversial)	DRUJ injury / persistent instability → ulnar styloid fixation, DRUJ pinning in **mid-supination**

Also decompress NV structures; volar ± dorsal approach

Ligament injury – short radiolunate

Loss of reduction after casting unfixed **volar lunate facet fragment → short radiolunate ligament →** volar carpal subluxation

✗ Wrong options: SL · RSC · LT · ulnolunate

6 Complications

– – Active space – –

Complication	Key fact
Acute CTS	Risk: delay → **CTR**
Stiffness	Loss **30–40%** of F/E arc; MUA, hardware removal
Post-traumatic OA	Non-anatomic articular reduction
Chronic radiocarpal instability	–
Late intercarpal disruption	Occult SL/LT injury

How it's asked

"Carpus subluxes volarly after casting – ligament on displaced fragment?"	**→ short radiolunate (volar lunate facet)**
"Ligament restraining ulnar translation?"	**→ RSC (radial styloid)**
"View for lunate facet fracture?"	**→ teardrop (10° proximal lateral)**
"Commoner nerve injury?"	**→ median**

– – Active space – –

RAYNAUD'S SYNDROME

– Hand · 75 / 103 –

1 Basics

Definitions & epidemiology

Raynaud's	exaggerated **vasoconstriction of digital arteries** to stimuli
Raynaud's phenomenon / syndrome (secondary)	**known underlying cause**
Raynaud's disease (primary)	**idiopathic**; young **premenopausal ♀ <40 y**
Prevalence	vasospastic disease **3–5%** of US population; **40–50%** develop painful digital ulcers

Associations of Raynaud's phenomenon

Condition	Raynaud's incidence
Scleroderma	**80–90%**
Dermatomyositis	30%
SLE	18–26%
RA	11%
CREST	Calcinosis, Raynaud's, oesophageal dysmotility, sclerodactyly, telangiectasia
Neurovascular compression	Thoracic outlet syndrome

2 Mechanism

Triggers	**cold** or **sympathetic stimuli** (pain, **emotional stress**)
Triphasic colour change	**white** (vasospasm) → **blue** (cyanosis, venous stasis) → **red** (rebound hyperaemia) → dysaesthesia



White phase

3 Presentation

	Raynaud's disease (primary)	Raynaud's phenomenon (secondary)
Age / sex	**<40, young** ♀	**>40**, occasional ♀ predominance
Distribution	**Bilateral**	**Asymmetric**, distal digits
Progression	**Slow**	**Rapid**
Pulses	Present	**Often absent**
Trophic changes / ulcer / gangrene	Rare	**Frequent**
Allen test	Normal	**Abnormal**
Labs	Normal	Often abnormal
Angiography / microangiology	Normal	Often abnormal



Digital gangrene

- - Active space - -

4 Diagnosis & Classification

Diagnostic criteria – Allen & Brown (Raynaud's disease)

Criterion
Intermittent attacks of acral discoloration
Bilateral involvement
No clinical arterial occlusion
Gangrene / trophic changes **rare**
Symptoms **>2 years**
♀ predominance
No other disease explaining findings

Differentiate by age · gender · time course of progression · severity of exam

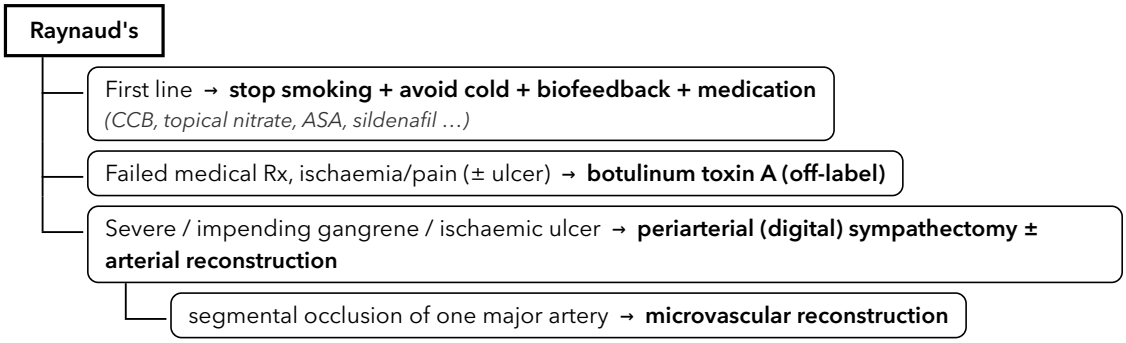
Key radiology findings

X-ray **normal, not indicated**

Angiography secondary: often abnormal (e.g. incomplete ulnar artery obstruction)

Doppler documents ↑ digital flow after BTX

5 Management



Medical management

Critical **smoking cessation + avoid cold**; thermal biofeedback; stress, protective garments

Commonest drug **calcium channel blocker** (nifedipine; oral only)

Other drugs ASA, **sildenafil** / PDE inhibitors, intra-arterial reserpine, dipyridamole, pentoxifylline, prostaglandins, ACE-I/ARB, TCAs, SSRIs, α-antagonists

Topical vasodilator mechanism

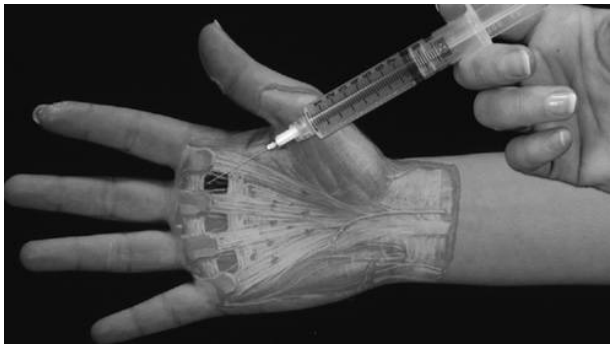
Topical nitrate **only topical drug** used; **↑ cGMP in vascular smooth muscle** → vasodilation

Not topical NSAID (COX), steroid (PLA2), lidocaine (Na^+), CCB (L-type Ca^{2+} – oral only)

Botulinum toxin – mechanism, outcomes & pre-treatment testing

Indication	**failed medical management** + ischaemia/pain; **ulcers not a contraindication**; **off-label**
Mechanism	**cleaves pre-synaptic SNAREs → ↓ ACh release → ↑** digital perfusion
Other proposed	blocks **α2 receptor recruitment** on vascular smooth muscle · **↓ substance P & glutamate** (pain) · inhibits **RhoA kinase → ↓** smooth-muscle constriction · may ↓ sympathetic tone
Outcomes	**all of**: ulcer healing, ↓ pain, ↓ cold intolerance, ↑ digital blood flow
Absolute contraindications	**pregnancy / breastfeeding → urine pregnancy test before injection** · anaphylactic allergy · active infection · aminoglycosides / anticholinergics
Caution	respiratory disease, myasthenia gravis

✗ Wrong options: **Traps** – proprioception (non-depolarising blockers), Ca^{2+} channel excitability (gabapentin/pregabalin), muscle strengthening



Botox injection method

Periarterial sympathectomy indications

Indication	**Raynaud's failed medical Rx + ischaemic digital ulcer** / impending gangrene; esp. collagen vascular / occlusive disease
Technique	**periarterial (digital)** = commonest; strip **adventitia + sympathetic fibres**; may extend to forearm
Not indicated	**calcific arterial disease** · Buerger's in **active smoker** · CRPS (stellate block instead) · diabetic Charcot · TOS
Thoracoscopic sympathectomy	**out of favour → rebound hypersympathetic response**
Microvascular reconstruction	rare: **small-segment occlusion of one major artery**, other spared

6 Complications

Digital ischaemia	ulceration, **gangrene → amputation**

How it's asked

"BTX mechanism in Raynaud's"	**→ cleaves pre-synaptic SNAREs → ↓ ACh → ↑ perfusion**
"BTX outcomes in vasospastic disorders"	**→ all of the above**

– – Active space – –

"Before repeat Botox in scleroderma patient"	→ **urine pregnancy test** ✗ Wrong options: HbA1c, sympathectomy, CXR
"Failed >1 y CCB + ARB + nitrate, no ulcer — next"	→ **botulinum toxin A** ✗ Wrong options: thoracoscopic sympathectomy, microvascular reconstruction
"Topical drug for scleroderma fingertip ulcer — mechanism"	→ **↑ cGMP** (nitrate)
"Periarterial sympathectomy indicated"	→ **Raynaud's, failed medical Rx, ischaemic ulcer** ✗ Wrong options: smoking Buerger's, CRPS, Charcot, TOS
"Why thoracoscopic sympathectomy abandoned"	→ **rebound hypersympathetic response**
"Incidence of Raynaud's in scleroderma"	→ **80–90%**

REPLANTATION

– Hand · 76 / 103 –

1 Basics

Epidemiology 90% of UL amputations traumatic; ♂:♀ **4:1**; most at **digit** level

Mechanisms **sharp** (best) · blunt · avulsion · crush

Assess timing, level, number of digits, preservation of part, **associated injuries**, PMH; stump: zone of injury, viability, contamination

4 Diagnosis & Classification

Indications & contraindications by level (thumb, multiple digits, zone)

Indications (primary / "absolute")	Relative indications
Thumb at any level (even proximal to FPL insertion)	Single digit **distal to FDS insertion (zone I)** – good function
Multiple digits	Ring avulsion (Urbaniak 2A)
Through palm	Through/above elbow
Wrist or proximal (distal forearm/wrist = excellent function)	–
Almost any part in a child	–

Primary contraindications	Relative contraindications
Severe vascular disease	**Single digit proximal to FDS insertion (zone II)**
Mangled / crush	Medically unstable
Segmental / multilevel	Disabling psychiatric illness
Ischaemia **>6 h with large muscle**	Contamination
–	Ischaemia **>12 h, no muscle**

Zone II single digit FDS insertion = mid-middle phalanx = distal border of zone II → **PIP stiffness**, ↓ sensation, FDP–FDS adhesions → patient **bypasses stiff index**, uses long finger

Zone II exceptions **thumb, multiple digits, children**

Distal limit need **~4 mm intact skin proximal to nail fold** for veins; distal to nail fold → vessels bi/trifurcate → do not replant (TO)

Proximal phalanx level (TO) poor PIP motion → do not replant (single digit)

Ring avulsion (Urbaniak) **2A** (vascular only) → repair · **2B** (bone/tendon/nerve) or **3** (degloving) → **amputation**

Not reasons to refuse age, workers' comp, malingering

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Ischaemia time criteria

Level	Warm	Cold
Proximal to carpus (muscle)	**<6 h**	**<12 h**
Digit (distal to carpus)	**<12 h** (Orthobullets/OB) · **≥8 h** (TO)	**<24 h** (Orthobullets/OB) · **up to 30 h** (TO)

Muscle necrosis (TO) irreversible after **6 h warm**; cooling to **4 °C** → 12 h

Wrist with 8 h warm **relative contraindication**

Forearm with 10 h warm do not replant (TO; good only if <6 h)

Prognostic factors – mechanism & ischaemia time

Most important factor **mechanism of injury** (sharp > crush/avulsion)

Favourable survival **sharp**, **female**, **age <13**, **non-smoker**, warm <12 h / cold <24 h (digit)

Unfavourable **crush**, avulsion, **ribbon sign** (corkscrew artery = intimal avulsion → needs segmental resection + vein graft)

Level DIP–PIP replants good · **PIP–MCP poor** (flexion contracture)

5 Management

Transport of amputated part

Part rinse; wrap in **gauze moist with RL/saline → sealed plastic bag → ice water**; cool to **~4 °C** (TO)
✗ Wrong options: direct ice · dry ice

Stump moist gauze, compression

Surgical sequence & major limb replantation

Shunt first proximal replants with large muscle (↓ warm ischaemia)

Source	Sequence
TO	**Bone → extensor → flexor → artery → nerve → vein → skin**
OB	Bone → extensor → flexor → arteries → nerves → veins (veins may precede nerves) → skin
Orthobullets	Bone ± shortening → extensor → artery → veins → flexor → nerve → skin ± fasciotomy

Always **bone first**; **arteries before veins**; vascular repair = greatest priority (TO); artery right after bone if ischaemia >3–4 h

Vein grafts use **generously**; bone shortening avoids grafts

Wrist level, ischaemia approaching 6 h (TO) **restore arterial flow first** (muscle mass, reperfusion risk) · **cold ischaemia better than warm** (TO)

Finger order **thumb → long → ring → small → index**

Multiple digits **structure-by-structure** (fastest) · **digit-by-digit = longest**

Palm amputation, digits intact, clean **anatomic replantation of hand**
✗ Wrong options: transpositional (for multidigit loss)

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TABLE 1. Guidelines Followed During Major Replantations[18]

Level of Amputation	Injury Time to Arrival at Hospital	Technical Guidelines
Proximal to mid-arm level	i. ≤3 h	Debride; fix bone; do artery, vein, and nerve repair; and then release clamps (regular sequence)
	ii. 3–4 h	Debride, fix bone, do artery repair first, release artery clamp to perfuse for 5–10 min, clamp artery, repair other structures, and then release arterial clamp only initially (venous clamp released after 5 min)
	iii. 4–5 h	Use preliminary arterial shunting on arrival and then do the sequence as in (ii). If any delay occurs during procedure, the artery clamp can be released once every 30 min for 5 min
	iv. 5–6 h	Gray zone for replanting. Replant only if fingers are freely passively mobile/replant and do a proximal below elbow amputation to gain length for prosthesis fitting
	v. >6 h	Do not replant
Lower third of arm and proximal forearm	i. ≤4 h	Regular sequence. Preserve elbow by shortening on nonjoint side
	ii. 4–6 h	Elbow arthrodesis—enables more muscle debridement. Do artery repair first after bone fixation, release clamp for perfusion for 5 to 10 min, clamp artery, and then repair other structures. Then release arterial clamp only initially (venous clamp released after 5 min)
	iii. 6–7 h	Consider preliminary arterial shunting. Do replant only if the thumb is passively mobile and follow sequence as in (ii)
	iv. >7 h	Do not replant
Mid-forearm to wrist level	i. ≤6 h	Regular sequence. Ensure excision of muscles attached to tendons in the amputated part
	ii. 6–8 h	Debride, fix bone, do artery repair first, release clamp, allow perfusion for 5–10 min, clamp artery, and then repair other structures
	iii. >8 h	Replant if thumb is freely passively mobile

Operative sequence

Ray resection vs replantation – function & palmar width

Ring finger amputated at proximal phalanx — **early 4th ray resection + 5th ray transposition** → best cosmesis & grip
✗ Wrong options: delayed → ↓ grip, pronation, cosmesis · revision amputation

Transposition — after **central ray** (long/ring) resection → prevents gap & **scissoring**

Ray resection — **↓ palmar width** → ↓ grip; good cosmesis; most **return to work**

Ray resected	Greatest loss
Long	**Grip**
Index	**Pinch**

Contaminated avulsion-crush small finger + ribbon sign — **ray resection**
✗ Wrong options: replant even in child

Postoperative monitoring – arterial vs venous insufficiency, leeches & antibiotics

Environment — warm room (**80 °F**); avoid caffeine, chocolate, nicotine; hydration; aspirin, dipyridamole, LMW dextran, heparin

Monitoring — **skin temperature most reliable**: drop **>2 °C in <1 h** or **<30 °C** · **pulse oximetry <94%**

	Arterial insufficiency	Venous congestion
Timing	**<12 h** post-op	**>12 h**
Cause	**Vasospasm → thrombosis** (commonest early failure)	Outflow failure
Treatment	**Release bandages**, **dependent position**, heparin, **stellate ganglion block**, early exploration	**Elevate**, **leeches**, heparin pledgets
Avoid	**Leeches**	–

Leeches — *Hirudo medicinalis* → **hirudin** (100× heparin) → venous congestion

Leech infection — ***Aeromonas hydrophila*** (Gram −ve rod, leech gut) → **ciprofloxacin** (**DNA gyrase inhibitor**) first line · alt **TMP-SMX / 3rd-gen cephalosporin**

Reperfusion injury – allopurinol

Mechanism — **hypoxanthine → xanthine** (xanthine oxidase)

– – Active space – –

Agent	**allopurinol** (xanthine oxidase inhibitor) improves outcome ✗ Wrong options: tPA, heparin, COX-2, leeches

WALANT anaesthesia

WALANT (TO)	wide-awake local anaesthesia **no tourniquet**; lidocaine/bupivacaine + **epinephrine**; no sedation; **Lalonde (2005)**
Purpose (TO)	proposed to ↓ **waiting time** for surgery
Advantages (TO)	↓ **adhesions** (immediate mobilisation; collagen from day 3) · **intraoperative active testing** (tendon transfer tension) · comorbid patients / tourniquet to avoid
Incorrect (TO)	"not suitable for **tendon transfers**" → it **is** suitable for complex procedures

6 Complications

Early failure	**arterial thrombosis from vasospasm** (<12 h)
Stiffness	replanted digit ~**50% total motion**; **tenolysis = commonest secondary surgery** (**47.2%** of secondary procedures)
Major limb	**myonecrosis**, **myoglobinuria → renal failure** (fatal), acidosis, **hyperkalaemia** (TO)
Others	infection, **cold intolerance**

How it's asked

"Best indication — healthy 35-y"	→ **wrist-level amputation** (TO) ✗ Wrong options: thumb distal to nail bed · index at PP · forearm 10 h warm
"Relative contraindication"	→ **ring finger through proximal phalanx shaft**
"Replantation indicated"	→ **thumb through proximal phalanx** ✗ Wrong options: multilevel, crush, Urbaniak 2B, long finger at PP
"Indications EXCEPT"	→ **long finger through proximal phalanx**
"Absolute indication"	→ **thumb proximal to FPL insertion**
"Absolute indications EXCEPT"	→ **single digit distal to FDS** (relative)
"Index cut proximal to FDS, 15 h cold — why not replant?"	→ **local anatomy (zone II stiffness)** ✗ Wrong options: delay · age
"Most important factor in outcome"	→ **mechanism of injury**
"Unfavourable for survival"	→ **crushed digit**
"Wrist amputation — true"	→ **8 h warm ischaemia = relative contraindication** ✗ Wrong options: veins before arteries · bone last

"Repair sequence"	→ **bone, extensor, flexor, artery, nerve, vein, skin** (TO)
"Increases surgical time"	→ **digit-by-digit**
"Clean palm amputation, digits intact"	→ **anatomic replantation of hand**
"Temp drop 33 → 29 °C, SpO_2 87% – EXCEPT"	→ **leeches**
"Leech prophylaxis MOA"	→ **DNA gyrase inhibition** (ciprofloxacin)
"Inhibits xanthine formation"	→ **allopurinol**
"Ring finger GSW at PP – best grip/cosmesis"	→ **early 4th ray resection + 5th ray transposition**
"Ray resection – true"	→ **↓ palmar width**
"Avulsion-crush small finger, ribbon sign, contaminated"	→ **ray resection**
"WALANT – incorrect"	→ **unsuitable for tendon transfers** (TO)

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SAGITTAL BAND RUPTURE

– Hand · 77 / 103 –

1 Basics

Epidemiology

Synonym	**"boxer's knuckle"**; **zone 5** extensor injury
Who	**pugilists** (pros: index/middle; amateurs: ring/little); **RA** (chronic, attritional)
Finger	**middle most common (48%)** · little 31% · index 14% · ring 7% (Orthobullets) · "middle or ring" (OB203)
Side	**radial SB : ulnar SB = 9:1** → tendon subluxes **ulnarly**
Association	MCP collateral ligament injury; often dorsal capsule torn too

Anatomy & biomechanics

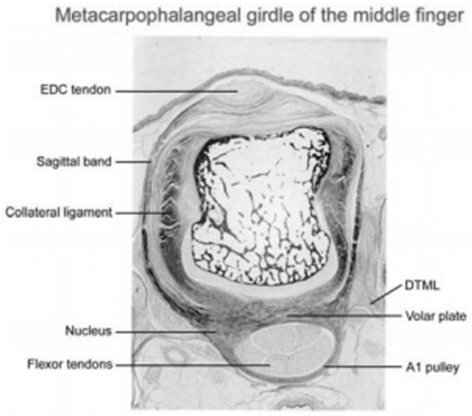
Sagittal band part of closed **girdle** around MC head with palmar plate; origin **volar plate + intermetacarpal ligament** at MC neck → inserts on extensor hood

Primary stabiliser of extensor tendon at MCP = **sagittal band**; **secondary = juncturae tendinum**

Function resists **ulnar deviation** of tendon (esp. MCP flexion); prevents **bowstringing** in MCP hyperextension

Sectioning	Result
Ulnar SB (partial/complete)	**No** dislocation
Radial SB distal part	No instability
50% of proximal radial SB	**Subluxation**
Complete radial SB	**Dislocation**

Instability greatest **wrist flexion** + MCP flexion; central digits > border; **middle least stable**, **little most stable** (junctura)



Sagittal band girdle

2 Mechanism

Traumatic	forceful resisted flexion or extension ("flicking"); **direct blow** to MCP (forced flexion); hood laceration
Atraumatic	**RA**; spontaneous

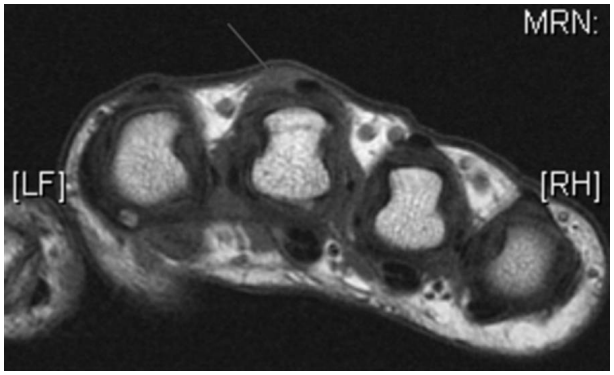
3 Presentation

Symptoms	MCP pain, focal swelling, **snapping** with extension; palpable dorsal defect on fist, resolves with MCP extension
Key finding	**cannot actively extend MCP from flexion, but CAN HOLD extension once passively placed** ✗ Wrong options: lag of 30°/60° · Bunnell intrinsic test · IP extension
IP extension	preserved (intrinsics)
Pseudo-triggering	snapping from subluxation/relocation → **avoid unnecessary trigger release**
Dislocation into intermetacarpal gully	most unstable MCP flexion + **wrist flexed**; least with wrist extended
Provocative	pain on resisted MCP extension with IPs extended
RA	ulnar deviation of digits at MCP

4 Diagnosis & Classification

Key radiology findings

Modality	Use / finding
X-ray	PA, lateral, oblique; **Brewerton** (AP, dorsal fingers on cassette, MCP 45° flexed); **stress view → collateral avulsion**
Dynamic US	When **swelling obscures exam**; EDC subluxation on MCP flexion
MRI	Axial at long MCP, **MCP flexed**; acute = poor definition, discontinuity, thickening
MRI direction	**Ulnar SB defect → tendon radial**; **radial SB defect → tendon into ulnar gully**

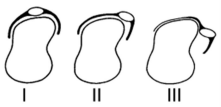


MRI: radial sagittal band rupture, tendon subluxed ulnarly

Differentials	collateral injury, EDC rupture, trigger finger, junctura disruption, congenital SB deficiency, MCP OA

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Rayan & Murray classification

Type	Description	Image
I	SB injury, **no** tendon instability	
II	SB injury + tendon **subluxation**	
III	SB injury + tendon **dislocation**	

5 Management

Sagittal band rupture

- Acute (<6 wk Orthobullets / <3 wk OB) → **extension / yoke / relative-motion splint 4–6 wk**
- Professional athlete → **open direct repair**
- Chronic (>6 wk) or failed splint
  - primary repair possible → **direct repair (Kettlekamp)**
  - not possible → **extensor centralisation**

Treatment – sagittal band splinting vs open repair

Splint — **MCP in extension, IPs free** (extension or **yoke** splint)
✗ Wrong options: volar finger splint · buddy tape · dorsal block · intrinsic-plus

Relative-motion splint (OB) — **MCP 30° hyperextension, PIP free**

Timing conflict — acute = **within 6 wk** (Orthobullets) · splint if seen **within 3 wk**, surgery if **>2–3 wk** or failed splint (OB203) – read the options

Pro athlete (NFL) — **open repair of sagittal band** → earlier rehab, quicker RTS
✗ Wrong options: junctura repair

Centralisation techniques

Technique	Key detail
Trapdoor	**Ulnar-based partial-thickness capsular flap**, tendon deep to flap
Kilgore	Distally based **radial** EDC slip around **radial collateral**
Carroll	Distally based **ulnar** EDC slip, deep to tendon, around **radial collateral**
McCoy	**Proximally** based EDC slip around **lumbrical insertion**
Watson	Distally based EDC slip under **deep transverse MC ligament**
Wheeldon	**Ring junctura** divided, brought over tendon, sutured to torn radial SB (middle finger)
Other	Fascial strips / free tendon graft

Rehab — 0–4 wk resting splint MCP + IP 0° · from 2 wk motion splint MCP 0°, IPs free · AROM 4–8 wk · strengthen >8 wk

6 Complications

MCP flexion contracture after nonop / delayed presentation → secondary **intrinsic tightness**

X vs Y

Test / finding	Diagnosis
Can't extend MCP, holds when placed	**Sagittal band rupture**
Can't hold MCP extension	EDC rupture/laceration, **PIN palsy**
Locking in flexion	Trigger finger (A1)
PIP laxity to valgus (PIP 30°, MCP 90°)	Collateral ligament injury
Rigid DIP with PIP flexed 90° over table	**Elson** – central slip
Can't flex DIP	Jersey finger (FDP)

How it's asked

"Key diagnostic finding of SB rupture?"	**→ holds MCP extension after passive extension**
"Finding present in SB injury?"	**→ can't actively extend MCP but maintains it after passive extension**
"Amateur boxer 2–3 wk – initial treatment?"	**→ extension / yoke splint (MCP extended, IPs free)**
"Punched wall 3 days ago, dorsal defect on fist – next step?"	**→ MCP extension splinting** ✗ Wrong options: trigger injection · hood reconstruction
"NFL player, tendon subluxing on MCP flexion?"	**→ open repair of sagittal band**
"Relative-motion splint position?"	**→ MCP 30° hyperextension, PIP free**
"Which band and which way?"	**→ radial SB → tendon subluxes ulnarly**

– – Active space – –

SCAPHOID FRACTURE

– Hand · 78 / 103 –

1 Basics

Epidemiology – most common carpal fracture

Most common carpal fracture (TO) **scaphoid (60%) > triquetrum > trapezium**; 15% of acute wrist injuries

Who ♂ 2:1, 3rd decade, contact sport

Location	Adult	Note
Waist	**65–70%**	Commonest
Proximal third	25% (5–10% TO)	**AVN / nonunion**
Distal third	10% (10–20% TO)	Heals well
Tubercle	5%	–

Children historically **distal pole/tubercle** (ossifies distal → proximal; 52% tubercle TO) · **now waist (middle ⅓) most common** (OB) → read options

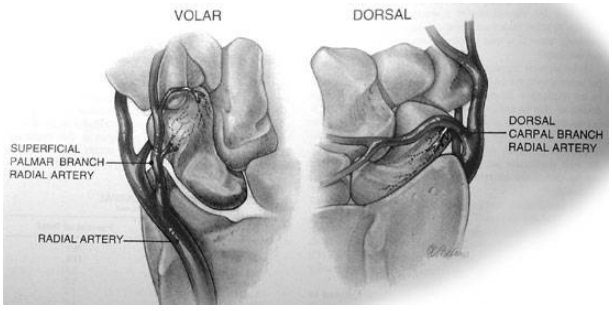
Anatomy & blood supply

Bone largest of proximal row; **>75% articular cartilage**; links proximal & distal rows

Major supply **dorsal carpal branch of radial artery** → enters **dorsal non-articular ridge** → **retrograde** to **proximal 80%**

Minor supply **superficial palmar arch (volar radial a.)** → **distal tubercle** → distal 20%

AVN by level **proximal ⅕ → 100%** · **proximal ⅓ → 33%** (untreated)



Retrograde scaphoid blood supply

2 Mechanism

Wrist loading biomechanics

Mechanism **axial load on hyper-dorsiflexed, pronated, ulnar-deviated wrist** (FOOSH)

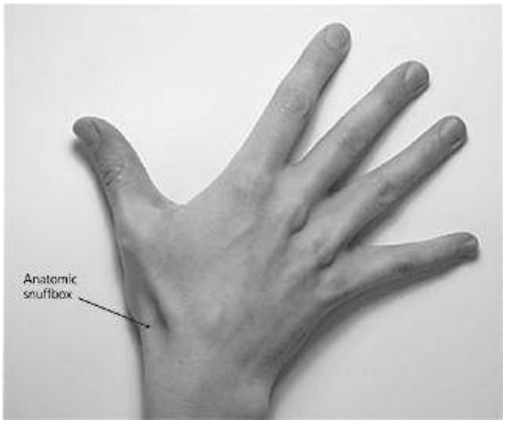
Force transmission neutral: scaphoid fossa ≈ lunate fossa (slight scaphoid predominance) · **extension → ↑ force through scaphoid fossa**

Stability	transverse > vertical/oblique
Humpback	distal fragment **flexes**, proximal **extends** (pulled by SLIL) → DISI

3 Presentation

Diagnosis – snuffbox tenderness, scaphoid view & MRI

Signs	**snuffbox tenderness** · **scaphoid tubercle tenderness** · **scaphoid compression** (axial thumb load) · pain on **resisted pronation**
All 3 positive in 24 h	sensitivity 87–100%, specificity 74%
Secondary indicators (TO)	radial wrist swelling · pain on **radial deviation** · tubercle tenderness most sensitive if examined **within hours** · **pain out of proportion → suspect occult fracture** · always check for associated **SL injury**
Stecher's view (TO)	**20° ulnar deviation, wrist pronated** · MRI = gold standard for occult fracture + AVN (TO) · CT best for fracture line, displacement, sclerosis, union



Anatomic snuffbox

4 Diagnosis & Classification

Key radiology findings

Imaging	Fact asked
X-ray series	PA, lateral, **45° semi-pronated oblique**, **scaphoid view**
Scaphoid view	**PA in 20° ulnar deviation** → waist best
Initial X-ray negative	**27% → repeat in 14–21 d** (immobilise meanwhile)
MRI	**Most sensitive for occult # <24 h** (~100%); **proximal pole vascularity (T1)**
Bone scan	Occult # at **72 h** (sens 100%, spec 98%)
CT (1 mm along scaphoid axis)	**Displacement, humpback, union/nonunion after fixation**; not for occult #

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Measurement / sign	Meaning
Displacement >1 mm	Unstable → fix
Intrascaphoid angle >35° / humpback	Flexion malalignment → ORIF (volar)
Radiolunate angle >15°	**DISI** → ORIF
Cysts, fracture widening	**Delayed union** (Herbert C)
Sclerosis / ↓ T1 signal proximal pole	**AVN** (MRI T1 best)

✗ Wrong options: **Wrong views** – 30° supinated oblique = **pisiform**; hyperextension axial = **carpal tunnel view**; clenched fist = **SL ligament**

High suspicion + negative X-ray + wants to RTP — **MRI**



Scaphoid view (20° ulnar deviation)

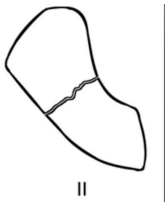
Herbert subtypes (TO) — A1 tubercle · A2 incomplete · **B1 distal oblique** · **B2 waist** · **B3 proximal pole** · **B4 trans-scaphoid perilunate** · **B5 comminuted** · C delayed union · D nonunion

Herbert & Fisher (stability)

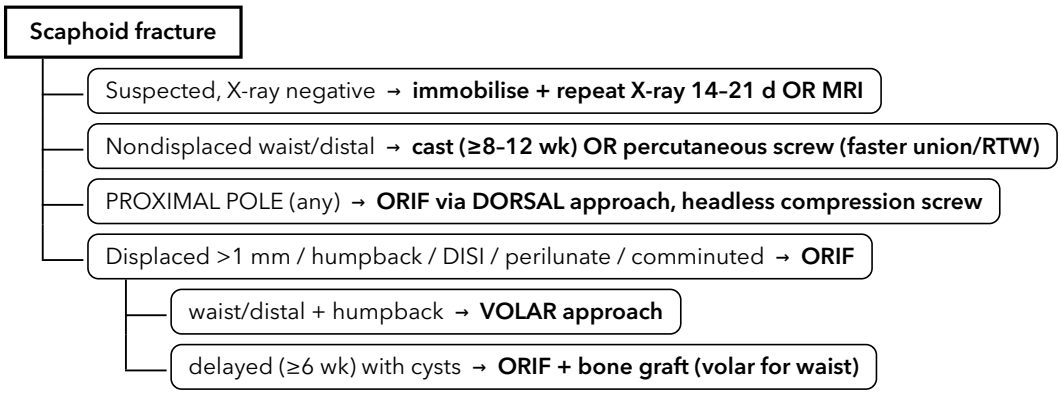
Type	Description
A	**Stable acute** (tubercle A1, incomplete waist A2)
B	**Unstable acute** – distal oblique, complete waist, **proximal pole**, trans-scaphoid perilunate
C	**Delayed union** (cysts, widening)
D	**Nonunion**

Mayo (location) — I distal tubercle · II distal articular · III distal ⅓ · IV middle ⅓ · V proximal ⅓

Russe (fracture line) — I **horizontal oblique** · II **transverse** · III **vertical oblique (least stable)**

Russe I	Russe II	Russe III
		

5 Management



Nonop union **<1 mm displacement → 90%** (88–95% if started <3 wk); ↑ nonunion if immobilisation delayed **>4 wk**

Cast duration by site (TO)	Weeks
Distal pole	**6–8 wk**
Waist	**10–12 wk**
Proximal pole	**4–6 months**

Cast type short-arm **thumb spica** (TO) · wrist alone sufficient (OB) – read the options

Operative indications (TO) displacement **> 1 mm** · humpback · comminution · proximal pole · SNAC · high demand · **retrograde screw preferred** in most

Cast **wrist alone sufficient** (thumb & elbow add nothing but stiffness – OB); long vs short arm: no consensus; ≥8 wk, up to 12–14 wk; RTP only when healed

PEMF may help delayed union

Operative vs nonoperative – percutaneous fixation of nondisplaced waist fractures

Percutaneous vs cast **↓ time to union (6–7 vs 10–12 wk)**, **faster RTW/sport**; similar total cost; no higher union rate; no better 2-y grip/ROM

Union after surgery 90–95%

Strongest reason to fix (stem) **fracture location (proximal pole)**
✗ Wrong options: age, job, dominance, chronicity at 1 wk

– – Active space – –

ORIF indications & approach (displacement, proximal pole, delayed presentation)

Operative criteria

Treatment	Criteria
Cast	Stable, **nondisplaced (<1 mm)** waist/distal; start **<3–4 wk**
Percutaneous screw	Nondisplaced waist (faster union/RTW) · displacement >1 mm **without** significant angulation
ORIF	**Displaced**, humpback, DISI, perilunate, comminuted, unstable vertical oblique, **proximal pole**
ORIF + graft	**Delayed presentation with cysts**
Vascularised graft	**Nonunion with cysts/AVN** only

ORIF indications **displacement >1 mm** · **intrascaphoid angle >35°** · **radiolunate angle >15° (DISI)** · humpback 15° · **perilunate** · comminuted · unstable vertical oblique · **proximal pole**

	Volar approach	Dorsal approach
Best for	**Waist & distal ⅓, humpback**	**Proximal pole**
Interval	**FCR – radial artery**	Limit to proximal half of scaphoid
Blood supply	**Spared** (TO ✓)	Risk to dorsal ridge vessels
At risk	–	**EPL**; proximal pole cartilage
Weakness	Reduction harder (TO); not for oblique/proximal pole (can't be perpendicular)	Poor distal ⅓ access
Must close	RSC ligament	–

TO: volar better because **(1) less vascular injury, (3) distal ⅓ easier**
✗ Wrong options: easier reduction · oblique fractures

Proximal pole **open reduction + headless differential-pitch screw via dorsal approach** (100% healing reported)
✗ Wrong options: cast · K-wires · percutaneous · vascularised graft acutely

K-wires only if proximal fragment too small for screw head

Displaced waist with cysts at 6 wk **ORIF + bone graft, volar**

Screw fixation technique – central axis position

Most important **long screw in central axis** (into subchondral bone) → max rigidity

Center-center on AP & lateral; **volar approach → excise proximal trapezium** to get dorsal-ulnar start point (else too volar & radial → nonunion)

Open vs percutaneous dorsal open: **↓ risk of screw prominence** above subchondral bone (direct view)

Don't violate STT cartilage



Percutaneous screw fixation

Paediatric scaphoid fractures

Most common now	**middle ⅓ (waist)**; unicortical up to 21%
Treatment	nondisplaced → cast **4–6 wk**; displaced → ORIF

6 Complications

Persistent pain after fixation – CT assessment

Nonunion definition	no union **6 mo** after cast/fixation
Pain months after fixation, hardware OK on X-ray	**fine-cut CT along scaphoid axis** ✗ Wrong options: MRI (metal) · bone scan (nonspecific) · clenched-fist view

Complication	Key fact
Nonunion	5–10% after cast; ↑ proximal pole, **vertical oblique**, displacement >1 mm, age, **smoking**
AVN	13–50%; proximal ⅕ ≈ 100%
Malunion	**Humpback** (flexed distal fragment)
Screw prominence	Subchondral penetration → arthrosis
SNAC	End result of nonunion

Vascularised bone graft (medial femoral condyle)

Indication	nonunion with **cysts / AVN / collapse**
MFC graft pedicle	**longitudinal branch of descending geniculate artery** (superomedial genicular in 11–20%)
MFT (medial femoral trochlea)	**transverse branch** of DGA → **proximal pole** (osteochondral)
1,2 ICSRA	distal radius pedicled; smaller graft
Not for acute fractures	vascularised graft only for nonunion

– – Active space – –

X vs Y

Imaging question	Answer
Occult fracture <24 h	**MRI**
Occult at 72 h (no MRI)	Bone scan
Displacement / union after fixation	**CT**
Proximal pole AVN	**MRI T1**

Graft	Pedicle	Use
MFC	**Longitudinal br. DGA**	**Waist nonunion with collapse**
MFT	**Transverse br. DGA**	**Proximal pole** (needs cartilage)
1,2 ICSRA	Radial artery branch	Smaller dorsal graft

How it's asked

"Most commonly fractured carpal bone?"	→ **scaphoid** (TO)
"Snuffbox tender, X-ray normal, wants to compete tomorrow"	→ **MRI**
"How to shoot a scaphoid view?"	→ **PA in 20° ulnar deviation**
"Percutaneous vs cast for nondisplaced waist"	→ **reduced time to union**
"Nonop – immobilise what?"	→ **wrist only** (OB)
"Most compelling reason for ORIF"	→ **location (proximal pole)**
"Volar better than dorsal because"	→ **less vascular injury + distal ⅓ easier** (TO)
"Proximal pole fracture"	→ **open reduction + headless screw via dorsal approach**
"Displaced waist, cysts at 6 wk"	→ **ORIF + graft, volar**
"Most important for fixation strength"	→ **central-axis screw into subchondral bone**
"Nonunion after volar screw – what would have helped"	→ **center-center position**
"Open vs percutaneous dorsal"	→ **↓ screw prominence**
"Pain 7–9 mo after fixation"	→ **CT along scaphoid axis**
"Force through scaphoid fossa ↑ with"	→ **wrist extension**
"MFC vascularised graft pedicle"	→ **longitudinal branch of descending geniculate**
"Commonest paediatric pattern"	→ **middle ⅓** (OB)

SCAPHOID FRACTURE NONUNION

– Hand · 79 / 103 –

1 Basics

Epidemiology & risk factors

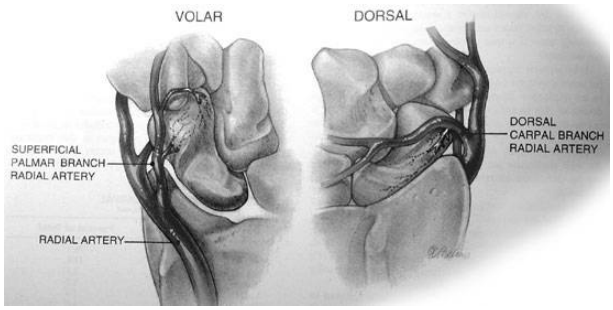
Incidence	**5–10% after immobilisation** (Orthobullets) · "up to 10%" (OB) · **~25% after surgical fixation** (some studies)
Demographics	♂:♀ **2:1**; 3rd decade
Risk factors	**proximal pole** fracture · **vertical oblique** pattern · **displacement >1 mm** · age · **nicotine** · **delayed diagnosis/treatment**
Why prone	retrograde supply, limited cancellous bone, few soft-tissue attachments, delayed Dx
Missed fracture	fracture line may need **1–2 wk** to appear on X-ray

Anatomy – blood supply

Shape	boat / twisted peanut; oblique to long axis; **largest proximal-row bone**; **>75% articular cartilage**
Articulations	radius, lunate, trapezium, trapezoid, capitate

Vessel	Entry	Supplies
Dorsal carpal branch of radial a. (major)	Dorsal non-articular ridge	**Proximal 80%, retrograde**
Superficial palmar arch (volar radial a.)	Distal tubercle	Distal 20%

AVN rate	**15–50%** of scaphoid fractures; ↑ proximally; **proximal 1/5 → ~100%**
AVN site	**proximal** pole ✗ Wrong options: distal



Scaphoid blood supply

2 Mechanism

Humpback deformity	proximal pole **extends** with lunate (SL ligament) · distal pole **flexes** (scaphotrapezial lig.) · hinge = **volar radioscaphocapitate ligament**
Natural history	untreated → carpal derangement, pain, cartilage loss → **SNAC**

– – Active space – –

3 Presentation

History	remote fall / "sprain"; get prior op notes & imaging
Symptoms	wrist pain with motion, ↓ grip; some asymptomatic
Exam	swelling, **snuffbox / fracture-site tenderness**, ↓ ROM

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray	PA, lateral, 45° semi-pronated oblique, **scaphoid view**; cysts, sclerosis, resorption, hardware failure, **humpback on lateral**, SNAC
CT	**Best to evaluate nonunion & plan surgery**; **1 mm cuts in scaphoid plane**; gap, angulation, prior technical error, SNAC cysts
MRI	**Osteonecrosis**: ↓ T1 signal **proximal fragment** (different signal between fragments); inconsistent utility, gadolinium helps · distal-fragment oedema also dark on T1 → compare FS images (TO)
Perfusion (dynamic contrast) MRI (TO)	**Zero proximal-pole enhancement = necrosis** → dead pole not simply fixed · viable pole → freshen + compression screw
CT vs MRI (TO)	Weeks-old fracture with pain: **CT for union, MRI for viability**

Intra-op punctate bleeding — **highly predictive** of vascular proximal pole; absent → vascularised graft



Scaphoid view

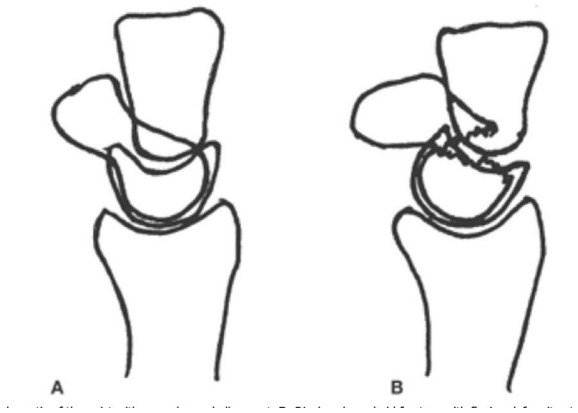
Diagnosis – nonunion, humpback deformity & proximal pole osteonecrosis

Diagnosis — history + exam + X-ray + **CT**; MRI for AVN

MRI picture	Diagnosis
Fracture line + **different signal between fragments**	**Scaphoid nonunion + osteonecrosis**
Most/all of scaphoid necrotic	**Preiser's**
Lunate	Kienböck
Fluid area with dense rim	Intraosseous ganglion
SL ligament injury, widened interval	SL dissociation

Sequela of nonop Rx **humpback deformity**

✗ Wrong options: distal pole AVN · capitolunate OA = SLAC III · radiolunate OA



Humpback deformity

Classification – stability

Type	Features
Stable	Length & alignment kept, **fibrous union**
Unstable	Loss of length/alignment, carpal instability or chondral change

5 Management

Scaphoid nonunion

- No AVN, no SNAC → **ORIF + non-vascularised graft + screw**
  - no humpback → **inlay (Russe)**
  - humpback / carpal collapse → **interposition wedge (Fisk), VOLAR approach**
- Proximal pole AVN / no punctate bleeding, no pancarpal OA → **vascularised graft**
  - pedicled distal radius (1,2-ICSRA dorsal · Mathoulin volar · Sotereanos)
  - free MFC (volar, humpback) · MFT if cartilage lost · iliac crest
- Advanced SNAC / arthritis → **salvage (PRC, 4-corner / scaphoid excision + intercarpal fusion, arthrodesis)**

Operative criteria

Situation	Treatment
No union by **6 mo**, technical error (screw position, implant failure, distraction)	**ORIF + graft**
Inadequate prior immobilisation / refuses surgery	Cast (up to 6 mo) ± **PEMF** – 69% union by 3 mo (fixed, no error)
Technical error / displacement	**no cast**

– – Active space – –

Without osteonecrosis – ORIF with non-vascularised graft (volar approach for humpback)

Standard	**ORIF + autologous bone graft**; union **92%**; best when failure = simple technical error ✗ Wrong options: cast · arthroscopy · 4CF / arthrodesis in young w/o SNAC
Humpback, no AVN, no prior surgery	**volar approach + standard (non-vascularised) graft + screw** ✗ Wrong options: K-wire · VBG · dorsal
Volar advantages	corrects humpback (K-wire joysticks), grafts volar defect, **preserves dorsal blood supply**
Dorsal approach	most acute waist fractures; proximal pole visualisation; ↑ **devascularisation risk**
Screw direction	volar: **distal → proximal** · dorsal: proximal → distal; plate = volar buttress; K-wire removable
Casting nonunion	4–6 mo, high failure; thumb spica / above-elbow **does not improve union**

Graft	Type	Indication	Union
Russe (inlay)	Non-vasc corticocancellous, volar (FCR–radial a.)	**No collapse/ humpback**	**92%**
Fisk (interposition wedge)	Non-vasc, distal radius (Fernandez: iliac crest)	**Humpback / carpal collapse**	72–95%
BMP / PRP	Adjunct	No SNAC; avoids free flap	High (series)

Proximal pole AVN – vascularised (MFC) graft

Indications (VBG)	waist nonunion + **proximal pole AVN** · **no punctate bleeding** · not with pancarpal OA / collapse · failed non-vasc graft
Relative contraindication	no intact cartilage shell · advanced collapse with OA (e.g. SNAC + scaphocapitate arthrosis → arthrodesis)

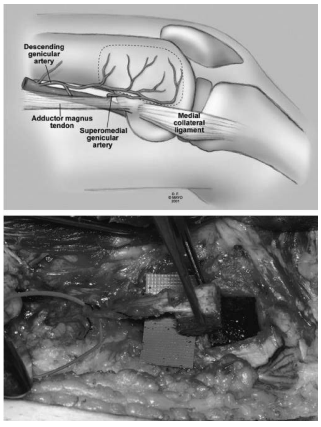
Graft	Pedicle / approach	Outcome
1,2-ICSRA (Zaidemberg)	Branch of **radial a.**, **dorsal** (1st–2nd compartments)	**71% nonunion / 50% AVN**; ↑ failure with DISI/ humpback
Mathoulin	**Volar carpal a.**, volar	Wedge for humpback
Sotereanos	Artery of **4th dorsal compartment**, dorsal capsular pedicle	Inlay
Local VBG overall	–	82% good/excellent
Free MFC	**Descending genicular a.** (longitudinal br., from **SFA**) or superomedial genicular; **volar**, anastomosis to **radial a.**	**100% by 13 wk** (1 study)
Free MFT	Osteochondral; DGA periosteal branches	AVN + **cartilage loss** of proximal pole; >90%
Free iliac crest	**Deep circumflex iliac a.**	76%

AVN + no bleeding + collapse	**free vascularised MFC via volar approach** ✗ Wrong options: non-vasc · 1,2-ICSRA dorsal = inferior union

VBG vs non-vasc – conflict	**88% vs 47%** union with proximal pole AVN (OB8865) · **no difference** in type/source on meta-analysis for waist nonunion (OB219567) · no clear superiority (Orthobullets) – read the options
Graft source	no single source superior (iliac crest = distal radius); young patients **85–100%** union
MFC uses	**proximal scaphoid AVN / nonunion** ✗ Wrong options: lunate OCD · acute humpback · SNAC with arthrosis · Kienböck stage I (4+5 ECA there)



1,2-ICSRA graft



MFC graft

6 Complications

Osteonecrosis	proximal fractures
Graft failure → SNAC	radioscaphoid (styloid) → capitolunate arthrosis

– – Active space – –



SNAC wrist

X vs Y

	Russe	Fisk	1,2-ICSRA	Free MFC
Vascularised	No	No	✓ pedicled	✓ free
Approach	Volar	Volar	**Dorsal**	**Volar**
Humpback correction	No	**Yes**	Poor	**Yes**
Proximal pole AVN	No	No	Yes (inferior)	✓ **best**

How it's asked

"MRI 18 mo after fall – fracture line, different fragment signal"	→ **scaphoid nonunion + osteonecrosis** ✗ Wrong options: Preiser, Kienböck
"Radiographic sequela of nonop Rx"	→ **humpback deformity**
"Waist nonunion 6 mo, young"	→ **ORIF + autologous bone graft**
"Humpback, no AVN – best surgery"	→ **volar, standard graft, screw**
"Army medic, nonunion >1 y, no SNAC"	→ **ORIF + bone graft** ✗ Wrong options: cast, PRC, fusion
"Delayed Dx humpback on CT"	→ **ORIF single screw, volar approach** ✗ Wrong options: dorsal · cast · 4CF
"MFC VBG + fixation best for"	→ **proximal scaphoid AVN**
"AVN, no bleeding at surgery, collapse"	→ **free vascularised femoral graft, volar approach**
"Waist nonunion with AVN – true statement"	→ **vascularised = non-vascularised union rates** ✗ Wrong options: dorsal corrects humpback · iliac crest superior

SCAPHOID LUNATE ADVANCED COLLAPSE (SLAC)

– Hand · 80 / 103 –

1 Basics

SL dissociation – pathomechanics

Definition	predictable degenerative arthritis from **chronic SL dissociation**; **most common form of wrist arthritis** (Watson & Ballet 1984)
Causes	SL ligament tear (most common cause of carpal instability); also **crystalline arthropathy**
SLIL	C-shaped: dorsal, proximal (fibrocartilage), volar; **dorsal = thickest (3 mm) & strongest → most critical for SL stability** (volar ~1 mm)
Not key SL stabilisers	RSC (sectioning with intact SLIL → no kinematic change), DRC, DIC, volar SLIL
SL dissociation	**scaphoid flexes + lunate extends** = **DISI**; SL angle **>70°**, lunate >10° extended
Why radiolunate spared	lunate stays **concentric** in spheroidal lunate fossa in all positions of rotation

2 Mechanism

Stages & sequence of arthritis (radial styloid, radioscaphoid, capitolunate; radiolunate spared)

Sequence	**radial styloid–scaphoid → entire radioscaphoid → capitolunate** (capitate migrates proximally into SL gap, subluxes dorsally on lunate)
First joint affected	**radioscaphoid** (radial styloid tip)
Spared	**radiolunate** (and lunotriquetral)

3 Presentation

Symptoms	pain at SL interval, difficulty weight-bearing through wrist, stiffness, progressive weakness
Signs	dorsal SL tenderness, ↓ ROM, ↓ grip
Watson scaphoid shift	pressure on **scaphoid tuberosity**, wrist **ulnar → radial deviation** → scaphoid subluxes dorsally (pain) → clunk on release; **positive early only** (arthritis stabilises scaphoid later); compare sides

4 Diagnosis & Classification

Key radiology findings

Views	PA + lateral; stress views and MRI unnecessary for staging

– – Active space – –

SL diastasis (PA)	**>3 mm** (Orthobullets) · **>2 mm** + asymmetric to other side (OB) – read the options
PA by stage	I: **radial styloid beaking**, sclerosis · II: narrowing whole scaphoid fossa · III: capitolunate narrowing, capitate migrates proximally
Lateral	**DISI**, capitate subluxed dorsally on lunate
MRI	thinning of proximal scaphoid / scaphoid facet / capitolunate cartilage, synovitis

Watson classification

Stage	Arthritis	Image
I	**Radial styloid – scaphoid**	
II	**Scaphoid – entire scaphoid facet** of radius	
III	**Capitate – lunate** (+ scaphocapitate)	
"IV" (controversial)	**Radiolunate / pancarpal**	–

Radiolunate spared	all Watson stages; **pancarpal ("stage IV")** → think other aetiology (RA, AVN)
Differential	**SNAC** (stage II = scaphocapitate; proximal pole preserved as extension of lunate)

5 Management

SLAC wrist

- Mild → **NSAIDs, splint, steroid injection**
- Stage I → **radial styloidectomy / AIN + PIN denervation**
- Stage II → **PRC or scaphoid excision + 4CF**
  (smoker / fewer complications → PRC)
- Stage III (capitolunate) → **scaphoid excision + 4CF or wrist fusion**
- Radiolunate / pancarpal → **total wrist fusion**

Operative criteria by joint status

Procedure	Needs intact	Contraindicated by
PRC	**Radiolunate facet + capitolunate (capitate head)** + **RSC**	**Capitolunate arthritis (stage III)**, incompetent RSC
Scaphoid excision + 4CF (capitate–lunate–triquetrum–hamate)	**Radiolunate facet**	**Radiolunate arthritis**
Total wrist fusion	–	– (best pain relief + grip, loses motion)

Not contraindications radioscaphoid arthritis, capitolunate arthritis (for 4CF), SL disruption, scaphoid osteonecrosis

Stage I – styloidectomy & denervation

Radial styloidectomy open or arthroscopic (1-2 portal); removes scaphoid–styloid impingement

Denervation **PIN + AIN** (distal branches = proprioception/capsule only) ± dorsal ulnar sensory, radial sensory, LABC, palmar cutaneous; **↓ pain, no ↑ function/ strength/ROM**; can add to bony procedures for II–III

Salvage – four-corner fusion vs PRC (capitolunate arthritis contraindication, RSC ligament)

PRC excise scaphoid + lunate + triquetrum; capitate sits in lunate fossa; **preserve RSC** = **primary stabiliser after PRC** (prevents ulnar translocation)
✗ Wrong options: DRC, DIC, ulnocapitate, ulnotriquetral

4CF + scaphoidectomy motion through radiolunate joint; **only option addressing all stage III sites**

PRC vs 4CF (stage II) similar long-term results; **PRC = fewer complications**, technically easier, no prolonged immobilisation → **preferred in smokers** (Tomaino)
✗ Wrong options: nonunion, hardware, dorsal impingement

Not for SLAC STT fusion (SL instability, STT OA, Kienböck) · scaphocapitate fusion (Kienböck) · **distal pole excision (SNAC)** · RSL fusion · **TWA** (young/active, e.g. farmer; OK for older low-demand / RA)

Total wrist fusion pancarpal / radiolunate arthritis, failed PRC; less suitable in a 65-year-old needing motion

Radioscapholunate fusion – distal scaphoid excision

RSL fusion radioscaphoid + radiolunate OA with **preserved midcarpal joint**

Add distal scaphoid pole excision (~20%) ↑ ROM (dart-thrower's motion), ↓ strain → **↑ fusion rate**; ± triquetrum excision

Without it nonunion, poor motion (esp. radial deviation), weak grip

– – Active space – –

X vs Y

	SLAC	SNAC
Cause	SL ligament dissociation	Scaphoid nonunion
Stage II	Entire radioscaphoid	**Scaphocapitate**
Proximal scaphoid	Arthritic	**Preserved** (moves with lunate)
Stage I extra option	–	**Distal pole excision**
Radiolunate	Spared	Spared

	PRC	Scaphoidectomy + 4CF
Stage	**II**	**II–III**
Capitolunate OA	**contraindicated**	Yes
Radiolunate OA	No	No
Complications	**Fewer**	Nonunion, hardware, impingement

How it's asked

"First joint affected in SLAC?"	→ **radioscaphoid** ✗ Wrong options: capitolunate · radiolunate
"Stage III SLAC joint?"	→ **capitolunate**
"Order of degeneration?"	→ **radial styloid → radioscaphoid → capitolunate**
"Stage III – arthritis absent between?"	→ **lunate and radius**
"Stage III, preserve motion?"	→ **scaphoid excision + 4CF** ✗ Wrong options: PRC · STT · neurectomy · wrist fusion
"Primary stabiliser after PRC?"	→ **radioscaphocapitate**
"Stage III SLAC treatment?"	→ **scaphoid excision + capitate-lunate-triquetrum-hamate arthrodesis**
"Stage II, smoker, lowest complication risk?"	→ **PRC** ✗ Wrong options: 4CF · styloidectomy · TWA
"Stage III, farmer, failed injections?"	→ **scaphoid excision + 4CF** ✗ Wrong options: scaphocapitate · distal pole excision · TWA
"Degeneration that contraindicates PRC?"	→ **capitate–lunate**
"Contraindication to 4CF?"	→ **radiolunate arthritis**
"RSL fusion nonunion + stiff – what step was missed?"	→ **distal scaphoid pole excision**
"Most critical structure for SL stability?"	→ **dorsal SLIL**

Scaphoid Lunate Advanced Collapse (SLAC)

"Result of SL dissociation?"

→ scaphoid flexion + lunate extension (DISI)

– – Active space – –

– – Active space – –

SCAPHOID NONUNION ADVANCED COLLAPSE (SNAC)

– Hand · 81 / 103 –

1 Basics

Scaphoid anatomy

Main blood supply	**dorsal carpal branch of radial a.** → dorsal nonarticular ridge → **proximal 80%, retrograde**
Minor supply	**superficial palmar arch branch** (volar radial) → distal tubercle → distal 20%
Motion	flexes with wrist flexion + radial deviation; extends with extension + ulnar deviation

2 Mechanism

Natural history – radiolunate preserved

Sequence	**radioscaphoid (radial styloid) first** → scaphocapitate/midcarpal → pancarpal
Radiolunate	**preserved** until very end stage → **least likely** in natural progression
Poor union predictors	nonunion **>5 y**, **proximal pole necrosis**, heavy work, prior radial styloidectomy, immobilisation duration

3 Presentation

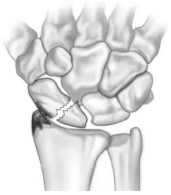
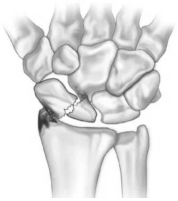
Symptoms	↓ grip & pinch, stiffness in **extension + radial deviation**
Exam	tender **radioscaphoid** joint; ↓ extension & radial deviation

4 Diagnosis & Classification

Key radiology findings

Views	**AP + lateral wrist**
Findings	scaphoid nonunion + radioscaphoid arthritis progressing per stage; **capitolunate & lunate fossa preserved** until late
Differential	SLAC wrist, scaphoid nonunion without arthrosis

SNAC classification

Stage	Orthobullets (radiographic)		OB7857 explanation
I	**Radial styloid + radial side of scaphoid**		Radial styloid
II	Stage I + **scaphocapitate**		Radioscaphoid
III	**Periscaphoid** (proximal lunate & capitate may be kept)		Midcarpal

TO staging 1 radial styloid · 2 radioscaphoid · **3 capitolunate** (matches OB7857)

(TO)	PRC	4-corner fusion
Pros	Faster recovery, simpler	Strong, stable, keeps some motion
Cons	**Not if capitolunate arthritis**	Technically demanding, nonunion risk

Source conflict stage II = scaphocapitate (Orthobullets) · radioscaphoid (OB7857) – read the options

5 Management

- **SNAC wrist**
  - Frail / low demand → **OBSERVATION**
  - Stage I → **RADIAL STYLOIDECTOMY + scaphoid repair/SL stabilisation, or DISTAL POLE EXCISION**
  - Stage II–III → **PRC / 4-CORNER FUSION / wrist arthrodesis**
    - capitate head arthritis → no PRC → **4-CORNER FUSION**
    - capitate head intact → **PRC or 4CF**

Union attempt only if **no arthrosis or stage I** (with styloidectomy); no arthroscopic debridement with arthrosis

– – Active space – –

Salvage – four-corner fusion, PRC & distal pole excision (contraindications, RSC ligament)

Procedure	Indication	Key facts / contraindications
Distal pole excision	Stage I (SNAC I–III not contraindications)	**Pre-existing DISI = contraindication** (↑ SL angle, destabilises proximal row); not for STT OA
PRC	Stage II–III	Needs intact **capitate head + lunate fossa**; **capitolunate/capitate head arthritis = contraindication**; ↓ motion & grip; **preserve RSC**
Scaphoid excision + 4-corner fusion (or capitolunate arthrodesis)	Stage II–III (incl. capitate arthritis)	Retains **60% motion, 80% grip**
Wrist arthrodesis	Stage II–III	Final salvage

RSC ligament — most **volar-radial** ligament; radial styloid → carpus obliquely, radial to long & short RL ligaments → excision at PRC → **ulnar translocation of carpus**



RSC ligament at PRC

62 y, stage III SNAC, failed splint/NSAIDs → scaphoid excision + 4-corner fusion
✗ Wrong options: debridement · ORIF · bone graft ± vascularised

6 Complications

PRC + RSC excised → ulnar translocation
✗ Wrong options: capitate AVN (that is a contraindication, not complication) · DRUJ instability

Distal pole excision → ↑ SL angle / DISI, worsening arthritis in half (Malerich)

Prognosis

Punctate bleeding at surgery — union **92%** obvious · **71%** questionable · **0%** none

X vs Y

	PRC	4-corner fusion
Needs capitate head cartilage	**Yes**	No
Stability depends on	**RSC ligament**	Fusion
Motion / grip	↓	**60% / 80%**

How it's asked

"Untreated scaphoid nonunion — least likely arthritis?"	→ **radiolunate**
"Stage III SNAC, failed nonop?"	→ **scaphoid excision + 4-corner fusion**
"Motion-sparing salvage on capitate head; most volar-radial ligament excised — outcome?"	→ **ulnar translocation of carpus**
"Contraindication to distal pole excision?"	→ **DISI** ✗ Wrong options: SNAC I–III arthrosis
"Arthritis between which bones contraindicates PRC?"	→ **capitate & lunate (capitate head)**
"Main blood supply of scaphoid?"	→ **dorsal carpal branch of radial artery (retrograde, proximal 80%)**

– – Active space – –

SCAPHOLUNATE LIGAMENT INJURY & DISI

– Hand · 82 / 103 –

1 Basics

Epidemiology

Acute	**10–30%** of intra-articular DRF / carpal fractures (Orthobullets) · **30%** of DRF (OB)
Degenerative	tears in **>50% of >80-y-olds**; incomplete > complete
Commonest carpal instability (TO)	**SL instability / DISI** = commonest form of **CID** (carpal instability dissociative)

Anatomy – dorsal SLIL is strongest

SLIL — **C-shaped**; **dorsal, proximal (membranous), volar** components

SLIL component	Thickness	Role
Dorsal	**3 mm**	**Strongest; greatest constraint to SL translation**; must tear for dissociation
Volar	1 mm	2nd
Proximal (fibrocartilaginous membrane)	–	**Minimal strength**

Yield strength	**dorsal > volar > proximal**
Intercalated segment (TO)	scaphoid, lunate, triquetrum: **no tendon insertions**
Balance	scaphoid wants to **flex** (oblique, distal row pull, RSC); triquetrum + lunate want to **extend**; SLIL = tether
Secondary stabilisers (TO)	DRCL, **DIC**, **scaphotrapezial (STT)**, **RSC**, scaphocapitate
SLIL alone cut	no static change; needs ≥1 secondary stabiliser (TO)

2 Mechanism

Injury	FOOSH with **wrist extension + ulnar deviation + carpal supination**; also repetitive (crutch walking) (TO)
Pathoanatomy	**scaphoid flexes + lunate extends → DISI** → untreated → **SLAC**

Secondary stabilisers & progression to static DISI

Stage (TO)	Ligaments	X-ray
Dynamic instability	SLIL torn/incompetent; secondary stabilisers intact	**Normal static**; abnormal on **stress / clenched fist**
Static (SL dissociation)	SLIL + secondary stabiliser	Abnormal static → DISI
SLAC	Attritional changes	Progressive arthritis

Primary secondary stabiliser (TO) — **STT ligament complex** (+ scaphocapitate) → failure = **dynamic → static DISI**
✗ Wrong options: DIC · space of Poirier · volar radiolunate

Static DISI = repair window passed (TO) — reconstruction / capsulodesis / salvage

Iatrogenic DISI after distal scaphoid excision

Distal pole scaphoid excision (for STT OA) — shortens scaphoid → triquetral extension moment dominates → **DISI** (SL angle 80°)

Do not cause DISI — STT fusion, trapezoid resection, LRTI, 4-corner fusion

STT procedure	Complication
Distal scaphoid excision	**DISI**; **flexor** tendon rupture (sharp remnant)
STT fusion	**Radial styloid impingement ~33%**; adjacent OA radiocarpal 42%, TMC 28%

STT OA — 2nd commonest carpal OA; linked to thumb CMC OA; options: distal pole excision, STT fusion, pyrocarbon

3 Presentation

Symptoms — **dorsoradial wrist pain**, ↑ with loading (**push-ups**), clicking, ↓ grip, giving way

Tenderness — snuffbox / **dorsal SL interval just distal (~1 cm) to Lister's tubercle**

Watson (scaphoid shift) test — thumb on **volar scaphoid tubercle**, wrist **ulnar → radial deviation** → proximal pole subluxes dorsally → **painful clunk** on release

4 Diagnosis & Classification

Diagnosis – clenched-fist views, scapholunate angle & DISI

Next step, normal PA after FOOSH with dorsal tenderness — **(bilateral) PA clenched-fist view** (capitate driven proximally)
✗ Wrong options: arthroscopy first · CT · EMG

Not radial deviation PA — **closes** SL gap

– – Active space – –

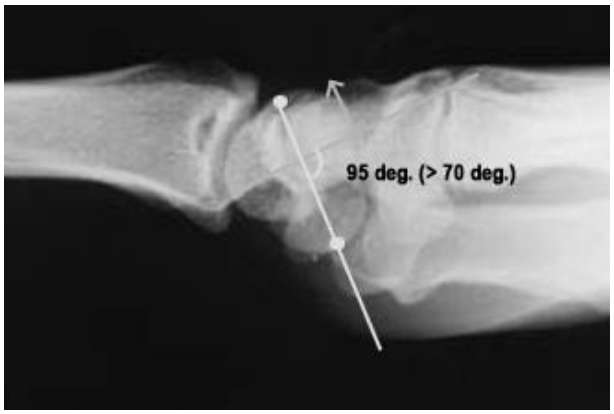


Clenched-fist view widening SL gap

Key radiology findings

View / sign	Threshold / meaning
Terry Thomas (PA / clenched fist)	**SL gap >3 mm**
Cortical ring sign	Flexed scaphoid seen end-on
Scaphoid	Shortening; **humpback** = unstable scaphoid fracture / malunion
SL angle (lateral)	Normal **30–60°** (TO) · **30–70°** (OB); **>70° = DISI / SL dissociation**; static DISI **>60° (often >80°)** (TO module)
Radiolunate angle (TO)	Normal **15° ext – 15° flex**; **>15° extension → DISI**
Capitolunate angle	**>20°** (Orthobullets) · **>15°** (TO)
Carpal height ratio	↓ as collapse progresses
Lunate shape on PA (TO)	DISI → **piece-of-pie** (TO – verify: piece-of-pie classically = palmar-rotated lunate in lunate dislocation) · VISI → **C-shaped / moon lunate** · mild palmar tilt → isosceles
PA with 10° ulnar angulation (TO)	Measures **mid-SL interval**
VISI	SL angle **<30°**

Modality	Fact
Arthrography	Screens; shows tear but **not size**; compare contralateral
MRI	Overused; **low sensitivity**
Arthroscopy	**Gold standard**



DISI: SL angle >70°



Terry Thomas + cortical ring sign

SL dissociation staging (TO)

Stage (TO)	Description	Treatment
I	**Occult**	Conservative
II	**Repairable**	**SL repair + dorsal capsulodesis**
III	**Dynamic DISI**	**Three-ligament tenodesis**
IV	**Reducible static**	Tenodesis
V	**Irreducible**	**Radiocarpal fusion + distal scaphoid excision**
VI	**SLAC wrist**	**Scaphoid excision + midcarpal fusion**

SLAC stages (TO)

Stage	Arthritis
I	**Radial styloid–scaphoid**
II	Whole radioscaphoid
III	**Capitolunate (midcarpal)**
IV	Pancarpal incl. radiolunate

Radiolunate joint (TO) **preserved** in SLAC; SL joint itself spared until late (OB)

– – Active space – –

Perilunate spectrum (rocking lunate, Mayfield I)

Mayfield stage (TO)	Injury
I	**SL dissociation / scaphoid fracture / scaphoid rotation** (scaphoid fracture **not always** present)
II	Lunocapitate → perilunate dislocation
III	**LT disruption** / triquetral dislocation
IV	**Lunate dislocation** (extruded)

Parameter	Stage I	Stage II	Stage III	Stage IV
Radiographic findings	Scaphoid rotation	Capitate dislocation	Malrotated triquetrum and scaphoid Triquetrolunate diastasis Dislocated triquetrum Volar triquetral fracture	Lunate dislocation
Joints disrupted	Scapholunate	Scapholunate Capitolunate	Scapholunate Capitolunate Triquetrolunate	Scapholunate Capitolunate Triquetrolunate Radiolunate
Ligaments torn or attenuated	Radioscaphoid Scapholunate interosseous Radiocapitate	Radioscaphoid Scapholunate interosseous Radiocapitate Radial collateral ligament	Radioscaphoid Scapholunate interosseous Radiocapitate Radial collateral ligament Palmar radiotriquetral +/−Ulnotriquetral	Radioscaphoid Scapholunate interosseous Radiocapitate Radial collateral ligament Palmar radiotriquetral +/−Ulnotriquetral +/−Dorsal radiocarpal

Mayfield stages

Rocking lunate (TO) **SL + LT ligament tears**

SL tear (TO) → ↑ SL angle (lunate follows triquetrum into extension) · **LT tear** → ↓ SL angle

5 Management

SLIL injury

- Acute undisplaced / chronic asymptomatic → **NSAIDs, splint, repeat imaging**
- Acute, reducible, repairable → **OPEN REPAIR + K-wires ± dorsal capsulodesis**
- Reducible, not repairable or >18 mo → **RECONSTRUCTION (FCR / Brunelli)**
- Fixed unreducible DISI or SLAC → **SALVAGE**
  - SLAC I-II → **PRC or scaphoid excision + 4-corner fusion**
  - SLAC III → **4-corner fusion**
  - SLAC IV → **total wrist arthrodesis**

Operative criteria

Indication	Treatment
Acute, no malalignment, repairable	**SL ligament repair**
Chronic **reducible**, **<18 mo** (Orthobullets) · **<6 wk** acute (TO)	Repair
Acute non-repairable or reducible **>18 mo**	**Reconstruction**
Pathoanatomy = scaphoid fracture	ORIF / CRPP ± arthroscopy
Rigid unreducible DISI / severe DJD	**STT or SLC fusion** (SL fusion alone → **highest nonunion**)

Acute repair

Acute SL dissociation (5 days)	**open repair + percutaneous pinning** ✗ Wrong options: cast · arthroscopic pinning · capsulodesis (for chronic missed)
Technique	incision distal to radial styloid (protect **radial sensory n.**); 2 parallel K-wires in scaphoid → **joystick scaphoid into extension/supination/ulnar dev, lunate into flexion/radial dev** → drive into lunate; suture anchors
Augment	**Blatt / Berger dorsal capsulodesis** (DIC advancement); alone for chronic when repair not feasible
Post-op	K-wires out **8–10 wk**; no heavy labour 4–6 mo
Casting alone	insufficient for most; may suit incomplete tears

Reconstruction

Options	**FCR transfer (modified Brunelli, direct reduction)**, ECRB tenodesis (indirect), **3LT**, SLAM, **RASL** (screw, sacrifices motion), bone-ligament-bone graft; **avoid weave** (late failure)
Modified Brunelli steps (TO)	**distally based 8 cm split FCR** → drill **dorsal → palmar** → tendon through scaphoid → **transverse trough in lunate** → suture anchors → tension with **radiotriquetral ligament as pulley** · standard reconstruction for **static SL instability**
Dorsal approach (TO)	**lazy-S over Lister's tubercle** · expose **compartments 2 + 4** · optional **PIN neurectomy** · best SL visualisation
Post-op (TO)	**thumb spica 2 wk** · **K-wires out at 8 wk** · detachable splint + ROM after 8 wk
Salvage (TO)	**4-corner fusion** (keeps ~50% flex-ext; preserves radiolunate) · **PRC** (needs intact **capitate head + lunate fossa**)

6 Complications

List	**SLAC progression**, arthritis, post-op pain, stiffness, fatigue, ↓ grip

– – Active space – –

X vs Y

	DISI	VISI
Ligament	**SLIL (dorsal)**	**LTIL** + dorsal radiotriquetral ± volar radiolunate
Lunate	**Extended**	**Flexed** (triquetrum extends)
SL angle	**>70°**	**<30°**
Pain	Dorsoradial	**Ulnar**, ↑ pronation + ulnar deviation
Tests	**Watson**	**LT shuck / Reagan ballottement, Kleinman shear**
Normal variant?	**Always pathological**	Can occur with **ligamentous laxity**
Frequency	Commoner	Less common; LT arthrodesis nonunion 20–40%

Carpal translation	Ligament
Ulnar translation of carpus	**RSC** (radiocarpal dislocation, PRC)
Volar translation	**Short radiolunate** (volar-ulnar critical corner)
Lunotriquetral fusion	Commonest **carpal coalition**

How it's asked

"True about SL dissociation EXCEPT"	→ **SL angle 45° in advanced cases** (TO)
"FOOSH, SL gap 3.7 mm, radiolunate 25° extension — Dx?"	→ **SL instability with DISI** (TO)
"SL dissociation, all true EXCEPT"	→ **VISI** (TO) (✓ DISI, Terry Thomas, cortical ring)
"Watson +, PA normal — best view?"	→ **PA clenched fist** ✗ Wrong options: radial deviation · carpal tunnel view
"6 d after rollerblading, dorsal tenderness, normal X-rays — next?"	→ **bilateral PA clenched-fist X-ray**
"Radiographic finding of DISI?"	→ **lunate extended relative to scaphoid** ✗ Wrong options: SL angle <30°
"Untreated SL injury — first carpal deformity?"	→ **scaphoid flexion** ✗ Wrong options: lunate flexion · humpback
"SL angle 80° — which ligament?"	→ **dorsal scapholunate**
"72-y SL gap after MVA — ligament?"	→ **dorsal SL interosseous**
"Acute SL dissociation, footballer, 5 d?"	→ **open repair + percutaneous pinning**
"DISI 2 y after STT surgery — index operation?"	→ **distal scaphoid resection**
"Counsel before distal pole scaphoid excision?"	→ **DISI**

"Secondary stabiliser whose failure → static DISI?"	→ **STT ligament complex** (TO)
"Mayfield stage I true?"	→ **scaphoid fracture not always present** (TO)
"Rocking lunate results from?"	→ **SL + LT ligament tear** (TO)
"Garcia-Elias stage V (irreducible) SL dissociation — treatment?"	→ **radiocarpal fusion + distal scaphoid excision** (TO)
"Modified Brunelli — tendon used?"	→ **distally based split FCR (8 cm)** (TO)

– – Active space – –

SEYMOUR FRACTURE

– Hand · 83 / 103 –

1 Basics

Definition & epidemiology

Definition	**displaced distal phalanx physeal fracture + nailbed injury** in **children** (TO) = **open fracture**
Not	adult thumb hyperextension sports injury (TO)
Physeal involvement	20–30% of paediatric phalangeal fractures
Commonest digit	**middle finger** (TO)

Fracture types (TO)	Nailbed injury types
SH I	Nailbed laceration
SH II	**Nail plate subluxation** (superficial to eponychial fold)
Juxta-epiphyseal metaphyseal, 1–2 mm distal to physis	**Interposed germinal matrix** at fracture

Anatomy – why it flexes

Extensor	inserts **dorsal epiphysis** (with volar plate) → extension force on epiphysis
FDP	inserts **volar metaphysis** → flexion force on shaft
Result	**flexion moment at physis** → **apparent mallet deformity** (TO)

2 Mechanism

Direct trauma / crush	**finger caught in door** (TO), hammer, heavy object, ball, sport

3 Presentation

Exam	apparent **mallet deformity**; swelling, ecchymosis, subungual haematoma; **nail plate superficial to eponychial fold**
Delayed	tender swollen DIP, infection



Seymour fracture: flexion at physis, nail plate displaced

4 Diagnosis & Classification

Key radiology findings

Views	PA (**may look normal**) + **lateral** (flexion at fracture)
Signs	**widened physis** (→ **interposed tissue**) or epiphysis–metaphysis displacement
vs mallet fracture	paediatric mallet = **SH III/IV** avulsion, line **enters DIPJ**; Seymour line **traverses physis**, not DIPJ

5 Management

- **Seymour fracture**
  - Closed, minimally displaced, no interposition → **CLOSED REDUCTION (hyperflexion) + SPLINT**
  - Closed, displaced/unstable → **CLOSED REDUCTION + PIN ACROSS DIPJ**
    - reduction difficult → assume interposition → **OPEN**
  - Open (nail plate above eponychial fold) → **I&D + REMOVE INTERPOSED TISSUE + REDUCE + NAILBED REPAIR ± K-wire**

Anatomy & management – nail bed repair, remove interposed tissue, reduce

Open Seymour	**nail plate removal + nailbed repair + removal of interposed soft tissue (germinal matrix) + open reduction**
Must include	**removal of interposed soft tissue** (pinning optional adjunct)
Pinning (conflict)	open reduction + **retrograde K-wire across fracture + DIPJ** (Orthobullets, Abzug) · pinning **not required**; nail plate replaced under fold = sufficient stability (OB)
Avoid DIPJ-sparing pin	poor purchase in epiphysis
Avoid closed reduction + pin in interposition → locks germinal matrix in fracture	
Avoid mallet splint in situ / buddy taping	growth & nail disturbance
Nailbed	repair every nailbed injury; absorbable (6-0) suture or Dermabond

– – Active space – –

Post-op	extension splint **4–6 wk**; IV Abx → short oral course
Open vs closed	**open management fewer complications**; Abzug: surgery for open **and** closed
ED management	89% managed in ED; OR mainly for failed closed reduction (Lin)
Antibiotics	**within 24 h → infection 6.9%** vs **>24 h → 76.5%** (Rask; overall 27.3%); **clindamycin** for delayed (bone penetration)
Delayed treatment	infection 0% (acute appropriate) · 15% (partial) · **45% (delayed)** (Reyes & Ho)



Seymour SH II lateral

6 Complications

Complication – osteomyelitis

Highest risk, delayed presentation	**osteomyelitis** ✗ Wrong options: growth arrest · septic arthritis · extension deformity
List	**nail dystrophy**, **growth disturbance / physeal arrest**, secondary displacement, **chronic osteomyelitis** (not treated as open), **flexion deformity**
Nail growth disturbance	11/24 despite full motion in 23/24 (Krusche-Mandl)

X vs Y

	Seymour fracture	Paediatric mallet fracture
Fracture	**SH I/II or juxta-epiphyseal**	**SH III/IV**
DIPJ involvement	No	Yes
Nailbed	**Injured (open)**	Usually intact
Key treatment	**Debride, remove matrix, repair nailbed**	Splint

How it's asked

"True of Seymour fracture EXCEPT"	→ **adult thumb hyperextension in sports** (TO)

"12-y crushed middle finger, mallet posture, subungual haematoma — next?"	→ **nail plate removal, nailbed repair, remove interposed tissue, open reduction** ✗ Wrong options: closed pin · mallet splint
"Surgery must include?"	→ **removal of interposed soft tissue** ✗ Wrong options: pinning
"Anatomic cause of deformity?"	→ **extensor on dorsal epiphysis, flexor on volar metaphysis**
"Delayed Seymour, highest-risk complication?"	→ **osteomyelitis**

– – Active space – –

– – Active space – –

SKIN GRAFTING

– Hand · 84 / 103 –

1 Basics

Definition	**avascular** dermal tissue covering a **well-perfused** bed; usually autologous
Nutrition	diffusion from wound bed (STSG & FTSG)
Goals	cover deep structures · bacterial barrier · restore function · prevent contracture
Indication	well-perfused bed over **muscle or subcutaneous tissue**
Contraindication	exposed **bone, tendon, nerve, vessel** → flap
Palmar vs dorsal skin	palm: **no hair or sebaceous glands**, **Meissner + Vater-Pacini** corpuscles; dorsum = like rest of body → STSG adequate

5 Management

STSG vs FTSG

	STSG	FTSG
Use	**Dorsal hand**; beds where contraction won't limit motion	**Volar hand & fingertips**
Donor	**Anterolateral thigh**	**Proximal forearm, medial upper arm, hypothenar eminence**
Content	Keratinocytes + variable dermis	**Full epidermis + dermis**, hair follicles, sweat glands; **no subcutaneous fat**
Contracture	More (thicker = less)	**Less**
Sensation / durability	Less	**Better reinnervation, more shear-resistant**

STSG thickness	thin **0.005–0.012 in** · intermediate **0.012–0.018** · thick **0.018–0.030**
Meshed STSG	↑ surface area, ↓ haematoma & infection → **better take**
Defat FTSG	fat = barrier to vascular ingrowth → graft failure
FTSG technique	gentle tension, multiple **tie-over sutures** (↓ shear), medicated gauze + moist cotton, dressing **5–7 d**
FTSG cons	haematoma/seroma → failure; revascularisation **2–3 d**



FTSG

Palmar defect coverage

Palm defect (post-resection)	**flap + full-thickness skin coverage** (e.g. posterior interosseous artery island flap)

(✗ Wrong options: **Traps** – VAC alone) · wet-to-dry alone · STSG alone · flap + meshed STSG



Palmar defect after mass resection

Other grafts

Allograft	**temporary** bed preparation for autograft
Xenograft	**biologic dressing**

How it's asked

"Palmar soft-tissue defect after mass excision – coverage?"	→ **flap with full-thickness skin** ✗ Wrong options: meshed STSG
"Graft for dorsal hand"	→ **STSG**
"Why remove fat from FTSG?"	→ **barrier to vascularisation**
"Meshing advantage"	→ **↓ haematoma/infection, better take**

– – Active space – –

SNAPPING EXTENSOR CARPI ULNARIS (ECU)

– Hand · 85 / 103 –

1 Basics

Epidemiology

Sports **tennis** (backhand), **golf** (club strikes ground), rugby; overuse (e.g. suturing)

Anatomy – 6th compartment & subsheath

Compartment	Contents	Condition
1	APL, EPB	De Quervain
2	ECRL, ECRB	**Intersection syndrome** (1st/2nd crossing)
3	EPL	–
4	EIP, EDC	–
5	EDM	**Vaughan-Jackson**
6	**ECU**	**Snapping ECU**

ECU inserts **base of 5th MC**; innervated by **PIN**

ECU subsheath fibro-osseous tunnel, **part of TFCC**, most critical to ECU stability

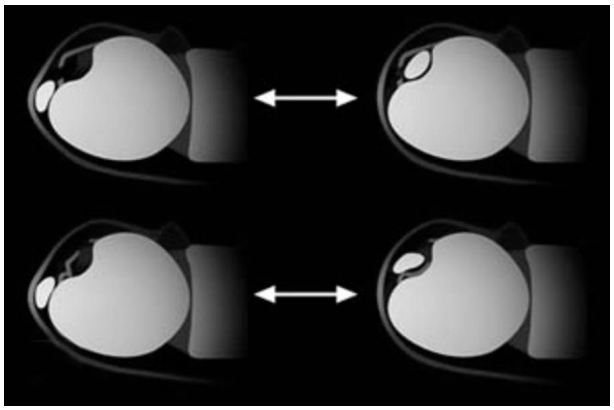
Kinematics pronation: straight course · **supination: exits 6th compartment at ~30°** → max tension on subsheath

Subluxation position **supination + wrist flexion + ulnar deviation**

2 Mechanism

Pathology **attenuation or rupture of ECU subsheath** → subluxation → snapping → **ECU tendonitis**

Lesion	Behaviour
Attenuation	Sheath intact but stripped ulnar/palmar → **false pouch**
Ulnar-sided tear	Subluxes in supination, **reduces in pronation**
Radial-sided tear	Subluxes in supination, lies **on top of torn subsheath** in pronation



Ulnar vs radial subsheath tears

3 Presentation

Symptoms	dorsoulnar wrist **pain + snapping/popping**
Exam	painful snap with **flexion + supination**; **reduces with pronation**
Negatives	no foveal pain with ulnar compression (TFCC) · no DRUJ instability

4 Diagnosis & Classification

Diagnosis – snapping with supination/flexion & examination tests

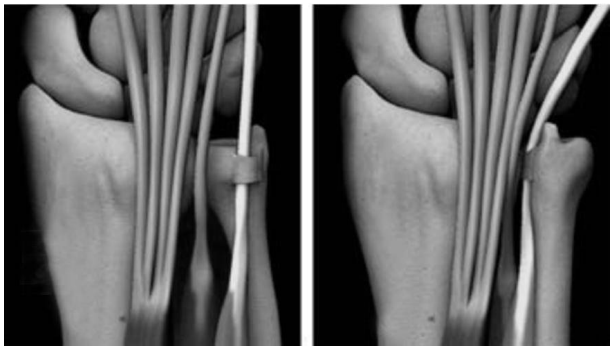
Diagnosis **clinical** (click with pathognomonic motion)

Test	Positive in
Flexion + supination → painful snap	**Snapping ECU**
ECU synergy test (elbow 90°, full supination, wrist neutral, fingers extended → resisted **thumb + middle finger abduction**)	**ECU tendonitis**
Ulnar → radial deviation + scaphoid pressure	Watson (SL)
Ulnar soft spot tenderness (styloid–FCU)	TFCC
Axial load + max ulnar deviation	Ulnar styloid impingement / ulnocarpal abutment
Piano key	DRUJ instability
Finkelstein / Eichhoff	De Quervain
Resisted wrist extension, elbow extended, pronated	Lateral epicondylitis

Key radiology findings

Modality	Fact
X-ray	**Unremarkable**
Ultrasound	**Dynamic** assessment of ECU stability
MRI	Tendonitis (**thickening + high T2 intrasubstance signal**), degenerative tears, TFCC pathology

– – Active space – –



ECU position in pronation vs supination

Differential TFCC tear · hook of hamate fracture · ulnar styloid fracture · ulnar styloid impaction · ulnar tunnel · pisotriquetral arthritis

5 Management

Snapping ECU

- First line → **SPLINT / LONG ARM CAST: PRONATION + EXTENSION + RADIAL DEVIATION**
- Fails nonop → **SUBSHEATH SURGERY ± WRIST ARTHROSCOPY**
  - Acute → **DIRECT SUBSHEATH REPAIR**
  - Chronic → **EXTENSOR RETINACULUM FLAP RECONSTRUCTION**

Nonoperative immobilisation position

Position **pronation, slight extension, radial deviation** → minimal ECU angulation vs ulna

Avoid supination / ulnar deviation lower threshold for subluxation

Arthroscopy **concurrent TFCC tears in 50%**

X vs Y

	Snapping ECU	TFCC tear	Ulnar styloid impaction	LT tear
Key feature	**Snap on supination**, reduces in pronation	**Deep foveal pain**, worse with ulnar compression	Pain with pronation, + UV	VISI, LT shuck pain
Snap/ pop	**Yes**	No	No	No

How it's asked

"Golfer, ulnar pain + snap pronation → supination, normal X-ray, no foveal pain?"

→ **ECU subluxation**
✗ Wrong options: TFCC · ulnar abutment

"Tennis backhand pop, dorsoulnar tenderness?"	→ **ECU subluxation** ✗ Wrong options: intersection (radial side)
"Exam finding in subluxing ECU?"	→ **pain with flexion + supination**
"MRI ECU thickening + T2 signal – positive test?"	→ **ECU synergy test**
"Immobilisation position?"	→ **pronation, extension, radial deviation**
"Chronic reconstruction?"	→ **extensor retinaculum flap**

– – Active space – –

– – Active space – –

SWAN NECK DEFORMITY

– Hand · 86 / 103 –

1 Basics

Definition — **PIP hyperextension + DIP flexion** ("double curve") from PIP force imbalance

Incidence — up to **50% of RA** patients

Pathomechanics – triangular ligament & lateral band tenotomy

Primary lesion — **lax volar plate** → PIP hyperextension; causes: trauma, generalised laxity, RA (chronic PIP synovitis)

Secondary lesion — **PIP extension force > flexion force**

Cause	Mechanism
Mallet finger	DIP extension force transferred to PIP
FDS rupture / laceration	Lost dynamic check-rein → unopposed PIP extension + volar plate failure
Intrinsic contracture	Lateral bands tethered (transverse retinacular lig.) → extension force goes to PIP, not terminal tendon
MCP volar subluxation	RA
Cerebral palsy	Extrinsic extensor overpull ± intrinsic spasticity

Cascade (TO) — PIP hyperextends → **lateral bands sublux dorsally** (above axis) → ↑ extensor lever arm → self-perpetuating; extensor "used up" at PIP → slack at DIP + FDP pull → **DIP flexion**

Triangular ligament — dorsal, base of middle phalanx just distal to central slip; **keeps lateral bands dorsal → contracture/tightness → swan neck**
✗ Wrong options: boutonniere: lateral bands volar

Not swan neck causes — volar plate **contracture**, sagittal band insufficiency (extensor subluxation at MCP)

3 Presentation

Symptoms — **snapping and locking** of fingers

Exam — resting PIP hyperextension + DIP flexion

Bunnell-Littler (intrinsic tightness) test (TO) — PIP flexion **more restricted with MCP extended** than flexed → **intrinsic tightness**
✗ Wrong options: capsule contracture · volar plate laxity · extrinsic adhesions



Swan neck deformity

4 Diagnosis & Classification

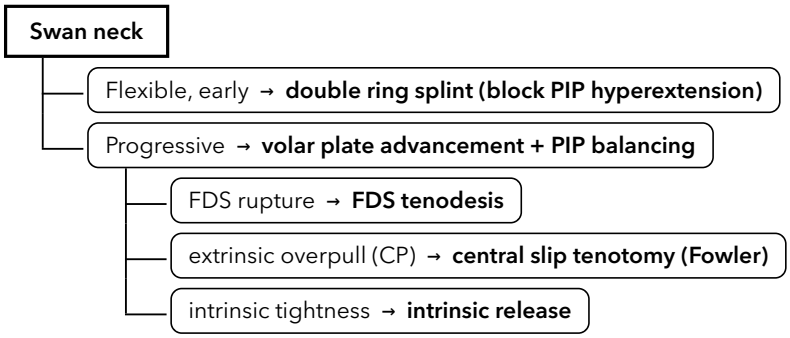
Key radiology findings

Radiographs AP + lateral hand; PIP joint destruction defines Nalebuff IV

Nalebuff classification (TO)

Type	Finding	Treatment principle
I	PIP flexible in all positions	Restore check-rein: **FDS tenodesis** (most widely used), dermodesis, **SORL / ORL reconstruction** (Brooks-Graner), volar plate repair, lateral band translocation
II	PIP flexion limited with **MCP extended** (+ Bunnell-Littler)	**Intrinsic release + type I procedure** (e.g. **FDS tenodesis + intrinsic release**)
III	PIP flexion limited in all MCP positions; **joint space preserved**	**MUA** (aim 80–90° flexion) → dorsal skin release / Z-plasty → **lateral band mobilisation**; check flexor gliding; then type I procedure ± intrinsic release
IV	Stiff PIP + **radiographic destruction**	**Arthrodesis** or **arthroplasty** (+ soft-tissue balancing)

5 Management



Operative procedures

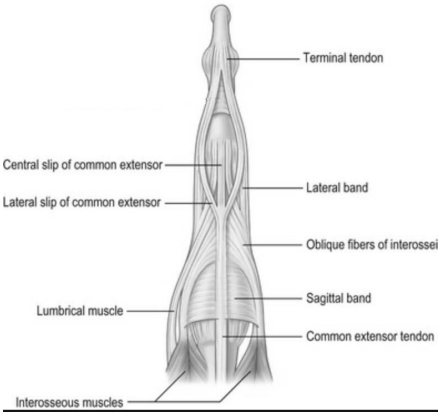
Nonop **double ring splint** prevents PIP hyperextension

Volar plate advancement addresses laxity

FDS tenodesis / sling (TO) volar zig-zag; window either side of **A3**; one FDS slip cut **~1.5 cm** from insertion, distal attachment intact; PIP flexed **20–30°**; fixed to proximal phalanx (anchor) or **distal edge of A2** → static tether

– – Active space – –

ORL reconstruction (TO)	ulnar lateral band or tendon graft; **SORL (Brooks-Graner)** → dynamic PIP + DIP stability
Volar plate repair (TO)	when volar plate disruption was initial injury; anchors/drill holes base of P2
Intrinsic release (Littler) (TO)	dorsoulnar incision; excise **triangular aponeurosis / oblique fibres** + lateral band at distal half of PIP; preserve **central slip, lateral slips, transverse fibres**
Central slip tenotomy (Fowler)	CP swan neck from **extrinsic extensor overpull** (structure C) ✗ Wrong options: terminal tendon A · lateral band B · lateral slip D · lumbrical E
CP alternatives	intrinsic lengthening for intrinsic spasticity; lateral band rerouted volar to PIP axis (Van Heest & House)
Type IV (TO)	**arthrodesis** most predictable: **40–50° ulnar digits**, less for index; **silastic arthroplasty** preferred ring & small (motion > pinch)



Extensor anatomy: C = central slip

Thumb swan neck (Nalebuff type III thumb) (TO)

Sequence (TO)	**CMC subluxation/adduction** → compensatory **MCP hyperextension** for pinch
Treatment (TO)	fix foundation first: **CMC arthroplasty (LRTI)** → then MCP capsulodesis or fusion

X vs Y

	Swan neck	**Boutonniere**
PIP	Hyperextension	Flexion
DIP	Flexion	Hyperextension
Lateral bands	**Dorsal**	**Volar**
Key structure	Lax volar plate; tight **triangular ligament**	Central slip rupture; attenuated triangular ligament
Trauma origin	Mallet, FDS rupture	Central slip injury

How it's asked

"Best procedure, type II swan neck?"	→ **FDS tenodesis + intrinsic release** (TO)
"PIP flexion worse with MCP extended?"	→ **intrinsic tightness** (TO)
"Triangular ligament tightness causes?"	→ **swan neck** ✗ Wrong options: boutonniere, sagittal band, volar plate contracture
"CP swan neck, failed splint – tenotomy of?"	→ **central slip (C)**
"Type IV (destroyed PIP)?"	→ **arthrodesis / arthroplasty** (TO)
"Thumb swan neck – address first?"	→ **CMC (LRTI)** (TO)
"Initial treatment?"	→ **double ring splint**

– – Active space – –

SYNDACTYLY

– Hand · 87 / 103 –

1 Basics

Epidemiology – most common web

Definition **most common congenital limb malformation**; adjacent digits webbed (Greek *syn* = together, *dactylos* = digit) (TO)

Incidence **1 in 2,000–2,500**; ♂ > ♀; Caucasians > African Americans

Web	%
Long–ring	**50** (most common (TO))
Ring–small	30
Index–long	15
Thumb–index	5

Genes & associations

Pathophysiology **failure of apoptosis** to separate digits

Genetics pure syndactyly **AD**, reduced penetrance, variable expression; **family history 10–40%**

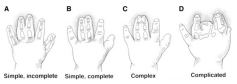
Associations **acrosyndactyly** (constriction ring / Streeter dysplasia) · **Poland** · **Apert** · **Carpenter** (acrocephalopolysyndactyly)

Acrosyndactyly digits fused **distally**, proximal **fenestrations**

4 Diagnosis & Classification

Diagnosis **clinical**

Classification (simple / complex / complicated)

Type	Definition	Image
Simple	**Soft tissue only**, no bony connection	
Complex	**Side-to-side fusion of adjacent phalanges**	
Complicated	**Accessory phalanges / abnormal bones** in fusion	
Complete	Skin to **fingertips**	
Incomplete	Skin short of fingertips	–

Temtamy & McKusick (TO)

Type	Features
I	**Most common**: long–ring (hand) / 2nd–3rd toes; AD, incomplete penetrance
II	**Synpolydactyly**: long–ring and/or 4th–5th toes + accessory digit in web
III	Ring–small
IV	Complete syndactyly **all fingers**
V	**Metacarpal/metatarsal fusion** ring–small / 3rd–4th toes

5 Management

Syndactyly

- Acrosyndactyly → **release in NEONATAL period**
- Border digits (ring–small, thumb–index) / multiple fusions → **release ~6 mo**
  (unequal lengths → tethering, angulation, rotation, shortening)
- Central (long–ring) → **~1 y / 12–18 mo (Orthobullets: up to 2 y)**
- Simple toe syndactyly, no functional issue → **nonop**

Timing & technique

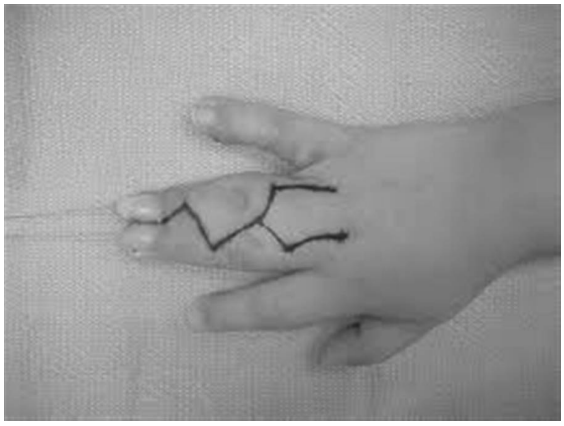
Timing – conflict **~1 y** (Orthobullets) · **12–18 mo**; delay **>18 mo ↓ scar contracture & web creep** (TO)

– – Active space – –

Multiple digits	**two stages, one side of a digit at a time** → protect digital artery
Sequence	release **digits of unequal length first** (border); middle–ring **6 mo later**; all **before school age**
Bilateral	simultaneous if **<18 mo**; staged if >18 mo
Goals (TO)	new **paronychial nail folds** · interdigital commissure skin · **new web space**
Incisions	**interdigitating zigzag flaps** (Cronin, dorsal + volar) → avoid longitudinal scar · **dorsal fasciocutaneous flap** for web · modified double reverse Z-plasty (TO)
Sutures	**absorbable (5-0 chromic catgut)**
Grafts	**FTSG preferred, less contracture** (TO) · FTSG ↑ web creep, hyperpigmentation, hair; STSG ↑ flexion/extension lag & graft breakdown (Deunk, no preference)
Technique (TO)	**volar anchor / triangular flap** · **Buck-Gramcko flaps for nail folds** · FTSG for residual defects
Stage interval	**6 mo** (Orthobullets) · **3–6 months** (TO) – read the options

	Syndactyly (TO)	**Symphalangism (TO)**
Involves	Adjacent fingers	Same finger
Fusion	Soft tissue / bone	Longitudinal phalangeal (PIP absent)
Rx	Surgery	Conservative

Metacarpal synostosis (TO) commonly **4th–5th** · carpal synostosis – conservative



Zigzag flaps

6 Complications

Complication – web creep

Web creep	**most common** (8–60%) = **distal migration of the web commissure**; warn parents pre-op; regular follow-up
Early creep	**necrosis of tip of dorsal quadrilateral flap** + loss of FTSG in web
Late creep	adolescence; **discrepant growth** of scar/graft vs surrounding tissue

Treatment reconstruct web with **local skin flaps**

Other complication (TO)	Fact
Infection	Rare, **1.6%**
Scar contracture	STSG, bone growth
Devascularisation / digital nerve injury	Operate **one side at a time**
Nail deformity	–



Web creep

How it's asked

"18-mo, middle–ring release – likely complication"	→ **web creep** (TO) ✗ Wrong options: nail deformity, infection, contracture
"Most common complication involves"	→ **web commissure**
"Most common finger involvement"	→ **long–ring** (TO)
"Release border digits when?"	→ **~6 mo**
"Acrosyndactyly release when?"	→ **neonatal period**
"Bony side-to-side fusion"	→ **complex**
"Early web creep cause"	→ **dorsal flap tip necrosis / FTSG loss**

– – Active space – –

TENDON TRANSFER PRINCIPLES

– Hand · 88 / 103 –

1 Basics

Principles of transfer

Strength	force ∝ **cross-sectional area** (max at resting length); amplitude ∝ **muscle length**; work = force × amplitude
Grade loss	**lose 1 MRC grade** after transfer → transfer **grade 5** donor (foot drop: 4/5 acceptable → ≥3/5 antigravity) · donor **≥ 4/5** (TO median/radial tip-to-toe) · **at least MRC 3** (TO radial palsy lecture) – read the options
Indications (TO)	**leprosy** · nerve not repairable · old age · comorbidities · prerequisites: good skin cover, protective sensation, full PROM, one pulley · junction by **Pulvertaft weave**, tension by **tenodesis effect**
Donor	**expendable**, similar excursion + power
One tendon → one function	
Synergistic	rehab easier (finger flexion + wrist extension; wrist flexion + finger extension)
Line of pull	**straight** (avoids tethering, late failure); pulley or tenodesis adjusts excursion
In-phase vs out-of-phase	in-phase best, but **out-of-phase acceptable** (often the only donor) – not a contraindication

Smith 3-5-7 rule	Excursion
Wrist flexors / extensors	**3 cm**
EDC, FPL, EPL	**5 cm**
FDS, FDP	**7 cm**

Selection	missing function → available units → options
Brachial plexus priorities	elbow flexion (musculocutaneous) → shoulder stabilisation (suprascapular) → brachiothoracic pinch (pectoral) → sensation C6–7 (lateral cord) → wrist extension + finger flexion

Prerequisites – supple joints

Precludes transfer	**target joint cannot be passively corrected to neutral** (static contracture / bony deformity) → release first
Not contraindications	full PROM · motion only in direction of correction · out-of-phase donor · no pulley
Nonop (therapy, splints, antispasticity)	↓ PROM, spasticity

3 Presentation

Nerve palsy patterns

Palsy	Low	High
Radial (vs PIN origin)	PIN syndrome (wrist extension spared via ECRL)	+ triceps, BR, **ECRL** lost
Median (vs AIN origin)	**Loss of opposition (APB)**	+ **FPL, FDP index/middle, FDS, PL** lost
Ulnar	Power pinch ↓, **Wartenberg**, **clawing marked**	+ **FDP ring/small lost** (key deficit) → **less clawing**

Radial nerve level (TO)	Motor supply
Main trunk	Triceps (3 heads), anconeus, **BR, ECRL**
Deep branch	**ECRB, supinator**
PIN (purely motor)	APL, EPB, EPL, ECU, EIP, EDC, EDM

Radial course (TO) purely **posterior cord** · axilla behind axillary a. → **triangular interval** (long + lateral triceps branches) → spiral groove (medial head) with profunda brachii → anterior to lateral epicondyle → deep + superficial · PIN pierces supinator (**arcade of Frohse**)

Radial palsy causes (TO) **humeral shaft fracture (commonest)** · **Saturday night / crutch palsy** (axilla / upper arm) · Wartenberg (SRN under BR) · PIN / supinator syndrome · key sensory area **1st dorsal web** · SRN forms very painful neuromas → protect

Median palsy sign (TO)	Meaning
Hand of benediction	**High** lesion: index + middle can't flex on making a fist (FDS + FDP lost)
Pointing index (clasping test)	Index stays straight when hands clasped
O sign (Kiloh-Nevin)	**AIN palsy**: pulp-flat pinch (FPL + index FDP)
Ape thumb	**Low / long-standing**: thenar wasting, thumb adducted + extended in plane of MCs
Pen test	**APB** – key muscle to test for median nerve in hand

Brachial plexus avulsion signs **Horner's (C8–T1)**, appears 2–3 d · severe pain in anaesthetic limb · **loss of rhomboids**

Studies sensory/motor evoked potentials > standard EMG/NCS

- - Active space - -

4 Diagnosis & Classification

Bouvier test (claw hand)

Technique wrist neutral, **block MCP hyperextension** → can PIP extend?

Result	Meaning	Transfer insertion
Positive (PIP extends)	**Simple claw**, central slip intact	MCP flexion only → **A1/A2 pulley or proximal phalanx**
Negative (PIP doesn't extend)	**Complex claw**	MCP flexion + PIP extension → **lateral bands**

(✗ Wrong options: **Distractors** – PIP 90° + rigid DIP = **Elson** (central slip)) · PIP flexes with MCP extended not flexed = **extrinsic tightness** · wrist tenodesis = tendon injury · thumb IP flexion on paper = **Froment**

5 Management

Timing

Timing	Indications
Early (3 wk–3 mo)	Total/near-total plexus injury, high energy
Late (3–6 mo)	Partial upper plexus, low energy

Post-op protect **3–4 wk** → ROM; protective splint 3–6 wk

Specific transfers

Goal	Donor	Recipient
Musculocutaneous – elbow flexion	Pec major, latissimus	Biceps
Elbow flexion	Common flexor mass	Proximal humerus (**Steindler flexorplasty**)
Radial – elbow extension	Deltoid, latissimus, biceps	Triceps
Wrist extension	**Pronator teres**	**ECRB**
Finger extension	**FDS, FCR or FCU**	EDC
Thumb extension	**PL or FDS**	EPL
Low median – opposition	**FDS ring** (FCU pulley, **Bunnell**)	Base P1 / APB
	EIP (around ulnar wrist)	APB
High median – thumb IP flexion	**BR**	FPL
Index/middle flexion	FDP ring/small (ulnar)	FDP index/middle (side-to-side)
Ulnar – thumb adduction	FDS or ECRB	Adductor pollicis
Index abduction	APL, ECRL, EIP	1st DI
Anti-claw	FDS, ECRL (**volar to transverse MC ligament**)	Lateral bands ulnar digits

Radial nerve/PIN palsy transfers

Most consistent radial palsy transfer — **PT → ECRB**
✗ Wrong options: **BR → ECRB**: BR is radial-innervated

Radial palsy ladder (TO) — primary repair → graft (gap **> 2–2.5 cm**, sural, 4–6 cables) → nerve transfer (before denervation) → tendon transfer → free functional muscle transfer

Internal splint (TO) — **PT → ECRB END-TO-SIDE at time of radial nerve repair** (ECRB insertion kept, reinnervation expected) → power grip + wrist stability, prevents extensor stretch · external alternative: cock-up splint

Radial palsy set (TO)	Wrist	Fingers	Thumb
Modified Robert Jones	PT → **ECRL + ECRB**	FCU → EDC 3–5	FCR → EDC 2, EIP, EPL
Standard FCU	PT → ECRB	FCU → EDC	PL → EPL (no PL → split FCU to EDC + EPL)
Brand (FCR)	PT → ECRB	FCR → EDC	PL → EPL
Boyes superficialis	PT → ECRB	**FDS ring → EDC 3–5; FDS middle → EIP + EPL**	**FCR → APL + EPB**

Set rules (TO) — **PT → ECRB = constant** in every set · FCU expendability debated → sacrifice only one wrist flexor · Boyes needs **intact FDP**, keeps FCR for thumb · aim = mass finger extension

Technique (TO) — PT harvested **with periosteal strip** (length, anchorage) · FCU routed round ulnar border → EDC just proximal to retinaculum · **EPL rerouted out of 3rd compartment** toward PL (abduction + extension) · Pulvertaft weave

Post-op splint (TO) — forearm **15–30° pronation** · wrist **~45° extension** · MCP **10–15° flexion** · thumb **max extension + abduction** · PIPs free · off at 4–6 wk → short-arm extension splint 2 wk

Free functional muscle transfer (TO) — extensor muscle loss (**Volkmann**, crush, burns) → **gracilis** to finger/wrist extensors, **PIN as motor nerve**, posterior interosseous a. (or radial recurrent) as vessels

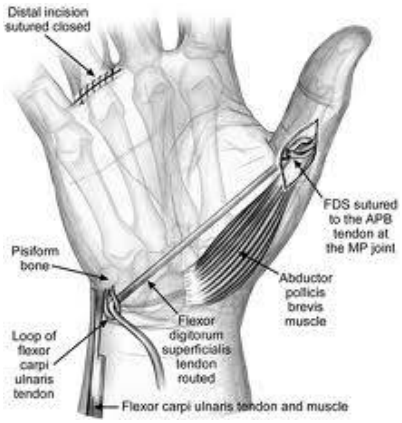
PL — not always present; FDS → APL/EPB unnecessary if EPL rerouted

PIN transected, no recovery at 1 y — **FCR → EDC + PL → EPL**
✗ Wrong options: sural graft at 1 y · wrist fusion + FCU · PT → wrist extensors (wrist extension preserved by ECRL)

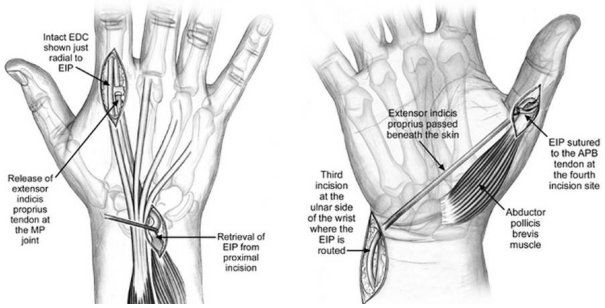
Median/ulnar palsy transfers – opponensplasty & claw correction

Opposition transfer	Note
Ring/long FDS (Bunnell, FCU pulley)	Low median; not high median (FDS denervated)
EIP → APB	**High median** (radial-innervated) ✓; also after median laceration
Camitz **PL**	not high median (denervated); **no pronation/MC flexion** → not true opposition
Huber **ADM**	Ulnar-innervated

– – Active space – –



Bunnell opponensplasty



EIP opponensplasty

High median from supracondylar spur + ligament of Struthers
decompress + FPL/FDP-I + **EIP → APB**
✗ Wrong options: FDS · Camitz · neurotisation = pre-ganglionic plexus · CMC fusion

Low median laceration (4 cm above wrist) after repair
sensation usually returns; thenar motor unpredictable → **EIP opponensplasty at 6 mo**
✗ Wrong options: FDP tenolysis · CTR · EDM → 1st DI (ulnar intact; neurotisation = distal AIN → median motor) · ulnar

High median transfers (TO)
BR → FPL: BR divided at radial styloid, freed proximally → ~30 mm excursion; tension **elbow 45° flexion** · **ECRL → FDP index + middle** (power): divided at 2nd MC base, routed subcutaneously round radial border; set **wrist 30° flexion**, fingertips touch palm on tenodesis · side-to-side FDP tagging = range, little power

Opponensplasty (TO)	Donor	Key points
Burkhalter	**EIP** (PIN)	Divided at insertion (**repair extensor expansion** → no lag); round **ulnar border** of wrist subcutaneously; POP in opposition 4 wk + splint 3–4 wk
Bunnell	**Ring FDS**	Divided **between A1 and A2**, looped through **FCU pulley** → APB · **strongest** (flexor to flexor) · not in combined ulnar palsy · risk **swan neck of ring finger**
Camitz	**PL + strip of palmar fascia**	**Elderly / CTS** (with release) · abduction > true opposition · PL may be absent
Miller	**ECU**	Round ulnar border; APB divided at origin, turned as distally based tendon
Huber	**ADM** islanded on NV bundle	**Children** (thumb hypoplasia, **Blauth 3**) · turned 180° like a page · **adds thenar bulk**

Opponensplasty setting (TO) pulley at **pisiform / FCU** · tension wrist neutral + thumb opposed · tenodesis check: **thumb abducts with wrist extension** · insertions: Brand, Littler, Riordan, **Royle-Thompson** (palmar aponeurosis pulley)

Burkhalter MCP flexor replacement for intrinsic loss → ↑ grip, ↓ clawing

Tensioning EIP-to-EPL transfer

EPL rupture after DRF EIP → EPL (synergistic)

Best tensioning **wide-awake local anaesthesia, functional testing** (max extension without losing flexion)
✗ Wrong options: maximum tension · tenodesis estimate

Tetraplegic (paralytic) hand (TO)

Segments (TO) C5 elbow flexion · **C6 wrist extension** · C7 elbow extension · C8 finger flexion · T1 intrinsics · only **grade ≥ 4** muscles transferable · plan after **1 yr** (recovery plateau)

Mittal group (TO)	Most distal grade-4 muscle
0	None below elbow
1	Brachioradialis
2	ECRL
3	ECRB
4	Pronator teres
5	FCR
6	EDC
7	EPL
8	FDP
9	FPL

Priorities (TO) **elbow extension first** → wrist extension → thumb pinch → grasp → thumb extension → finger extension · rest by **tenodesis**

Transfers (TO) **posterior deltoid → triceps** (or biceps rerouting) · **PT → ECRB** (wrist extension essential for tenodesis grip) · claw: **FDS tenodesis** across wrist · contraindications: **fixed contracture** (absolute), poor passive ROM, recovery < 1 yr, poor motivation · neuroprosthesis = adjunct

6 Complications

Adhesions aggressive therapy ± **tenolysis**

Prognosis **age = leading factor** (worse >30 y); **distal > proximal** lesions

– – Active space – –

X vs Y

	Low ulnar palsy	High ulnar palsy
FDP ring/small	Intact	**Lost**
Clawing	**More**	**Less** (ulnar paradox)

	Low median	High median
Deficit	Opposition	Opposition + thumb IP + index/middle flexion
Opposition donor	**FDS ring** or EIP	**EIP** (FDS, PL denervated)

How it's asked

"Precludes tendon transfer?"	→ **joint not passively correctable to neutral** ✗ Wrong options: out-of-phase donor · no pulley
"PIN transected, no function at 1 y?"	→ **FCR → EDC + PL → EPL**
"Most consistent radial palsy transfer?"	→ **PT → ECRB** ✗ Wrong options: BR → ECRB
"High median palsy (Struthers) — restore opposition?"	→ **EIP → APB** ✗ Wrong options: ring FDS · Camitz
"Median laceration 4 cm proximal to wrist — secondary op at 6 mo?"	→ **EIP opponensplasty**
"Ulnar claw — test to choose insertion site?"	→ **Bouvier: block MCP extension, PIP fails to extend** ✗ Wrong options: Elson · Froment
"Tension EIP → EPL?"	→ **awake under local, functional testing**
"Excursion of FDS/FDP?"	→ **7 cm**
"Strength after transfer?"	→ **loses one grade**
"Leading prognostic factor?"	→ **age (worse >30)**
"Internal splint in radial nerve palsy?"	→ **PT → ECRB end-to-side at nerve repair** (TO)
"Boyes transfer — finger extension donor?"	→ **FDS ring → EDC; FDS middle → EIP + EPL** (TO)
"Strongest opponensplasty?"	→ **Bunnell ring FDS** (TO)
"Opponensplasty for child with Blauth 3 hypoplasia?"	→ **Huber ADM transfer** (TO)
"Single muscle to test for median nerve in the hand?"	→ **APB (pen test)** (TO)

THROMBOANGIITIS OBLITERANS (BUERGER'S DISEASE)

– Hand · 89 / 103 –

1 Basics

Definition & epidemiology

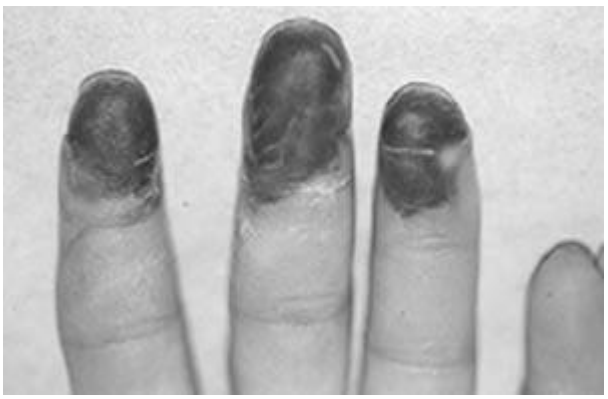
Definition	**non-atherosclerotic, segmental, inflammatory** disease of **small + medium vessels** of hands and feet
Demographics	12.6/100,000 (US); ♂:♀ **3:1**; **<45 y**
Risk factors	**smoking**, chewing tobacco
Pattern	**distal vessels first**, proximal last

2 Mechanism

Phase	Histology
Acute	Thrombus with **neutrophils + giant cells** occludes lumen, **wall spared**
Subacute	Progressive thrombus organisation
Chronic	Inflammation gone; organised thrombus + vascular fibrosis

3 Presentation

Early	**intermittent claudication** (feet, legs, hands, arms), numbness/tingling
Late	critical ischaemia → **rest pain**; frequently **painful** ✗ Wrong options: painless
Exam	ulcers, **necrotic distal digits**, large erythematous superficial vessels, cold hands/feet, ↓ pulses, sensory findings up to 70%
Allen test	**positive in young smoker with digital ischaemia** → suggestive



Necrotic digits

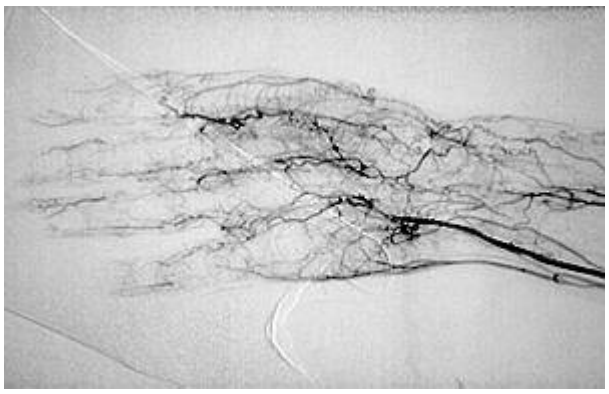
– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Arteriography — **study of choice** if diagnosis uncertain; excludes mimics

Signs — **"corkscrew" vessels** (diagnostic in right setting) · collaterals → **"spider leg"**



Corkscrew arteriogram

Echocardiogram — exclude **proximal embolic source**

Labs — exclude alternative diagnoses (autoimmune/vascular mimics)

5 Management

Buerger's disease

- All tobacco users → **smoking cessation + symptomatic care**
- Refractory pain / digital ischaemia → **sympathectomy (controversial)**
- Gangrene / non-healing ulcer / refractory pain → **amputation**

Smoking cessation — **only treatment that ↓ amputation risk** (education, pharmacotherapy, groups)

Symptomatic — avoid cold, gentle exercise, daily **aspirin**, vasodilators
✗ Wrong options: shown to ↓ amputation or halt progression

Sympathectomy — cut nerves to affected area; controversial

No prophylactic amputation of unaffected digits – not described

Prognosis — **quit → 94% avoid amputation**; continue → **43% amputation within 8 y**

How it's asked

"45-y male smoker, digital ulcers – true statement?"	→ **arteriography useful for diagnosis** ✗ Wrong options: aspirin ↓ amputation · prophylactic amputation · painless · proximal first
"Only treatment that reduces amputation?"	→ **smoking cessation**
"Arteriogram sign?"	→ **corkscrew vessels**

"Acute-phase histology?"

→ **occlusive thrombus, vessel wall spared**

– – Active space – –

– – Active space – –

THUMB CMC DISLOCATION

– Hand · 90 / 103 –

1 Basics

Epidemiology

Incidence **<1% of hand injuries**; **dorsal (dorsoradial) » volar**

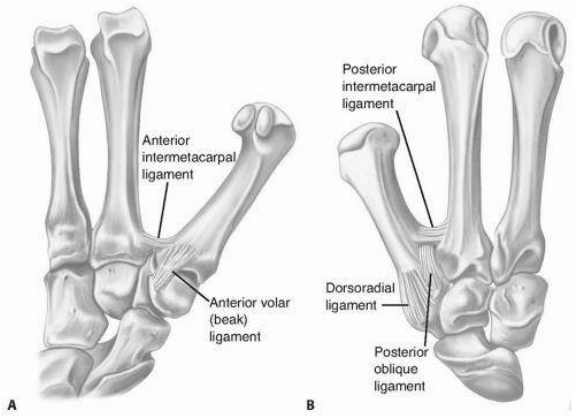
Stabilisers – dorsoradial ligament & anterior oblique ligament

Ligaments 16 total; **7 TMCJ stabilisers** (sAOL, dAOL, DRL, POL, UCL, IML, DIML) + 9 trapezial

4 key restraints **AOL**, POL, IML, **DRL**

Ligament	Fact asked
Dorsoradial (DRL)	**Primary stabiliser vs dorsal/dorsoradial subluxation** in all positions; strongest & thickest; "check rein"; taut first; joint stays reduced if all others cut; **torn in dorsal dislocation**; lax in stage IV OA
Superficial AOL	does not stabilise in flexion or prevent dorsal subluxation; laxity allows pronation
Deep AOL (beak)	Volar trapezial tubercle → volar beak of MC; **pivot for pronation**; taut only in **hitchhiker** position; **stripped/peeled off MC base but continuous** in dislocation; stays on volar fragment in Bennett/Rolando
Posterior oblique (POL)	Purely dorsal, ulnar to DRL; not primary
Intermetacarpal / dorsal IML	**Restrains radial translation** of 1st MC
Trapeziotrapezoid	Stabilises TT joint, not CMC

Historical vs current AOL once thought primary; **DRL now accepted primary** (biomechanical studies)



Thumb CMC ligaments labelled

2 Mechanism

Commonest	**axial force on flexed thumb**
Less common	dorsal force into 1st web (motorcycle handlebar)
Pathoanatomy	**DRL torn**; **AOL stripped from MC base, remains continuous**

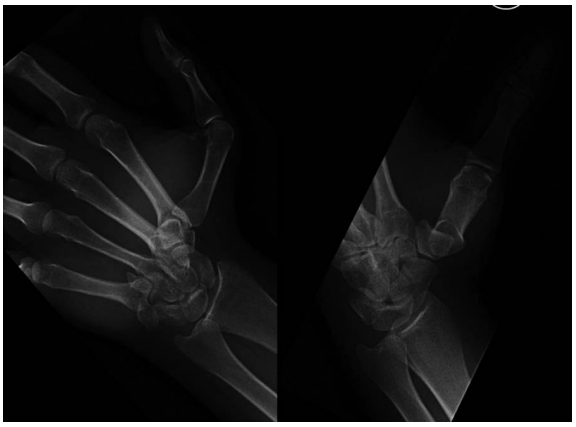
3 Presentation

Findings	thenar pain, swelling, bruising; cannot make a fist

4 Diagnosis & Classification

Key radiology findings

Modality	Fact
X-ray PA + lateral	Joint-space widening, **dorsoradial shift of MC**
Roberts view	Base-of-thumb fractures
MRI	Persistent/recurrent instability; plan reconstruction



Dorsoradial CMC dislocation

5 Management

Thumb CMC dislocation

- Stable after reduction (AOL intact) → **CLOSED REDUCTION + SPICA IN EXTENSION + PRONATION**
- Incongruent / unstable → **CRPP ± open capsulorrhaphy**
- Grossly unstable (AOL possibly torn) → **LIGAMENT RECONSTRUCTION (TENDON AUTOGRAFT) + TEMPORARY PINS**

Ligament reconstruction with FCR

Recommended for gross instability	reconstruction + pinning → **better abduction & pinch** than CRPP

– – Active space – –

Structure reconstructed	**Orthobullets**: dorsal capsuloligamentous complex · **OB (Eaton-Littler, FCR graft)**: **volar oblique ligament** — read the options
Eaton-Littler	FCR graft; better results in **non-arthritic thumbs**; APL is used but is not the structure reconstructed

6 Complications

List	anterior osteophyte often visible · **low recurrence**

X vs Y

	CMC dislocation	**Bennett / Rolando fracture**
AOL	**Peeled off MC base, continuous**	**Attached to volar fragment**
DRL	**Torn**	–
MC translation	Ligamentous	Through fracture

How it's asked

"Dorsal or volar? Which ligament torn?"	→ **dorsal; DRL torn (AOL intact, peeled off)**
"Stabiliser in opposition besides AOL?"	→ **dorsoradial ligament**
"Incongruent after reduction — primary restraint to dorsal subluxation?"	→ **dorsoradial ligament**
"Primary stabiliser vs dorsoradial subluxation?"	→ **DRL** ✗ Wrong options: volar oblique (old teaching) · dorsal IML (radial translation)
"Eaton-Littler FCR reconstruction rebuilds?"	→ **volar oblique ligament** ✗ Wrong options: APL

THUMB COLLATERAL LIGAMENT INJURY

– Hand · 91 / 103 –

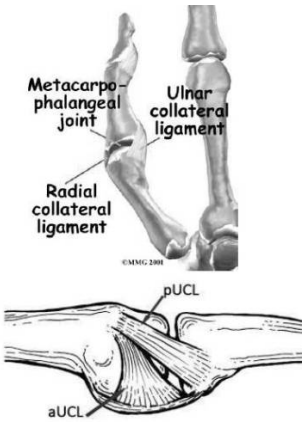
1 Basics

Epidemiology

UCL vs RCL	**UCL 10× more common**; UCL = **86%** of athletic thumb injuries; most common injury at thumb MCP (TO)
Skier's thumb	**acute** (ski pole); most common UE skiing injury (7–32%)
Gamekeeper's thumb	**chronic** attenuation from repetitive valgus (Scottish gamekeepers breaking rabbits' necks) (TO)

UCL anatomy – proper vs accessory ligament & testing positions

Ligament	Course	Taut / restrains radial stress in
Proper UCL (pUCL)	**Dorsal third of MC head → volar P1**	**Flexion** (+ dorsal capsule)
Accessory UCL (aUCL)	Palmar to pUCL → **volar plate**	**Extension** (+ volar plate)



Proper vs accessory UCL

Static stabilisers	bony shape (flat MC condyles), collaterals, volar plate, dorsal capsule
Dynamic stabilisers	**adductor pollicis** (inserts P1, **superficial to UCL**, ulnar n.); EPL, EPB, FPL, APB, FPB
RCL side	covered by **abductor aponeurosis** → Stener-like RCL lesion rare; RCL tear → **subluxation** rather than overt instability

2 Mechanism

UCL	**radially directed force → hyperabduction / extension** = **valgus** (ski pole, FOOSH)
RCL	**axial load** (adduction)

– – Active space – –

Site of UCL rupture — **distal (P1 base)** in most; bony avulsion **20–30%** (TO)
✗ Wrong options: **avulsion from MC head** (TO)

RCL rupture site — evenly proximal / midsubstance / distal

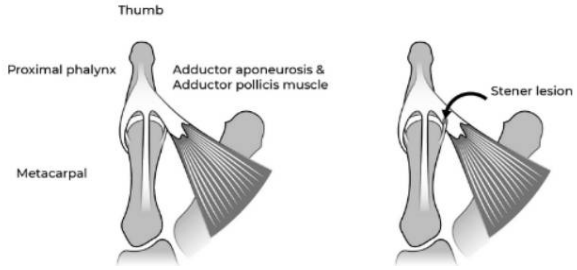
Stener lesion – adductor aponeurosis

Stener lesion — completely torn UCL (pUCL + aUCL) retracts **proximal + superficial to adductor aponeurosis** → interposed aponeurosis **blocks reduction & healing → surgery**

Blocking structure — **adductor pollicis aponeurosis** (**ulnar n.**)
✗ Wrong options: opponens, FPB, APB (median thenars) · volar plate · FPL · lumbricals · anterior oblique ligament

Incidence — **65–85%** of complete ruptures (OB) · **64–87%** of grade III (TO)

Second bony fragment — may mark distal UCL end → Stener lesion



Stener lesion

✗ Distractor	Where it blocks
Volar plate, sesamoids, FPL	Thumb MCP **dorsal dislocation**
Lumbricals	Finger MCP (2–5) dorsal dislocation
Anterior oblique (beak) ligament	Holds **Bennett** fragment; CMC arthritis/instability

3 Presentation

UCL — **ulnar MCP pain** worse with **pinch/grasp**; tender **distally**; **palpable ulnar mass = Stener**; haematoma

RCL — **radial MCP pain**; tender **proximally**; presents late (**>10 wk**) vs UCL (2–10 wk)

Stress examination

Technique — X-ray **first** (don't displace avulsion); stress in **extension and 30° flexion**; prevent P1 rotation; ± local anaesthetic

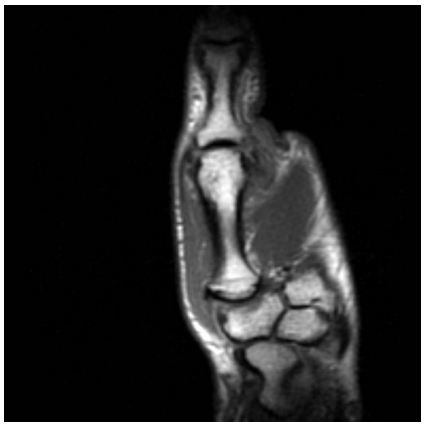
Laxity	Injured
30° flexion only	**Isolated proper UCL** (torn off **P1**) – partial
Extension (and flexion)	**Proper + accessory UCL ± volar plate** – complete

RCL — anterior/posterior drawer → volar subluxation, no endpoint

4 Diagnosis & Classification

Key radiology findings

X-ray	PA, lateral, oblique; **avulsion at ulnar base of P1** / condylar fx; **sag sign** (P1 **supination**, UCL) · P1 **pronation** (RCL) · **volar P1 subluxation** on lateral (dorsal capsule/extensor injury)
Stress views	controversial; only after bony avulsion excluded
MRI	equivocal exam; **gold standard**, **100% sensitivity + specificity** for tear/ Stener (TO)
US	operator-dependent; **76–88% sens, 81–83% spec**, 81% accuracy



Stener lesion MRI

UCL/RCL instability grading

Grade	Findings	Tear
1	Sprain, no instability	Incomplete
2	Asymmetric laxity, **endpoint present**	Incomplete
3	**No endpoint**; **30–35° opening** or **10–15° > contralateral**	**Complete**

Complete tear thresholds	**>35° opening and/or >15° vs other side** (TO) · 30–35° (Orthobullets) · criteria range 15–45°; many use **>35°** (Newland, OB) – read the options

5 Management

- **Thumb UCL injury**
  - Grade 1–2 / <15° side difference → **immobilise 4–6 wk (spica)**
  - Grade 3 / Stener / displaced fragment → **acute repair**
    - ± adductor advancement
  - Chronic (>3–8 wk) / poor tissue → **tendon graft reconstruction**
    - failed / arthritic → **MCP fusion 15° flexion**

– – Active space – –

Operative criteria

Nonoperative (4–6 wk; TO partial: spica 4 wk)	Operative repair
Partial tear (unstable in flexion only)	**Complete tear** (unstable flexion + extension)
<15° side-to-side difference	**>15° side-to-side**
Bony skier's thumb, **stable + undisplaced** fragment	**>30–35° opening**
	Stener lesion
	Fragment **displaced >1 mm**, **malrotated**, or **>⅓ joint surface** (TO)

Not a surgical indication (TO) **ipsilateral metacarpal fracture**

Reconstruction chronic **>3 wk** (TO) / **>3–8 wk** (Orthobullets); **PL graft** through bone tunnels (screw, button, anchor); 92% satisfaction

Adductor advancement with acute repair (100% RTS) or chronic salvage

MCP fusion salvage; **15° flexion**

Repair technique

UCL S/chevron incision; transosseous sutures / anchors; immobilise **IP free**; strength 4–6 wk; >90% excellent; RTP ~100%

RCL straight radial incision; may resect **abductor aponeurosis**; spare **dorsal radial sensory branches**; anchors/pull-out; ± K-wire; **96% good–excellent**

RCL vs UCL – delayed repair

RCL **more amenable to primary repair even with delayed presentation**

	UCL	RCL
Frequency	**10×**	1×
Mechanism	**Abduction moment**	Axial load / adduction
Rupture site	**Distal**	Even proximal/mid/distal
Overlying aponeurosis	**Adductor** → Stener; may need removal	Abductor → Stener rare
Presentation	2–10 wk	**>10 wk**
Delayed repair	Reconstruct if chronic	**Still repairable**
X-ray	P1 supination (sag)	P1 pronation

6 Complications

Complication	Fact
Stiffness (MCP, IP)	**Most common** after repair
Persistent instability	**15%** of grade 3 treated nonop → reconstruct
Skin necrosis	Pull-out suture technique
Superficial radial neurapraxia	Observe
↓ pinch strength	

How it's asked

"Role of proper vs accessory UCL?"	→ **proper: radial stress in flexion; accessory: radial stress in extension**
"Skier, laxity in 30° flexion but not extension – MRI?"	→ **proper UCL torn off proximal phalanx**
"Skier's thumb – all EXCEPT"	→ **avulsion from 1st MC head** (TO; it's P1 base)
"Not an indication for surgery in gamekeeper's thumb?"	→ **ipsilateral metacarpal fracture** (TO)
"Structure blocking UCL reduction – muscle & nerve?"	→ **adductor pollicis, ulnar nerve**
"Lettered figure – what blocks reduction?"	→ **adductor aponeurosis (E)**
"Palpable ulnar nodule, UCL retracted on MRI – blocking structure?"	→ **adductor aponeurosis** ✗ Wrong options: anterior oblique ligament · volar plate · FPL · lumbricals
"RCL vs UCL – true of RCL?"	→ **more amenable to delayed repair** ✗ Wrong options: adductor removal · abduction moment · distal rupture · presents <2 wk
"Most common complication after repair?"	→ **stiffness**

– – Active space – –

THUMB HYPOPLASIA

– Hand · 92 / 103 –

1 Basics

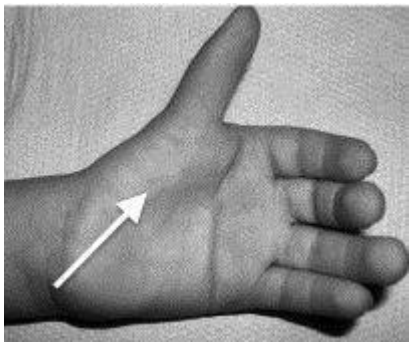
Incidence	**1/100,000** live births; ♂ = ♀
Laterality	**bilateral ~60%**; right > left
Association	often partial/complete **absence of radius**
Embryology	AER, progress zone, **FGF** signalling abnormalities

Associated syndromes & lab markers

Associated anomalies	**>80%**: **VACTERL**, **Holt-Oram**, **TAR**, **Fanconi anaemia**, Cornelia de Lange, Okihiro; radial aplasia, thrombocytopenia, renal/cardiac/CNS
Work-up	cardiac (auscultation, **echo**), renal + abdominal **US**
Labs	**FBC + peripheral smear** → Fanconi
Chromosomal challenge (breakage) test	**detects Fanconi anaemia before marrow failure**

3 Presentation

Exam	thenar hypoplasia, absent skin creases (muscle/tendon anomaly), excessive MCP abduction, **UCL laxity**, **web-space tightness**
Pollex abductus	**FPL** attaches to normal insertion **and extensor tendon**; frequent in **type IIIA**
Also	variable EPL insertion



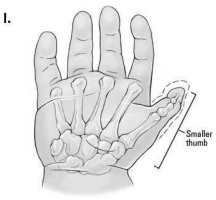
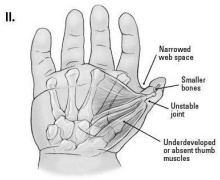
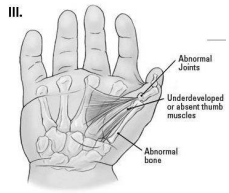
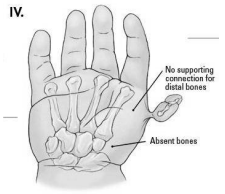
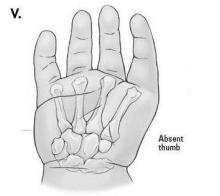
Thenar hypoplasia

4 Diagnosis & Classification

Key radiology findings

Views	**bilateral hand, wrist and forearm** films → musculotendinous vs osseous deficiency

Blauth classification

Type	Description	Treatment	Image
I	**Minor**: all components present, small	None (opponensplasty if ↓ abduction)	
II	All bones present; **MCP UCL instability**, thenar hypoplasia, **adduction contracture**	MCP stabilisation + 1st web release + opponensplasty	
IIIA	Musculotendinous + osseous deficiency; proximal ⅓ MC aplasia, **CMC intact**; no active MCP/IP motion	**Reconstruct** (as II)	
IIIB	Basal (proximal ⅔) MC aplasia, **deficient CMC**; no active MCP/IP motion	**Amputate + pollicise**	
IV	**Floating thumb** (skin + NV pedicle only)	Amputate + pollicise	
V	**Absent thumb**	Pollicise	

IIIA vs IIIB **CMC joint stability**

✗ Wrong options: osseous completeness · musculotendinous integrity · MCP stability · EPL (all deficient in both)

5 Management

Hypoplastic thumb – CMC joint stable?

- Yes (I, II, IIIA) → **reconstruct**
  - I, adequate abduction → **observe**
  - I, ↓ abduction → **opponensplasty**
  - II / IIIA → **web release + opponensplasty + MCP stabilisation**
- No (IIIB, IV, V) → **thumb amputation + index pollicisation**

– – Active space – –

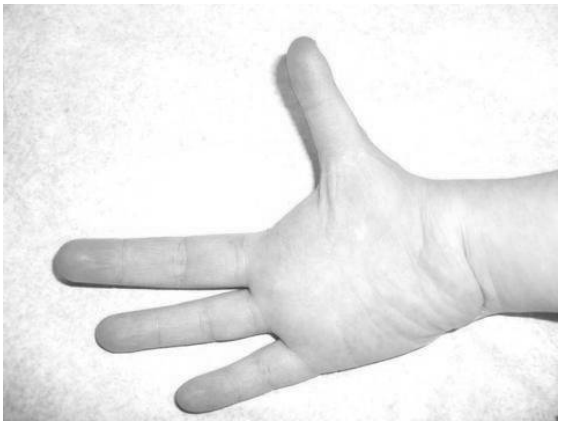
Reconstruction vs pollicisation – CMC joint stability

Deciding factor — **CMC stability** (resists grasp/pinch; full circumduction arc)
✗ Wrong options: contralateral thumb · thenar function · EPL · MCP stability · web contracture · absent IP joint

Antiposition — abducted, slightly extended, pronated

Procedure	Technique
Opponensplasty	**FDS** or **ADM** · Blauth II–IIIA: **FDS ring** transfer or **Huber (ADM) opponensplasty** (TO)
1st web deepening	**Z-plasty**
MCP stabilisation	**Fusion** · UCL reconstruction with **FDS** · UCL with free graft
Pollicisation	Index on NV bundles; detach 1st dorsal + palmar interossei; shorten by removing **index MC + physis**; stabilise index MCP; rebalance tendons; plan skin to avoid grafts · **12–18 mo** · **index MP → thumb CMC** · EIP → EPL · EDC → APL · 1st DI → APB · 1st PI → adductor pollicis (TO)

9-month-old, basal MC aplasia, deficient CMC, no MCP/IP motion → pollicisation
✗ Wrong options: observation · opposition transfer alone · web release + transfer + MCP fusion · UCL reconstruction



Post-operative pollicisation

How it's asked

"Most important variable, reconstruct vs ablate?"	**→ CMC joint stability**
"Key difference Blauth IIIA vs IIIB?"	**→ CMC joint stability**
"Adopted 3-year-old with hypoplastic thumb — deciding factor?"	**→ CMC stability**
"Deficient CMC, absent MCP/IP motion — definitive treatment?"	**→ pollicisation**
"Detect Fanconi before marrow failure?"	**→ chromosomal breakage (challenge) test**
"Pollex abductus?"	**→ FPL inserts normally + into extensor; type IIIA**

TRIANGULAR FIBROCARTILAGE COMPLEX (TFCC) INJURY

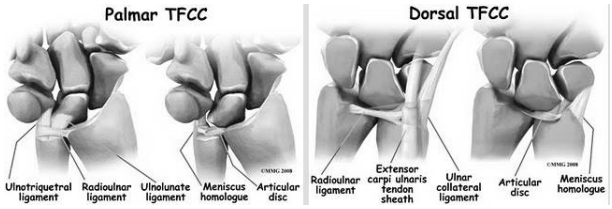
– Hand · 93 / 103 –

1 Basics

Anatomy – components, foveal insertion & radioulnar ligaments

Function (TO) transmits **~20% of wrist load**; **main stabiliser of DRUJ**; cushion between ulnar head and ulnar carpus

TFCC components	Not components
Dorsal + volar **radioulnar ligaments** (superficial + deep limbs)	**Extensor digiti minimi (EDM) sheath** (TO)
Central **articular disc**	LT interosseous ligament
Meniscus homologue	RSC ligament
Ulnar collateral ligament	FCU sheath
ECU subsheath (6th compartment)	
Origin of **ulnolunate + ulnotriquetral** ligaments	



TFCC components

Radioulnar ligaments **primary restraint to dorsal/palmar translation of radius on (fixed) ulna**; span sigmoid notch margins ↔ ulna

Deep fibres **ligamentum subcruentum → ulnar fovea** (key DRUJ stabiliser; repair these) · **superficial → ulnar styloid**

Articular disc load bearing, almost no stabilising role

UL / UT / ECU subsheath stabilise carpus to ulna; minimal DRUJ stability

Functional zone (TO)	Role	Components
Distal	Ulnocarpal stability	TFC, meniscus homologue, UL, UT
Proximal	**DRUJ stability**	Radioulnar ligaments
Functional UCL	Dynamic stabiliser	**ECU sheath** + thickened ulnar capsule

– – Active space – –

Reciprocal RULs (TO) **supination → palmar RUL tight · pronation → dorsal RUL tight** · UL/UT tighten reciprocally in radio-ulnar deviation

Ulnocarpal ligament (TO)	Origin → insertion	Key feature
Ulnocapitate	Fovea → palmar-ulnar capitate	**Most superficial**
Ulnotriquetral	Palmar RUL + styloid → triquetrum	Oblique
Ulnolunate	TFC disc → palmar lunate	**Motion coupling, not stability**

Ligamentum subcruentum (TO) vascular connective tissue between RUL limbs at styloid → **aids healing**

Meniscus homologue (TO) C-shaped, **highly vascular + elastic**, round styloid tip · **prestyloid recess** between MH + TFC: type 1 narrow-long, type 2 wide-short, type 3 no ulnocarpal opening (connects DRUJ)

ECU sheath (TO) **taut in pronation → prevents dorsal dislocation of ulnar head** · relaxes in supination

Volar RUL injury → dorsal subluxation of ulna · dorsal RUL injury → volar subluxation of ulna

Secondary DRUJ stabiliser dorsal oblique band of IOM

Blood supply periphery vascular (10–40%); **central disc avascular**

Concepts of TFCC structure

Concept	Author
2-D structure	**Palmer & Werner** (1981)
"Hammock" / suspension (3 zones: foveal, distal hammock, UCL)	**Nakamura**
"Iceberg" (proximal foveal pc-TFCC = submerged, functional; distal dc-TFCC = visible tip, shock absorber)	**Atzei & Luchetti** (TO)

Iceberg (TO) pc-TFCC not seen through radiocarpal portals → indirect **hook sign** / DRUJ portals; pc-TFCC tear → DRUJ instability, ↓ rotation, ↓ grip, high reoperation

2 Mechanism

Most common mechanism **fall on extended wrist + forearm pronation**
✗ Wrong options: supination · flexion · axial load ulnar deviation

Type 1 traumatic; traction on ulnar wrist

Type 2 degenerative; **positive ulnar variance**, ulnocarpal impaction; central tears with repetitive activity

3 Presentation

Diagnosis – mechanism & examination

Symptoms ulnar-sided wrist pain, clicking, painful **turning a door key**

Fovea sign tender **soft spot between ulnar styloid and FCU**, volar ulnar head → pisiform; **95% sensitive, 87% specific** for foveal / UT injury

Fovea sign interpretation (TO) positive = **foveal avulsion (DRUJ unstable)** · negative with ulnar pain = **UT ligament tear (DRUJ stable)**

Dimple sign (TO) hand flat, ulnar head shifts palmarly → dorsal dimple = **multidirectional DRUJ instability**

Palpation (TO)	Site	Condition
Full pronation	Dorsal radial metaphysis	EDC tenosynovitis
Full pronation	Proximal-ulnar corner of lunate	Ulnocarpal impaction / Kienböck
Full pronation	Dorsal ulnar head	DRUJ arthritis
Full supination	**Ulnar snuffbox / fovea**	**TFCC tear / DRUJ instability**
Palmar	Pisiform, FCU, hook of hamate	ECU tendinopathy / pisotriquetral OA

Provocation pain with **ulnar deviation** (compression) or radial deviation (tension), resisted ulnar deviation

Piano key sign (TO) ballotable distal ulna = **DRUJ instability**: **TFCC tear, Galeazzi (nonunion), Madelung**; **not isolated ulnar styloid fracture**

(✗ Wrong options: **Distractors** – ECU subluxation (snap in supination + flexion + ulnar deviation; ECU tenderness)) · hook of hamate (CT / carpal tunnel view) · ulnar styloid fracture (X-ray) · Guyon's (interossei weakness, ulnar numbness)

4 Diagnosis & Classification

Key radiology findings

X-ray usually **normal**; **zero-rotation PA → ulnar variance**; pronated grip PA (dynamic); widened DRUJ / dorsal ulna on true lateral if unstable

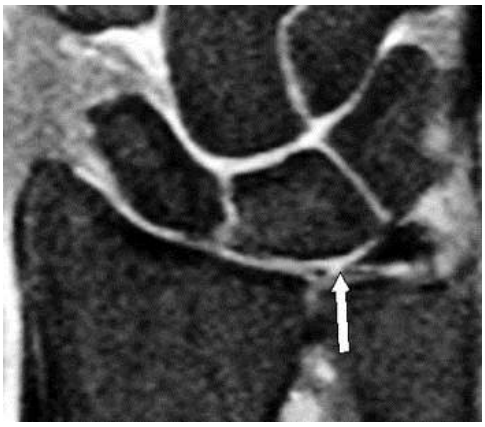
Arthrography extravasation (largely replaced by MRI)

MRI clues (TO) **DRUJ effusion** · **fluid in prestyloid recess** · **"floppy" disc** · read **coronal** (axials poor) · small tears missed at 1.5 T → 3 T or MR arthrography

MRI / MR arthrogram **sensitivity 74–100%**; lunate ulnar-side tear = ulnocarpal impaction

CT (TO) best for **DRUJ subluxation** + bony detail · **MR arthrography** = highest sensitivity for perforations (TO)

Arthroscopy **gold standard / most accurate**; after several months failed splinting



MRI: TFCC tear

– – Active space – –

Palmer classification

Class 1 – Traumatic	Lesion
1A	**Central perforation**
1B	**Ulnar avulsion** (± styloid fx; Orthobullets: without)
1C	Distal avulsion (UL/UT origin)
1D	Radial avulsion

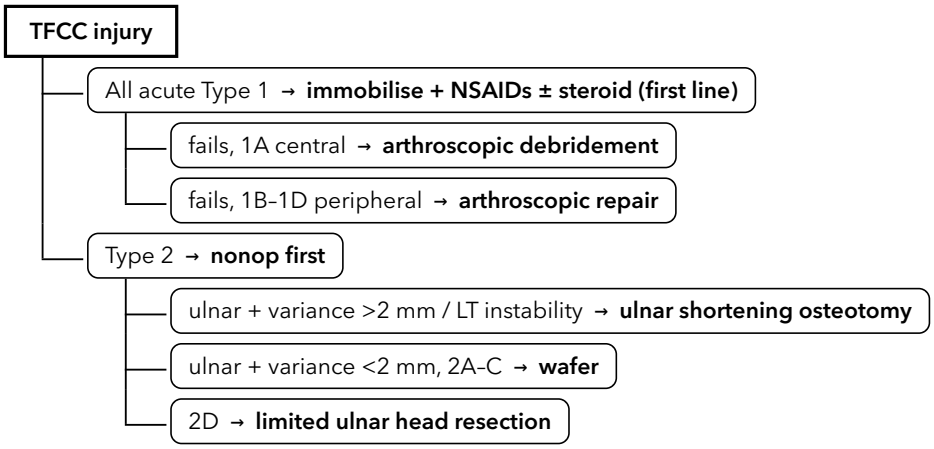
Class 2 – Degenerative	Lesion
2A	Wear / thinning
2B	2A + lunate and/or ulnar chondromalacia
2C	2B + perforation
2D	2C + ligament (LT) disruption
2E	2D + ulnocarpal + DRUJ arthritis



Palmer classification

Defect adjacent to ulna, stable DRUJ (TO)	**Class 1B**
Need for surgery ↑ (TO)	associated fractures, instability, DRUJ injury; most isolated tears no early surgery
Ulnar wrist pain differential (TO)	TFCC · ulnocarpal impaction · ECU · hook of hamate · LT instability · ulnar styloid impaction · **TILT (triquetral impingement ligament tear)** · DRUJ OA
Differentials	ulnocarpal abutment, ulnar styloid impaction, ECU tendonitis, hook of hamate fracture, ulnar tunnel, pisotriquetral OA

5 Management



Treatment – immobilisation, debridement & wafer procedure with ulnar-positive variance

Acute 1B, DRUJ stable, 5 days — **immobilisation + NSAIDs**
✗ Wrong options: arthroscopy yet · repair · shortening · wafer

Conservative (TO) — **above-elbow cast 4–6 wk**, NSAIDs, steroid or **PRP** injection, proprioceptive + grip PT

Surgery (TO)	Procedure
Central 1A	Arthroscopic debridement
Peripheral 1B / 1C	Suture anchor / transosseous repair
Ulnar styloid nonunion	**Excision + TFCC reattachment**
Chronic DRUJ instability	**Foveal reattachment of RUL** / ulnar shortening
Approach	**Palmar for UL/UT** tears · **dorsal for RUL / foveal** repair

6U portal (TO) — soft spot between **UT ligament and ECU sheath** → ulnar recess, **foveal repair**

Rehab (TO) — 0–4 wk immobilise neutral rotation · 4–8 wk ROM + pronosupination · 8–12 wk strengthening · full recovery **3–4 months**

Debridement — 1A; 3-4 + 6R portals; **keep 2 mm peripheral rim**; 80% pain relief; **not effective with ulnar-positive variance**

Repair — 1B–1D (peripheral, vascular); acute athletic best; outside-in / inside-out / transosseous; immobilise **6 wk**; acute (<3 mo) → **80% motion + grip**; risk ECU tendonitis (knot), dorsal sensory nerve

DRUJ (Mulford) — mild acute → immobilise 4–6 wk in stable position; complex reducible + styloid fx → ORIF; without fx → TFCC repair + pin DRUJ

Ulnar diaphyseal shortening — variance **>2 mm**, **LT instability** (tightens ulnocarpal ligaments); plate; nonunion, hardware removal

Wafer — removes **2–4 mm** distal ulnar head (open or arthroscopic through central defect); ulnar cortex intact, not into DRUJ; no plate

Wafer threshold — variance **<2 mm** (Orthobullets) · **2 mm positive variance + ulnocarpal impingement** = indication with debridement (OB) – read the options

Darrach — **contraindicated** (ulnar stump instability, ECU subluxation); salvage only

6 Complications

Repair — ECU tendonitis from knot, dorsal ulnar sensory nerve injury

Ulnar shortening — nonunion, hardware irritation

Positive variance — worse debridement/repair outcomes

– – Active space – –

X vs Y

	Ulnar shortening osteotomy	Wafer
Variance	**>2 mm**	**<2 mm** (2A–C)
Fixation	Plate	None
Extra benefit	Tightens UL/UT → **LT instability**	Arthroscopic via central tear
Complications	Nonunion, hardware	Technically difficult

How it's asked

"Components of TFCC EXCEPT?"	→ **EDM sheath** (TO)
"Anatomical component of TFCC?"	→ **ECU tendon sheath** ✗ Wrong options: LTIL · RSC · FCU sheath
"Main restraint to dorsal/palmar radius translation?"	→ **dorsal + palmar radioulnar ligaments** ✗ Wrong options: disc
"Deep radioulnar fibres insert on?"	→ **ulnar fovea** ✗ Wrong options: ulnar styloid = superficial
"Deep-fibre injury → dorsal ulnar subluxation — which structure?"	→ **volar radioulnar ligament**
"Fall on pronated extended wrist, tender distal to ulnar styloid, no ECU tenderness?"	→ **TFCC tear**
"Normal X-ray + CT, foveal tenderness, no snapping?"	→ **TFCC tear** ✗ Wrong options: ECU subluxation · hamate
"Most common mechanism?"	→ **wrist extension + forearm pronation**
"Indication to add wafer to debridement?"	→ **2 mm positive ulnar variance + ulnocarpal impingement**
"Acute 1B, stable DRUJ, 5 days?"	→ **immobilization with NSAIDs**
"Defect adjacent to ulna, stable DRUJ — Palmer?"	→ **Class 1B** (TO)
"Tight radioulnar ligament in supination?"	→ **palmar RUL** (TO)
"Positive foveal sign indicates?"	→ **TFCC foveal avulsion, DRUJ unstable** (TO)
"Iceberg concept given by?"	→ **Atzei & Luchetti** (TO) ✗ Wrong options: Palmer & Werner · Nakamura
"Piano key sign NOT seen in?"	→ **ulnar styloid fracture** (TO) ✗ Wrong options: TFCC tear · Galeazzi · Madelung

TRIGGER FINGER

– Hand · 94 / 103 –

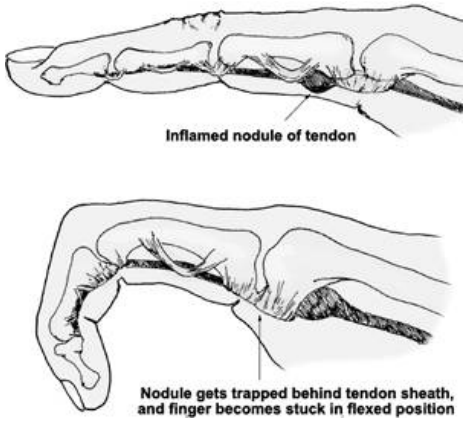
1 Basics

Epidemiology

Incidence	**2–3%** general population · **10% of diabetics**
Who	♀ **>50 y**, **diabetics** (bilateral, multiple digits)
Adult digits	**ring & long** most common
Associations	RA, calcific tendinitis, septic tenosynovitis, **CTS (>60%** of trigger digits have clinical/EDS CTS), congenital trigger thumb · DM, amyloidosis, hypothyroidism, sarcoidosis, gout, pseudogout

Pathology – A1 pulley

Site	**stenosing tenosynovitis at A1 pulley** (overlies MCP) → tendon–pulley size mismatch ✗ Wrong options: Grayson's, ORL, TCL, extensor
Histology	**fibrocartilaginous metaplasia** of tendon/pulley; chondrocyte proliferation; **↑ type III collagen**; hyperglycaemia → collagen cross-links
Nodule	on **FDP** (FDS often unaffected)
Trigger thumb	**Av (variable annular) pulley** causes stenosis in up to **75%**



A1 pulley

3 Presentation

Symptoms	progressive pain at A1 (distal palm), clicking, catching, **locking in flexion at PIP**; referred dorsal MCP/PIP pain
Exam	tender A1, **palpable nodule**, triggering on flexion/extension

4 Diagnosis & Classification

Diagnosis	**clinical**; X-ray not needed

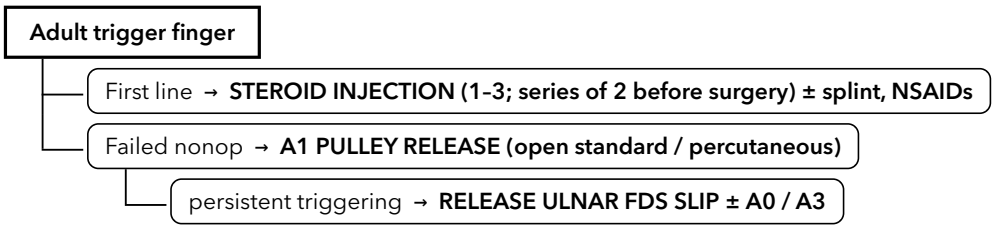
– – Active space – –

Green classification

Grade	Feature
I	Palm pain, tenderness at A1
II	**Catching**
III	**Locking, passively correctable**
IV	**Fixed locked digit**

Differentials **lumbrical plus** (paradoxical extension on flexion) · joint contracture (can't passively extend) · **pyogenic flexor tenosynovitis** (Kanavel, ↑ markers) · **MCP locking** (collateral/volar plate caught on MC head osteophyte)

5 Management



Nonoperative

Splint immobilise **MCP or DIP only**; PIP free; relief 40–97%

Pathology (TO) **thickened A1 pulley, not the tendon** (volume–capacity mismatch) · inflammation → granulation "**rice bodies**" · worse in **morning** · associations DM, RA, hypothyroidism, pregnancy · diagnosis clinical; US only if atypical

Injection **best initial treatment** · **70–80% success in non-diabetics** (TO) · complications incl. **skin depigmentation** (TO); relief **60–90%** (up to 90% non-diabetics); in/superficial to sheath

Diabetics less relief (Orthobullets classic) · recent literature: success not influenced by DM – read options; surgery cost-effective in DM

Poorer response longer symptom duration

Injection complications fat atrophy, **tendon rupture**, **transient hyperglycaemia** (diabetics)

Infection risk – timing after steroid injection

Rule injection **within 90 days of surgery → ↑ postoperative infection**

Last injection 2 mo ago → schedule surgery in **1.5 mo** (>90 days)

Operative

	Open release	Percutaneous release
Status	**Standard**; easier to confirm release	18–19 G needle, bevel parallel to tendon
Success	>90%	>90% (↑ with **US**)
RTW	–	Earlier
Avoid in	–	**Thumb & little finger** (nerve proximity)

Recurrence	↑ in **diabetics**
Percutaneous technique (TO)	**16 or 18G needle**, release **A1 only**, test active flexion/extension · avoid in thumb (radial digital n.)
Open release (TO)	**gold standard** · incision over A1 · release A1, **preserve A2**
WALANT / local	allows intraoperative check of triggering; post-op early active ROM 4×/day
Extra releases	**A0** (aponeurosis proximal to A1) · one/both FDS limbs · A3
RA	**FDS slip excision without A1 release** → spares A1 → does not worsen **ulnar drift**
Ulnar FDS slip	resected to help prevent MCP ulnar drift

Percutaneous thumb release – radial digital nerve risk

Most at risk	**radial digital nerve of thumb** (superficial, **oblique** course across A1)
Distance	mean **2.9 mm** from release point; within 2 mm in 5/10 hands (Bain) · 2.7 mm (Orthobullets pulley data)
Little finger	variable nerve crossing → also avoid percutaneous

Paediatric trigger finger/thumb – additional releases & digital nerve risk

Trigger thumb	**10× more common** than trigger finger in children; A1 constriction; **IP flexion contracture**; **>60% resolve spontaneously** (Baek); surgery → **radial digital nerve** crosses field
Paediatric trigger finger	**Notta's node proximal to A1**, flexion contracture, triggering; causes: proximal **FDS decussation**, FDS/FDP nodules, thick A2, **tight A3**
A1 release alone	often fails (6/18, Cardon) → **release one/both FDS limbs** ± A3 (Bae: all → A1 + single FDS slip resection)
Timing	operate **2–4 y** (prevent IP contracture)
Revision (persistent triggering)	**extensile incision**; order per OB answer: **A3 release → then 1 FDS slip** · per OB explanation: ulnar FDS slip → then A3 / remaining slip
Avoid in children	complete **A2 release** (bowstringing) · **FDP debulking** (adhesions) · horizontal incision (no prox/distal exposure)



Paediatric trigger thumb

6 Complications

Radial digital nerve injury	**thumb release** (superficial, oblique)

– – Active space – –

Bowstringing	**A2/A4 damage** → pulley reconstruction · after trigger release done elsewhere → **proximal part of A2 damaged** (TO)
Others	stiffness, scar tenderness, dehiscence, haematoma, **tendon scoring** (percutaneous), infection

How it's asked

"Diabetic, ring finger locking, distal palm tender – structure?"	→ **A1 pulley**
"Child – release in addition to A1"	→ **one or both limbs of sublimis (FDS)** ✗ Wrong options: A4
"Paediatric trigger thumb surgery – at risk"	→ **radial digital nerve**
"Persistent paediatric triggering after A1 release"	→ **extensile incision, A3 release, then 1 FDS slip** ✗ Wrong options: A2 release · FDP debulking
"Percutaneous thumb release – consent for"	→ **radial digital nerve of thumb**
"Injected 2 mo ago – when to operate?"	→ **in 1.5 mo (>90 days)**
"RA trigger finger"	→ **FDS slip excision, preserve A1**

ULNAR STYLOID IMPACTION SYNDROME

– Hand · 95 / 103 –

1 Basics

Epidemiology & pathoanatomy

Definition	**impaction of ulnar styloid tip on triquetrum** → ulnar-sided wrist pain; **long ulnar styloid** or **ulna-positive** wrist
Demographics	common; **Asians > Whites** (more positive variance)
Causes of ulnar positivity	normal variant · trauma / **radial malunion** · congenitally short radius · **premature radial physeal closure** · **Madelung deformity** (volar-ulnar distal radial physis dyschondrosis)
Ulnar impaction (OB)	degenerative; excess ulnocarpal load → TFCC degeneration → chondromalacia lunate/triquetrum/ulnar head → **LT instability** → ulnocarpal & DRUJ OA; mostly ulna +ve, may occur neutral/negative

Anatomy – ulnocarpal load

Ulnocarpal load	**~20%** of wrist load
Pronation & grip	**↑ ulnar variance** & load

Ulnar variance

Variance	Ulna – radius	Radius load	Ulna load	Image
Neutral	0 (<1 mm)	**80%**	**20%**	
Positive	**+2.5 mm**	60%	**40%**	
Negative	−2.5 mm	95%	5%	

+2.5 mm ulnar load – conflict **40%** (Orthobullets table) · **up to 50%** (Orthobullets anatomy text)

– – Active space – –

3 Presentation

Symptoms	ulnar wrist pain, **worse with pronation / grip**
Exam	swelling, tenderness **ulnar styloid / TFCC**, painful ↓ ROM
Ulnar stress test	**max ulnar deviation + axial load + supination → pronation** → reproduces pain

4 Diagnosis & Classification

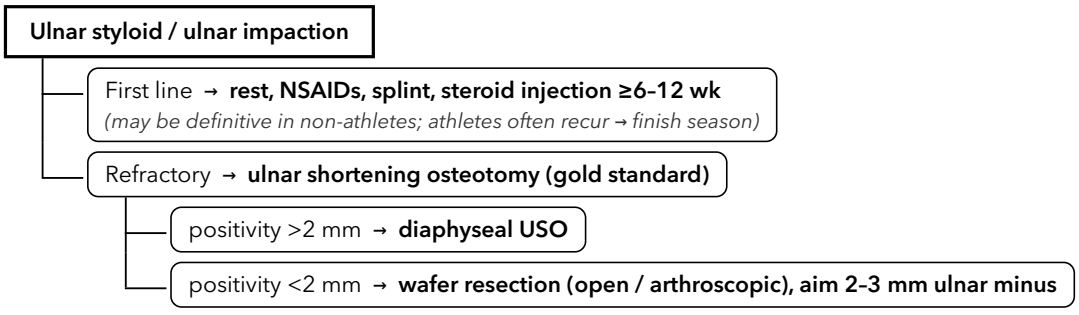
Key radiology findings

View / test	Fact
PA wrist	**Positive ulnar variance** + **subchondral sclerosis / cysts** ulnar styloid ± triquetrum
Styloid process index	(ulnar styloid length − ulnar variance) ÷ ulnar head width; **<0.22 normal**
Pronated grip PA	**Dynamic** ulnar variance
Contralateral views	Comparison
MRI	**TFCC** and **LTIL**

Diagnosis & failed treatment – positive ulnar variance, styloid impaction

Failed OT + steroid + isolated TFCC debridement	cause = **unaddressed positive ulnar variance** ✗ Wrong options: inadequate debridement · SL tear · scaphoid fracture
TFCC debridement alone	**90%** improve & return to work (when variance not the problem)
Differentials	TFCC injury · **ulnocarpal abutment** · DRUJ injury · pisotriquetral OA · ECU tendinitis · LT tear

5 Management



Ulnar shortening osteotomy – nonunion risk

USO	**gold standard**; preferred in athletes after failed cast + PT
Wafer (partial ulnar head/styloid resection)	<2 mm positivity; encouraging early results, **not proven superior**; post-op immobilisation ~6 wk

USO nonunion **4–8%; ~6% in smokers**

Factor	Nonunion effect
Smoking	**↑ (≈6%)**
Low BMD	**↑**
Transverse vs oblique	**No difference** (4.16% vs 3.86%)
Volar vs ulnar plate	**No difference**
BMI >30	**No association**

6 Complications

USO complications **nonunion** · tendon rupture · **persistent pain / hardware irritation** · infection

How it's asked

"Failed OT, injection, TFCC debridement – why still pain?"	→ **positive ulnar variance** (needs USO)
"Martial artist, failed cast + PT, USO recommended – pathology"	→ **ulnar impaction** (MRI: ulnar lunate/ triquetral changes)
"USO – true statement"	→ **nonunion ~6% in smokers** ✗ Wrong options: oblique worse · BMI · plate position · BMD irrelevant
"Gold-standard operation"	→ **ulnar shortening osteotomy**
"Load through ulna in neutral variance"	→ **20%**
"View for dynamic variance"	→ **pronated grip PA**

– – Active space – –

ULNAR TUNNEL SYNDROME

– Hand · 96 / 103 –

1 Basics

Epidemiology

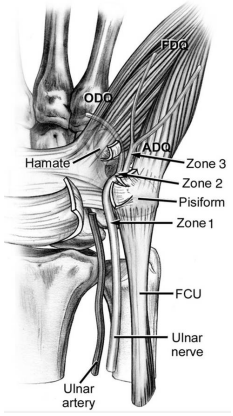
Incidence	**less common than cubital tunnel**
Risk	**cyclists → "handlebar palsy"**; manual workers

Anatomy – Guyon's canal

Length	**~4 cm**; proximal TCL edge → **aponeurotic arch of hypothenar muscles**
Contents	ulnar nerve + artery; nerve **bifurcates** into superficial sensory + deep motor

Boundary	Structure
Floor	**Transverse carpal ligament**, hypothenar muscles
Roof	**Volar carpal ligament**
Ulnar (medial)	**Pisiform**, pisohamate ligament, abductor digiti minimi
Radial (lateral)	**Hook of hamate**

Deep motor branch	**all interossei**, **lumbricals 3 & 4** (Orthobullets) · "4th & 5th lumbricals" (TO, as printed) · hypothenars · **adductor pollicis** · **deep head FPB**
Dorsal ulnar cutaneous branch	arises **~5 cm proximal** to ulnar styloid / Guyon's canal → **dorsal ulnar hand spared** in ulnar tunnel
Innervated proximal to Guyon (spared)	**FCU**, **FDP ring/little**; ECU = PIN; FPL = AIN



Guyon's canal zones

2 Mechanism

Causes – ganglion & imaging

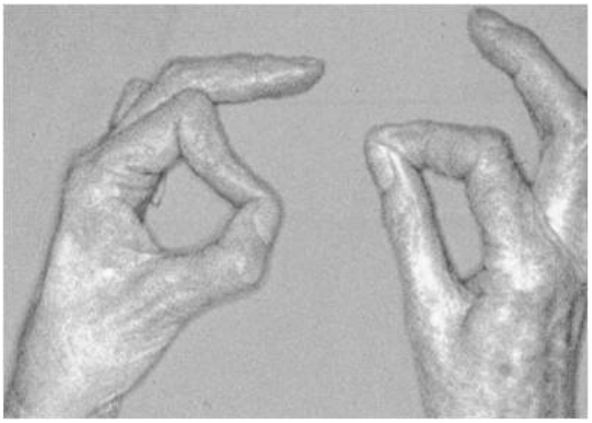
Commonest	**ganglion cyst = 80% of non-traumatic cases** (often from **pisohamate joint**)
Others	lipoma · repetitive trauma · **ulnar artery thrombosis/aneurysm (hypothenar hammer)** · **hook of hamate fracture/nonunion** · pisiform dislocation · inflammatory arthritis/RA · anomalous muscle (ADM), fibrous bands · palmaris brevis hypertrophy · idiopathic
Ulnar artery occlusion (TO)	may cause **Raynaud in ulnar 3 digits** (sympathetic fibres travel with artery)

3 Presentation

Symptoms	pain/paraesthesia **volar ulnar 1½ digits**; weak intrinsics, grip, pinch
Spared	**dorsal ulnar hand sensation** · **FDP ring/little** strength · median territory

Sign	Mechanism
Claw ring + little	Loss of intrinsics (MCP flexion, IP extension)
Froment (book test)	Weak **adductor pollicis** → compensatory **thumb IP flexion by FPL** (AIN); thumb + index IPJ flexion when pinching paper
Jeanne	Thumb **MCP hyperextension** (EPL) with IP flexion
Wartenberg sign	**Abducted little finger** – weak **3rd palmar interosseous** (ulnar) + unopposed **EDQ/EDM** (PIN)
Weak pinch	Loss of thumb adduction → **up to 70% pinch lost**
Weak grasp	Loss of MCP flexion
Allen test	**Ulnar artery thrombosis**
Difficulty crossing index/middle	Interossei weakness

Wartenberg distractors	EDQ (unopposed, not weak) · ADM weakness → adducted · dorsal interossei → adducted · **Wartenberg syndrome** = superficial radial nerve (cheiralgia paraesthetica)



Froment sign

4 Diagnosis & Classification

Guyon's canal zones & motor vs sensory findings

Zone	Location	Typical cause	Deficit
1	**Proximal to bifurcation**	**Ganglion, hook of hamate fracture**	**Mixed motor + sensory**
2	**Deep motor branch** (radial)	**Ganglion, hook of hamate fracture**	**Motor only**; **most commonly affected (TO)**
3	**Superficial sensory branch** (ulnar)	**Ulnar artery thrombosis / aneurysm**	**Sensory only**

No zone 4 3 zones only

Lesion in Guyon's canal (TO) can give **both** motor + sensory loss (motor, sensory or mixed by zone)

Key radiology findings

Modality	Fact
X-ray / carpal tunnel view / CT	**Hook of hamate fracture**
MRI wrist	**Ganglion** (T2 hyperintense) → **study that identifies causative lesion**; gradient echo shows aneurysm
Doppler US / arteriogram / MR angiography	**Ulnar artery thrombosis / aneurysm**
US	Ganglion (well-circumscribed, homogeneous → no biopsy)



MRI: ganglion in Guyon's canal

Electrodiagnostics

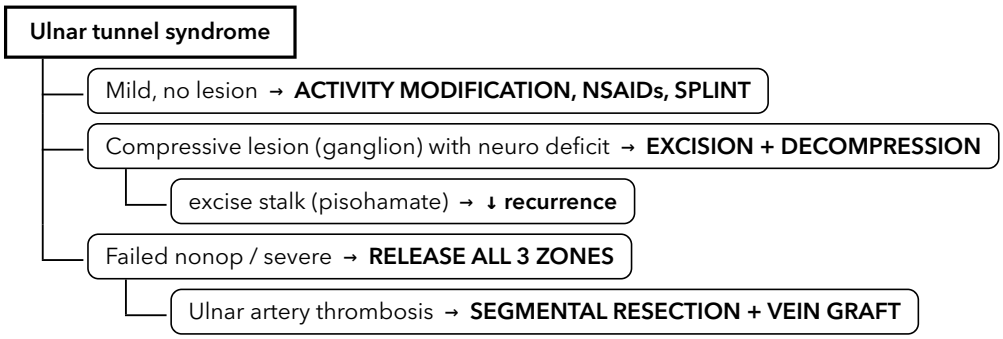
NCS/EMG diagnosis + prognosis; **CV <50 m/s across elbow**; ↓ SNAP & CMAP amplitudes (Orthobullets, TO)

EMG localisation fibrillations in **adductor pollicis**
✗ Wrong options: FCU · FDP little · ECU · FPL

Pure motor lesion normal sensory conduction, **delayed motor to 1st dorsal interosseous**

EMG/NCV confirms neuropathy but **does not identify cause** (→ MRI)

5 Management



Local decompression release hypothenar origin · excise ganglion · **resect hook of hamate** · vascular repair · **explore all three zones**

Ulnar artery thrombosis (TO) **segmental resection + vein graft** preferred when feasible

With CTS add **carpal tunnel release**

Tendon transfers

Goal	Transfer
Correct claw	ECRL / ECRB / PL graft, **volar to transverse MC ligament**, 2- or 4-tailed to **A2 pulley** ring + little
Power pinch	**Smith transfer**: ECRB or **FDS ring**
Little finger adduction (Wartenberg)	**Ulnar insertion of EDM** → A1 pulley / RCL little finger

6 Complications

Recurrence main complication (ganglion stalk left)

X vs Y

	Ulnar tunnel (Guyon)	Cubital tunnel
Dorsal ulnar hand sensation	**Spared**	**Lost**
Clawing	**More**	Less
FCU / FDP ring-little	Normal	Weak
Tinel	Wrist	Elbow; + elbow flexion test
Compression site	**Ganglion** in canal	Osborne's ligament / FCU heads / anconeus epitrochlearis

– – Active space – –

Look-alike	Differentiator
Limburg-Comstock	FPL–FDP index anomalous connection (~20%); cannot flex thumb IP without index DIP
Parsonage-Turner	Viral brachial neuritis; shoulder pain; cannot flex thumb IP + index
C8–T1 radiculopathy	Neck; dermatomal
AIN (Gantzer muscle)	Loss FPL, FDP index, PQ; no sensory loss

How it's asked

"Lesion in Guyon's canal leads to?"	→ **both motor + sensory loss** (TO)
"Ring/little numbness, no weakness, ulnar artery aneurysm?"	→ **zone 3 compression** (TO) ✗ Wrong options: normal EDS · Jeanne · cervical
"Intrinsic atrophy, normal sensation, lesion at Guyon — zone?"	→ **zone 2**
"Ganglion in Guyon + weak 1st DI — additional finding?"	→ **Froment sign** ✗ Wrong options: dorsal ulnar numbness
"Dorsal sensation normal, Froment + — EMG fibrillations in?"	→ **adductor pollicis**
"Abducted little finger — weak muscle?"	→ **3rd palmar interosseous**
"Ganglion between TCL and volar carpal ligament — sensory area?"	→ **volar ulnar 1½ digits only** ✗ Wrong options: dorsal ulnar
"Weak key pinch, volar ulnar numbness, dorsal normal — cause?"	→ **ganglion in Guyon's canal**
"Pure motor, ganglion on MRI/US — next step?"	→ **cyst excision** ✗ Wrong options: biopsy · hamate excision
"Which study identifies the causative lesion?"	→ **MRI wrist** ✗ Wrong options: EMG/NCV

ULNAR VARIANCE

– Hand · 97 / 103 –

1 Basics

Definition & demographics

Ulnar variance (UV) length of ulna vs radius at the wrist; **neutral (ulnar zero) = difference <1 mm**

Demographics UV **lower in males**; **UV ↑ with age**

Child gymnasts **positive UV** (distal radial physeal arrest)

Load distribution & radiographic measurement

Ulnar variance	Ulna – radius	Load through radius	Load through ulna	Image
Positive	**+2 mm**	60%	**40%**	
Positive	+1 mm	70%	30%	
Neutral	0 (<1 mm)	**80%**	**20%**	
Negative	−1 mm	90%	10%	
Negative	**−2 mm**	**95%**	**5%**	

Palmer & Werner +2.5 mm → ulnocarpal load **18% → 42%**; −2.5 mm → **4.3%**

Dynamic UV **more positive in pronation** and **with grip**; **↓ in supination**

- - Active space - -

2 Mechanism

Causes

Cause	Positive UV	Negative UV
Congenital	**Madelung deformity**	Reverse Madelung
Trauma	DRF with radial shortening, **growth arrest** (SH injury), **Galeazzi, Essex-Lopresti**	Ulnar fracture with shortening
Iatrogenic	**Radial head resection**	Joint-levelling (ulnar shortening / radial lengthening)

Carpal instability adaptive (CIA) after radius malunion

CIA (TO) carpal malalignment **secondary to extrinsic deformity**; carpal bones + intrinsic/extrinsic ligaments **normal**

Commonest cause (TO) **DRF malunion with dorsal tilt** (normal volar tilt ≈ 11°) → **lunate extends (tilts dorsally)** to realign with radial surface → **adaptive DISI**
✗ Wrong options: lunate flexes · ulnar translocation

Key difference from CID (TO) in CIA **intrinsic interosseous ligaments intact**; CID = primary intrinsic ligament failure

Other causes (TO) **scaphoid humpback malunion** · metacarpal malunion · **Madelung (↑ volar tilt → adaptive VISI)**

Clinical (TO) pain, stiffness, ↓ grip, **↓ extension**, midcarpal prominence; **Watson + ballottement negative**

X-ray (TO) DR malunion (↓ volar tilt, ↓ inclination, shortening, DRUJ incongruity); **↑ SL angle but SL gap normal**

Treatment (TO) **correct skeletal deformity**: **opening wedge DR osteotomy + structural graft + volar locking plate** → carpus resets; scaphoid malunion → volar wedge graft (iliac crest / vascularised MFC) + headless screw; **no ligament reconstruction or intercarpal fusion**

4 Diagnosis & Classification

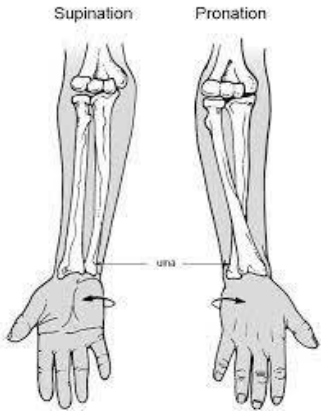
Key radiology findings

View **PA wrist, shoulder abducted 90°, elbow flexed 90°, neutral rotation**

PA vs AP UV **more positive on PA** (pronation); more negative on AP (supination)
✗ Wrong options: radial inclination differs · AP better for scaphoid · DRUJ/ lunate facet better on one view

Measurement line tangential to **ulnar articular surface** ⊥ shaft vs line tangential to **lunate fossa of radius** ⊥ shaft; ulnar line distal = positive, proximal = negative; normal 0 mm

MRI estimates but **cannot quantify** UV (wrist position not reproducible)



UV changes with forearm rotation

5 Management

Principle treat the condition: ulnar abutment, TFCC tear, Kienböck

6 Complications

Associated conditions

Positive UV	Negative UV
Ulnar abutment (impaction) syndrome	**Kienböck disease**
TFCC tears	**Ulnar impingement** (ulna hits radius proximal to sigmoid notch)
LT ligament tears, SLD	
Arthrosis ulnar head / lunate / triquetrum	



Positive UV with ulnocarpal abutment

- - Active space - -

X vs Y

	CIA	CID
Intrinsic ligaments	**Intact**	Torn (e.g. SL)
SL gap	Normal	Widened
Watson test	Negative	Positive
Treatment	**Correct osteotomy** → carpus resets	Ligament repair / reconstruction / fusion

How it's asked

"–2.5 mm UV – load distribution?"	→ **95% radius / 5% ulna**
"Neutral UV load?"	→ **80% radius / 20% ulna**
"PA vs AP view?"	→ **UV more positive on PA**
"UV more positive in?"	→ **pronation and grip**
"Negative UV associated with?"	→ **Kienböck**
"CIA after dorsally tilted DR malunion – lunate?"	→ **extends (dorsal tilt)** (TO)
"CIA vs CID fundamental difference?"	→ **intrinsic ligaments intact in CIA** (TO)

ULNOCARPAL ABUTMENT SYNDROME

– Hand · 98 / 103 –

1 Basics

Load transmission

Ulnar variance	Ulna load	Radius load
Neutral	**20%**	**80%**
+2 mm	**40%**	60%

Definition excessive impact between **ulnar head and carpus (primarily lunate)** → common ulnar-sided wrist pain

Positive variance causes/associations SL dissociation, **TFCC tears**, LT tears, **radial shortening after Colles**, **distal radius physeal fracture** in adolescence (TO)

3 Presentation

Symptoms ulnar/dorsal DRUJ pain; ↑ with **ulnar deviation**, **axial load**, rotation (jar opening, wringing towel, door opening, punching, weight bearing on extended wrist); relief with rest

Test	Technique / meaning
Ballottement	Dorsal–palmar ulna displacement in ulnar deviation → pain
Nakamura ulnar stress	Pronated wrist ulnar deviated + axial load, flex/extend → pain
Fovea test	Palpate between styloid + FCU → **TFCC / ulnotriquetral ligament tear**

4 Diagnosis & Classification

Key radiology findings

Required view **PA/AP in neutral rotation** (zero rotation) to measure variance

Pronated grip view **↑ impaction** (dynamic positive variance)

X-ray signs **positive ulnar variance · sclerosis/cysts of lunate + ulnar head**

MRI TFCC tear (changes treatment); T1/T2 lunate signal change

Arthrography TFCC + LT tears

– – Active space – –

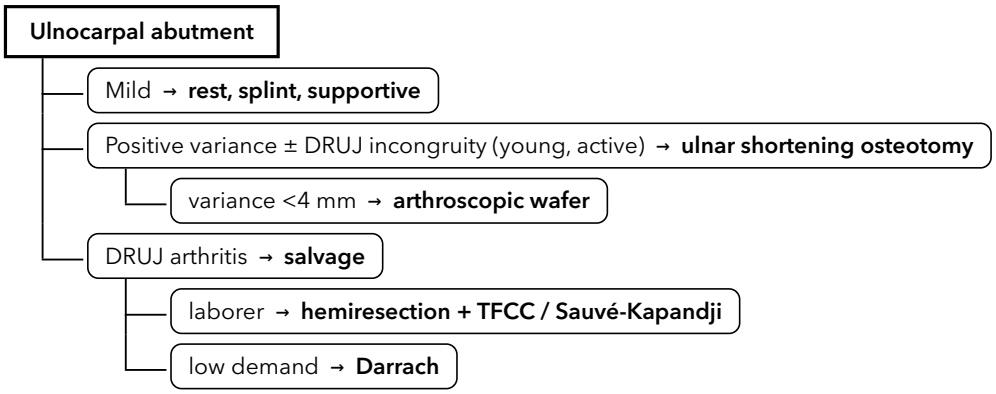


Ulnar impaction

Differential – benign variant

Differentials	ulnar styloid impaction, DRUJ instability/arthritis, TFCC tear, LT tear, pisotriquetral arthritis, ECU tendinitis/instability, **LT coalition**
Lunotriquetral coalition	**uncommon normal variant, no significant sequelae**; incidental; commoner in African descent; persistent pain → arthrodesis ✗ Wrong options: takedown for ROM · scaphoid fracture risk · abnormal TFCC · inflammatory arthritis
Coalition syndromes	arthrogryposis, diastrophic dwarfism, Turner, Ellis-van Creveld

5 Management



Ulnar shortening osteotomy vs wafer – indications & outcomes

Ulnar shortening osteotomy	**treatment of choice** for most positive variance + **DRUJ incongruity** in young adults (TO: MMA student after adolescent DRF → **distal ulnar osteotomy**; ✗ Wrong options: TFCC repair alone · interposition · Darrach)
USO advantages	preserves ulnar dome cartilage + DRUJ; **tightens TFCC + ulnocarpal ligaments**; may improve DRUJ congruity; **carpal cysts can resolve**
USO approach	**subcutaneous border of ulna**; ± arthroscopic TFCC repair/débridement
USO pitfalls	protect **dorsal sensory branch of ulnar n.**; **dorsal plate → ECU tendinitis**
USO contraindications	**DRUJ arthrosis** · **reverse oblique DRUJ inclination** (↑ radioulnar contact) ✗ Wrong options: "results encouraging with DRUJ degeneration"

Wafer arthroscopic removal of **2–4 mm** under TFCC; variance **<4 mm**

– – Active space – –

At 1 y	USO vs wafer
Outcome scores, grip, ROM, pain	**Equal**
Reoperation / complications	**↑ USO** (symptomatic hardware)
Return to work	**Wafer faster**

DRUJ arthritis – hemiresection arthroplasty

Procedure	Indication / key fact
Hemiresection arthroplasty	Post-traumatic DRUJ arthritis; **needs intact or reconstructed TFCC** (prevents ulnar instability); **heavy laborer**
Sauvé-Kapandji	DRUJ fusion + proximal ulnar **pseudarthrosis**; **manual laborers**; preserves ulnocarpal support
Darrach (ulnar head resection)	**Low-demand / elderly**; risk **painful proximal stump instability**
Ulnar head replacement	Severe arthrosis; **salvage failed Darrach**

Laborer with DRUJ arthritis + TFCC tear **hemiresection + TFCC repair/reconstruction**
✗ Wrong options: Darrach · complete head resection · isolated TFCC reconstruction · styloid repair



Sauvé-Kapandji

How it's asked

"MMA puncher, positive variance after adolescent DRF – best treatment?"	→ **distal ulnar (shortening) osteotomy** (TO)
"USO most indicated for?"	→ **young, ulnar positive + DRUJ incongruity** ✗ Wrong options: neutral/negative · elderly with DRUJ arthritis
"USO – all true EXCEPT"	→ **good results despite DRUJ arthritis**
"USO vs wafer at 1 y?"	→ **↑ reoperation with USO** ✗ Wrong options: better ROM/grip · earlier RTW

– – Active space – –

"Construction worker, DRUJ arthritis + TFCC tear?"	→ **hemiresection arthroplasty + TFCC repair**
"Fused lunate–triquetrum on X-ray?"	→ **benign variant, no sequelae**
"Load through ulna with +2 mm variance?"	→ **40%**

WARTENBERG'S SYNDROME

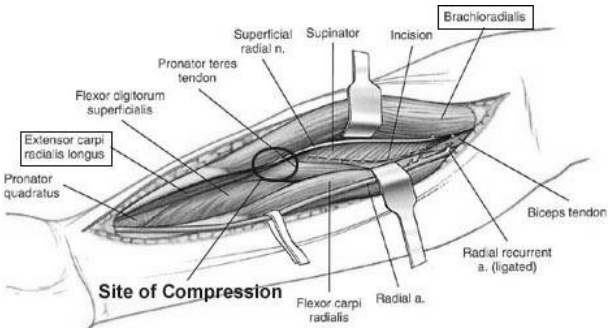
– Hand · 99 / 103 –

1 Basics

Definition	**"cheiralgia paresthetica"** = compressive neuropathy of the **superficial sensory radial nerve (SRN)** (TO); **no motor deficit**
Incidence	rare, **<1/100,000/yr**
Demographics	♂:♀ **1:4**; 20–70 y
Association	**De Quervain's in 20–50%**

Anatomy – superficial radial nerve

Origin	radial nerve divides anterior to lateral epicondyle → **PIN (motor)** + **SRN (sensory)** (TO)
Course	deep to **brachioradialis** in forearm
Emerges subcutaneous	between **brachioradialis & ECRL**, **9 cm** proximal to radial styloid (Orthobullets, TO) · **~8 cm** (OB explanation) – read the options
Dorsal branch	1–3 cm radial to **Lister's tubercle** → 1st & 2nd web spaces
Palmar branch	within 2 cm of 1st dorsal compartment, over EPL → dorsolateral thumb
Overlap	with **LACN** → hypoaesthesia may be absent



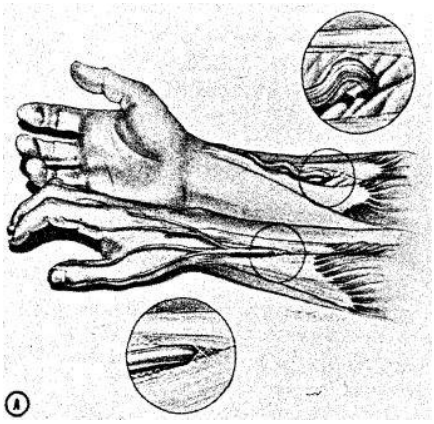
SRN emerging between brachioradialis and ECRL

2 Mechanism

Compression site & sensory-only presentation

Site	**scissoring of brachioradialis + ECRL tendons in forearm PRONATION** (TO) ✗ Wrong options: supination · ECRL/ECRB · APL/EPB
Other	**fascial bands** at subcutaneous exit
Causes	forearm fracture, **handcuffs**, tight **wristwatch**/bracelet/cast; repetitive circumduction (knitting)

– – Active space – –



Scissoring of brachioradialis and ECRL in pronation

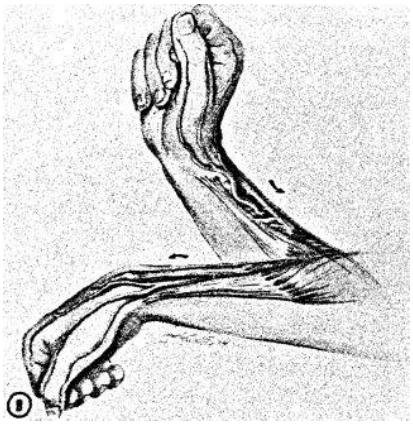
3 Presentation

Symptoms burning, ill-defined pain + paresthesia/numbness **dorsoradial hand**, dorsal 1st web & thumb; dislikes watch

Aggravated by (TO) **wrist flexion + ulnar deviation + pronation**

Motor **none** (TO)

Exam	Fact
Tinel over SRN	**Most common finding**
Wrist flexion + UD + pronation 1 min	Provocative
Finkelstein	↑ symptoms in **96%** (nerve traction), even without 1st compartment tenosynovitis (TO)
Diagnostic wrist block	Temporary relief



Provocative test: flexion, ulnar deviation, pronation

4 Diagnosis & Classification

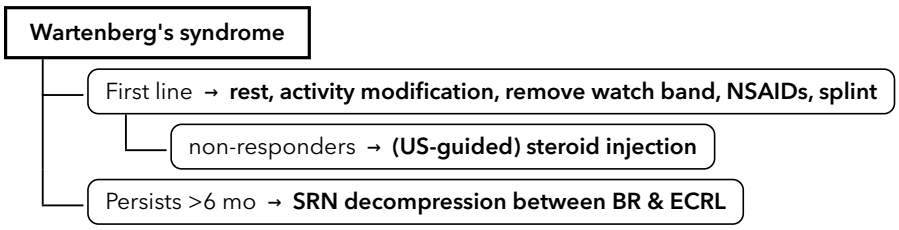
Diagnosis clinical

Key radiology findings

X-ray limited value; old forearm fracture

EMG/NCS limited value; often **normal**

5 Management



Radial sensory nerve decompression

Nonop most improve (TO); **spontaneous resolution common**; steroid injection evidence limited

Surgery **after 6 mo** of failed nonop (Orthobullets); OB MCQ answered decompression after 6 wk failed nonop

Procedure **SRN decompression in BR–ECRL interval**: neurolysis + release of fascia
✗ Wrong options: LACN · ECRL–ECRB · BR–ECRB · APL–EPB

Incision longitudinal, **volar to Tinel** → protects **LACN**, avoids scar tethering over SRN

Outcome **74%** success

6 Complications

Complications failed decompression, persistent pain/numbness, wound dehiscence, infection

X vs Y

	Wartenberg's syndrome	De Quervain's	Intersection syndrome	LACN neuritis
Structure	SRN	1st compartment (APL, EPB)	1st crossing 2nd compartment	LACN
Site	BR–ECRL, ~8–9 cm above styloid	Radial styloid	**4–6 cm proximal to Lister's** (TO)	Lateral forearm
Pronation ↑ pain	**Yes**	**No**	No	–
Sign	Tinel over SRN	Finkelstein	**"Wet leather" crepitus**, swelling	Tinel over LACN (mimics)

Wartenberg's sign (TO) **abducted little finger** in **ulnar nerve palsy**: unopposed radial-innervated EDM/EDC; more with fingers extended
✗ Wrong options: Wartenberg's syndrome

– – Active space – –

How it's asked

"Wartenberg syndrome NOT characterised by?"	→ **compression of motor branch of radial nerve** (TO)
"Compression takes place between?"	→ **brachioradialis & ECRL in pronation** (TO)
"Failed nonop – decompress?"	→ **radial sensory nerve between BR & ECRL**
"Most common exam finding?"	→ **Tinel over SRN**
"Finkelstein positive in?"	→ **96%**
"Incision placed where?"	→ **volar to Tinel** (protect LACN)
"Abducted little finger in ulnar palsy?"	→ **Wartenberg's sign** (TO)

WRIST ARTHRITIS

– Hand · 100 / 103 –

1 Basics

Patterns by location

Pattern	Rank / key fact
SLAC	**Most common**
STT arthrosis	**2nd most common**
SNAC	Scaphoid nonunion
DRUJ arthrosis	Post-traumatic, RA
Pisotriquetral arthrosis	Ulnar-sided pain

Causes primary OA · **post-traumatic → SLAC/SNAC/DRUJ** · RA · congenital (**Madelung**) · idiopathic (**Kienböck, Preiser**)

2 Mechanism

SLAC SL tear → **palmar rotary subluxation of scaphoid** → radiocarpal OA → **capitolunate** OA; **radiolunate spared**

SNAC proximal pole stays with lunate, distal scaphoid flexes → early **radial styloid–distal scaphoid** OA; radiolunate spared

RA wrist **supinated, palmarly dislocated, radially deviated, ulnarly translocated**; early DRUJ loss → **caput ulnae** (dorsal ulna)

Golf – trailing-wrist dorsal rim impaction

Wrist (TO)	Motion	Injuries
Trailing	Extends in backswing, flexes in follow-through	**Dorsal rim impaction syndrome**
Leading (more vulnerable)	Radial → ulnar deviation	Hamate stress injury, **hook of hamate #**, **ECU synovitis/subluxation**, TFCC

Dorsal rim impaction (TO) repetitive **axial load in dorsiflexion** → pinpoint radiocarpal pain on **loaded extension**; MRI: synovitis ± loose bodies; Rx training modification, strengthening, brace limiting dorsiflexion → arthroscopy if fails
✗ Wrong options: passive

4 Diagnosis & Classification

Key radiology findings

Standard hand series + joint-specific views

– – Active space – –

Pisotriquetral **supinated oblique, 30° supination** (profile of PT joint); **Garraud view** = 30° supination + slight extension

View	Used for
30° supinated oblique	**Pisotriquetral arthritis**
Clenched-fist PA	**SL widening**
Carpal tunnel view	**Hook of hamate #**
Standard AP / lateral	not preferred for PT joint



Pisotriquetral arthritis

5 Management

First line **NSAIDs, bracing, intra-articular steroid**

Radioscapholunate fusion – distal scaphoid & triquetrum excision

Post-traumatic radiocarpal OA (after DRF) **RSL fusion + excise triquetrum + distal pole scaphoid** → frees midcarpal joint → ↑ radial deviation + F/E arc; alternative to total wrist fusion

(✗ Wrong options: **Traps** – AIN neurectomy (volar; fusion is dorsal)) · capitolunate interposition (capitolunate preserved) · dorsal intercarpal ligament section · ulnar shortening (unnecessary once triquetrum excised)

Ulnar-sided wrist arthritis

Pisotriquetral refractory → **pisiform excision**

DRUJ arthrosis / abutment	Note
Darrach	Distal ulna resection
Sauvé-Kapandji	–
Partial resection + interposition	–
Ulnar head replacement	Primary or **salvage of failed Darrach**; early results promising



Ulnar head replacement

X vs Y

	SLAC	SNAC
Cause	SL ligament tear	Scaphoid nonunion
First joint	Radiocarpal → capitolunate	Radial styloid–**distal** scaphoid
Radiolunate	**Spared**	**Spared**

How it's asked

"Add to RSL fusion for post-DRF arthrosis?"	→ **excise triquetrum + distal pole scaphoid**
"Best image for pisotriquetral arthritis?"	→ **30° supinated oblique view** ✗ Wrong options: lateral · clenched fist · AP · carpal tunnel view
"Trailing-wrist injury in golf?"	→ **dorsal rim impaction syndrome** (TO) ✗ Wrong options: ECU synovitis · hook of hamate · scaphoid #
"Joint spared in SLAC/SNAC?"	→ **radiolunate**

– – Active space – –

WRIST ARTHROSCOPY

– Hand · 101 / 103 –

1 Basics

Indications

Diagnostic + therapeutic	**TFCC** · **SL / LT interosseous ligaments** · reduction aid (**distal radius, scaphoid**, perilunate (TO)) · **ulnocarpal impaction** / distal ulnar resection · chondral debridement · loose bodies · **synovectomy** · **dorsal ganglion** · SNAC/SLAC · **septic wrist I&D** · unexplained mechanical pain

Set-up

Position	supine, **elbow 90°**, traction tower **10 lb** to fingers
Landmarks	Lister's tubercle, scaphoid, lunate, DRUJ, ECU
Scope	**2.7 mm, 30°**
Portal creation	**skin incision only → blunt (haemostat) dissection**

Portals & nerves at risk

Naming by adjacent **extensor compartments**; **≥11 dorsal**: 5 radiocarpal, 4 midcarpal, 2 DRUJ (TO: 9 radiocarpal/intercarpal + 2 DRUJ) + volar portal radial to FCR at proximal crease (TO)

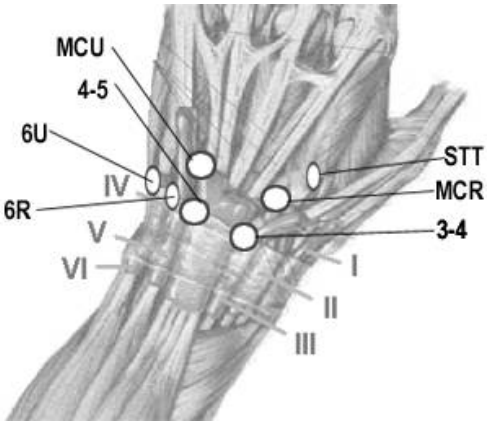
Radiocarpal portal	Location / use	At risk
3-4	**1 cm distal to Lister's**, EPL–EDC; **first portal, primary viewing ("workhorse")**	EPL, EDC; distal **PIN** (TO)
4-5	Ring MC line, EDC–EDM; instrumentation, **TFCC view**	EDC, EDM; distal PIN (TO)
6R	**Radial to ECU**; TFCC, UL/UT/LT ligaments, SLIL (TO)	**Dorsal sensory branch ulnar n.** (**~8 mm**)
6U	**Ulnar to ECU**; ulnar TFCC repair, irrigation	**Dorsal sensory branch ulnar n.** (± ulnar n./a.)
1-2	**APL–ECRB**, dorsal snuffbox; radial styloid, inflow, styloidectomy	**SBRN + radial artery**

– – Active space – –

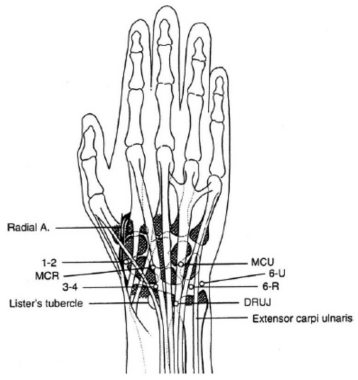
Midcarpal / CMC portal	Location / use	At risk
MCR	1 cm distal to 3-4, radial border middle MC, ECRB–EDC; **SL, SC, STT** joints	ECRB, EDC
MCU	1 cm distal to 4-5, ring MC, EDC–EDM; **LC, LT, TH** joints	EDC, EDM
STT	Index MC axis, ulnar to EPL	ECRB, ECRL
TH (TO)	Ulnar, distal to triquetrum; inflow/outflow	–
1U / 1R	Ulnar / radial to **EPB** at thumb CMC; CMC DJD debridement	**SBRN** (portals straddle 1st branch); dorsal radial a.

Midcarpal portals needed for full carpal view & **instability** assessment

SBRN distance from 3-4 – conflict **16 mm** (Orthobullets/OB) · **9 mm** closest branch (Kiliç cadaver study, OB) → superficial incision + blunt dissection



Wrist portals



Portal map and nerves

Arthroscopic anatomy – visible structures

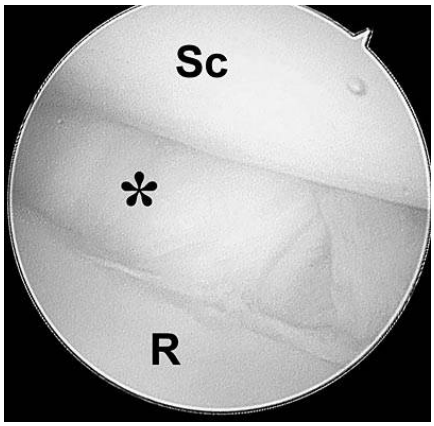
Seen (radiocarpal) **SL & LT ligaments** · **superficial TFCC** over ulnar head · **radioscaphocapitate** (discrete volar capsular band)

Not seen **ECU tendon** (extra-articular) · **dorsal intercarpal lig.** (dorsal)

RSC ligament radial styloid → scaphoid waist → proximal radial capitate; **prevents ulnar translocation**; seen from 3-4 looking volar-radial

Palmar oblique ligament 1st–2nd MC bases
✗ Wrong options: wrist

– – Active space – –



RSC from 3-4 portal

6 Complications

Overall	**1–2%**
Most common overall (TO)	**failure to achieve the procedure**
Most common set-up (TO)	finger traction → oedema, transient digital nerve neurapraxia
Nerves	**dorsal sensory ulnar** (6R/6U, esp. 6U for TFCC sutures → small open incision before knot tying) · **SBRN** (1-2, 1U, 1R) · **distal PIN** (3-4, 4-5; less mobile, 4th compartment floor) ✗ Wrong options: **motor branch of ulnar n.** (TO)
Dorsoradial wrist/ thumb hyperaesthesia after scope	**1-2 portal** (SBRN)
Radial artery	1-2 portal (styloidectomy); dorsal branch with CMC portals
Tendon injury – conflict	**EDC** (index & small) most common (TO) · **EPL & EDM** from improper portal placement (Orthobullets); late **EPL rupture**, ECRL tenosynovitis (TO)
Other	chondral injury, portal infection, stiffness, **MCPJ pain (over-distraction)**

Misfiled – kinematic TKA balancing

Kinematic alignment (TO) restores **pre-arthritic** alignment & joint lines; **ligament-sparing** (bone cuts, no releases); femur positioned first, tibia adjusted

KA imbalance (TO)	Correction
Tight extension & flexion	Remove more tibia
Tight extension, stable flexion	Remove posterior osteophytes, strip posterior capsule, ↓ posterior slope
Tight flexion, stable extension	↑ posterior tibial slope
Extension: tight medial, loose lateral	**Remove medial osteophytes · recut tibia 2° more varus · thicker liner** (✗ Wrong options: **medial release**)
Extension: tight lateral, loose medial	Remove lateral osteophytes · recut 2° more valgus · thicker liner
Fixed valgus	Release arcuate complex + popliteus · recut 2° more varus

How it's asked

"Most commonly injured tendon"	→ **EDC** (TO)
"Nerve injury NOT common"	→ **motor branch of ulnar nerve** (TO)
"Dorsoradial wrist/thumb hyperaesthesia after scope – portal?"	→ **1-2** ✗ Wrong options: 6U, 6R, MCU, 4-5
"Initial portal"	→ **3-4, 1 cm distal to Lister's**
"Incorrect portal–nerve pair"	→ **4-5 ; dorsal sensory ulnar** ✗ Wrong options: correct: 1U/1-2 SBRN, 6U/6R DSBUN
"Structure seen from 3-4 looking volar-radial"	→ **radioscaphocapitate ligament**
"Cannot be seen at radiocarpal arthroscopy"	→ **extensor carpi ulnaris tendon**
"KA TKA tight medial / loose lateral in extension – NOT recommended"	→ **medial release** (TO)

– – Active space – –

WRIST LIGAMENTS & BIOMECHANICS

– Hand · 102 / 103 –

1 Basics

Wrist motion

Joints radiocarpal + intercarpal; axes: flexion–extension, radial–ulnar deviation, prono-supination (TO)

Motion	Normal	Functional	Contribution
Flexion	65°	10°	**40% radiocarpal / 60% midcarpal**
Extension	55°	35°	**66% radiocarpal / 33% midcarpal**
Radial deviation	15°	10°	**90% midcarpal**
Ulnar deviation	35°	15°	50 / 50

Carpal rows (TO)

PCR (TO) scaphoid, lunate, triquetrum, pisiform; mobile between bones, moves as a row; **intercalated segment**
✗ Wrong options: tendon attachments

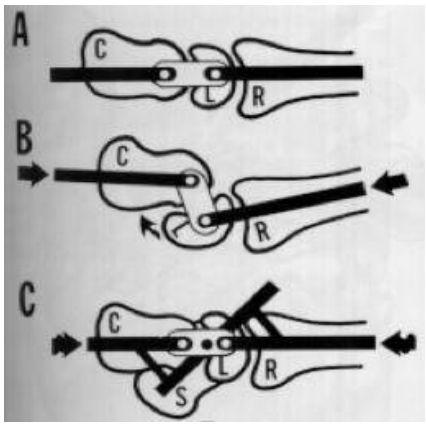
DCR (TO) trapezium, trapezoid, capitate, hamate; tightly bound → **single functional unit**

Lunate–capitate (TO) **no ligamentous connection**

Biomechanic concepts

Concept	Key points
Link	Radius–lunate–capitate chain; **capitate head = centre of rotation**; ✓ efficient, strong volar ligaments; (✗ Wrong options: more links → instability)
Column	**Lateral (mobile)**: scaphoid, trapezium, trapezoid; **central (FE)**: lunate, capitate, hamate (lunocapitate = centre); **medial (rotation)**: triquetrum + distal row
Row	Proximal + distal rows; **scaphoid bridges** rows
Others (TO)	Row-column theory, **ring theory**

Scaphoid resting force / **radial deviation → scaphoid flexes**, triquetrum extends; **ulnar deviation → scaphoid extends**



Link concept

Carpal relationships

Ratio	Normal
Carpal height / MC3 length (collapse)	**0.54**
Ulna–capitate / MC3 (ulnar translation)	**0.30**

Load **radius 80% / ulna 20%**; ulnar **lengthening ↑** ulnar load, **shortening ↓**

Lunate alignment (PA) ≥ **2/3 of lunate** over radius (lunate fossa)

Ligament overview

Extrinsic bridge carpus to radius / metacarpals; volar + dorsal

Intrinsic carpal-to-carpal; most important **SLIL + LTIL**

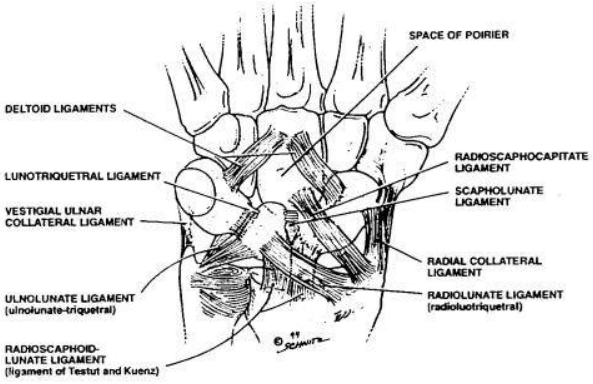
Extrinsic vs intrinsic failure (TO) extrinsic: **stiffer, lower yield strength → rupture midsubstance** · intrinsic: **avulse** (not rupture)

No dorsal ulnocarpal ligaments (TO) extrinsics mostly volar · **ulnocapitate arises from ulnar styloid** · **scaphocapitate + triquetrocapitate cross the midcarpal joint**

Distal row instability (TO) occurs **only as axial / longitudinal** instability (distal row = fixed unit)

Volar vs dorsal **volar stronger**; volar = secondary stabilisers of scaphoid; **dorsal converge on triquetrum**

Space of Poirier weak area, **floor of carpal tunnel at proximal capitate**, between **RSC and long radiolunate**; ↑ in dorsiflexion, disappears in palmar flexion; perilunate: distal row separates from lunate · **lunate dislocation: lunate escapes into it**



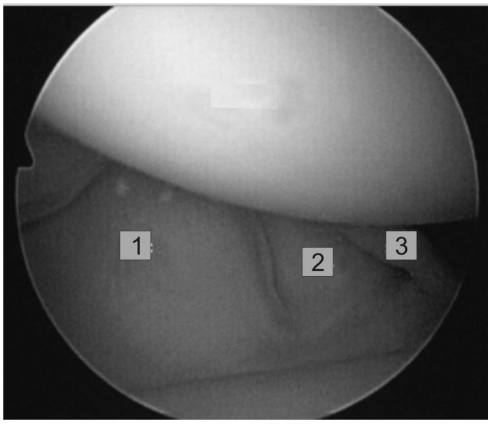
Space of Poirier

– – Active space – –

Extrinsic ligaments – radioscaphocapitate & radiocarpal dislocation

Ligament	Key fact
Radial collateral	Volar radiocarpal
Radioscaphocapitate (RSC)	Radial styloid → capitate; **sling for scaphoid waist**; **main restraint to ulnar translation**; **preserve in PRC** (primary stabiliser, prevents ulnar drift); at risk with large radial styloidectomy
Long radiolunate (radiolunotriquetral, volar RL)	Counteracts ulnar-distal lunate translocation; **abnormal in Madelung = Vickers ligament**
Radioscapholunate (Testut & Kuentz)	**Neurovascular conduit only**; no mechanical strength; mesocapsule with **AIN branches + radial arcade** rami (TO); extrinsic (TO)
Short radiolunate	Stabilises lunate; from **volar-ulnar "critical corner" → primary restraint to volar carpal translation**; disrupted in stage IV perilunate
Volar ulnocarpal	Ulnotriquetral, ulnolunate, ulnocapitate
Dorsal radiotriquetral (DRC)	Must be torn **with LTIL** for **VISI**
Dorsal intercarpal (DIC)	With DRC forms lateral **"V"** → secondary stabiliser vs **DISI**
Dorsal radiolunate, radioscaphoid	Dorsal extrinsic

Arthroscopy from 3-4 portal (EPL–EDC) volar extrinsics **radial → ulnar: RSC, long RL, short RL**; RSL (Testut) just radial to SLIL



Arthroscopic volar extrinsic ligaments 1–3 radial to ulnar

Radiocarpal dislocation rare, high-energy; **dorsal > volar**; ORIF essential

Ligament not repaired	Late deformity
RSC	**Ulnar translocation** of carpus (multidirectional instability)
Short RL (critical corner)	**Volar subluxation** of carpus
SLIL	DISI (Terry Thomas sign)

Surgical steps (Ilyas) provisional reduction → NV decompression → joint debridement → intercarpal injuries → fracture fixation / soft-tissue repair

SL injury suspicion radial styloid fracture exiting at **scaphoid–lunate fossa junction**

Classifications **Moneim** (intercarpal dissociation; type II & open → worse) · **Dumontier** (radial styloid involvement)

RSC reconstruction **brachioradialis** tendon

– – Active space – –

Intrinsic & extrinsic ligament function (LT, RSL)

SLIL part	Role
Dorsal	**Thickest, strongest**; prevents **translation**; primary SL stabiliser
Volar	Prevents **rotation**
Proximal	No significant strength

LTIL part	Role
Volar	**Strongest**
Dorsal / proximal	Weaker

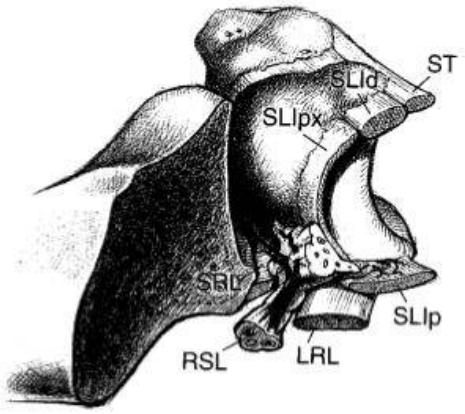
SLIL torn scaphoid flexes, lunate extends → **DISI**

LTIL torn (+ dorsal radiotriquetral) lunate flexes with normal scaphoid → **VISI**; **volar LTIL** = most important contributor (TO)

Isolated DRC or RSC injury (TO) no VISI

Distal row trapeziotrapezoid, trapeziocapitate, capitohamate

Palmar midcarpal STT, scaphocapitate, triquetrocapitate, triquetrohamate



SLIL components

Six features of carpal instability (TO)

Feature	Categories
Chronicity	**Acute 0–1 wk · subacute 1–6 wk · chronic > 6 wk**
Aetiology	Traumatic · inflammatory
Location	Radiocarpal · proximal row · midcarpal · distal intercarpal
Severity	**Predynamic** (pain, no malalignment) · **dynamic** (malalignment on stress views) · **static** (malalignment on plain X-ray)
Direction	DISI · VISI · dorsal · volar · radial / ulnar translation
Pattern	**CID** (SL, LT, scaphoid fracture, Kienböck) · **CIND** (radiocarpal, midcarpal) · **CIC** (perilunate, axial dislocations)

– – Active space – –

View (TO)	Purpose
PA neutral + lateral	Neutral confirmed by **clear 3rd CMC joint space**
Clenched fist	Load → SL gap (Terry Thomas)
PA with **10° angulation from ulna**	Mid-SL interval
Radial / ulnar deviation PA	Scaphoid flexion / extension
Flexion / extension lateral	Lunate tilt
Stress (volar / dorsal drawer)	**Midcarpal instability**

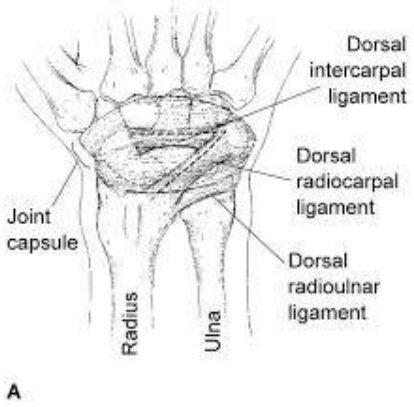
Secondary stabilisers against DISI – dorsal intercarpal ligament

SL instability — **most common carpal instability**; FOOSH

Secondary SL stabilisers — **DIC + DRC** (lateral V) → missed injury → persistent DISI → **SLAC**

DIC — assess and repair/reconstruct with SLIL surgery (Mitsuyasu)

SLIL cut alone — <2 mm translation; need SLIL + volar or dorsal ligaments for significant translation



Dorsal intercarpal and dorsal radiocarpal ligaments

X vs Y

	DISI	VISI
Intrinsic	**SLIL (dorsal)**	**LTIL (volar)**
Extrinsic needed	DIC / DRC secondary	**Dorsal radiotriquetral**
Lunate	Extended	Flexed
Test	Watson scaphoid shift	Ballottement / shuck

Ligament	Classic MCQ association
RSC	PRC preservation · ulnar translocation
Long RL	Madelung (Vickers)
Short RL	Critical corner · volar subluxation · stage IV perilunate
RSL (Testut)	Neurovascular conduit
Dorsal radiotriquetral	VISI with LTIL
DIC	Secondary DISI stabiliser

How it's asked

"Volar extrinsic disrupted → ulnar translation after radiocarpal dislocation?"	→ **RSC** ✗ Wrong options: short RL, SLIL, radiotriquetral, RSL
"Arthroscopic structures 1, 2, 3 from 3-4 portal?"	→ **RSC, long RL, short RL**
"RSC repaired — without surgery what deformity?"	→ **ulnar translocation of carpus** (Figure D) ✗ Wrong options: volar subluxation = short RL · SLIL · perilunate · Madelung
"PRC — preserve which ligament?"	→ **RSC**
"Secondary stabiliser preventing DISI?"	→ **dorsal intercarpal ligament**
"VISI, positive ballottement — ligament?"	→ **volar lunotriquetral interosseous** (TO)
"RSL all except?"	→ **provides mechanical strength** — false (TO)
"Ligament abnormal in Madelung?"	→ **long radiolunate (Vickers)**
"Lunate escapes into?"	→ **space of Poirier**
"Radial deviation is mostly?"	→ **midcarpal (90%)**
"Extrinsic vs intrinsic wrist ligaments — mode of failure?"	→ **extrinsic rupture midsubstance; intrinsic avulse** (TO)
"Subacute carpal instability — time window?"	→ **1–6 weeks** (TO)

– – Active space – –

WRIST TRAUMA RADIOGRAPHIC EVALUATION

– Hand · 103 / 103 –

1 Basics

Carpal ossification sequence

Order counterclockwise (right hand) from capitate: **capitate → hamate → triquetrum → lunate → scaphoid → trapezium → trapezoid → pisiform**

First **capitate** (largest carpal bone), **1–3 months**, often with hamate

Last **pisiform** (only **sesamoid** carpal bone)

Bone	Ossifies (Orthobullets table)	OB explanation	Fusion
Capitate	**1 y**	**1–3 mo**	14–16 y
Hamate	2 y	shortly after capitate	14–16 y
Triquetrum	3 y	–	14–16 y
Lunate (Kienböck)	4 y	~5 y	14–16 y
Scaphoid (Preiser)	5 y	~6 y	14–16 y
Trapezium (thumb MC base)	5–6 y	~6 y	14–16 y
Trapezoid	6–7 y	–	14–16 y
Pisiform	**9–10 y**	~12 y	14–16 y

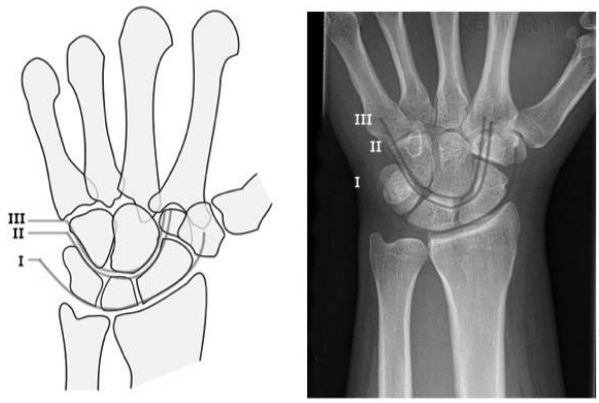
Conflict ages differ between sources (above) – read the options; order is the same



Carpal ossification order

Gilula's arcs

Arc	Line
Proximal	Proximal scaphoid + lunate + triquetrum
Middle	Distal scaphoid + lunate + triquetrum
Distal	Proximal capitate + hamate



Gilula's carpal arcs

4 Diagnosis & Classification

Amsterdam wrist rules (X-ray criteria)

X-ray if any	Suspected distal radius #	Suspected carpal #
↑ age, swelling, deformity, TTP distal radius	Pain on **palmarflexion**, **supination**, painful **radioulnar ballottement**; pain on ulnar deviation = ↓ risk	**Male**, **snuffbox swelling**, pain on **radial deviation**; painful thumb axial compression = ↓ risk

– – Active space – –

Key radiology findings – views

View	Position / beam	Best for
AP / PA	Shoulder abd 90°, elbow 90°; supinated (AP) / pronated (PA)	**Carpal joints: AP > PA**; DRF
Lateral	**Shoulder adducted**, elbow 90°, forearm neutral; 10° cephalad	Alignment, carpal instability; adequate = pisiform volar cortex within central 1/3 of volar scaphoid
IR oblique (pronated 45°)		**Radial carpus**; **best for scaphoid waist + distal pole**
ER oblique (supinated 45°)		**Ulnar carpus**; **pisiform, hamate body**
Ulnar deviation	Aim at scaphoid	Scaphoid #, lateral wrist; scaphoid/lunate **extend**
Radial deviation		Medial wrist, palmar triquetral #; scaphoid/lunate **flex**
Carpal tunnel	Wrist extended; 2.5 cm distal to MC3 base, 25–30° cephalad	**Hook of hamate (best)**, pisiform, trapezium
Carpal bridge	4 cm proximal to wrist bend, 45° caudad	Pisiform, trapezium, hamate hook
Clenched fist	Supinated, ulnar dev, fist (capitate driven proximally); bilateral	**SL injury**
Scaphoid view	**30° extension + 20° ulnar deviation**	Scaphoid #
Clements-Nakayama	Pronated 45°, ulnar dev; at snuffbox (20° cephalad if can't deviate)	**Trapezium #**

Why adduct shoulder for lateral ulna does not rotate with pronation/supination

Key radiology findings – signs

Sign	View	Meaning
Pronator quadratus fat pad	Lateral	Wrist effusion (bowed PQ fascia)
Scaphoid fat pad	Ulnar deviation	Effusion / scaphoid #
Piece-of-pie (triangular lunate)	PA	Lunate dislocation
Spilled teacup	Lateral	**Stage IV**: lunate volar to radius
Pooping duck	Lateral / IR oblique	**Dorsal triquetral cortical #**
Absent "eye" (cortical ring) + sclerosis	PA	**Hook of hamate #**

Wrist MRI technique & normal variants (TO)

Positioning (TO) prone **"superman"** position, hand overhead at magnet isocentre, **dedicated wrist coil** (head coil substitute; never body coil) · small FOV, high resolution

Planning (TO) axials ⊥ forearm · **coronals along the RADIUS axis** · sagittals ⊥ coronals (most pulsation artefact) · slices **3 mm**, 0.3–0.5 mm gap

Sequence (TO)	Use
Coronal PDFS	**Single most useful**: oedema, fluid, **TFCC + SL + LT in one image**
Coronal GRE	Trabeculae, cartilage
Coronal T1	Marrow fat (AVN, sclerosis), chip fractures hidden by oedema
Axial T1/T2 + FS	Tendons, median + ulnar nn., carpal tunnel
Sagittal FS	Axes: radius, lunate, capitate, MC3 collinear
T2 mapping (3 T)	Early cartilage dehydration (early lunatomalacia)

Normal, not pathology (TO) **heterogeneous signal at ligamentum subcruentum** · **multiple APL slips** · thin radial cartilage beside TFCC disc · fovea subosseous cysts · striated SLIL

Measure ulnar variance on X-ray, not MRI (TO) changes slice to slice · SL angle easier on lateral X-ray

Radial styloid (TO) high inclination / elongation (post-Colles) → **radioscaphoid arthritis** · **carpal boss** = bony hypertrophy at 3rd CMC (boxers)

Scaphoid fracture

Location	%
Waist	**65**
Proximal 1/3	25
Distal 1/3	10 (**commonest in children**)

PA proximal pole; waist/distal pole distorted (scaphoid flexed)

Negative X-ray + high suspicion thumb spica, **repeat X-ray 14–21 d** (re-evaluate 12–21 d)

ORIF criteria (instability)	Threshold
Displacement	**>1 mm** any view
Humpback	**15°**
SL angle	**>60°**
Radiolunate angle (DISI)	**>15°**
Intrascaphoid angle	**>35°**
Other	Proximal pole, perilunate, comminuted, vertical/oblique unstable, non-displaced waist (listed)

Lunate / perilunate dislocation

PA **break in Gilula's arc**, lunate–capitate overlap, **piece-of-pie**

– – Active space – –

Lateral **loss of radius–lunate–capitate colinearity**; stage III capitate dorsal to lunate; **stage IV spilled teacup**; SL angle **>70°**

Timing	Treatment
Acute **<8 wk**	Emergent closed reduction/splint → **open reduction, ligament repair, fixation ± CTR** (✗ Wrong options: nonop)
Chronic **>8 wk**	**PRC**
Chronic + degenerative	**Total wrist arthrodesis**



Piece-of-pie sign: triangular lunate



Spilled teacup sign

Other carpal fractures

Fracture	Best view	Nonop	Surgery
Triquetrum	Lateral / IR oblique (dorsal); radial dev (palmar)	Stable dorsal/palmar cortical, undisplaced body	ORIF if unstable / displaced body
Hook of hamate	**Carpal tunnel**	Most	**Excise chronic nonunion**
Hamate body	**ER oblique**	Extra-articular undisplaced (rare)	ORIF displaced / intra-articular
Pisiform	**ER oblique or carpal tunnel**	Most	**Pisiformectomy**: displaced symptomatic, painful nonunion
Trapezium	Clements-Nakayama, carpal tunnel		

Normal variants (accessory ossicles)

Ossicle	Location
Lunula	In TFCC between ulnar styloid + triquetrum
Os triangulare	Ulnar styloid + lunate + triquetrum
Os hamuli proprium	Unfused hook of hamate ossification centre (mimics #)
Os centrale carpi	Dorsal, scaphoid + trapezoid + capitate
Os epilunate	Dorsal lunocapitate joint
Os styloideum	Between MC2 & MC3 bases
Os paratrapezium	Radius side / base of MC1
Os radiostyloideum	Near radial styloid, lateral to mid-scaphoid

How it's asked

"Single most useful wrist MRI sequence?"	→ **coronal proton-density fat-suppressed** (TO)
"Ossification centre appears at 1–3 months?"	→ **largest carpal bone = capitate** ✗ Wrong options: scaphoid/Preiser, lunate/Kienböck, trapezium, pisiform
"Last carpal bone to ossify?"	→ **pisiform**
"Best view for hook of hamate?"	→ **carpal tunnel view**
"Best view for scaphoid waist + distal pole?"	→ **IR (pronated) oblique**
"Scaphoid view position?"	→ **30° extension + 20° ulnar deviation**
"Suspected scaphoid #, normal X-ray?"	→ **thumb spica, repeat X-ray 14–21 d**
"Stage IV perilunate lateral sign?"	→ **spilled teacup**
"Pooping duck sign?"	→ **dorsal triquetral cortical fracture**